

International Series in  
Operations Research & Management Science

David L. Olson  
Özgür M. Araz

# Data Mining and Analytics in Healthcare Management

Applications and Tools



 Springer

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# Preface

Healthcare management involves many decisions. Physicians typically have a variety of potential treatments to select from. There are also important decisions involved in resource planning, such as identifying hospital capacity and hiring requirements. Healthcare decisions discussed in the book include:

- Selection of patient treatment
- Design of facility capacity
- Utilization of resources
- Personnel management
- Analysis of hospital stay
- Annual disease progression
- Comorbidity patterns
- Patient survival rates
- Disease-specific analysis
- Overbooking analysis

The book begins with a discussion of healthcare issues, discussed knowledge management, and then focused on analytic tools useful in healthcare management.

Lincoln, NE

David L. Olson  
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# Contents

- 1 Urgency in Healthcare Data Analytics . . . . . 1**
  - 1.1 Big Data in Healthcare . . . . . 1
  - 1.2 Big Data Analytics . . . . . 3
  - 1.3 Tools . . . . . 3
  - 1.4 Implementation . . . . . 4
  - 1.5 Challenges . . . . . 5
  - References . . . . . 6
- 2 Analytics and Knowledge Management in Healthcare . . . . . 7**
  - 2.1 Healthcare Data Analytics . . . . . 8
  - 2.2 Application Fields . . . . . 8
    - 2.2.1 Disaster Management . . . . . 9
    - 2.2.2 Public Health Risk Management . . . . . 10
    - 2.2.3 Food Safety . . . . . 10
    - 2.2.4 Social Welfare . . . . . 11
  - 2.3 Analytics Techniques . . . . . 11
  - 2.4 Analytics Strategies . . . . . 12
    - 2.4.1 Information Systems Management . . . . . 12
    - 2.4.2 Knowledge Management . . . . . 12
    - 2.4.3 Blockchain Technology and Big Personal Healthcare Data . . . . . 13
  - 2.5 Example Knowledge Management System . . . . . 13
  - 2.6 Discussion and Conclusions . . . . . 15
  - References . . . . . 18
- 3 Visualization . . . . . 21**
  - 3.1 Datasets . . . . . 21
    - 3.1.1 Healthcare Hospital Stay Data . . . . . 21
    - 3.1.2 Patient Survival Data . . . . . 24

|          |                                                     |           |
|----------|-----------------------------------------------------|-----------|
| 3.1.3    | Hungarian Chickenpox Data . . . . .                 | 30        |
| 3.2      | Conclusions . . . . .                               | 33        |
|          | References . . . . .                                | 34        |
| <b>4</b> | <b>Association Rules . . . . .</b>                  | <b>35</b> |
| 4.1      | The Apriori Algorithm . . . . .                     | 36        |
| 4.1.1    | Association Rules from Software . . . . .           | 37        |
| 4.1.2    | Non-negative Matrix Factorization . . . . .         | 38        |
| 4.2      | Methodology . . . . .                               | 38        |
| 4.2.1    | Demonstration with Kaggle Data . . . . .            | 38        |
| 4.2.2    | Analysis with Excel . . . . .                       | 42        |
| 4.3      | Review of Applications . . . . .                    | 44        |
| 4.3.1    | Korean Healthcare Study . . . . .                   | 44        |
| 4.3.2    | Belgian Comorbidity Study . . . . .                 | 49        |
| 4.4      | Conclusion . . . . .                                | 51        |
|          | References . . . . .                                | 51        |
| <b>5</b> | <b>Cluster Analysis . . . . .</b>                   | <b>53</b> |
| 5.1      | Distance Metrics . . . . .                          | 53        |
| 5.2      | Clustering Algorithms . . . . .                     | 54        |
| 5.2.1    | Demonstration Data . . . . .                        | 55        |
| 5.2.2    | $K$ -means . . . . .                                | 57        |
| 5.2.3    | EWKM . . . . .                                      | 59        |
| 5.3      | Case Discussions . . . . .                          | 60        |
| 5.3.1    | Mental Healthcare . . . . .                         | 60        |
| 5.3.2    | Nursing Home Service Quality . . . . .              | 62        |
| 5.3.3    | Classification of Diabetes Mellitus Cases . . . . . | 63        |
| 5.4      | Conclusion . . . . .                                | 66        |
|          | References . . . . .                                | 68        |
| <b>6</b> | <b>Time Series Forecasting . . . . .</b>            | <b>69</b> |
| 6.1      | Time Series Forecasting Example . . . . .           | 69        |
| 6.2      | Classes of Forecasting Techniques . . . . .         | 70        |
| 6.3      | Time Series Forecasts . . . . .                     | 70        |
| 6.4      | Forecasting Models . . . . .                        | 71        |
| 6.4.1    | Regression Models . . . . .                         | 72        |
| 6.4.2    | Coincident Observations . . . . .                   | 72        |
| 6.4.3    | Time . . . . .                                      | 73        |
| 6.4.4    | Lags . . . . .                                      | 73        |
| 6.4.5    | Nonlinear Data . . . . .                            | 73        |
| 6.4.6    | Cycles . . . . .                                    | 74        |
| 6.5      | OLS Regression . . . . .                            | 74        |
| 6.6      | Tests of Regression Models . . . . .                | 75        |
| 6.6.1    | Sum of Squared Residuals (SSR) . . . . .            | 75        |

|          |                                                                        |            |
|----------|------------------------------------------------------------------------|------------|
| 6.7      | Causal Models . . . . .                                                | 76         |
| 6.7.1    | Multicollinearity . . . . .                                            | 77         |
| 6.7.2    | Test for Multicollinearity . . . . .                                   | 77         |
| 6.8      | Regression Model Assumptions . . . . .                                 | 78         |
| 6.8.1    | Autocorrelation . . . . .                                              | 78         |
| 6.8.2    | Heteroskedasticity . . . . .                                           | 79         |
| 6.9      | Box-Jenkins Models . . . . .                                           | 81         |
| 6.10     | Conclusions . . . . .                                                  | 85         |
|          | References . . . . .                                                   | 85         |
| <b>7</b> | <b>Classification Models . . . . .</b>                                 | <b>87</b>  |
| 7.1      | Basic Classification Models . . . . .                                  | 87         |
| 7.1.1    | Regression . . . . .                                                   | 87         |
| 7.1.2    | Decision Trees . . . . .                                               | 88         |
| 7.1.3    | Random Forest . . . . .                                                | 88         |
| 7.1.4    | Extreme Boosting . . . . .                                             | 88         |
| 7.1.5    | Logistic Regression . . . . .                                          | 88         |
| 7.1.6    | Support Vector Machines . . . . .                                      | 89         |
| 7.1.7    | Neural Networks . . . . .                                              | 89         |
| 7.2      | Watson Healthcare Data . . . . .                                       | 90         |
| 7.2.1    | Initial Decision Tree . . . . .                                        | 92         |
| 7.2.2    | Variable Selection . . . . .                                           | 92         |
| 7.2.3    | Nurse Data . . . . .                                                   | 96         |
| 7.3      | Example Case . . . . .                                                 | 97         |
| 7.4      | Summary . . . . .                                                      | 100        |
|          | Reference . . . . .                                                    | 104        |
| <b>8</b> | <b>Applications of Predictive Data Mining in Healthcare . . . . .</b>  | <b>105</b> |
| 8.1      | Healthcare Data Sources . . . . .                                      | 105        |
| 8.2      | Example Predictive Model . . . . .                                     | 107        |
| 8.3      | Applications . . . . .                                                 | 108        |
| 8.3.1    | General Hospital System Management . . . . .                           | 108        |
| 8.3.2    | Disease-specific Applications . . . . .                                | 108        |
| 8.3.3    | Genome Research . . . . .                                              | 111        |
| 8.3.4    | Internet of Things Connectivity . . . . .                              | 111        |
| 8.3.5    | Fraud Detection . . . . .                                              | 112        |
| 8.4      | Comparison of Models . . . . .                                         | 112        |
| 8.5      | Ethics . . . . .                                                       | 114        |
| 8.6      | Summation . . . . .                                                    | 115        |
|          | References . . . . .                                                   | 115        |
| <b>9</b> | <b>Decision Analysis and Applications in Healthcare . . . . .</b>      | <b>117</b> |
| 9.1      | Selection Criteria . . . . .                                           | 117        |
| 9.2      | Decision Tree Analysis . . . . .                                       | 121        |
| 9.3      | Decision Analysis in Public Health and Clinical Applications . . . . . | 123        |
| 9.4      | Decision Analysis in Healthcare Operations Management . . . . .        | 124        |
|          | References . . . . .                                                   | 124        |

|           |                                                        |     |
|-----------|--------------------------------------------------------|-----|
| <b>10</b> | <b>Analysis of Four Medical Datasets</b>               | 127 |
| 10.1      | Variable Selection                                     | 127 |
| 10.2      | Pima Indian Diabetes Dataset                           | 128 |
| 10.3      | Heart UCI Data                                         | 134 |
| 10.4      | India Liver Data                                       | 137 |
| 10.5      | Watson Healthcare Data                                 | 142 |
| 10.6      | Conclusions                                            | 145 |
|           | References                                             | 149 |
| <b>11</b> | <b>Multiple Criteria Decision Models in Healthcare</b> | 151 |
| 11.1      | Invasive Breast Cancer                                 | 151 |
| 11.2      | Colorectal Screening                                   | 153 |
| 11.3      | Senior Center Site Selection                           | 155 |
| 11.4      | Diabetes and Heart Problem Detection                   | 156 |
| 11.5      | Bolivian Healthcare System                             | 157 |
| 11.6      | Breast Cancer Screening                                | 158 |
| 11.7      | Comparison                                             | 159 |
|           | References                                             | 159 |
| <b>12</b> | <b>Naïve Bayes Models in Healthcare</b>                | 161 |
| 12.1      | Applications                                           | 161 |
| 12.2      | Bayes Model                                            | 162 |
| 12.2.1    | Demonstration with Kaggle Data                         | 163 |
| 12.3      | Naïve Bayes Analysis of Watson Turnover Data           | 165 |
| 12.4      | Association Rules and Bayesian Models                  | 170 |
| 12.5      | Example Application                                    | 170 |
| 12.6      | Conclusions                                            | 174 |
|           | References                                             | 174 |
| <b>13</b> | <b>Summation</b>                                       | 177 |
| 13.1      | Treatment Selection                                    | 177 |
| 13.2      | Data Mining Process                                    | 179 |
| 13.3      | Topics Covered                                         | 180 |
| 13.4      | Summary                                                | 181 |
|           | <b>Name Index</b>                                      | 183 |
|           | <b>Subject Index</b>                                   | 189 |

# Chapter 1

## Urgency in Healthcare Data Analytics



**Keywords** Healthcare · Big data · Analytics

Healthcare quality and disease prevention are ever-more critical. Data access and management is critical in improving the health services delivery. Accurate and timely available data is necessary for effective implementation of analytics projects and to improve quality in health services and programs design (Strome 2013). Health promotion and disease prevention efforts have become one of the core issues in service design in healthcare by many healthcare actors, including public health policy makers, hospitals, and financial organizations. Razzak et al. (2021) listed fundamental challenges in health promotion:

- Reduction in the growing number of patients through effective disease prevention;
- Curing or slowing down progression of disease;
- Reduction in healthcare cost through improving care quality.

They contended that information technology, especially in the form of big data analytics, could maximize identification and reduction of risk at earlier stages. In addition to population health analytics around the health promotion and disease prevention programs, hospitals and health insurance companies have been recently developing systems to benefit from big data and analytics. In this chapter we review most recent studies on big data in healthcare, big data analytics and tools, analytics implementation process, and potential challenges.

### 1.1 Big Data in Healthcare

Wang et al. (2019) gave a thorough literature review of big data analytics in healthcare. Galetsi and Katsaliaki (2019) described big data analytics as the application and tools giving more knowledge to improve the information used in healthcare decision-making. They cited studies noting that spending on information

and communication technology would reach \$5 trillion US dollars by 2020, with over 80% of the growth coming from platform technologies such as mobile technology, cloud services, social media, and big data analytics. The healthcare industry has required faster turnaround times, increased rates of utilization of facilities, and dealing with issues such as data quality and analysis effectiveness. Automation and data management are becoming increasingly more important through clinical and operational information systems, electronic health records, and laboratory information systems. Benefits expected are improved and faster delivery of care, cost reduction, more effective drugs and devices. The healthcare industry is highly data intensive, with a growing role for healthcare analytics. Some of the types of data involved are:

- Clinical data from electronic health records involving data on patients, hospitals, diagnosis and treatment, and genomics;
- Sentiment data on patients collected from wearable sensors, as well as patient behavioral data from social media;
- Administrative and cost data to include operational and financial performance;
- Research and development data on drugs, vaccines, and sera from pharmaceutical companies.

Groves et al. (2013) emphasized the five following capabilities of big data analysis tools:

1. Monitoring—to collect and analyze what is happening in real time;
2. Prediction/simulation—modeling to gain insight about what might happen under various policies;
3. Data mining—to extract and categorize what happened;
4. Evaluation—testing performance of models seeking to understand why things happened;
5. Reporting—collecting model output and organizing it to gain insight.

Liu and Kauffman (2021) addressed information system support to community healthcare through chronic disease management. Short-term functions include monitoring patient compliance, promptly suggesting rapid feedback should readjustment in treatment be appropriate. In the longer term, systems can remind patients of regular checkups and provide social support. Electronic health records provide a shared platform for monitoring test results. Analysis of population group issues can lead to development of educational and preventive programs.

Data mining tools consist of models capable of different kinds of analysis, depending upon the nature of data available. Descriptive models such as clustering and association rules are useful in exploratory analysis. Predictive models such as classification for categorical outcomes, or forecasting for continuous outcomes, provide output intended to predict what will happen. Prescriptive models go a step further by seeking optimal (or at least improved) solutions.

## 1.2 Big Data Analytics

Groves et al. (2013) noted capabilities of big data analytics to include:

- Monitoring to measure current operations;
- Prediction of future outcomes;
- Data mining to measure what has happened;
- Evaluation to explain outcomes;
- Reporting to collect knowledge and make it available in a systematic manner.

Fontana et al. (2020) distinguished between predictive analysis (taking existing data and developing a model to predict outcomes) and etiological analysis (analyzing data with the intent of understanding causation).

Jovanović Milenković et al. (2019) described big data features of healthcare analytics to include:

- Combining financial and administrative data;
- Identifying ways to simplify operations leading to cost reduction;
- Monitoring, evaluating, and analyzing new solutions to healthcare problems;
- Access to clinical data concerning treatment effectiveness;
- Improved doctor understanding of patient response to treatment;
- Optimizing resource allocation;
- Streamlined management to track claims, clients, and insurance premiums.

Quality and efficiency for healthcare organizations can be enhanced through the use of big data in the following ways (Luna et al. 2014):

- Generation of new knowledge;
- Disseminating knowledge through clinical decision-support systems;
- Implementing personalized medicine;
- Empowering patients by better informing them of their health status;
- Improving epidemiological surveillance by tracking highly prevalent or deadly diseases.

## 1.3 Tools

Some of the tools available include advanced technology in the form of artificial intelligence (AI). Wang and Alexander (2020) reviewed three branches of AI implementation in healthcare. Text mining retrieves information from postings such as e-mail, blogs, journal articles, or reports. This form of data is unstructured, but a great deal of benefit to the biomedical industry has been obtained through text mining aiding knowledge discovery. Evidence-based medicine taps various forms of structured and unstructured data that can be aggregated and analyzed to predict outcomes for patients at risk.

Machine learning is a broader concept, involving inference from known facts to predict. Machine learning techniques have proven valuable in applications such as strengthening medical signal/image processing systems, in diagnosis, and in support of robotic surgery. Forms of data mining can be applied to aid public health through targeting vaccines, predicting patients at risk, and aiding in crisis prevention and in providing services to large populations.

Intelligent agents are a higher form of AI, applying autonomous systems for healthcare data management problems such as scheduling, automatic management, and real-time decision-making. Vast quantities of data from in-home or in-hospital devices can be gathered for applications such as identification of those who would benefit from preventive care. This includes genomic analysis.

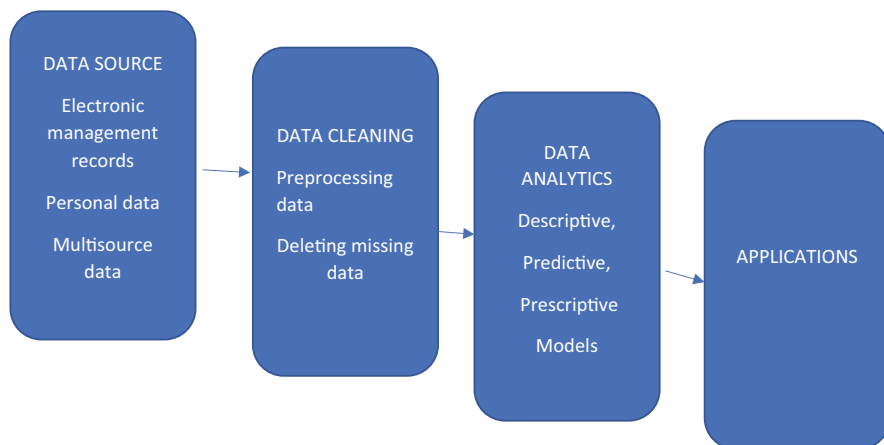
Hernandez and Zhang (2017) discussed human and technical resources needed for effective predictive analytics in healthcare. On the human side, expertise is needed in the form of data analysts to manage and create effective models. They should have expertise in electronic health records, claims data, genomics data, and data from wearable devices or social media. Expertise in pharmaceuticals is needed to design studies, defining predictor variables and interpreting results. Computer scientists are needed to provide access to programming languages and advanced predictive modeling. All of these resources need coordination by a system administrator. On the technology side, computing resources with adequate storage and security protection are required. Options include rent or purchase of data warehouses.

Because of the need for broad integration of healthcare systems, cloud computing is invaluable. Tools include data management platforms such as Apache Hadoop, an open-source (and thus free) system, with many useful capabilities, but not always appropriate for real-time analytics. Apache Spark Streaming permits both on-line and batch operations but requires expensive hardware with large RAM capacity. IBM InfoSphere Platforms are commercially available and can integrate with open-source tools such as Hadoop. Simpler (less expensive) systems such as Tableau, QlikView, TIBCO Spotfire, and other platforms offer visualization tools.

## 1.4 Implementation

Wang and Alexander (2020) gave the following predictive analytic steps:

1. Data gathering and dataset aggregation: Combining data from sources such as laboratory, genetic, insurance claim, medical records, records of medications, and other electronic healthcare information need to be aggregated. This aggregation can be by patient, service provider, or geographic location.
2. Identification of samples: Selection of observations and variables of interest for specific studies.
3. Dimension reduction: Not all available data is pertinent for a specific study, either due to irrelevance or lack of statistically useful content. This step involves



**Fig. 1.1** Healthcare data analysis (reformatted from Razzak et al. 2021)

reducing data dimensionality by focusing on useful and critical variables and observations.

4. Random split of data: Divide dataset into training, validation, and test sets for sound statistical practice.
5. Training models: Use training set to build models—usually applying multiple models such as regression, decision trees, neural networks, support vector machines, and others for prediction, clustering, and other algorithms for more exploratory research.
6. Validation and model selection: Apply models to the validation set as an intermediate step to fine-tune models leading to final selection.
7. Testing and evaluation: Test models on the test set and measure fit.

Razzak et al. (2021) offered an architecture for healthcare data analysis (Fig. 1.1):

## 1.5 Challenges

Big data analytics nevertheless face challenges. Jovanović Milenković et al. (2019) cited challenges that threaten the potential value. Because cloud computing is usually involved to integrate healthcare systems, cybersecurity is an issue. Specific challenges exist. First, healthcare big data comes in large volumes, without natural structure. Second, transferring and storing it generates high cost. Third, healthcare big data is also susceptible to data leaks. Fourth, as new technologies become available, users need continuous education. Fifth, healthcare data is not standardized, causing transmission problems. Lastly the sixth, there are unresolved legal issues.

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## Chapter 2

# **Analytics and Knowledge Management in Healthcare**



**Keywords** Knowledge management · Applications · Blockchain

Globalization has revolutionized healthcare, as it has almost every other sector of society. Growing populations face new health threats in the form of pandemics. At the same time, computer technology has been drastically changing how society operates. The world faces many new problems in political and environmental conditions. Other major risks with large-scale global impact include shortages or maldistribution of food and water and the spread of infectious diseases. The global impact of various disasters has changed over time, and the number of deaths caused by disasters such as epidemics and floods has decreased. However, some types of disasters are now seen more frequently than ever in history (Ritchie and Roser 2019). These dynamic conditions require fast processing of massive data for all purposes of analytics (including descriptive, diagnostic, predictive, and prescriptive) to support healthcare decision-making.

Advances in information technologies have given organizations and enterprises access to an unprecedented amount and variety of data (Choi and Lambert 2017). Big data refers to conditions where the scale and characteristics of data challenge the available resources for machine computation and human understanding. Zweifel (2021) noted that information technology has been expected to revolutionize healthcare for quite some time, but the medical field seems to be one of the slower sectors to adopt it. Human activity has increased in complexity, with needs to collect and process massive quantities of data in order to identify problems, and either support human decision-making more quickly or replace it with automatic action.

There are numerous data collection channels in current health systems that are linked by wireless sensors and internet connectivity. Business intelligence supported by data mining and analytics is necessary to cope with and seize the opportunities in this rapidly changing environment (Choi et al. 2017). Shafqat et al. (2020) gave a survey of sources of big data analytics integrated into healthcare systems:

- Social media platforms such as clickstream, interactive data from Facebook, Twitter, on-line blogs, and websites;
- Machine-to-machine data to include sensor readings and vital sign devices;

- Wearable health tracking devices, e.g., Fitbit, Apple Watch, and many others;
- Big transaction data from healthcare claims records and billings;
- Biometric data such as fingerprints, retinal scans, genetics, handwriting, and vital statistic measurements;
- Human generated data, from electronic medical records, physician notes, e-mails, and paper documents.

A considerable number of evolving analytical techniques have been applied to healthcare management. Big data-related techniques include optimization methods, statistical computing and learning, data mining, data visualization, social network analysis, and methods of data processing in information sciences (Araz et al. 2020). There are challenges related to capturing data, [data storage](#), data analysis, and data visualization. Araz et al. (2020) pointed out methods including granular computing, [cloud computing](#), bio-inspired computing, and [quantum computing](#) to overcome this data deluge. Challenges for big data analytics include complex data representation, unscalable computational ability, the curse of dimensionality, and ubiquitous uncertainty. The data can flow from many sources, including the web, social media, and the cloud. As a result of this flow, big data can be accumulated. The processing of these data is a major task, and the processing schemes include old batch processing, newer streaming, and even interactive processing.

## 2.1 Healthcare Data Analytics

We review data analytics from the perspective of healthcare. We seek to describe the relationship of published literature in applying business intelligence, knowledge management, and analytics to deal with operational problems based on the applications, techniques, and strategies used. Table 2.1 outlines these topics with the classification scheme used. For each of the categories, we state the purposes and tools used in implementation of the analytics. For example, risk assessment and management tools are used in healthcare management. Other methods used in this application area are system security, risk mitigation, and preparedness and response. Applications are classified as disaster management, public health, food safety, and social welfare, in which the use of analytics is emerging rapidly.

Note that the classification in Table 2.1 relates to the timeliest areas of healthcare analytics. In the following sections, we elaborate on these categories.

## 2.2 Application Fields

In healthcare, disaster management, humanitarian operations, public health risk management, food safety, and social welfare are important application areas in which business analytics has evolved the most with new tools and technologies.

**Table 2.1** Categories considered in healthcare data analytics for risk management with tools, methods, and purposes

| Classification categories | Purposes                                                                                                                                                                                            | Used methods and tools                                                                                                                                                                   |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Application fields        | <ul style="list-style-type: none"><li>• Disaster management</li><li>• Public health and clinical decisions</li><li>• Healthcare operations</li><li>• Food safety</li><li>• Social welfare</li></ul> | <ul style="list-style-type: none"><li>• Risk assessment and management</li><li>• Risk mitigation</li><li>• Preparedness and response</li><li>• System reliability and security</li></ul> |
| Analytics techniques      | <ul style="list-style-type: none"><li>• Descriptive</li><li>• Diagnostic</li><li>• Predictive</li><li>• Prescriptive</li></ul>                                                                      | <ul style="list-style-type: none"><li>• Statistics</li><li>• Visualization</li><li>• Simulation</li><li>• Machine learning</li><li>• Data mining</li><li>• Optimization</li></ul>        |
| Analytics strategies      | <ul style="list-style-type: none"><li>• Information systems management</li><li>• Knowledge management</li></ul>                                                                                     | <ul style="list-style-type: none"><li>• Blockchain technology</li><li>• Big data tools</li></ul>                                                                                         |

Identifying and defining the problem in the healthcare domain is an important step for any implementation of analytics. Here we review the application fields of analytics for the purposes of these practical problems.

2.2.1 Disaster Management

The many disasters of the past few decades have led to a shortage of healthcare resources and induced a need to change healthcare operations (Tippong et al. 2022). UN Office for Disaster Risk Reduction has reported over 7000 disaster events in the period 1998 through 2017, causing over 1.3 million deaths. Healthcare disaster response relies heavily on hospitals. Disaster impacts were categorized by Tippong et al. into four groups:

1. Hospitals have to allocate staff to shelters for initial treatment, meaning that they need a sufficient inventory of staff;
2. Hospitals have to provide evacuation service calling for ambulatory inventory;
3. The sudden surge of emergency patients taxes all healthcare resources, lowering hospital performance;
4. Admission and discharge protocols need to be modified to increase the ability to accept the surge of emergency patients.

This all means that contingency plans need to be developed. Analytic models have been widely applied in such contingency planning. These include risk analysis for pre-positioning of disaster relief assets, evacuation plans, and communication systems to coordinate relief efforts.

### ***2.2.2 Public Health Risk Management***

The global spread of infectious diseases and utilization of biological agents in mass casualty incidences (e.g., anthrax) continue to pose major threats to global public health and safety. Other than disease outbreaks, floods, earthquakes, hurricanes, and other natural disasters can affect mass populations; thus, they need to be considered and assessed for public safety. Over the years, the level of sophistication of these decision-support systems has been increasing with higher analytical capabilities embedded for complex decision-making situations, including visualization, real-time decision-making, and optimal resource management (He and Liu [2015](#)).

Electronic health records are evolving, but they offer great potential to aid in achievement of public health goals such as research, healthcare delivery, public health surveillance, and personal health management. They contain a wide variety of data types, to include notes, radiological output, vital measurements over time, etc. Wang and Alexander ([2020](#)) demonstrated the value of electronic health records in applications such as drug recommendation systems, risk factor identification, heart failure prediction, and fulfillment of personalized medical treatments.

Another major public health issue in the USA is the cost of healthcare. Politicians have looked to automated healthcare record systems to reduce these costs for over 25 years, although in practice very little reduction seems to occur. As public health continues to be challenged with newly emerging infectious diseases and the cost of care continues to rise, the use cases of analytics will continue to increase in related academic literature.

### ***2.2.3 Food Safety***

Foodborne illness causes a great deal of expense in the form of medical care, lost productivity, loss of life, and pain and suffering. Scharff ([2020](#)) developed a cost estimation model of foodborne illnesses from 29 pathogens related to meat and poultry products regulated by the US Department of Agriculture. Food attribution models took data by food category and pathogen. Output was combined with that of an illness model using pathogen data. These were combined to produce illness incidence estimates for meat and poultry products by pathogen and food category. Results found that a meat and poultry were vectors for over 30% of foodborne illnesses in the USA and over 46% of costs.

Among many applications of business analytics reported in food safety blockchain applications in food supply chains promise great advances for risk identification and mitigation. As an information management enabler, blockchain applications will be discussed later in the paper.

### 2.2.4 Social Welfare

Big data analysis can be used to reduce healthcare cost by identifying healthcare resource waste, provide closer monitoring, and increasing efficiency. Wu et al. (2017) examine how big data and analytics have impacted privacy risk control and healthcare efficiency to include tradeoffs and risks. Recently there are more analytics applications seen in the literature for social welfare, in the context of risk. There is great potential for measuring and improving social welfare using big data analytics, in part due to the big data generated by wearable technologies. Issues raised include privacy, as the vast amounts of concentrated data become attractive to hackers. The higher variety of information makes it harder to protect.

Prescriptive analytics applications are by far the most seen in the literature, incorporating uncertainty into optimization models with predictive analytics would provide more robust solutions. In the next section, we discuss these different categories of analytics techniques.

## 2.3 Analytics Techniques

Analytics covers means to analyze and apply the data generated. Business analytics tools can be used for four main purposes of analysis. *Descriptive* analytics focuses on reports, with statistics and data visualization playing a major role. There are descriptive modeling tools, often involving unsupervised learning of data mining, as in cluster analysis or association rule mining. These models usually do not aim prediction, but rather attempt to provide clues to data structure and relationships. *Diagnostic* analytics would commonly include automatic control systems, replacing human control with computer control for the sake of speed and enabling better dealing with complexity. *Predictive* analytics can provide capabilities of predicting the outcome of a behavior as a class or a series of values for a target variable as forecasts over time. Finally, *prescriptive* analytics involve optimization models, seeking better solutions under a variety of conditions which can also include decision-making under uncertainty. If sense-and-respond kinds of operations can be implemented using technologies such as blockchain, with the identification of different risk attitudes of the involved agents and/or customers, better operational efficiencies can be achieved. Therefore, determining how techniques in these four purposes of analytics can be used to solve the problem(s) discussed earlier is the next phase.

Emerging data sources and technologies have recently increased the application of analytical methods in many industries. For example, predicting geospatial spread of diseases (predictive) for better resource allocation (prescriptive) is critical for pandemic response (Araz et al. 2013); understanding emerging trends in consumer behavior during natural disasters such as hurricanes is critical. Other examples can be found in analyzing social networks and predicting the role of social media on public behavior, managing traffic flows during catastrophic events, and optimizing

location of relief facilities for maximum coverage and safety (Salman and Yücel 2015; Choi et al. 2018; Battara et al. 2018).

## 2.4 Analytics Strategies

We present strategic purposes the data analytics tools have evolved for applications. The right deployment strategy and the technology are the critical last phase of the data-driven ORM process. We review these strategies that enable intelligence via analytics for information systems management and knowledge management purposes within ERP systems, using big data analytics tools or newly emerging blockchain technologies.

### 2.4.1 *Information Systems Management*

Information systems (IS) provide means to obtain, organize, store, and access data. IS have developed with relative independence, although its interface with the operations management has a long history and continues to develop rapidly. There are several aspects of information systems' impact on operational use of data analytics. One of them is decision-support systems and another one is ERP systems. These systems unify data, seek improved business processes, and automate much of business. Big data extracted from both internal and external sources enable identification of what has happened (descriptive analytics), what might happen (predictive analytics), and, in some cases, optimized solutions (prescriptive analytics).

Recently, centralized ERP systems with a centralized database are challenged by decentralized ledger-based systems in the form of blockchain technology. Compared to ERP systems, the blockchain technology-based systems can keep permanent, traceable, and reliable data in a decentralized manner. In the future, we expect to see more research on blockchain technology-based risk analysis.

### 2.4.2 *Knowledge Management*

Knowledge management is an overarching term concerning identification, storage, and retrieval of knowledge. Knowledge identification requires gathering information from data sources. The storage and retrieval of knowledge involve designing and operating databases within management information systems.

Human, technological, and relationship assets are needed to successfully implement knowledge management. Knowledge management is characterized by the presence of big data (Olson and Lauhoff 2019). There are several big data analytics techniques used in the literature for analytics implementation to develop and support knowledge management.

### **2.4.3 *Blockchain Technology and Big Personal Healthcare Data***

Proper knowledge management is critical for competitive advantage in any industry. Human, technological, and relationship assets are needed to successfully implement knowledge management and current advancements allow deeper understanding of the data and information while posing some challenges in process development and management. Blockchain technology has been viewed as a potentially knowledge-based network for health log data management, capable of dealing with big data from mobile service (Chung and Jung 2022). Precision medical technology in the form of wearable systems linked to the Internet of Things enables personalized healthcare (concierge medicine). User data can be continuously uploaded for analysis and prediction of problems. Security is gained as blockchain networks cannot easily be forged or falsified by the user. Health log data can include personal, medical, location, and environmental information. The disadvantage is that blockchain systems are slow and use significant amounts of computer time which translates into electricity usage. The more complete the block chain system data upload, the slower and more expensive it will be.

## **2.5 Example Knowledge Management System**

Faria et al. (2015) proposed a clinical support system focused on quality-of-life for evaluation of medical treatment options for patients. Quality-of-life considers emotional, social, and physical aspects of patients. The treatment with the greatest medical probability of success is not always best. Patient pain, suffering, and financial condition need to be considered. The clinical support system Faria et al. proposed was intended to allow patients to make informed decisions.

There has been a growing pressure on the health sector to provide greater transparency, accountability, and access to the efficiency/effectiveness of available treatments. A clinical decision-support system applies information technology to healthcare by integrating knowledge management systems including robust models capable of converting information into knowledge by providing relevant data and prediction models. They help health professionals cope with the progressive increase in data, information, and knowledge. Areas of intervention include prevention (immunization), diagnosis (identification of patients with similar systems), and treatment (such as drug interaction alerts). They can provide measures of quality adjusted life years considering:

- Analysis of clinical decision predicted consequences;
- Economic evaluation in terms of cost of available treatments;
- Comparison with other patients with similar conditions.

This quality-of-life years considers tolerance for pain, emotional state, and impact on functional performance or social interaction. Economic and cost management include consideration of:

- Drug/technology use;
- Rate and duration of hospital admissions;
- Hospital costs;
- Prevention programs;
- Epidemiological knowledge;
- Pharmacoeconomic knowledge.

Quality-of-life years uses evidence-based medicine, focusing on patient satisfaction. They integrate clinical information with patient health status. This gives healthcare professionals a set of tools to systematically measure patient quality-of-life and to turn tacit knowledge (patient perceptions) into explicit knowledge.

Actual systems would include a Web server linked to patients and doctors, as well as a quality-of-life On-Line Transactional Processing (OLTP) system supported by a database system. Data mining algorithms and statistical models would enable:

- Evaluation of quality-of-life;
- Measurement of health gains and functional health status;
- Assessments of disease consequences;
- Categorization of patients through data mining classification algorithms;
- Analysis of deviant observations;
- Prediction of health, survival, and quality-of-life.

Measurement of patient quality-of-life is accomplished through on-line survey responses.

Such a system was tested on a target population of 3013 cancer patients with head and neck cancers. The initial analysis was conducted through descriptive statistics. Multiple linear regressions were then run to generate a predictive quality-of-life model with the independent variables given in Table 2.2. Nominal variables were converted to dummy variables for regression:

Significant predictors at the 0.05 level were years of smoking and size. Other variables near that level of significance were educational level, tracheostomy, liters of wine per day, and presence of a voice prosthesis. The overall model was significant with an  $F$  value of 3.85,  $p < 0.001$ . Data mining predictive models were then run. Table 2.3 shows accuracy of the four predictive models run.

The overall best fit was obtained with the support vector machine model, but the Naïve Bayes model was very close, and with the more robust model based on variables with better significance measures, Naïve Bayes was better.

The Quality-of-Life system gave a tool for patients and healthcare providers to consider quality-of-life issues in assessing alternative cancer treatments. This system gives a view of how knowledge management can be incorporated into computer models.

**Table 2.2** Independent variables for quality-of-life system

| Variable                    | Type    | Options   | <i>n</i> | Min   | Max      | <i>p</i> -value | VIF   |
|-----------------------------|---------|-----------|----------|-------|----------|-----------------|-------|
| Educational level           | Ordinal | 10 levels | 3013     | Grade | PhD      | 0.091           | 1.238 |
| Marital status              | Nominal | Alone?    | 2942     | 0     | 1        | 0.274           | 1.220 |
| Years smoking               | Ordinal | 6 levels  | 1785     | Not   | >40      | 0.049           | 2.428 |
| Cigarettes/day              | Ordinal | 6 levels  | 1826     | None  | >60      | 0.902           | 2.131 |
| Years drinking              | Ordinal | 6 levels  | 1811     | None  | >20      | 0.959           | 1.927 |
| Liters of beer/day          | Ordinal | 6 levels  | 1343     | None  | >10      | 0.705           | 2.217 |
| Liters hard alcohol/day     | Ordinal | 6 levels  | 1268     | None  | >10      | 0.443           | 2.471 |
| Size                        | Ordinal | 6 levels  | 1679     | TX    | T4       | <0.001          | 2.048 |
| Local metastasis            | Ordinal | 5 levels  | 1677     | NZ    | N3       | 0.172           | 1.755 |
| Metastasis distance         | Ordinal | 3 levels  | 2350     | MX    | M1       | 0.382           | 1.420 |
| Histopathological diagnosis | Ordinal | 0–9       | 2851     | 0     | 9        | 0.309           | 1.781 |
| Tracheostomy                | Nominal | yes/no    | 2857     | Yes   | No       | 0.067           | 3.363 |
| Type of feed                | Nominal | 3 inputs  | 1715     |       |          | 0.063           | 3.476 |
| Liters wine/day             | Ordinal | 6 levels  | 2591     | None  | >10      | 0.057           | 2.818 |
| Smoking                     | Nominal | 3 levels  | 1390     | 0     | 2        | 0.955           | 1.344 |
| Years ex-smoker             | Ordinal | 5 levels  | 3013     | 0–1   | >15      | 0.560           | 1.501 |
| Last appointment            | Ordinal | 11 levels | 1535     | 1st   | >5 years | 0.201           | 1.257 |
| Voice Prosthesis            | Nominal | Yes/no    | 3013     | Yes   | No       | 0.095           | 1.725 |

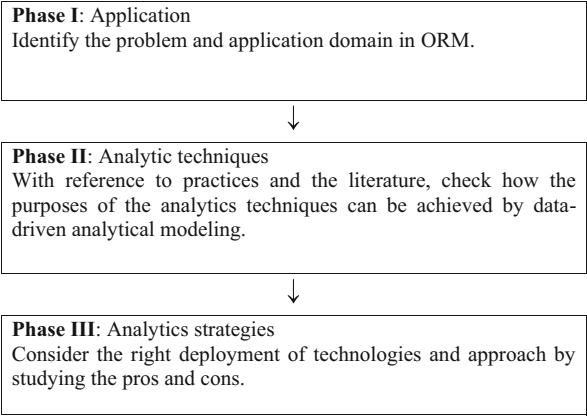
**Table 2.3** Model accuracies

| Algorithm                  | Accuracy      | Accuracy                    |
|----------------------------|---------------|-----------------------------|
|                            | All variables | Variables significant < 0.1 |
| <i>K</i> -nearest neighbor | 55.92%        | 39.83%                      |
| Naïve Bayes                | 63.06%        | <b>63.79%</b>               |
| Decision tree              | 42.51%        | 40.26%                      |
| SVM                        | <b>65.58%</b> | 58.65%                      |

## 2.6 Discussion and Conclusions

Recent technological developments and the advancements in technology-based tools for social and economic life provide organizations and enterprises access to unprecedented amounts and varieties of data. Since it is easier than ever to store and process these data in platforms such as Apache Hadoop, etc., there are higher expectations in processing velocity for service delivery and related products. Developments and integration of new data collection channels in existing industrial systems, linked by the Internet of Things (IoT) via wireless sensors and internet connectivity, provide new problems and solutions for providing better and robust services. Business intelligence supported by these analytics and big data tools are now at the heart of

**Fig. 2.1** A three-phase process for data-driven ORM



this rapidly changing business environment. In this discussion, we investigate and present trends in operational risk management (ORM) related to various types of natural and man-made disasters that have been challenging the world’s social and economic routines in recent years. We also emphasize the need for analytics tools for monitoring systems, and the integration of these tools, including predictive modeling, into operational decision-making. In Fig. 2.1, a three-phase process of implementing data-driven ORM is proposed based on our findings in this review.

In summary, organizations seek incorporation of data analytics by leveraging real-time emerging data and surveillance systems, predicting future impact and reactions, optimizing strategic and logistical decisions for effective response planning and execution. This field will continue to evolve as information technologies enable greater data streaming and processing and various risk factors continuously challenge the supply chain management world. For example, geo-mobile technologies and crowdsourcing methods are opening up new opportunities for prescriptive analytics applications in the field of disaster management. Public health applications also seek integration of data collection and processing tools for real-time decision-making. Food safety is recently gaining more attention and blockchain applications in food supply chains promise great advancements in risk identification and mitigation. In transportation, prescriptive analytics applications dominate the literature, however, incorporating uncertainty into optimization models with predictive analytics would advance the field with more robust solutions. While social welfare in the context of risk is still immature, there is great potential of measuring and improving social welfare using analytics with big data technologies. Application of information systems has evolved into enterprise resource planning (ERP) systems, seeking integration of organizational reporting and offering single-sourced real-time data. Human involvement in operational protocol development along with artificial intelligence tools needs more investigation.

Our conclusions are summarized in Table 2.4, following Araz et al. (2020). The integration of predictive and prescriptive modeling in decision-support systems for ORM is a trend in practical applications. Systems integration and real-time data

**Table 2.4** Knowledge management features

| Classification categories | Major findings                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Future directions                                                                                                                                                                                                                                                                                                                                                                                                              | Key research questions                                                                                                                                                                                                                                                                                                                                                                  |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Application Fields        | <ul style="list-style-type: none"> <li>• Disaster Management: Mobile technologies and crowdsourcing with prescriptive analytics in disaster management operations</li> <li>• Public health risk management: Integration of data collection and processing tools for real-time decision-making</li> <li>• Food safety: Blockchain applications in food supply chain offer improved risk identification and mitigation</li> <li>• Social welfare: While in early development stages, there is high potential in measuring and improving social welfare using analytics with big data technologies</li> <li>• Transportation: Prescriptive analytics dominate applications</li> </ul> | <ul style="list-style-type: none"> <li>• Real-time data streaming and processing are becoming factors in all sectors</li> <li>• Operational risk assessment tools will evolve for disaster management, public health, food security, social welfare, public and commercial transportation applications</li> <li>• Incorporating uncertainty into optimization models with predictive analytics for robust solutions</li> </ul> | <ul style="list-style-type: none"> <li>• How do real-time data streaming and processing technologies support healthcare?</li> <li>• What are new risk assessment schemes in public health and other disaster management situations?</li> <li>• How to establish predictive analytics for robust solutions incorporating uncertainty into optimization models for healthcare?</li> </ul> |
| Analytics techniques      | <ul style="list-style-type: none"> <li>• High interest in leveraging real-time emerging data</li> <li>• Need of more surveillance systems for predicting future impact</li> <li>• Many studies on prescriptive analytics for strategic and logistical decisions for effective planning</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                  | <ul style="list-style-type: none"> <li>• Data streaming and processing in predictive and prescriptive analytics</li> <li>• More surveillance systems</li> <li>• Data-driven operational risk analysis</li> </ul>                                                                                                                                                                                                               | <ul style="list-style-type: none"> <li>• What are the critical data-driven techniques that can be applied? What are their impacts?</li> </ul>                                                                                                                                                                                                                                           |
| Analytics strategies      | <ul style="list-style-type: none"> <li>• Human-technology relationship tools are critical for successfully implementing knowledge management.</li> <li>• Information systems have evolved into ERP systems</li> <li>• More tools seeking integration of</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                 | <ul style="list-style-type: none"> <li>• Research on human involvement in operational protocol development along with Artificial Intelligence tools</li> <li>• High demand for systems integration and real-time data processing tools for risk analysis</li> </ul>                                                                                                                                                            | <ul style="list-style-type: none"> <li>• What can Artificial Intelligence bring? What is the role of humans?</li> <li>• How to integrate existing systems with real-time data processing?</li> <li>• What are the values and impacts of</li> </ul>                                                                                                                                      |

(continued)

**Table 2.4** (continued)

| Classification categories | Major findings                                             | Future directions                                                                | Key research questions                                |
|---------------------------|------------------------------------------------------------|----------------------------------------------------------------------------------|-------------------------------------------------------|
|                           | organizational reporting and single-sourced real-time data | • The deployment strategies of new and disruptive technologies, e.g., blockchain | deploying disruptive technologies (e.g., blockchain)? |

processing tools will be in higher demand for operational management. The specific areas of ORM that data analytics can help deserve deeper explorations. In addition, real-time data streaming and processing are becoming the major interests of all sectors and operational risk assessment tools will continue to evolve for disaster management, public health, food security, social welfare, and public and commercial transportation applications. The deployment of new technologies, such as blockchain, will see more use in the future.

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# Chapter 3

## Visualization



**Keywords** Visualization tools · Excel graphics

Tools to communicate data are evolving with new technologies developed and gaining significant attention, particularly when discussing large quantities of data. This significance is also broad in practice as audiences interested in the data analytics and insights becoming larger with more diverse backgrounds. We can use text, numbers, tables, and a variety of graphs for presenting data. The tool we use must be a good fit to the content of information we present. For example, we use graphs when there are a large number of data points or categories, and details are not as important as the overall trend or share of categories in the data pool. For one or two numbers, or observations that can be summarized in one or two numbers, text will work better.

Microsoft Excel offers many visualization tools, including tables, graphs, and charts. Data tables demonstrate data one cell at a time. Highlighting rows, columns, or cells can aid in communication. There are many types of charts, quite a few available in Excel. Excel contains chart tools to format and design charts. We will demonstrate some of these with three healthcare datasets obtained from [www.kaggle.com](http://www.kaggle.com), a data mining site with many publicly accessible datasets on many topics.

### 3.1 Datasets

#### 3.1.1 *Healthcare Hospital Stay Data*

Intensive Care Units (ICUs) often lack verified medical histories for incoming patients. A patient in distress or a patient who is brought in confused or unresponsive may not be able to provide information about chronic conditions such as heart disease, injuries, or diabetes. Medical records may take days to transfer, especially for a patient from another medical provider or system. Knowledge about chronic

**Table 3.1** Hospital stay averages by country

| ID | Country        | AvgStay | MRI   | CT    | Beds  |
|----|----------------|---------|-------|-------|-------|
| 1  | Hungary        | 7.1     | 2.25  | 6.27  | 2.25  |
| 2  | Russia         | 11.8    | 2.29  | 5.9   | 2.29  |
| 3  | Israel         | 5.5     | 2.78  | 7.97  | 2.78  |
| 4  | Czech Republic | 7.7     | 4.96  | 11.88 | 4.96  |
| 5  | Canada         | 7.9     | 5.12  | 11.13 | 5.12  |
| 6  | Poland         | 7.1     | 5.31  | 13.98 | 5.31  |
| 7  | Lithuania      | 7.5     | 5.88  | 16.24 | 5.88  |
| 8  | Great Britain  | 6.6     | 6.03  | 7.88  | 6.03  |
| 9  | Austria        | 5.6     | 6.32  | 39.5  | 6.32  |
| 10 | Slovakia       | 6.8     | 6.54  | 14.5  | 6.54  |
| 11 | France         | 5.7     | 6.76  | 11.5  | 6.76  |
| 12 | Turkey         | 4.5     | 7.38  | 11.2  | 7.38  |
| 13 | Portugal       | 8.4     | 8     | 26.56 | 8     |
| 14 | Latvia         | 6.5     | 8.18  | 28.18 | 8.18  |
| 15 | Denmark        | 3.6     | 8.32  | 13.52 | 8.32  |
| 16 | Slovenia       | 6.3     | 9     | 12.99 | 9     |
| 17 | Netherlands    | 6.9     | 9.3   | 10.65 | 9.3   |
| 18 | Estonia        | 5.9     | 9.33  | 15.41 | 9.33  |
| 19 | Belgium        | 7.2     | 9.85  | 16.94 | 9.85  |
| 20 | New Zealand    | 5.7     | 10.62 | 15.15 | 10.62 |
| 21 | Luxembourg     | 7.4     | 11.65 | 23.63 | 11.65 |
| 22 | Ireland        | 5.9     | 12.4  | 16.34 | 12.4  |
| 23 | Finland        | 6.6     | 13.55 | 15.87 | 13.55 |
| 24 | Spain          | 6.1     | 15.21 | 17.67 | 15.21 |
| 25 | South Korea    | 9.7     | 15.28 | 29.71 | 15.28 |
| 26 | Australia      | 7.1     | 16.23 | 28.26 | 16.23 |
| 27 | Italy          | 6.9     | 18.13 | 28.27 | 18.13 |
| 28 | Greece         | 5.5     | 20.21 | 31.13 | 20.21 |
| 29 | Iceland        | 5.6     | 20.93 | 38.91 | 20.93 |
| 30 | Germany        | 8.4     | 24.7  | 31.01 | 24.7  |
| 31 | USA            | 5.6     | 27.85 | 35.54 | 27.85 |
| 32 | Japan          | 21.7    | 39.14 | 95.52 | 39.14 |

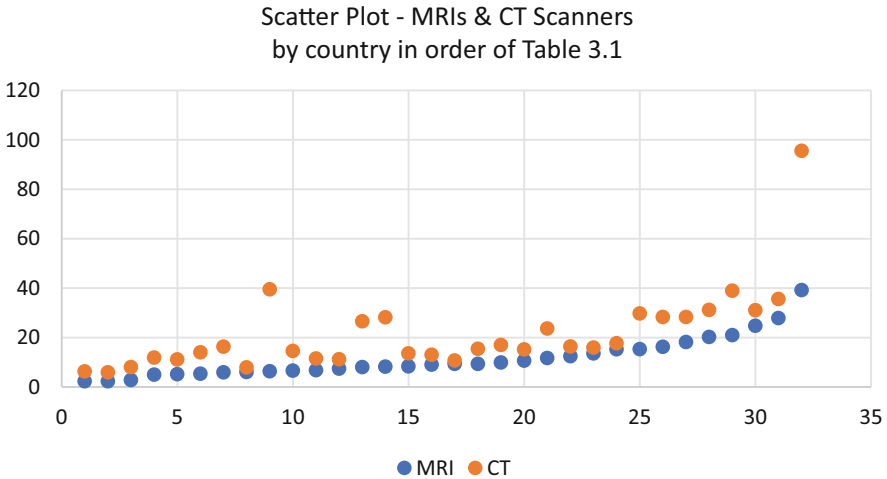
conditions can inform clinical decisions about patient care and ultimately improve patient's survival outcomes.

This dataset concerns the days patients stay in hospitals in OECD countries. It contains country, year (between 1992 and 2018, some yearly data missing for some countries), average hospital stay in days, and the average number of MRI units, CT scanners, and hospital beds. The source is <https://www.kaggle.com/datasets/babyoda/healthcare-investments-and-length-of-hospital-stay>. Table 3.1 gives the average data for each country:

Table 3.1 is organized by average hospital stay in ascending order. This is obtained in Excel by selecting the data and sorting. The relationship with MRIs,

**Table 3.2** Correlation of hospital stay average data

|         | AvgStay | MRI   | CT    | Beds  |
|---------|---------|-------|-------|-------|
| AvgStay | 1.000   |       |       |       |
| MRI     | 0.489   | 1.000 |       |       |
| CT      | 0.687   | 0.838 | 1.000 |       |
| Beds    | 0.489   | 1.000 | 0.838 | 1.000 |



**Fig. 3.1** Scatterplot

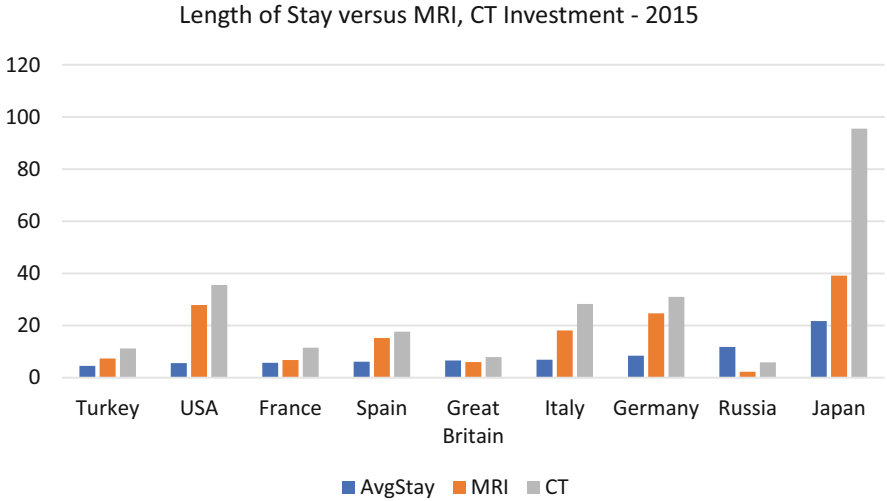
CT scanners, and hospital beds is not apparent from the table. Statistically, a first step might be to obtain correlations among the numeric variables. Table 3.2 shows this Pearson correlation:

This indicates that the number of CT scanners has the highest correlation with hospital stay, while MRIs and beds, with a weaker but still significant relationship with stay, have a perfect correlation with each other. This leads us to glance again at Table 3.1, where we see the data is identical. Obviously, the dataset has a flaw—either MRI data overwrote the Beds data, or vice versa. We will focus now on average hospital stay versus MRI and CT scan equipment.

Along with correlation, you can run scatter plots to visualize relationships. The scatter plot for the two types of equipment is given in Fig. 3.1:

The correlation between MRI and CT equipment is visually apparent. The full set of 32 countries is too large to see much in the way of charts. Thus we will focus on nine countries that might be of interest. A bar chart displays multiple variables by country (Fig. 3.2):

Note that you do not want to have too many countries listed in the plot, otherwise you will lose information with the visualization. But this chart shows the relative average stay in order, ranging from Turkey’s 4.5 days to Japan’s 21.7 days. MRI investment is highest in Japan, followed by USA and Germany. CT scan investment is very high in Japan and moderately high in the USA, Germany, and Italy. Figure 3.2



**Fig. 3.2** Bar chart of selected country hospital stays, MRIs, and CTs

is not as good at relating CT scan investment with length of stay—Table 3.1 is much better for that. Graphs and tables can be used with each other to provide a good picture of what is going on in a dataset.

**3.1.2 Patient Survival Data**

This dataset also comes from the [https://journals.lww.com/ccmj/Citation/2019/01001/33\\_THE\\_GLOBAL\\_OPEN\\_SOURCEW\\_SEVERITY\\_OF\\_ILLNESS.36.aspx](https://journals.lww.com/ccmj/Citation/2019/01001/33_THE_GLOBAL_OPEN_SOURCEW_SEVERITY_OF_ILLNESS.36.aspx) site (Raffa et al. 2019). There are sixteen variables, all categorical. Table 3.3 gives variables and types:

Initial examination of data is supported by most software in the form of comparative counts. The Excel bar chart of the 17 admission departments is shown in Fig. 3.3:

Clearly the analyst can see that most admissions to the hospital were to the emergency room. Figure 3.4 shows the type of ICUs where patients went:

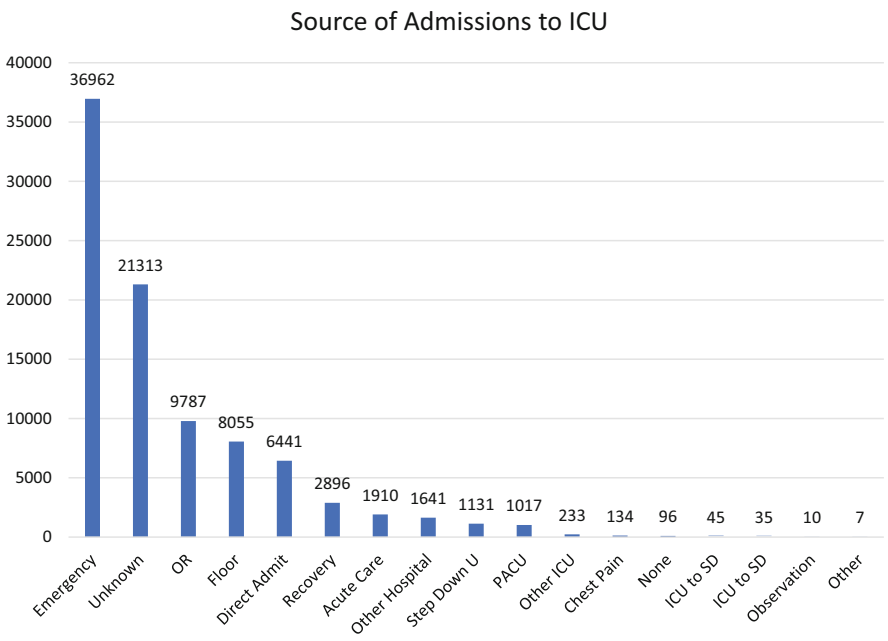
Figure 3.4 shows that the majority of patients went to the medical-surgery ICU. Table 3.4 measures the death rates by ICU:

Visualization involves exploring various graphical displays. But by manipulating the Excel file, we can obtain a clearer picture in Fig. 3.5 via an MS Excel spider chart:

Figure 3.6 shows from which departments = patients arrived and admitted to ICU, in other words, the sources of patients to ICU:

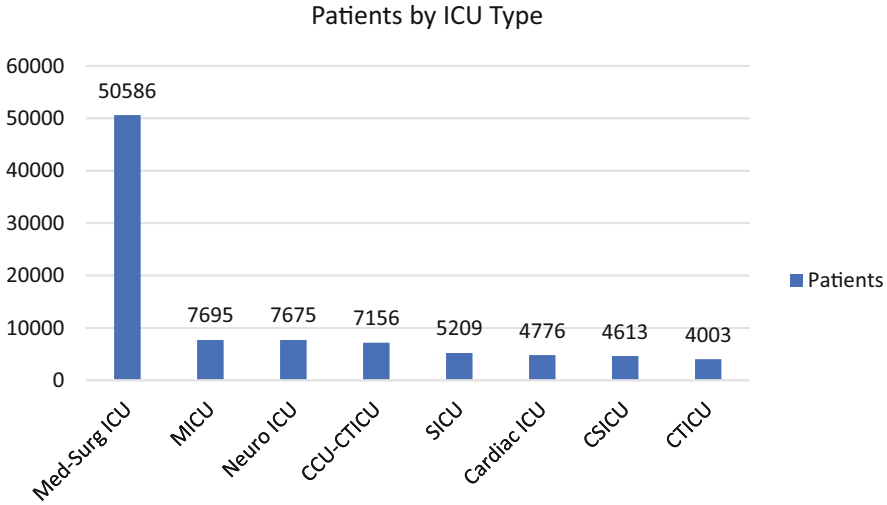
**Table 3.3** Patient survival dataset variables

| Variable                    | Categories               | Description                          |
|-----------------------------|--------------------------|--------------------------------------|
| Died                        | Yes/No                   | Died within hospital stay            |
| Elect                       | Yes/No                   | Admitted for elective surgery        |
| Ethnicity                   | Seven categories         |                                      |
| Gender                      | Female, Male             |                                      |
| Hospital Admit              | 17 Departments           | Location prior to hospital admission |
| ICU Admit                   | 5 Departments            | Location prior to ICU admission      |
| ICU Stay Type               | Admit, Transfer, Readmit |                                      |
| ICU Type                    | 8 ICU Specialties        |                                      |
| AIDS                        | 793 yes, 90,920 no       |                                      |
| Cirrhosis                   | 2143 yes, 89,570 no      |                                      |
| Diabetes Mellitus           | 21,207 yes, 89,570 no    |                                      |
| Hepatic Failure             | 1897 yes, 89,816 no      |                                      |
| Immunosuppression           | 3096 yes, 88,617 no      |                                      |
| Leukemia                    | 1358 yes, 90,355 no      |                                      |
| Lymphoma                    | 1091 yes, 90,622 no      |                                      |
| Solid tumor with metastasis | 2593 yes, 89,120 no      |                                      |



**Fig. 3.3** Counts of hospital admits

It is often good to utilize tabular data along with graphic displays. Table 3.5 shows more complete information relative to ICU patients by where they were admitted from.



**Fig. 3.4** Counts of ICU type: Medical intensive care unit, Cardiothoracic intensive care unit, Surgical intensive care unit, Cardiac surgery intensive care unit

**Table 3.4** ICU outcomes

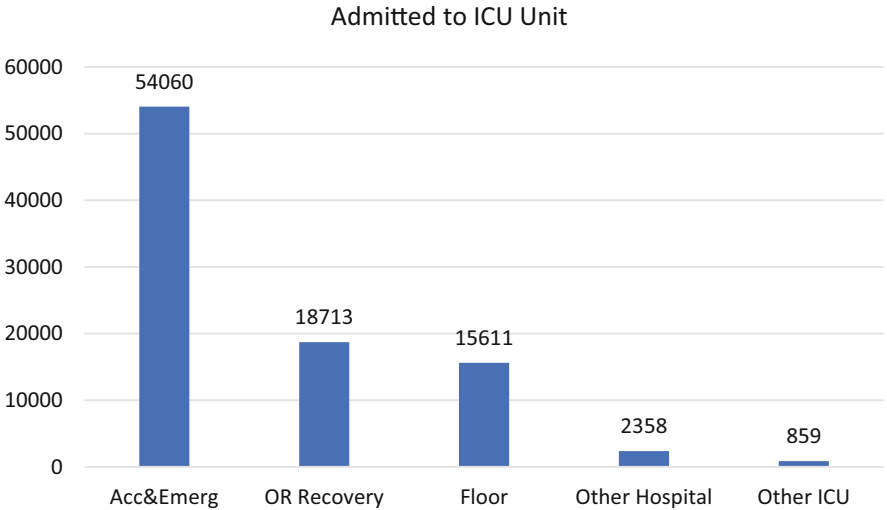
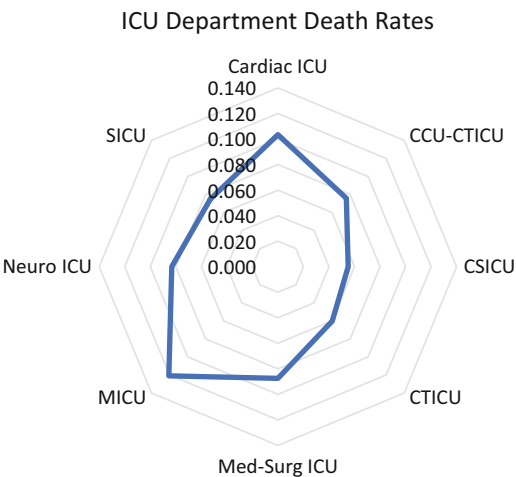
| ICU Departments                                              | Patients | Died | Death Rate |
|--------------------------------------------------------------|----------|------|------------|
| Cardiac ICU                                                  | 4776     | 494  | 0.103      |
| Critical care cardiothoracic intensive care unit (CCU-CTICU) | 7156     | 542  | 0.076      |
| Cardiac surgery intensive care unit (CSICU)                  | 4613     | 254  | 0.055      |
| Cardiothoracic intensive care unit (CTICU)                   | 4003     | 241  | 0.06       |
| Med-Surg ICU                                                 | 50,586   | 4426 | 0.087      |
| Medical intensive care unit (MICU)                           | 7695     | 930  | 0.121      |
| Neuro ICU                                                    | 7675     | 638  | 0.083      |
| Surgical intensive care unit (SICU)                          | 5209     | 390  | 0.075      |
| Total                                                        | 91,713   | 7915 | 0.086      |

Figure 3.7 displays the admitted column from Table 3.5:  
Figure 3.8 displays the death rate column from Table 3.5:  
A question of interest might be relative death rates by ethnicity or by gender. Through Excel custom sorts and counts, Table 3.6 is obtained:  
The data in Table 3.6 is sufficient to see that males had a slightly lower death rate than females. The formula for the  $z$ -test of difference is:

$$z = \frac{\text{prob(Female death)} - \text{prob(Male death)}}{\sqrt{p\text{bar} * (1 - p\text{bar}) \left( \frac{1}{\text{Female total}} + \frac{1}{\text{Male total}} \right)}}$$

where  $p\text{bar} = \frac{\text{Female death} + \text{Male death}}{\text{Female total} + \text{Male total}}$

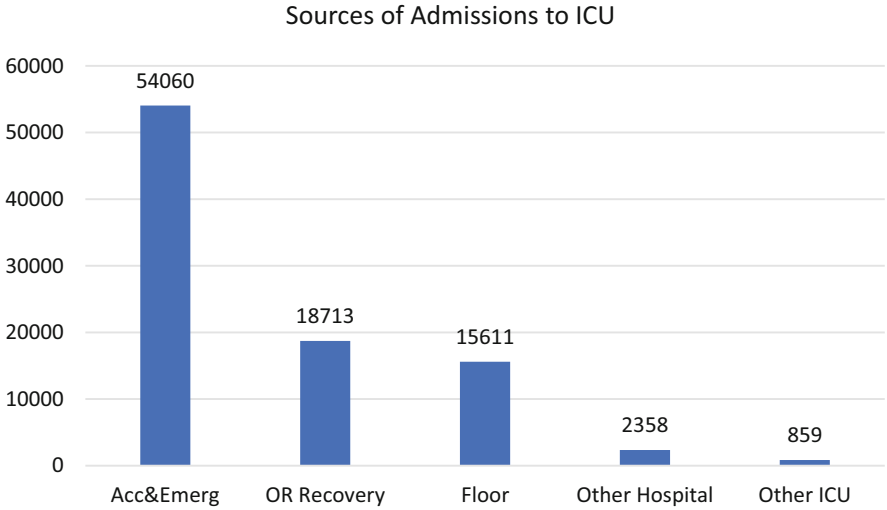
**Fig. 3.5** Death rates by ICU



**Fig. 3.6** ICU patient sources

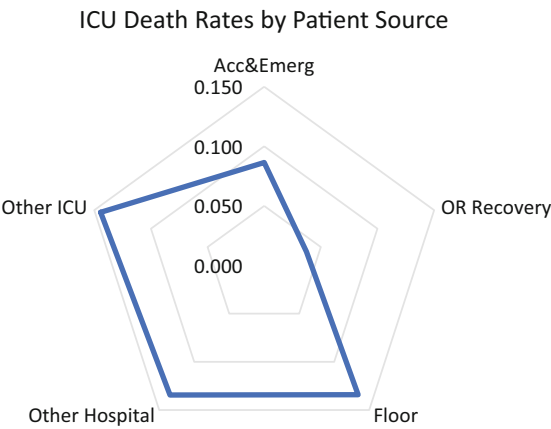
**Table 3.5** Death rates by ICU patient source

| ICU unit       | Admitted | Died | Death Rate |
|----------------|----------|------|------------|
| Acc&Emerg      | 54,060   | 4670 | 0.086      |
| OR Recovery    | 18,713   | 698  | 0.037      |
| Floor          | 15,611   | 2094 | 0.134      |
| Other Hospital | 2358     | 317  | 0.134      |
| Other ICU      | 859      | 124  | 0.144      |



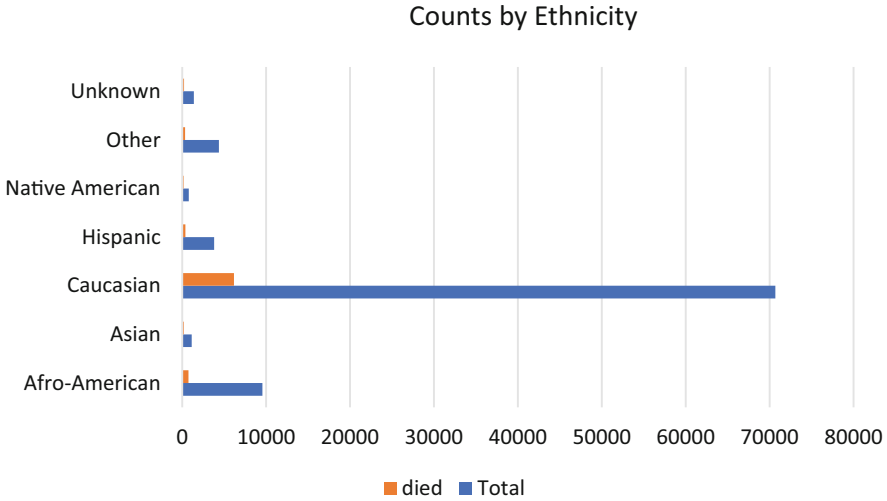
**Fig. 3.7** Bar chart of ICU patient source

**Fig. 3.8** Death rates by ICU unit patient source



**Table 3.6** Patient survival data death rates

| Ethnicity | Total  | Died | Death rate |
|-----------|--------|------|------------|
| Afro      | 9547   | 750  | 0.0786     |
| Asian     | 1129   | 93   | 0.0824     |
| Cauc      | 70,682 | 6168 | 0.0873     |
| Hispan    | 3798   | 376  | 0.0990     |
| Native    | 788    | 70   | 0.0888     |
| Other     | 4374   | 347  | 0.0793     |
| Unknown   | 1395   | 107  | 0.0767     |
| F         | 42,221 | 3731 | 0.0884     |
| M         | 49,469 | 4178 | 0.0845     |



**Fig. 3.9** Bar chart of deaths and total patients by ethnicity

Here  $pbar = 0.086258$ , and  $z = 2.103$ , yielding a probability of  $0.9822$  that there is a difference, which is significant beyond  $0.05$ , the conventional significance cutoff.

A graph might better show relative differences by ethnicity (Fig. 3.9).

In Fig. 3.9, total numbers of patients are nicely displayed. But they are on a much larger scale than deaths, which almost disappear for Unknown, Other, Native American, Hispanic, and Asian. In this case you might need a bar chart for each of the two measures. Died displayed alone is shown in Fig. 3.10.

But Fig. 3.10 doesn't reveal much, due to the varying number of patients by ethnicity (as seen in Fig. 3.9). It is more revealing to manipulate the data to divide deaths by total patients, and plot rates, as in Fig. 3.11:

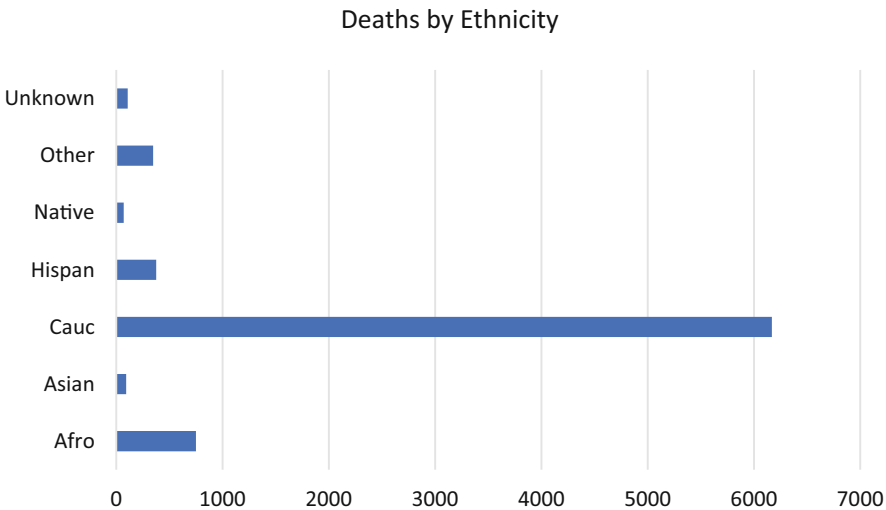
Figure 3.11 shows the relative death rates in a much clearer fashion. Hispanics experience the highest rate—about 25% greater than Afro-Americans.

We can also explore the dataset's contents concerning specific diseases. Table 3.7 extracts patient counts by ethnicity and disease, while Table 3.8 gives death counts:

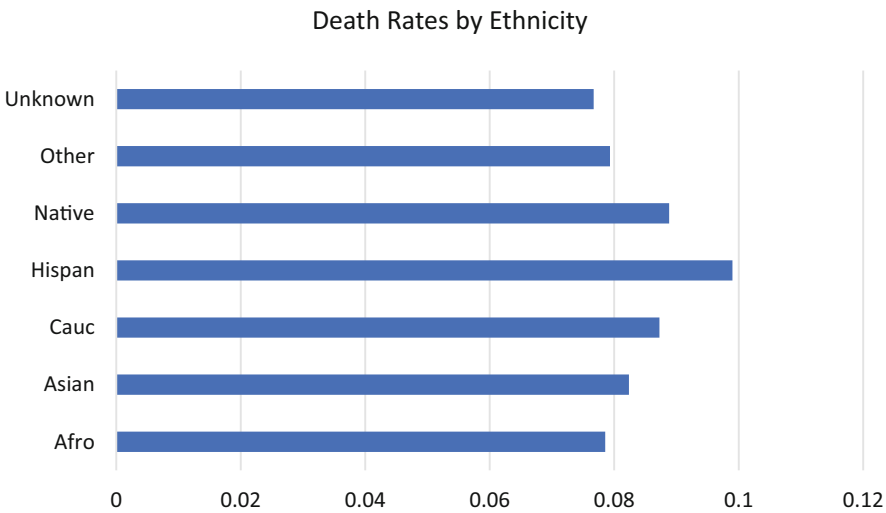
An obvious next step is to identify relative rates of death, given in Table 3.9, which divides Table 3.8 values by Table 3.7 values:

A radar chart (Fig. 3.12) gives a useful tool to compare Table 3.9 data, as the data is on a common scale:

In conjunction with prior figures, we might have clues as to why Hispanics experienced higher death rates in this system. Figure 3.12 shows that they had higher rates of AIDS, lymphoma, and leukemia in this dataset. The point is that by exploring the data, interesting questions can be identified.



**Fig. 3.10** Bar chart of deaths by ethnicity



**Fig. 3.11** Death rates by ethnicity

**3.1.3 Hungarian Chickenpox Data**

The Hungarian chickenpox data is a time series taken from [www.kaggle.com/datasets/rafatashrafjoy/hungarian-chickenpox-cases](https://www.kaggle.com/datasets/rafatashrafjoy/hungarian-chickenpox-cases). It consists of weekly counts of chickenpox cases in each of 20 Hungarian counties over the period 03/01/2015 to

**Table 3.7** Total cases by ethnicity/disease

| Ethnicity | AIDS | cirr | diab   | hepa | Immuno | leuk | lymph | tumor |
|-----------|------|------|--------|------|--------|------|-------|-------|
| Afro      | 114  | 170  | 2539   | 163  | 312    | 143  | 107   | 274   |
| Asian     | 3    | 21   | 301    | 19   | 31     | 9    | 9     | 35    |
| Cauc      | 589  | 1609 | 15,687 | 1437 | 2468   | 1071 | 871   | 2051  |
| Hispan    | 37   | 130  | 975    | 112  | 115    | 62   | 41    | 110   |
| Native    | 7    | 70   | 271    | 61   | 26     | 11   | 9     | 15    |
| Other     | 36   | 130  | 1234   | 94   | 124    | 51   | 45    | 83    |
| Unknown   | 7    | 13   | 200    | 11   | 20     | 11   | 9     | 25    |
| Totals    | 793  | 2143 | 21,207 | 1897 | 3096   | 1358 | 1091  | 2593  |

**Table 3.8** Total deaths by ethnicity/disease

| Ethnicity | AIDS | cirr | diab | hepa | Immuno | leuk | lymph | tumor |
|-----------|------|------|------|------|--------|------|-------|-------|
| Afro      | 9    | 22   | 172  | 17   | 51     | 24   | 10    | 43    |
| Asian     | 0    | 4    | 17   | 3    | 4      | 1    | 2     | 5     |
| Cauc      | 72   | 251  | 1264 | 227  | 376    | 155  | 119   | 342   |
| Hispan    | 9    | 22   | 84   | 23   | 13     | 16   | 10    | 19    |
| Native    | 0    | 19   | 29   | 15   | 1      | 1    | 0     | 2     |
| Other     | 3    | 10   | 100  | 11   | 18     | 5    | 5     | 15    |
| Unknown   | 2    | 5    | 14   | 3    | 6      | 2    | 2     | 6     |
| Totals    | 95   | 333  | 1680 | 299  | 469    | 204  | 148   | 432   |

**Table 3.9** Death rates by ethnicity/disease

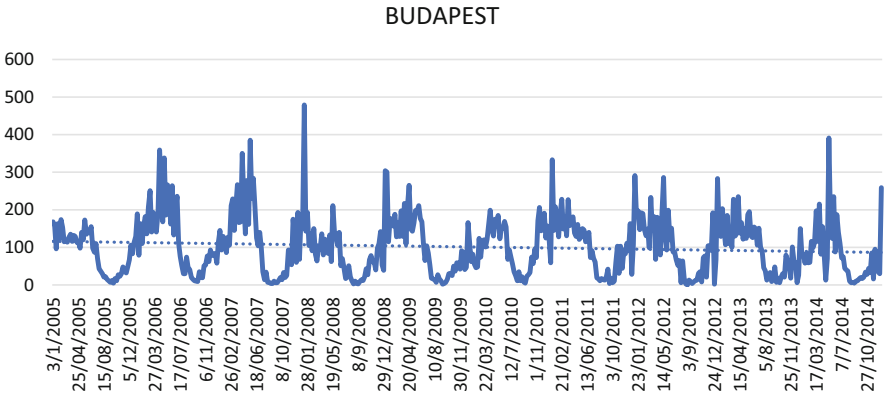
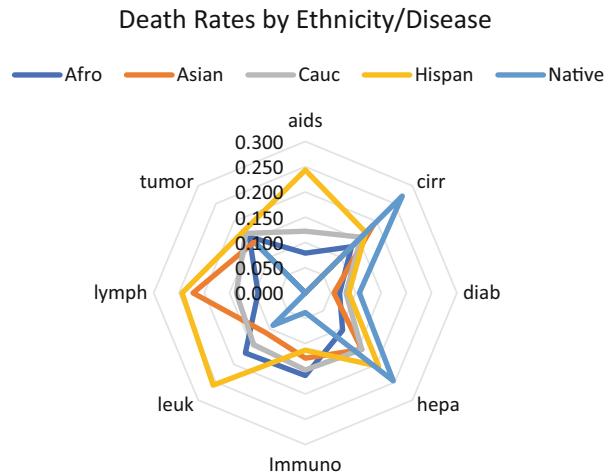
| Ethnicity | AIDS  | cirr  | diab  | hepa  | Immuno | leuk  | lymph | tumor |
|-----------|-------|-------|-------|-------|--------|-------|-------|-------|
| Afro      | 0.079 | 0.129 | 0.068 | 0.104 | 0.163  | 0.168 | 0.093 | 0.157 |
| Asian     | 0.000 | 0.190 | 0.056 | 0.158 | 0.129  | 0.111 | 0.222 | 0.143 |
| Cauc      | 0.122 | 0.156 | 0.081 | 0.158 | 0.152  | 0.145 | 0.137 | 0.167 |
| Hispan    | 0.243 | 0.169 | 0.086 | 0.205 | 0.113  | 0.258 | 0.244 | 0.173 |
| Native    | 0.000 | 0.271 | 0.107 | 0.246 | 0.038  | 0.091 | 0.000 | 0.133 |
| Other     | 0.083 | 0.077 | 0.081 | 0.117 | 0.145  | 0.098 | 0.111 | 0.181 |
| Unknown   | 0.286 | 0.385 | 0.070 | 0.273 | 0.300  | 0.182 | 0.222 | 0.240 |
| Average   | 0.120 | 0.155 | 0.079 | 0.158 | 0.151  | 0.150 | 0.136 | 0.167 |

29/12/2014. The dates were formatted since they contained different datetime format. The source was Rozemberczki et al. (2021):

The first thing one is inclined to do with time series data is plot it over time. Figure 3.13 shows the time series over the entire period for Hungary as a whole:

This data is a good example of time series data with seasonality and trend. Clearly there is a strong cycle, 52 weeks in duration. Inferences drawn from such graphs need to make sense, and a 52-week cycle for a seasonal disease does make sense. There also appears to be a slight trend downward, verified by the dotted trendline superimposed. There are two anomalies—one in late 2007, where there is a spike, and another in mid-2014. (Smaller spikes are also present.) A good analytic

**Fig. 3.12** Radar chart of death rates by ethnicity



**Fig. 3.13** Chickenpox cases by week in Hungary 2005–2014

approach would be to investigate what might have happened to cause those spikes, or other anomalies.

We might also be interested in the relative performance by county. Plotting the time series by week from 2005 through 2014 would be too compressed. So we might focus on year 2005. Figure 3.14 shows an overlay for the year 2005 for all counties.

Figure 3.14 is clearly too cluttered. We can retreat to more focused charts, such as the single county of Budapest shown in Fig. 3.15.

Figure 3.15 is more revealing than Fig. 3.13. Chickenpox is strong through June, then drops off in the summer. About October it picks back up again. A rational inference might be that there is little difference in relative cases for Budapest compared to the rest of the country. But Fig. 3.15 is clearer than Fig. 3.14. More information is not always more illuminating. As far as analytic tools, ARIMA forecasting is a good candidate for modeling and forecasting.

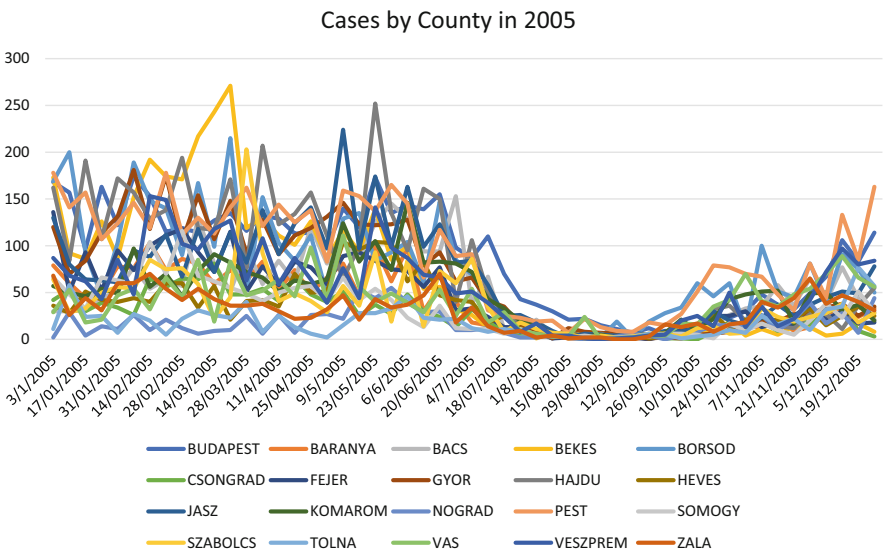


Fig. 3.14 Hungarian chickenpox cases in 2005 overlay

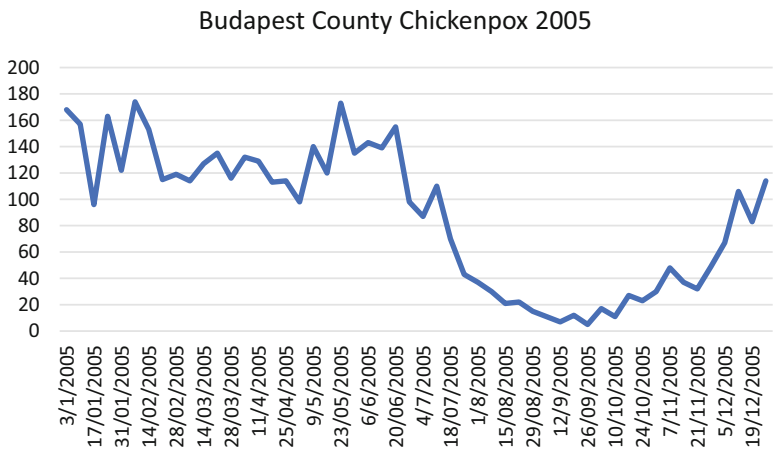


Fig. 3.15 Weekly chickenpox cases—Budapest County 2005

### 3.2 Conclusions

Data visualization is essential initial process in analytics as it provides humans an initial understanding of data. This is of course important in healthcare as it is in every other area of life. We have demonstrated basic MS Excel graphical tools on three healthcare datasets, in the high-level categories of graphs, tables, and charts. Excel

provides some useful basic tools that you as a user can select from to better communicate the story that your data is telling.

Graphic plots can be highly useful in identifying what type of analysis is appropriate. A first consideration is the type of data. Numeric data, as found in the Healthcare Hospital Stay dataset, can be explored by correlation, displayed through scatter charts and/or bar charts. Categorical, or nominal, data (data in words) as found in the Patient Survival dataset, can't be graphed other than by counting. Bar charts are usually appropriate. After manipulation of data, rates can be displayed with radar charts to compare differences. More formal hypothesis analysis can be applied. If numeric data is available over time, time series analysis is an obvious choice. The Hungarian Chickenpox data displayed common features of time series data. Initial exploration is usually supported with line charts.

Some data mining techniques offer other graphics, specific to their analyses. For instance, clustering analysis is supported by discriminant plots. Association rule algorithms may offer network plots showing the relationship between variables.

## References

- Raffa J, Johnson A, Celi LA, Pollard T, Pilcher D, Badawi O (2019) The global open source severity of illness score (GOSSIS). *Crit Care Med* 47(1):17. <https://doi.org/10.1097/01.ccm.0000550825.30295>
- Rozemberczki B, Scherer P, Kiss O, Sarkar R, Ferenci T (2021) Chickenpox cases in Hungary: a benchmark dataset for spatiotemporal signal processing with graph neural networks. <https://archive.ics.uci.edu/ml/datasets/Hungarian+Chickenpox+Cases#>

## Chapter 4

# Association Rules



**Keywords** Association rules · Affinity analysis · Apriori algorithm

Association rules seek to identify combinations of things that frequently occur together (**affinity analysis**). Association rules apply a form of machine learning, the most common of which is the Apriori algorithm. Structured data is saved in fixed fields in databases and traditionally focused on quantitative data that could be analyzed by classical statistics. Data comes in many forms, to include text, video, voice, image, and multimedia. Association rules are typically applied to data in non-quantitative form.

Association rules can provide information that can be useful in a number of ways. In the field of healthcare, many applications can be seen with clinical and population health data, for example they have been used to:

- Assess patterns of multimorbidity between hypertension, cardiovascular disease, and affective disorders in elderly patients;
- Analyze electronic health record databases to identify comorbidities;
- Monitor common diseases in particular areas and age groups to identify shortages of doctors;
- Identify patient problems when monitoring real-time sensor data to analyze abnormality in physiological conditions.

Association rules are of the IF-THEN type. Unfortunately, the more variables (products), the more combinations, which makes algorithms take a lot longer and generate a lot more output to interpret. Some key concepts in association rule mining include coverage (**support**), which is the number of instances occurring in the dataset, and accuracy (**confidence**), correct prediction in terms of the proportion of instances to which the rule applies. The third measure is **Lift**, i.e., the propensity to occur relative to average. Pairs of attributes are called item sets (which in grocery terms could be products purchased together). Association rule analysis seeks item sets above a specified support and confidence. The general approach is to identify item sets with required coverage and turn each into a rule or set of rules with accuracy above another specified level. Association rules are hard to control, i.e.,

one of the unsupervised machine learning techniques—some item sets will not produce any rules, others may produce many.

Association rules can provide information that can be useful in a number of different ways. In the field of marketing, for instance, they can provide:

- Identification of products to be placed together to attract customers for cross-selling;
- Targeting customers through marketing campaigns (coupons, mailings, e-mailings, etc.) seeking to get them to expand the number of products purchased;
- Drive recommendation engines in on-line marketing.

It is machine learning that has proven very useful to Amazon in recommending purchases based upon past purchases, as well as retail stores seeking to attract customers to purchase products based on current purchases. Thus grocery stores locate things like bacon close to eggs and orange juice, or diapers with milk.

Outside of retailing, there are other uses for association rules. Classically, they were applied to retail transaction analysis, akin to market basket analysis. With the emergence of big data, the ability to apply association rules to streams of real-time data is highly useful, enabling a great deal of Web mining for many applications, including e-business retail sales. Association rule mining is one of the most widely used data mining techniques. This can be applied to target marketing, by customer profile, space allocation strategy within stores, but can also be extended to business applications such as international trade and stock market prediction. In science and engineering applications, remotely sensed imagery data has been analyzed to aid precision agriculture and resource discovery (to include oil). It has been used in manufacturing to analyze yield in semiconductor manufacturing. It has been used to improve efficiency of packet routing over computer networks. In medicine it has been used for diagnosis of diseases. They also could be used in human resources management and other places where pairing behavior with results is of interest (Aguinis et al. 2013). In this chapter, we will focus on applications in healthcare, to follow the theme of the book.

## 4.1 The Apriori Algorithm

The apriori algorithm is credited to Agrawal et al. (1993) who applied it to market basket data to generate association rules. Association rules are usually applied to binary data, which fits the context where customers either purchase or don't purchase particular products. The apriori algorithm operates by systematically considering combinations of variables, and ranking them on either support, confidence, or lift at the user's discretion.

The apriori algorithm operates by finding all rules satisfying minimum confidence and support specifications. First, the set of frequent 1-itemsets is identified by scanning the database to count each item. Next, 2-itemsets are identified, gaining some efficiency by using the fact that if a 1-itemset is not frequent, it can't be part of

a frequent itemset of larger dimension. This continues to larger-dimensioned item sets until they become null. The magnitude of effort required is indicated by the fact that each dimension of item sets requires a full scan of the database. The algorithm is:

```

To identify the candidate itemset  $C_k$  of size  $k$ 
1. Identify frequent items  $L_1$ 
For  $k = 1$  generate all itemsets with support  $\geq \text{Support}_{\min}$ 
If itemsets null, STOP
Increment  $k$  by 1
For itemsets of size  $k$  identify all with support  $\geq \text{Support}_{\min}$ 
END
2. Return list of frequent itemsets
3. Identify rules in the form of antecedents and consequents from the
frequent items
4. Check confidence of these rules
If confidence of a rule meets  $\text{Confidence}_{\min}$  mark this rule as strong

```

The output of the apriori algorithm can be used as the basis for recommending rules, considering factors such as correlation, or analysis from other techniques, from a training set of data. This information may be used in many ways, including in retail where if a rule is identified indicating that purchase of the antecedent occurred without that customer purchasing the consequent, then it might be attractive to suggest purchase of the consequent.

The apriori algorithm can generate many frequent itemsets. Association rules can be generated by only looking at frequent itemsets that are strong, in the sense that they meet or exceed both minimum support and minimum confidence levels. It must be noted that this does not necessarily mean such a rule is useful, that it means high correlation, nor that it has any proof of causality. However, a good feature is that you can let computers loose to identify them (an example of machine learning).

### 4.1.1 Association Rules from Software

The R statistical programming software allows setting support and confidence levels, as well as the minimum rule length. It has other options as well. We will set support and confidence (as well as lift, which is an option for sorting output) below. The data needs to be put into a form the software will read. In Rattle, that requires data be categorical rather than numerical. The rules generated will be positive cases (IF you buy diapers, THEN you are likely to buy baby powder) and negative cases are ignored (IF you **didn't** buy diapers, THEN you are likely to do whatever). If you wish to study the negative cases, you would need to convert the blank cases to No. Here we will demonstrate the positive case.

Association rule mining seeks all rules satisfying specified minimum levels. Association rules in R and WEKA require nominal data.

### ***4.1.2 Non-negative Matrix Factorization***

There are advanced methods applied to association rule generation. Non-negative matrix factorization (NMF) was proposed by Lee and Seung (1999) as a means to distinguish parts of data for facial recognition as well as text analysis. Principal component analysis and vector quantization learn holistically rather than breaking down data into parts. These methods construct factorizations of the data. For instance, if there is a set of customers  $N$  and a set of products  $M$ , a matrix  $V$  can be formed where each row of  $V$  represents a market basket with one customer purchasing products. This can be measured in units or in dollars. Association rules seek to identify ratio rules identifying the most common pairings. Association rule methods, be they principal component analysis or other forms of vector quantization, minimize dissimilarity between vector elements. Principal components allow for negative associations, which in the context of market baskets does not make sense. NMF imposes non-negativity constraints into such algorithms.

## **4.2 Methodology**

Association rules deal with **items**, which are the objects of interest. We will demonstrate using patient survival data taken from [Kaggle.com](https://www.kaggle.com) (original source: [https://journals.lww.com/ccmjournal/Citation/2019/01001/33\\_THE\\_GLOBAL\\_OPEN\\_SOURCEW\\_SEVERITY\\_OF\\_ILLNESS.36.aspx](https://journals.lww.com/ccmjournal/Citation/2019/01001/33_THE_GLOBAL_OPEN_SOURCEW_SEVERITY_OF_ILLNESS.36.aspx)). On [www.Kaggle.com](https://www.kaggle.com), it is the Patient Survival Prediction Dataset.

### ***4.2.1 Demonstration with Kaggle Data***

Intensive Care Units (ICUs) often lack verified medical histories for incoming patients. A patient in distress or a patient who is brought in confused or unresponsive may not be able to provide information about chronic conditions such as heart disease, injuries, or diabetes. Medical records may take days to transfer, especially for a patient from another medical provider or system. Knowledge about chronic conditions can inform clinical decisions about patient care and ultimately improve patient's survival outcomes.

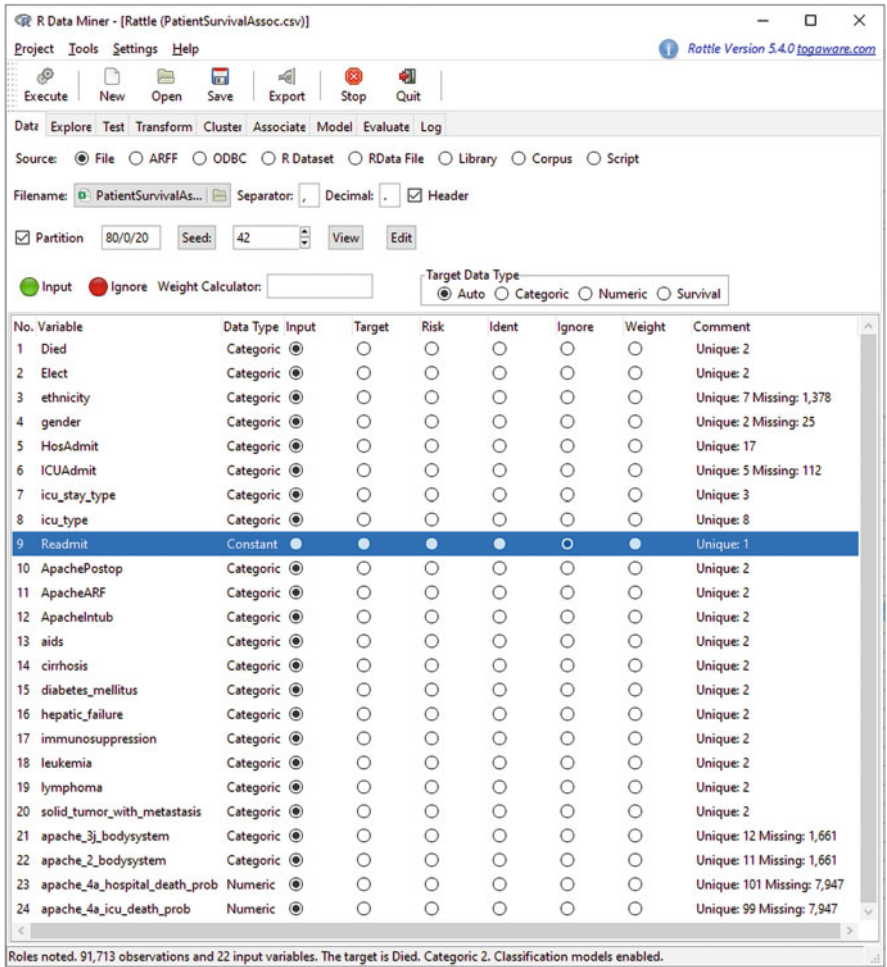


Fig. 4.1 Rattle screen for patient survival data

This data is shown in the Rattle screen (Fig. 4.1):

Note that variable Readmit had only one value, so Rattle disregarded it. For association rules we have no target. The last two variables were continuous and are also ignored as association rules need alphabetic data. This leaves 21 categorical variables. Association rules have a dimensionality issue with respect to the number of variables. Table 4.1 shows the number of rules obtained using 0.1 confidence and various levels of support:

The first 20 rules (of 1217) for support of 0.9 and confidence 0.1 are shown in Table 4.2.

**Table 4.1** Rules versus support for 21 categorical variables

| Support Level | Rules Obtained |
|---------------|----------------|
| 0.9           | 1217           |
| 0.8           | 5147           |
| 0.7           | 20,146         |
| 0.6           | 60,935         |
| 0.5           | 147,757        |
| 0.4           | 320,797        |
| 0.3           | 838,256        |
| 0.2           | 2,214,368      |
| 0.1           | 5,117,030      |
| 0.01          | 6,630,968      |

Table 4.2 shows the kind of results that can be generated. Note that all nine rules have no as the result for both the antecedent and the consequent. This may have some value—in the medical context, it might be useful to know that if you were diagnosing for AIDS, the presence of lymphoma might alleviate concern for AIDS (although lymphoma isn’t a good condition either). Weeding out positive results is sometimes what is needed and can cause quite a bit of work to isolate.

Note that all of the conditions in Table 4.2 are “no.” There may be some reason for being interested, but most of the useful rules have some “yes” conditions. The inference of rule [1] in Table 4.2 is that if you don’t have lymphoma, you don’t have AIDS, with a confidence of 0.9992. But two negatives aren’t usually interesting. “Yes” results tend to be rare, and thus at the bottom of the list, and as Table 4.1 shows, that list can be inordinately long. Therefore, it is wise to limit the number of variables. Table 4.3 shows the number of rules obtained using four variables: [1] Died; [2] Elective; and various diseases taken one at a time.

In all cases, the rules containing “yes” were at the bottom of the list (by support). Table 4.4 gives all 16 rules for AIDS using 3 variables (Died, Elective, and AIDS):

The inference of rule [1] is that if the patient lived through the hospitalization, they didn’t have AIDS at 0.9917 confidence. Rule [2] gives a 0.9140 confidence that patients without AIDS lived through the hospitalization.

For the next series, the data was partitioned randomly assigning 80% of the 91,713 observations (64,199 observations) to the training set, holding the other 17,514 for testing.

Figure 4.2 displays the association rule screen in Rattle.

Rattle uses the R arule procedure. In Fig. 4.2, the minimum support was set to 0.97, as was the minimum confidence level. These are very high but lowering them yields many rules. Usually applying association rules requires experimenting with a number of support/confidence settings. Here confidence was not a factor with the support levels used.

**Table 4.2** Rules for initial run

|      | Lhs                              |     | Rhs                  | Support | Confidence | Coverage | Lift   | Count  |
|------|----------------------------------|-----|----------------------|---------|------------|----------|--------|--------|
| [1]  | {lymphoma=no}                    | = > | {aids=no}            | 0.9873  | 0.9992     | 0.9881   | 1.0079 | 90,548 |
| [2]  | {aids=no}                        | = > | {lymphoma=no}        | 0.9873  | 0.9959     | 0.9914   | 1.0079 | 90,548 |
| [3]  | {leukemia=no}                    | = > | {aids=no}            | 0.9843  | 0.9991     | 0.9852   | 1.0079 | 90,277 |
| [4]  | {aids=no}                        | = > | {leukemia=no}        | 0.9843  | 0.9929     | 0.9914   | 1.0079 | 90,277 |
| [5]  | {leukemia=no}                    | = > | {lymphoma=no}        | 0.9813  | 0.9960     | 0.9852   | 1.0080 | 89,997 |
| [6]  | {lymphoma=no}                    | = > | {leukemia=no}        | 0.9813  | 0.9931     | 0.9881   | 1.0080 | 89,997 |
| [7]  | {leukemia=no,lymphoma=no}        | = > | {aids=no}            | 0.9805  | 0.9992     | 0.9813   | 1.0079 | 89,923 |
| [8]  | {aids=no,leukemia=no}            | = > | {lymphoma=no}        | 0.9805  | 0.9961     | 0.9843   | 1.0081 | 89,923 |
| [9]  | {aids=no,lymphoma=no}            | = > | {leukemia=no}        | 0.9805  | 0.9931     | 0.9873   | 1.0080 | 89,923 |
| [10] | {hepatic_failure=no}             | = > | {aids=no}            | 0.9785  | 0.9992     | 0.9793   | 1.0079 | 89,740 |
| [11] | {aids=no}                        | = > | {hepatic_failure=no} | 0.9785  | 0.9870     | 0.9914   | 1.0079 | 89,740 |
| [12] | {cirrhosis=no}                   | = > | {aids=no}            | 0.9758  | 0.9992     | 0.9766   | 1.0079 | 89,497 |
| [13] | {aids=no}                        | = > | {cirrhosis=no}       | 0.9758  | 0.9843     | 0.9914   | 1.0079 | 89,497 |
| [14] | {hepatic_failure=no}             | = > | {lymphoma=no}        | 0.9753  | 0.9959     | 0.9793   | 1.0079 | 89,446 |
| [15] | {lymphoma=no}                    | = > | {hepatic_failure=no} | 0.9753  | 0.9870     | 0.9881   | 1.0079 | 89,446 |
| [16] | {hepatic_failure=no,lymphoma=no} | = > | {aids=no}            | 0.9745  | 0.9992     | 0.9753   | 1.0079 | 89,374 |
| [17] | {aids=no,hepatic_failure=no}     | = > | {lymphoma=no}        | 0.9745  | 0.9959     | 0.9785   | 1.0079 | 89,374 |
| [18] | {aids=no,lymphoma=no}            | = > | {hepatic_failure=no} | 0.9745  | 0.9870     | 0.9873   | 1.0079 | 89,374 |
| [19] | {cirrhosis=no}                   | = > | {lymphoma=no}        | 0.9726  | 0.9959     | 0.9766   | 1.0079 | 89,201 |
| [20] | {lymphoma=no}                    | = > | {cirrhosis=no}       | 0.9726  | 0.9843     | 0.9881   | 1.0079 | 89,201 |

**Table 4.3** Rules versus variables by disease

| Disease           | 3 Variables Total | 3 Variables Yes | 4 Variables Total | 4 Variables Yes |
|-------------------|-------------------|-----------------|-------------------|-----------------|
| AIDS              | 16                | 6               | 66                | 19              |
| Cirrhosis         | 16                | 7               | 66                | 19              |
| Diabetes mellitus | 23                | 14              | 76                | 29              |
| Hepatic failure   | 16                | 7               | 66                | 19              |
| Immunosuppression | 16                | 7               | 62                | 15              |
| Leukemia          | 16                | 7               | 66                | 19              |
| Lymphoma          | 16                | 7               | 66                | 19              |
| Tumor metastasis  | 16                | 7               | 62                | 15              |

Experimenting with different levels of support yielded the following results:

Support 0.99, Confidence 0.95, 0 rules

Support 0.98, Confidence 0.95, 9 rules

Support 0.97, Confidence 0.95, 35 rules

Support 0.96, Confidence 0.95, 105 rules

Support 0.95, Confidence 0.95, 193 rules

Table 4.5 displays the rules obtained at the 0.98 support level:

Rattle has a Plot button that yields Fig. 4.3 for the 0.98 Support rule set:

Lowering support to 0.97 yielded more rules, as shown in Table 4.6:

Note that the first nine rules were identical, and the Support column makes it easy to see how the rule set was expanded. Other parameters might be correlated with Support, but clearly are different.

### 4.2.2 Analysis with Excel

Association rules are a machine learning means to initially explore data. Deeper analysis requires human intervention. For instance, we can sort the data file with Excel and glean pertinent information relative to what we are interested in. If we wanted to know the proportion of patients in this dataset that died by disease, digging through the association rules would take far too long. Excel sorting yields Table 4.7:

Note that there are many deaths here not accounted for by the eight diseases listed, and of those eight, there were many comorbidities.

**Table 4.4** Rules for three variables for AIDS

|      | Lhs                 | Rhs         | Support | Confidence | Coverage | Lift   | Count  |
|------|---------------------|-------------|---------|------------|----------|--------|--------|
| [1]  | {Died=no}           | {aids=no}   | 0.9061  | 0.9917     | 0.9137   | 1.0003 | 83,100 |
| [2]  | {aids=no}           | {Died=no}   | 0.9061  | 0.9140     | 0.9914   | 1.0003 | 83,100 |
| [3]  | {Elect=no}          | {aids=no}   | 0.8077  | 0.9895     | 0.8163   | 0.9981 | 74,077 |
| [4]  | {aids=no}           | {Elect=no}  | 0.8077  | 0.8147     | 0.9914   | 0.9981 | 74,077 |
| [5]  | {Elect=no}          | {Died=no}   | 0.7356  | 0.9012     | 0.8163   | 0.9864 | 67,468 |
| [6]  | {Died=no}           | {Elect=no}  | 0.7356  | 0.8051     | 0.9137   | 0.9864 | 67,468 |
| [7]  | {Died=no,Elect=no}  | {aids=no}   | 0.7281  | 0.9898     | 0.7356   | 0.9984 | 66,778 |
| [8]  | {Elect=no,aids=no}  | {Died=no}   | 0.7281  | 0.9015     | 0.8077   | 0.9866 | 66,778 |
| [9]  | {Died=no,aids=no}   | {Elect=no}  | 0.7281  | 0.8036     | 0.9061   | 0.9845 | 66,778 |
| [10] | {Elect=yes}         | {aids=no}   | 0.1836  | 0.9995     | 0.1837   | 1.0082 | 16,843 |
| [11] | {aids=no}           | {Elect=yes} | 0.1836  | 0.1853     | 0.9914   | 1.0082 | 16,843 |
| [12] | {Elect=yes}         | {Died=no}   | 0.1781  | 0.9691     | 0.1837   | 1.0606 | 16,330 |
| [13] | {Died=no}           | {Elect=yes} | 0.1781  | 0.1949     | 0.9137   | 1.0606 | 16,330 |
| [14] | {Died=no,Elect=yes} | {aids=no}   | 0.1780  | 0.9995     | 0.1781   | 1.0082 | 16,322 |
| [15] | {Elect=yes,aids=no} | {Died=no}   | 0.1780  | 0.9691     | 0.1836   | 1.0606 | 16,322 |
| [16] | {Died=no,aids=no}   | {Elect=yes} | 0.1780  | 0.1964     | 0.9061   | 1.0690 | 16,322 |

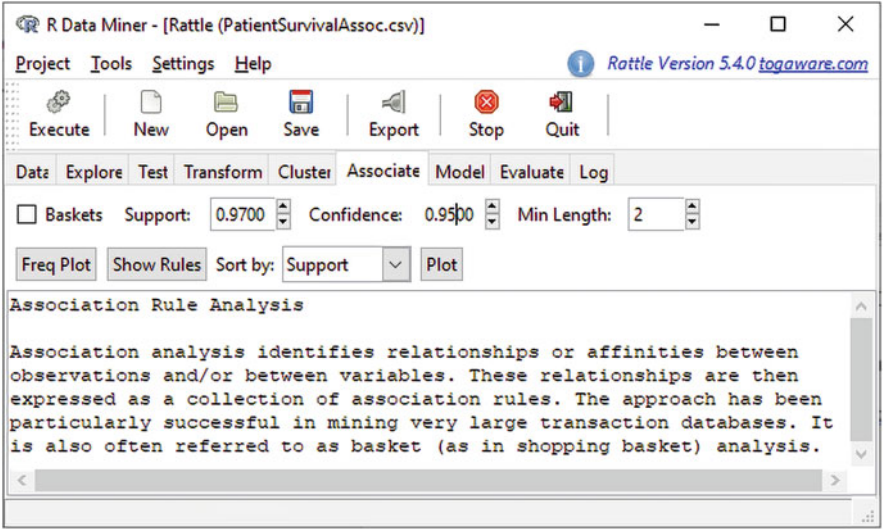


Fig. 4.2 Rattle association rule screen

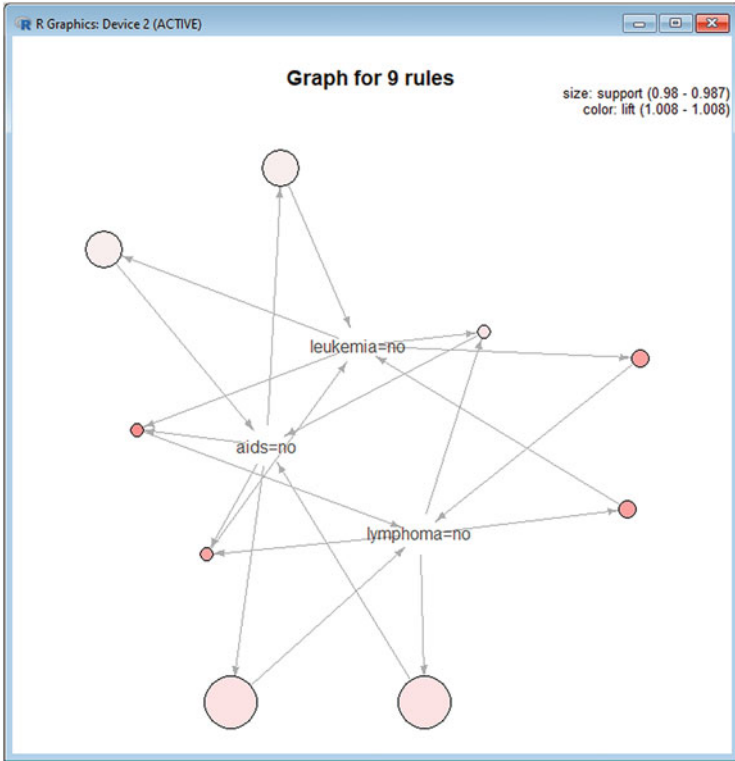
Table 4.5 Rules using 0.98 support

| Antecedent                    | Consequent    | Support | Confidence | Coverage | Lift  | Count  |
|-------------------------------|---------------|---------|------------|----------|-------|--------|
| {lymphoma=no}                 | {aids=no}     | 0.987   | 0.999      | 0.988    | 1.008 | 72,422 |
| {aids=no}                     | {lymphoma=no} | 0.987   | 0.996      | 0.991    | 1.008 | 72,422 |
| {leukemia=no}                 | {aids=no}     | 0.984   | 0.999      | 0.985    | 1.008 | 72,216 |
| {aids=no}                     | {leukemia=no} | 0.984   | 0.993      | 0.991    | 1.008 | 72,216 |
| {leukemia=no}                 | {lymphoma=no} | 0.981   | 0.996      | 0.985    | 1.008 | 71,984 |
| {lymphoma=no}                 | {leukemia=no} | 0.981   | 0.993      | 0.988    | 1.008 | 71,984 |
| {leukemia=no,<br>lymphoma=no} | {aids=no}     | 0.980   | 0.999      | 0.981    | 1.008 | 71,923 |
| {aids=no,<br>leukemia=no}     | {lymphoma=no} | 0.980   | 0.996      | 0.984    | 1.008 | 71,923 |
| {aids=no,<br>lymphoma=no}     | {leukemia=no} | 0.980   | 0.993      | 0.987    | 1.008 | 71,923 |

4.3 Review of Applications

4.3.1 Korean Healthcare Study

Today many documents in public health (as well as other areas) are digitized. The Internet of Things links data from wearable and smarty devices, yielding a massive amount of data in addition to the vast amounts from electronic medical records and personal health records. This enables text mining to classify, cluster, extract, search, and analyze data for patterns using less structured natural language documents. Kim



**Fig. 4.3** Plot output for 0.98 support rules

and Chung (2019) presented a method for association mining of healthcare big data drawn from the Korean Health Insurance Review & Assessment Service. This data includes information on medical resources, cost, and drugs. The first step was Web scraping, followed by data cleaning to include stop-word removal, tagging, and classification of words that have multiple meanings. The method used was term frequency applied to terms in a common theme applying inverse document frequency (TF-C-IDF). With this system word importance decreased if there were many documents using a common theme and thus the same words. Then word importance was identified, yielding a set of keywords. The Apriori algorithm was applied to the resulting database.

The first step was to gather raw data from health documents, followed by preprocessing. Ten thousand health documents were extracted from HTML5-based URLs. Of these, 2191 were excluded as having low relevance and low confidence. Of the remaining 7809 documents, 1000 were reserved as a test set, leaving a training set of 6809 documents. The training set Web pages were scraped using the *rvest* package in R using version R 3.4.1. Keywords were identified using frequency. A candidate corpus was created using the R studio *tm* package. This included a stop-word dictionary of 174 stop words such as “me,” “be,” “do,” etc.

**Table 4.6** Rules using 0.97 support

| Antecedent                        | Consequent           | Support | Confidence | Coverage | Lift  | Count  |
|-----------------------------------|----------------------|---------|------------|----------|-------|--------|
| {lymphoma=no}                     | {aids=no}            | 0.987   | 0.999      | 0.988    | 1.008 | 72,422 |
| {aids=no}                         | {lymphoma=no}        | 0.987   | 0.996      | 0.991    | 1.008 | 72,422 |
| {leukemia=no}                     | {aids=no}            | 0.984   | 0.999      | 0.985    | 1.008 | 72,216 |
| {aids=no}                         | {leukemia=no}        | 0.984   | 0.993      | 0.991    | 1.008 | 72,216 |
| {leukemia=no}                     | {lymphoma=no}        | 0.981   | 0.996      | 0.985    | 1.008 | 71,984 |
| {lymphoma=no}                     | {leukemia=no}        | 0.981   | 0.993      | 0.988    | 1.008 | 71,984 |
| {leukemia=no, lymphoma=no}        | {aids=no}            | 0.980   | 0.999      | 0.981    | 1.008 | 71,923 |
| {aids=no, leukemia=no}            | {lymphoma=no}        | 0.980   | 0.996      | 0.984    | 1.008 | 71,923 |
| {aids=no, lymphoma=no}            | {leukemia=no}        | 0.980   | 0.993      | 0.987    | 1.008 | 71,923 |
| {hepatic_failure=no}              | {aids=no}            | 0.978   | 0.999      | 0.979    | 1.008 | 71,790 |
| {aids=no}                         | {hepatic_failure=no} | 0.978   | 0.987      | 0.991    | 1.008 | 71,790 |
| {cirrhosis=no}                    | {aids=no}            | 0.976   | 0.999      | 0.977    | 1.008 | 71,601 |
| {aids=no}                         | {cirrhosis=no}       | 0.976   | 0.984      | 0.991    | 1.008 | 71,601 |
| {hepatic_failure=no}              | {lymphoma=no}        | 0.975   | 0.996      | 0.979    | 1.008 | 71,543 |
| {lymphoma=no}                     | {hepatic_failure=no} | 0.975   | 0.987      | 0.988    | 1.008 | 71,543 |
| {hepatic_failure=no, lymphoma=no} | {aids=no}            | 0.974   | 0.999      | 0.975    | 1.008 | 71,483 |
| {aids=no, hepatic_failure=no}     | {lymphoma=no}        | 0.974   | 0.996      | 0.978    | 1.008 | 71,483 |
| {aids=no, lymphoma=no}            | {hepatic_failure=no} | 0.974   | 0.987      | 0.987    | 1.008 | 71,483 |
| {cirrhosis=no}                    | {lymphoma=no}        | 0.973   | 0.996      | 0.977    | 1.008 | 71,353 |
| {lymphoma=no}                     | {cirrhosis=no}       | 0.973   | 0.984      | 0.988    | 1.008 | 71,353 |
| {hepatic_failure=no}              | {leukemia=no}        | 0.972   | 0.993      | 0.979    | 1.008 | 71,341 |
| {leukemia=no}                     | {hepatic_failure=no} | 0.972   | 0.987      | 0.985    | 1.008 | 71,341 |
| {cirrhosis=no, lymphoma=no}       | {aids=no}            | 0.972   | 0.999      | 0.973    | 1.008 | 71,294 |
| {aids=no, cirrhosis=no}           | {lymphoma=no}        | 0.972   | 0.996      | 0.976    | 1.008 | 71,294 |
| {aids=no, lymphoma=no}            | {cirrhosis=no}       | 0.972   | 0.984      | 0.987    | 1.008 | 71,294 |
| {hepatic_failure=no, leukemia=no} | {aids=no}            | 0.971   | 0.999      | 0.972    | 1.008 | 71,278 |

|                                   |                             |       |       |       |       |        |
|-----------------------------------|-----------------------------|-------|-------|-------|-------|--------|
| {aids=no,hepatic_failure=no}      | {leukemia=no}               | 0.971 | 0.993 | 0.978 | 1.008 | 71,278 |
| {aids=no,leukemia=no}             | {hepatic_failure=no}        | 0.971 | 0.987 | 0.984 | 1.008 | 71,278 |
| {cirrhosis=no}                    | {hepatic_failure=no}        | 0.971 | 0.995 | 0.977 | 1.016 | 71,272 |
| {hepatic_failure=no}              | {cirrhosis=no}              | 0.971 | 0.992 | 0.979 | 1.016 | 71,272 |
| {solid_tumor_with_metastasis=no}  | {aids=no}                   | 0.971 | 0.999 | 0.972 | 1.008 | 71,222 |
| {aids=no}                         | {solid_tumor_metastasis=no} | 0.971 | 0.979 | 0.991 | 1.008 | 71,222 |
| {cirrhosis=no,hepatic_failure=no} | {aids=no}                   | 0.971 | 0.999 | 0.971 | 1.008 | 71,211 |
| {aids=no,cirrhosis=no}            | {hepatic_failure=no}        | 0.971 | 0.995 | 0.976 | 1.016 | 71,211 |
| {aids=no,hepatic_failure=no}      | {cirrhosis=no}              | 0.971 | 0.992 | 0.978 | 1.016 | 71,211 |

**Table 4.7** Death rates from Excel

| Disease           | Total  | Died | Ratio |
|-------------------|--------|------|-------|
| AIDS              | 793    | 95   | 0.120 |
| Cirrhosis         | 2143   | 333  | 0.155 |
| Diabetes mellitus | 21,207 | 1680 | 0.079 |
| Hepatic failure   | 1897   | 299  | 0.158 |
| Immunosuppression | 3096   | 469  | 0.151 |
| Leukemia          | 1358   | 204  | 0.150 |
| Lymphoma          | 1091   | 148  | 0.136 |
| Tumor metastasis  | 2593   | 432  | 0.167 |
|                   | 91,713 | 7915 | 0.086 |

Stop words were removed from the corpus. The candidate corpus was the set of remaining words (all passing minimum support specified at 2) sorted by frequency by document, assuming that more frequent words were more important. To eliminate commonly used but unimportant words, TF-IDF was applied. IDF was the inverse number of the rate of documents in which a word was found at least once. If  $tf(x,y)$  is the frequency of word  $x$  in document  $y$ ,  $N$  is the size of the collected document set, and  $df_x$  is the number of documents in which word  $x$  is found at least once.

$$idf(x,y) = \left( \log \frac{N}{df_x} \right)$$

$$TF-IDF(x,y) = tf(x,y) \times idf(x,y)$$

The weight of word  $x$  that is scanned  $n$  times was calculated:

$$w_x = \left( 1 + \frac{t_{x,corpus}}{N} \right)$$

TF-C-IDF( $x,y$ ) is  $tf(x,y)$  times weight times IDF.

$$TF-C-IDF = tf(x,y) \times \left( 1 + \frac{t_x}{N} \right) \times \log \frac{N}{df_x}$$

The higher TF-C-IDF, the more important the word is. Thus Kim and Chung (2019) used this variable as the basis for identifying how important that word is. Health transactions were saved using .csv format for association analysis with the Apriori algorithm. Example rules included:

$$IF\{fatigue\&\&insomnia\} THEN \{depression\}$$

The consequent terms were then ranked by TF-C-IDF to focus on keyword importance, with associated antecedent terms from rules. For instance, the term {depression} was associated with antecedents {fatigue&&insomnia},

**Table 4.8** Shi et al. (2021) data for multimorbidity

| Group          | Total Number | Multimorbid Patients | Multimorbidity |
|----------------|--------------|----------------------|----------------|
| Male           | 45,796       | 29,712 (45.1%)       | 64.9%          |
| Female         | 52,836       | 36,227 (54.9%)       | 68.6%          |
| 2015 Age Group |              |                      |                |
| 40–49          | 13,400       | 6719 (10.2%)         | 50.1%          |
| 50–59          | 22,412       | 13,017 (19.7%)       | 58.1%          |
| 60–74          | 29,764       | 20,482 (31.1%)       | 68.8%          |
| ≥75            | 33,056       | 25,721 (39.0%)       | 77.8%          |

{fatigue&&mental}, and {mental}. Thus if the term depression is present for a case, the related maladies can be inferred. Kim and Chung (2019) evaluated the models using F-measure and efficiency of methods using simply TF, compared to TF-IDF, and TF-C-IDF. Efficiency was defined as:

$$\text{efficiency} = \frac{W_n - \text{Stop}W_n}{W_n} \times 100$$

where  $W_n$  is the total number of keywords and  $\text{Stop}W_n$  is the count of stop words involved in the extracted associative feature information. Precision, recall, F-measure, and efficiency improved consistently moving from TF to TF-IDF to TF-C-IDF.

### 4.3.2 Belgian Comorbidity Study

A Belgian database of patients with 100 chronic conditions was extracted from general practitioner Electronic Health Records (EHR) (Shi et al. 2021). The focus was on patients over 40 years of age with multiple diagnoses between 1991 and 2015. There were 65,939 such cases. The intent was to identify more than one chronic condition. About 67% of the patients had multimorbidity.

Markov chains were applied to identify probabilities of suffering a condition after experiencing another. Weighted association rule mining was applied to identify the strongest pairs, allowing the sequence of morbidities to be modeled. In traditional association rule mining, if cases with low frequency have a high comorbidity, they will rank high in the list of rules. But they are uninteresting due to their low frequency. Weighted association rule mining weights co-occurrence with the same items differently if the sequence changes.

The Intego database from the Flanders region (Truyers et al. 2014) contains longitudinal data of about 300,000 patients, a representative 2.3% of the Flemish population. Those patients 40 years of age and older were analyzed. Of those patients with multimorbidity, the average duration between first and last diagnosis

**Table 4.9** Shi et al. (2021) association rules

| Antecedents                    | Consequents                      | Support | Confidence | Lift |
|--------------------------------|----------------------------------|---------|------------|------|
| Suicide/suicide attempt        | Depressive disorder              | 0.00139 | 0.505      | 3.38 |
| Retinopathy                    | Diabetes non-insulin dependent   | 0.00236 | 0.521      | 2.92 |
| Retinopathy & hypertension     | Diabetes non-insulin dependent   | 0.00129 | 0.476      | 2.67 |
| Anxiety disorder/anxiety state | Depressive disorder              | 0.00373 | 0.297      | 1.99 |
| Acquired deformity of spine    | Back syndrome w/o radiating pain | 0.00106 | 0.120      | 1.99 |
| Somatization disorder          | Depressive disorder              | 0.00268 | 0.264      | 1.77 |
| Somatization disorder          | Irritable bowel syndrome         | 0.00136 | 0.134      | 1.74 |
| Rheumatoid/seropositive arth   | Osteoarthritis other             | 0.00207 | 0.193      | 1.70 |
| Dermatitis contact/allergic    | Dermatitis/atopic eczema         | 0.00521 | 0.136      | 1.65 |
| Diabetes insulin dependent     | Diabetes non-insulin dependent   | 0.00155 | 0.292      | 1.64 |
| Chronic alcohol abuse          | Depressive disorder              | 0.00405 | 0.231      | 1.55 |
| Chronic bronchitis             | Asthma                           | 0.00132 | 0.153      | 1.51 |

was 8.29 years. Diagnosis dates were used to determine sequence. Table 4.8 gives multimorbidity data:

As expected, patient count and multimorbidity rates both increased with age. The database contains all coded data registered in general practices. It contains clinical parameters, laboratory tests, disease diagnoses, and prescriptions. Along with medical diagnosis, data included complaints, lifestyle factors, and risk factors. The data was coded by four general practitioners and an epidemiologist, classifying cases as acute or chronic (expected duration of the years or longer). This yielded 105 chronic conditions (see Table 4.9 for examples). Markov chain models were applied to study sequences of condition development. Table 4.9 gives the resulting rules with lift  $\geq 1.5$  obtained from the weighted association rules mining.

A visualization in the form of a heatmap (transition probability matrix using Markov chain output) was developed. This graphically identified patterns of high prevalence. Chronic conditions with the highest prevalence were hypertension, depressive disorder, diabetes, and lipid disorder.

The apriori algorithm generates a large number of rules and does not guarantee efficiency and value of the knowledge created. Sornalakshmi et al. (2020) presented a method based on sequential minimal optimization combined with enhancement based on context ontology in the form of a hierarchical structure of the conceptual clusters of domain knowledge. The apriori algorithm does not identify transactions with identical itemsets, thus consuming unnecessary resources. The sequential minimal optimization regression can spot anomalies in physiological parameters, thus reducing cost by identifying patients more at risk and enabling earlier treatment before complications arise.

They were dealing with an application to analyze wireless medical sensors attached to patient bodies that collected physiological parameters in order to identify

physiological condition abnormalities. This information was provided to physicians, nurses, and caretakers of non-emergency patients in their homes. Raw input data was transmitted to a data repository, which was analyzed with association rule mining. An ontology was used to state context. Example ontology structure elements included rules such as:

If blood pressure, heart rate, pulse rate, respiratory level, or oxygen saturation exceeded a threshold value, abnormality is detected

The enhanced apriori algorithm found frequent itemsets directly, eliminating the infrequent subsets generated by the standard apriori algorithm. A sequential minimal optimization regression algorithm was used to predict abnormality detection, splitting the large number of potential rules into a smaller set.

## 4.4 Conclusion

Association rules are very useful in the form of providing a machine learning mechanism to deal with the explosion of large datasets and even big data. This can be for good or bad, as it is the case in any data mining application. Real-time automatic trading algorithms have caused damage in stock markets, for instance. However, they provide great value not only to retail analysis (to serve customers better), but also in the medical field to aid in diagnosis, in agriculture and manufacturing to suggest greater efficient operations, and in science to establish expected relationships in complex environments.

Implementing association rules is usually done through the apriori algorithm, although refinements have been produced. This requires software for implementation, although that is available in most data mining tools, commercial or open source. The biggest problem with association rules seems to be sorting through the output to find interesting results.

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# Chapter 5

## Cluster Analysis



**Keywords** Cluster analysis · Algorithms · Rattle software

The idea of clustering is to group the data into sets that are distinct from each other. Having the data points within clusters to be similar, or close to each other in the data space is desired. On the other hand, clusters are wanted to be dissimilar or have large distances between them. Accomplishing that is quite arbitrary, however, and it is hard to come up with clearly distinct clusters.

### 5.1 Distance Metrics

First, we will seek to describe available metrics. You are probably familiar with Euclidean geometry, where distance on a surface is defined as the square root of the sum of squared dimensional differences. Euclidean distance is a second power function (L2), where you take the square root of the sum of squares. For instance, if point A is at a grid point of 3 on the X-axis and 5 on the Y-axis, while point B is at grid point 7 on the X-axis and 2 on the Y-axis, the Euclidean distance would be:

$$\text{Euclidean} = \sqrt{(3 - 7)^2 + (2 - 5)^2} = \sqrt{16 + 9} = 5$$

Manhattan distance is a first power function (L1)—where distance is defined as the sum of absolute values. Thus, for the two points given above, the Manhattan distance would be:

$$\text{Manhattan} = |3 - 7| + |2 - 5| = 7$$

You can extend this idea to any power function—the third power function (L3) would be the cube root of the sum of cubed differences (absolute differences—no minus signs).

$$\text{Cubic} = \sqrt[3]{|3-7|^3 + |2-5|^3} = 4.5 \text{ (roughly)}$$

Although there is no “right” power to use, Manhattan distances tend to be less impacted by outliers, because higher powers magnify large differences. The standard is Euclidean just because we tend to think of distance as how the crow flies. There is one other interesting distance metric—the Farthest First metric, which is mathematically equivalent to the infinite root of the sum of distances taken to the infinite power ( $L_\infty$ ). (You don’t have to visualize that—all it means is that it converges to focusing on the greatest difference in the set).

$$\text{Farthest First distance} = \text{MAX}[|3-7|, |2-5|] = 4$$

These are interesting in cluster analysis because Euclidean and Manhattan metrics are available on Rattle, the R package for data mining, while those two plus Farthest First are available on WEKA.

Zumel and Mount (2020) gave distances in terms of data types. If your data is numerical and continuous, they suggest Euclidean distance. If our data is categorical and you want to compare lists expressed in these categories, they suggest Hamming distance, defined as the sum of mismatches.

If you have data in the form of rows of document text counting, you can use a cosine similarity which measures the smallest angle between two vectors. *R* will calculate this metrics for you, and you should let *R* do that for more than  $L1$ ,  $L2$ , or  $L_\infty$ .

## 5.2 Clustering Algorithms

There are several clustering algorithms available. Rattle offers four: *K*-Means, EWKM, Hierarchical, and BiCluster. WEKA (more research focused) has a list of eight optional models including SimpleKMeans, Hierarchical, and Farthest First.

*K*-Means is the standard algorithm for clustering. It requires data to be numerical. It works better statistically if the data is continuous (doesn’t include binary, or ideally doesn’t include discrete, data). But often understanding the clusters is improved by including discrete outcomes.

EWKM stands for Entropy weighted *K*-means. Weights are generated giving relative importance of each variable for each cluster. EWKM has relative advantages for high-dimensional data (lots of variables).

The hierarchical algorithm starts with two clusters, then iterates splitting one of the clusters to obtain three clusters and continues from there. The within cluster sum of squares (WSS) is used to assess which *K* is appropriate. As the number of clusters increases, WSS has to drop. But the rate of drop tends to form an “elbow,” where the improvement in WSS starts to have less impact. When the rate of improvement in WSS starts to decline, that *K* is inferred as highly useful (the ability to distinguish

cluster difference seems best). In Rattle, the use of the Iterate option provides the same information within the  $K$ -means algorithm.

Farthest First is a  $K$ -means algorithm but using the maximum difference distance as opposed to Euclidean. In Rattle, this is available under the term “Maximum” in the hierarchical metric.

### 5.2.1 Demonstration Data

The Healthcare Hospital Stay dataset described in Sect. 3.11 was taken from Kaggle (<https://www.kaggle.com/datasets/babyoda/healthcare-investments-and-length-of-hospital-stay>), which includes variables:

Location—code for hospital location (identifier, not used in clustering)

Time—year of event (identifier, not used in clustering)

Hospital\_Stay—days of patient stay

MRI\_Units—number of MRIs in hospital

CT\_Scanners—number of CT scanners in hospital

Hospital\_Beds—patient capacity in beds

There are a number of data features not ideal for  $K$ -means clustering (binary data; widely varying scales). Here we don’t have any of those, as all variables have ranges over 100. If there are key variables present important to understanding cluster differences, it is more important to include them than to attain statistical purity. But here, we are fortunate. Figure 5.1 shows the data included:

Figure 5.2 shows the screen for the first analysis conducted—to examine clusters using the elbow method. The elbow method is quick and dirty, but widely used. And it is easy to run multiple  $K$  values around the  $K$  indicated by the elbow method. There are many other methods that have been proposed to select the  $K$  value, but they are not necessarily worth the extra calculations.

Some of the parameters available are:

Seed—can be used to get different starting points

Runs—can be used to replicate algorithm to get more stable clusters

Re-Scale—if not checked, will run on standardized data to get rid of scale differences. If checked, uses data numbers as given (which will usually involve different scales, impacting resulting clusters). EWKM can be used as well to adjust for scalar differences.

Note that the Iterate Clusters box is checked. This gives Fig. 5.3, a plot of sum (within sum of squares) and its first derivative.

The blue line (circles) is the within cluster sum of squares. The red line ( $x$ ) gives the difference between this WSS and the prior WSS (the derivative, or rate of change). When the  $x$ -line stabilizes, that would be a good  $K$  to use. The elbow method is a rough approach to identify the value of  $K$  that is most likely useful. A widely used rule of thumb is that when the derivative line crosses below the sum

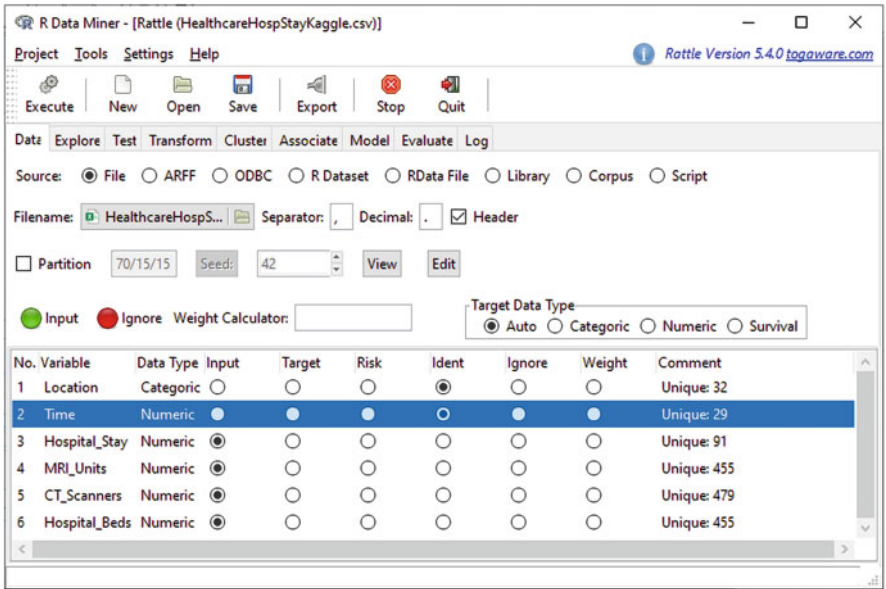


Fig. 5.1 Data screen for HealthcareHospitalStayKaggle.csv

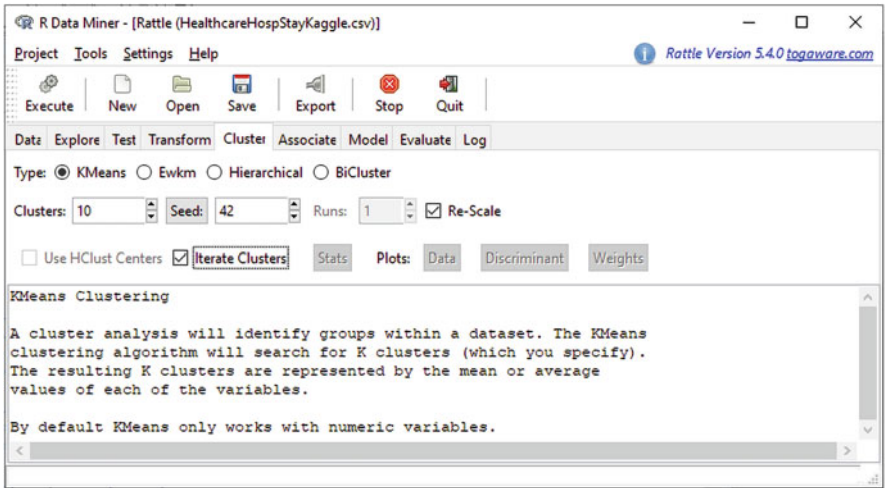
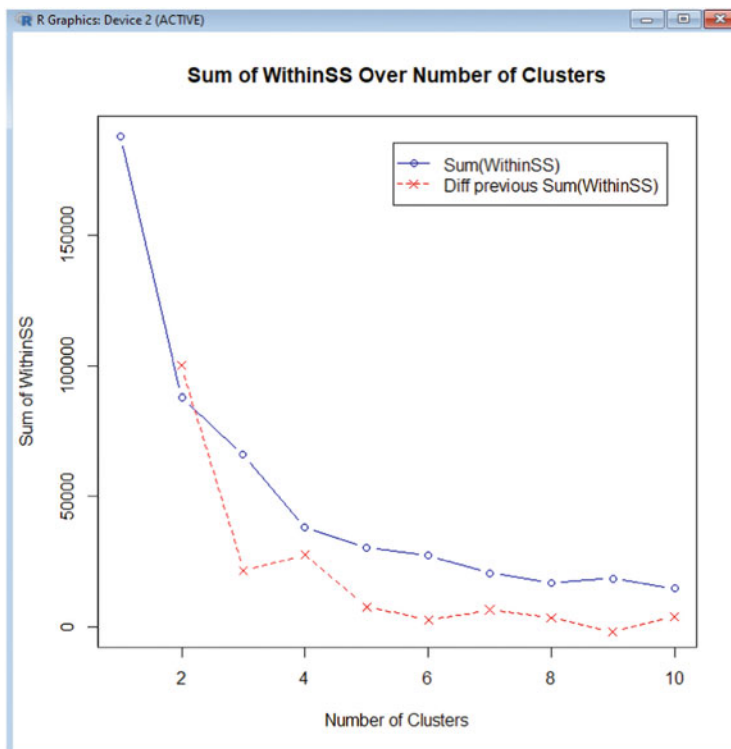


Fig. 5.2 Exploration of  $K$

(withinSS) line, that would be a good value for  $K$ . Here, that would indicate  $K = 3$ . This is a bit weird as the derivative is bouncing around. That might be because  $K = 4$  might not be very good, but  $K = 5$  might be useful.



**Fig. 5.3** Sum of squares by cluster number

### 5.2.2 K-means

Note that a thorough analysis might try multiple values and see which output is the most useful. In our case, our sample size isn't great, and our purpose is demonstration. So, we will look at  $K = 3$  and  $K = 5$ . Figure 5.4 gives the output for  $K = 3$ . Note that the seed is set at its default value—it can be changed in advanced analysis. You have a choice to leave the Re-Scale box unchecked, which will report the output in real terms (clustering does convert the data to adjust for scale). You can check the re-scale box and get the report in terms of proportions, which might yield a more robust set of clusters, but it would be harder for users to interpret.

Here there is little difference in hospital stay duration among the three clusters. Cluster 2 contains larger hospitals, cluster 1 smaller. Clicking on the Discriminant button yields a discriminant plot, shown in Fig. 5.5:

The discriminant plot converts the four variables into two dimensions, using eigen values. This enables plotting on two scales, although the eigen values themselves are not that easy to use. For instance, we really can't count the number of icons, so we cannot really see which cluster is which in this case, although we can see that the three clusters are fairly distinct. There were seven observations that were

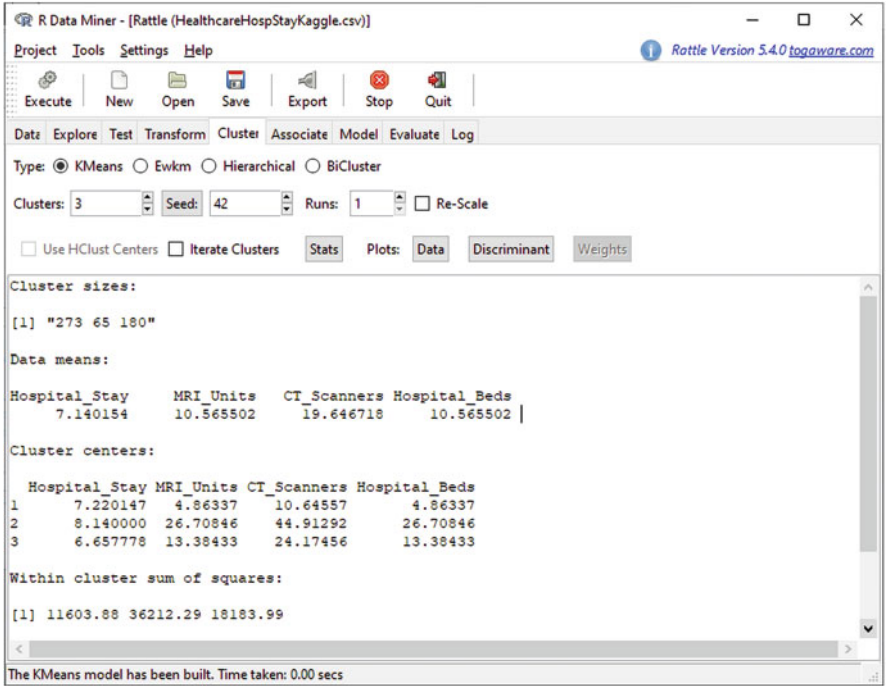


Fig. 5.4 Cluster screen in Rattle

radically different. The 93.1% of point variability explained is a function of the data and doesn't change with clusters.

To get the  $K$  values for data, you can utilize the Evaluate tab shown in Fig. 5.6: Note that the All button needs to be checked for this report to include the input variable values. The first 20 results for the model with  $K = 3$  are shown in Table 5.1:

We can now look at the other clustering model that the elbow method indicated would be interesting. Figure 5.7 gives the cluster model for  $K = 5$  without re-scaling:

Here cluster 2 contains the 7 largest hospitals, which clearly have longer average patient stays. Figure 5.8 shows the discriminant plot, a bit easier to see.

While the seven large hospitals in cluster 2 stand out, the other four clusters have a lot of overlap. Looking at cluster centers, cluster 4 is the next largest group of hospitals after cluster 2. Clusters 3 and 4 are very similar, although cluster 5 has more CT scanners. Cluster 1 has the smaller hospitals.

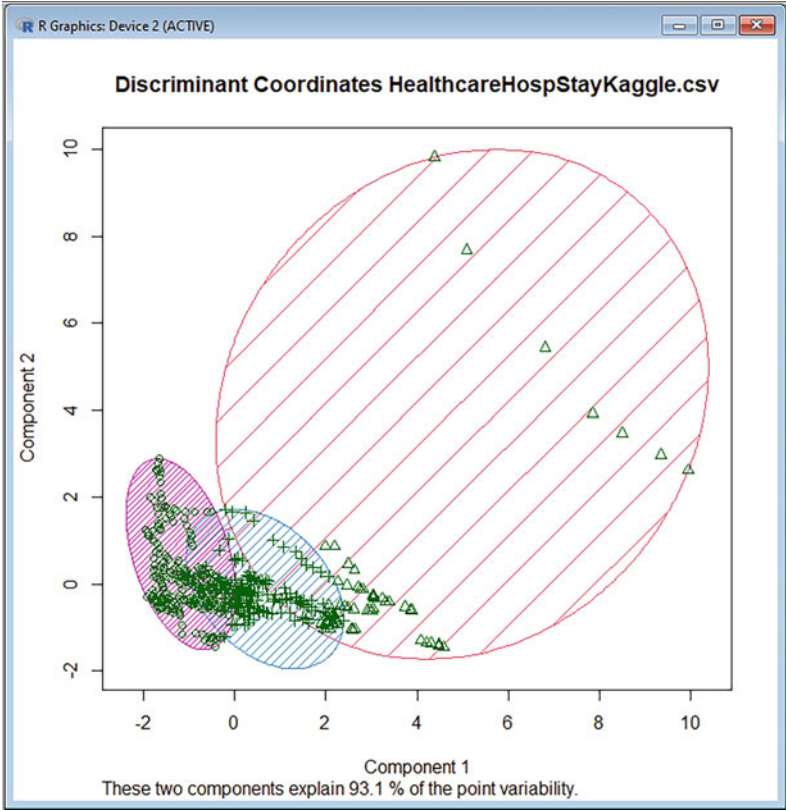


Fig. 5.5 Discriminant plot for  $K = 3$

5.2.3 EWKM

EWKM weights inputs and might be useful if scales are radically different. Here they are different, so EWKM might be useful. Output from running this clustering model for  $K = 3$  yields the output in Fig. 5.9:

Here cluster 1 includes the 7 largest hospitals. Cluster 3 consists of the smallest hospitals, although they have more MRI, CT scan, and hospital bed resources. Figure 5.10 shows the discriminant plot for this model:

We ran clusters with  $K$  of 2 to 6 for purposes of demonstration (Table 5.2). The initial pair of clusters ( $K = 2$ ) splits hospitals into two fairly large groups—with cluster 2 having more resources.  $K = 3$  isolates 65 hospitals out of the 123 larger and creates an intermediate cluster 3. Cluster 4 is where the 7 largest hospitals are isolated, remaining as a distinct set for  $K = 4, 5$ , and 6. Beginning with  $K = 4$ , a second largest set of hospitals emerges in cluster 4, identifiable for  $K = 5$  and 6 as well. This series of clusters demonstrates how cluster modeling can work.

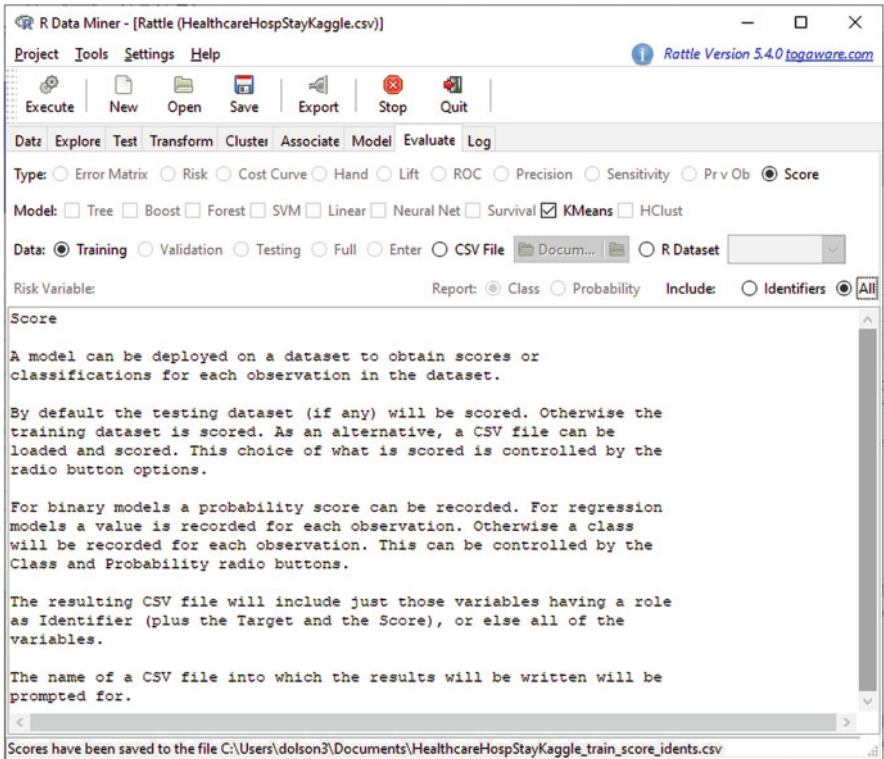


Fig. 5.6 Rattle evaluation tab

## 5.3 Case Discussions

We present two short and simple applications of clustering to healthcare issues. Both applied  $K$ -means clustering, and addressed the problem of selecting  $K$ .

### 5.3.1 Mental Healthcare

Bridges et al. (2017) were part of a group dealing with reduction of barriers to mental healthcare. A local federally qualified health center served about 35,000 patients per year, half of them children, over 90% at or below the poverty level. Mental health patients came with a variety of issues. Some with newly emerging symptoms needed brief and focused interventions. Others had long-standing concerns, difficulties accessing health services, or had barriers to seeking help of a financial, linguistic, or cultural nature. Referral was driven largely by informal cataloging of patients according to behavioral health needs. Providers asked the university research team to

**Table 5.1** Cluster classifications for  $K = 3$ 

| Location | Time | Hospital_Stay | MRI_Units | CT_Scanners | Hospital_Beds | kmeans |
|----------|------|---------------|-----------|-------------|---------------|--------|
| AUS      | 1992 | 6.6           | 1.43      | 16.71       | 1.43          | 1      |
| AUS      | 1994 | 6.4           | 2.36      | 18.48       | 2.36          | 1      |
| AUS      | 1995 | 6.5           | 2.89      | 20.55       | 2.89          | 1      |
| AUS      | 1996 | 6.4           | 2.96      | 21.95       | 2.96          | 1      |
| AUS      | 1997 | 6.2           | 3.53      | 23.34       | 3.53          | 1      |
| AUS      | 1998 | 6.1           | 4.51      | 24.18       | 4.51          | 3      |
| AUS      | 1999 | 6.2           | 6.01      | 25.52       | 6.01          | 3      |
| AUS      | 2000 | 6.1           | 3.52      | 26.28       | 3.52          | 3      |
| AUS      | 2001 | 6.2           | 3.79      | 29.05       | 3.79          | 3      |
| AUS      | 2002 | 6.2           | 3.74      | 34.37       | 3.74          | 3      |
| AUS      | 2003 | 6.1           | 3.7       | 40.57       | 3.7           | 3      |
| AUS      | 2004 | 6.1           | 3.76      | 45.65       | 3.76          | 3      |
| AUS      | 2005 | 6             | 4.26      | 51.54       | 4.26          | 3      |
| AUS      | 2006 | 5.9           | 4.89      | 56.72       | 4.89          | 2      |
| AUS      | 2009 | 5.1           | 5.72      | 39.14       | 5.72          | 3      |
| AUS      | 2010 | 5             | 5.67      | 43.07       | 5.67          | 3      |
| AUS      | 2011 | 4.9           | 5.6       | 44.32       | 5.6           | 3      |
| AUS      | 2012 | 4.8           | 5.5       | 50.5        | 5.5           | 3      |
| AUS      | 2013 | 4.7           | 13.84     | 53.66       | 13.84         | 2      |
| AUS      | 2014 | 4.7           | 14.65     | 56.06       | 14.65         | 2      |

help identify appropriate treatment (brief intervention, specialty referral, case management).

A subject group of 104 patients, 18 years of age or older who had been served by the clinic over a 3.5-month period. Data was gathered by questionnaire on perceived need, service utilization, and reasons for not seeking health. Demographic and chronic health conditions were also gathered.

SPSS clustering was applied. First, a hierarchical cluster analysis was applied on standardized scores using three cluster variables. This was used to identify an appropriate value for  $K$ . The analysis indicated three clusters were appropriate. Then  $k$ -means with  $K = 3$  was applied. Results are shown in Table 5.3:

Cluster 1 had the 40 patients experienced few barriers and thus utilized many available services. Cluster 2 (21 patients) included those with high perceived need for behavioral health services but low rates of use. This cluster reported the highest barriers to service. Cluster 3 (43 patients) had the lowest perceived need, low levels of prior use, and low barriers to service.

Cluster analysis in this case aided the clinic in identifying those patients who needed extra assistance in accessing needed mental healthcare. The key to the analysis was obtaining data through a simple questionnaire.

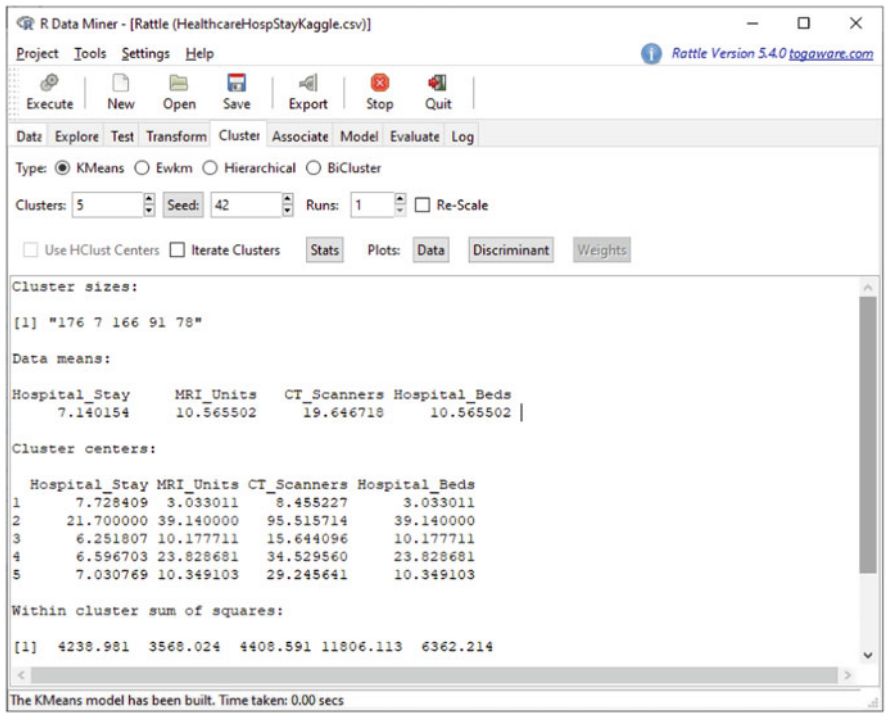


Fig. 5.7 Clusters for  $K = 5$

5.3.2 Nursing Home Service Quality

Nursing homes are growing in importance as populations are growing older. Healthcare service quality is very important, both for patients and staffing retention rates. Boakye-Dankwa et al. (2017) conducted a cross-sectional analysis of the relationships between long-term care work environments, satisfaction of employees and residents, workplace safety, and patient care quality. Data was obtained from a network of skilled nursing facilities in multiple eastern USA, all owned or managed by a single company. Data obtained included Medicare and Medicaid facility ratings, workers’ compensation claims, staffing levels and annual retention rates, employee and resident satisfaction survey results, and annual rates of the adverse events of pressure ulcers, falls, and unexplained weight loss, serious problems often found in nursing care facilities.

There were 26 variables available, but Wilcoxon nonparametric tests (or Chi-squared for binary variables) eliminated all but 10. Clustering was performed on the remaining ten variables.

K-means clustering was applied using SAS software seeking to identify groups. Preliminary analysis was accomplished using  $K = 2$ ,  $K = 3$ , and  $K = 4$ . The authors used the  $F$ -statistic as a basis for selecting  $K$ . This approach has been widely used,

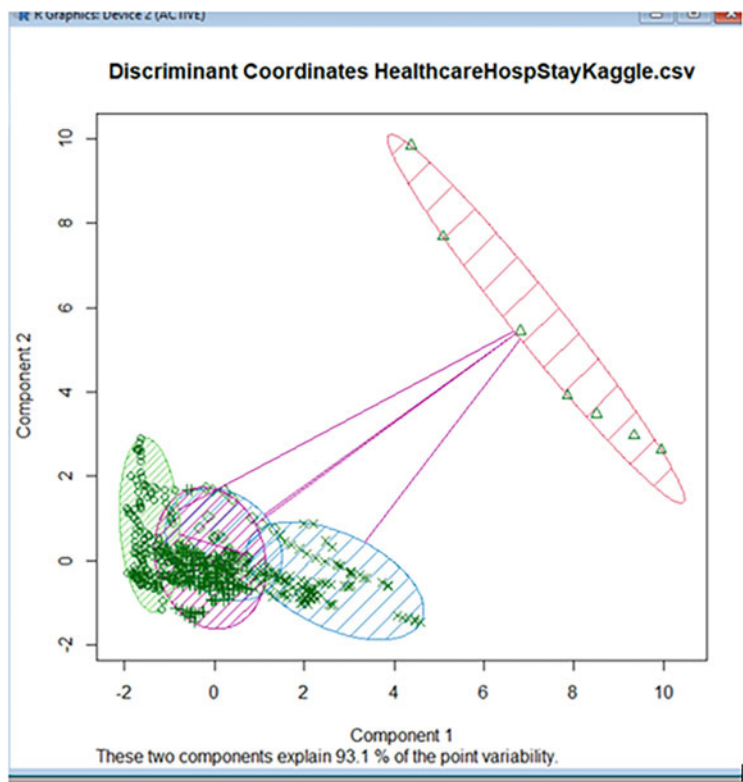


Fig. 5.8 Discriminant plot for  $K = 5$

but it is not necessarily the best. The  $F$ -statistic indicated that  $K = 2$  was best. The scatterplot provided in the article indicated some overlap, indicating that  $K = 3$  might have been better, but cluster results for  $K = 3$  were not reported. Table 5.4 gives results:

Cluster 1 had marginally less union representation. Cluster 1 had higher positive measures, to include survey ratings, as well as slightly lower negative rates.

5.3.3 Classification of Diabetes Mellitus Cases

Diabetes mellitus is one of the leading causes of death globally. Diabetes is categorized by two types. Type I arises from autoimmune reaction early in life. Type II develops slowly, related to lifestyle, especially inactivity and excess weight. Late detection of Type I diabetes usually results in delay of treatment. Overlapping clinical features, variable autoimmunity, and beta-cell loss complicate diagnosis, and it can be difficult to differentiate between Type I and Type II diabetes. Type II

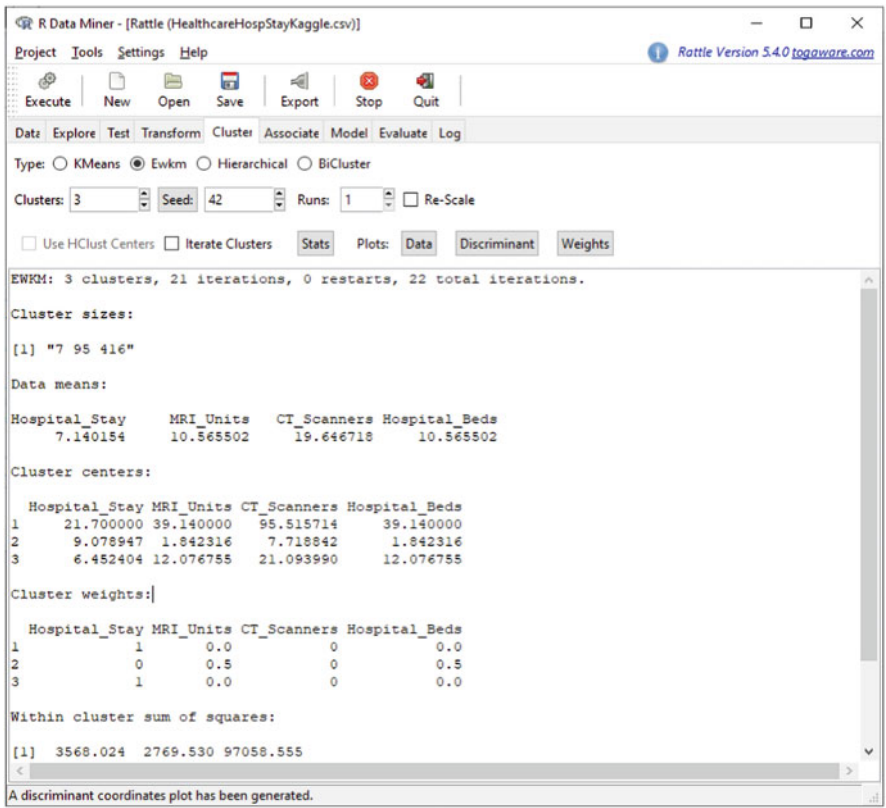


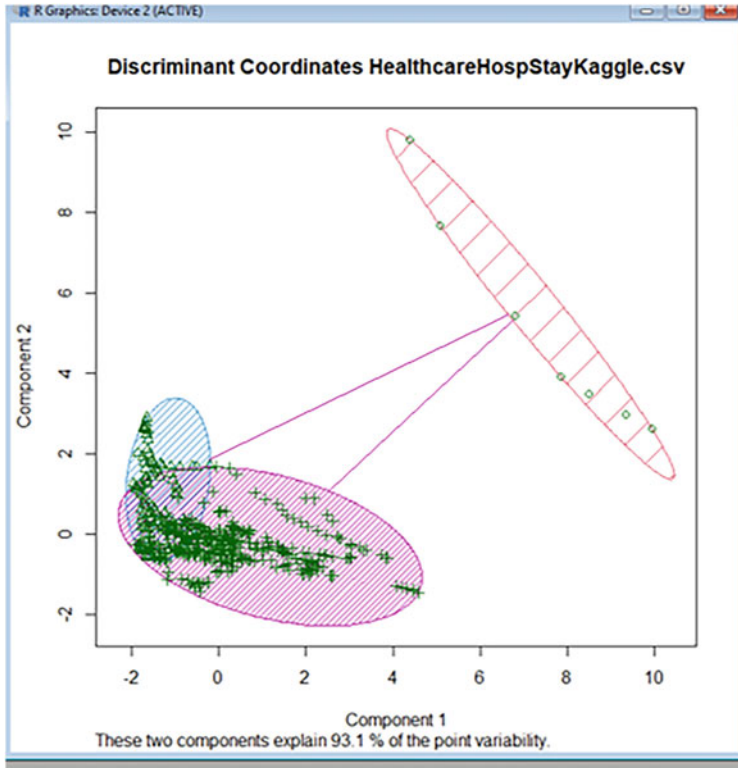
Fig. 5.9 EWKM clusters for  $K = 3$

diabetes also has been found to have subtypes with distinct clinical characteristics. This calls for personalized treatment, or precision medicine as each individual can respond differently to medications, each may have a different rate of disease progression, and different complications.

Omar et al. (2022) synthesized papers classifying subtypes of diabetes mellitus. They reviewed 26 papers that applied cluster analysis to classify diabetes for subtyping in search of personalized treatments. Applications were for subtyping as well as prediction.

The process used consisted of:

1. Data preparation involving cleaning, transformation and reduction, as clustering requires numerical data (and works best on non-binary data);
2. Identification of similarity metrics;
3. Selection of clustering algorithms;



**Fig. 5.10** EWKM with  $K = 3$  discriminant plot

- (a)  $K$ -means clustering partitions data into a selected number of clusters, easy to implement and relative fast, but limited by not including binary variables, and having lower efficiency with a large number of variables;
  - (b) Hierarchical clustering, often used to determine the number of clusters, but slow and harder to interpret;
  - (c) Density-based clustering (DBSCAN), which has worked well when nonlinear shapes are present, but slow and complex to interpret;
  - (d) Model-based clustering, to include self-organizing maps (neural network-based algorithm);
  - (e) Soft computing clustering, incorporating fuzzy sets.
4. Algorithm evaluation and validation—confusion matrices for classification models, area under the curve, and other tests for statistical approaches.

Table 5.5 is a comparison of advantages and disadvantages of algorithms given by Omar et al. (2022).

**Table 5.2** Clusters obtained with  $K = 2$  through 6

| Size    | Stay         | MRI          | CT           | Beds         | High          | Low           |
|---------|--------------|--------------|--------------|--------------|---------------|---------------|
| $K = 2$ |              |              |              |              |               |               |
| 395     | 7.06         | 6.93         | 13.84        | 6.93         |               | MRI, CT, Beds |
| 123     | 7.40         | <b>22.24</b> | <b>38.30</b> | <b>22.24</b> | MRI, CT, Beds |               |
| $K = 3$ |              |              |              |              |               |               |
| 273     | 7.22         | 4.86         | 10.65        | 4.86         |               | MRI, CT, Beds |
| 65      | 8.14         | <b>26.71</b> | <b>44.91</b> | <b>26.71</b> | MRI, CT, Beds |               |
| 180     | 6.66         | 13.38        | 24.17        | 13.38        |               |               |
| $K = 4$ |              |              |              |              |               |               |
| 151     | 7.63         | 3.31         | 8.81         | 3.31         |               | MRI, CT, Beds |
| 7       | 21.7         | <b>39.14</b> | <b>95.52</b> | <b>39.14</b> | MRI, CT, Beds |               |
| 211     | 6.53         | 10.35        | 18.98        | 10.35        |               |               |
| 109     | 6.53         | 21.85        | 35.06        | 21.85        | MRI, CT, Beds |               |
| $K = 5$ |              |              |              |              |               |               |
| 176     | 7.73         | 3.03         | 8.46         | 3.03         |               | MRI, CT, Beds |
| 7       | <b>21.70</b> | <b>39.14</b> | <b>95.52</b> | <b>39.14</b> | MRI, CT, Beds |               |
| 166     | 6.25         | 10.18        | 15.64        | 10.18        |               |               |
| 91      | 6.60         | 23.83        | 34.53        | 23.83        | MRI, CT, Beds |               |
| 78      | 7.03         | 10.35        | 29.25        | 10.35        |               |               |
| $K = 6$ |              |              |              |              |               |               |
| 121     | 8.08         | 2.18         | 6.92         | 2.18         |               | MRI, CT, Beds |
| 7       | <b>21.70</b> | <b>39.14</b> | <b>95.52</b> | <b>39.14</b> | MRI, CT, Beds |               |
| 126     | 6.68         | 6.26         | 13.63        | 6.26         |               | MRI, CT, Beds |
| 90      | 6.59         | 23.87        | 34.68        | 23.87        | MRI, CT, Beds |               |
| 98      | 6.11         | 12.31        | 16.31        | 12.31        |               |               |
| 76      | 7.05         | 10.41        | 29.41        | 10.41        |               |               |

**Table 5.3** Bridges et al. (2017) cluster means

| Characterization    | Well-served | Underserved | Subclinical |
|---------------------|-------------|-------------|-------------|
|                     | $n = 40$    | $n = 21$    | $n = 43$    |
| Variable            |             |             |             |
| Perceived need      | 5.15        | 4.24        | 2.16        |
| Service utilization | 1.63        | 0.67        | 0.81        |
| Barriers to service | 3.15        | 8.00        | 2.14        |

5.4 Conclusion

Cluster analysis is a tool for initial data analysis. It works best with continuous data, although it is widely applied using categorical ratings or binary data (converted to numerical form). If the problem being analyzed calls for using such data, that should override the statistical purity concerns. Clustering can be useful as an initial exploratory analysis. But the results are not controllable and may not emphasize

**Table 5.4** Boakye-Dankwa et al. (2017) cluster means

| Domain                 | Variable                             | Cluster 1 | Cluster 2 |
|------------------------|--------------------------------------|-----------|-----------|
|                        |                                      | $n = 118$ | $n = 85$  |
| Employees              | Sick hours                           | 39.74     | 40.17     |
|                        | Employee foundation                  | 0.46      | 0.37      |
|                        | MIS Employee satisfaction            | 1.93      | 1.78      |
|                        | Certified nursing aid retention rate | 0.75      | 0.68      |
|                        | Clinical staffing rate               | 4.76      | 3.75      |
| Residents and Services | Rate of pressure ulcers              | 0.02      | 0.04      |
|                        | Rate of falls                        | 0.17      | 0.21      |
|                        | Rate of unexplained weight loss      | 0.02      | 0.03      |
|                        | Satisfaction survey                  | 2.34      | 2.19      |
| CMS                    | Survey rating                        | 3.36      | 1.87      |

**Table 5.5** Clustering algorithm advantages/disadvantages

| Algorithm                    | Advantages                                                                                               | Disadvantages                                                                                                     |
|------------------------------|----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| <i>K</i> -means clustering   | Scalable, simple<br>Good at separating datasets with spherical cluster shapes                            | Need to specify $k$<br>Sensitive to outliers, noise, and initialization<br>Limited to numeric data                |
| <i>K</i> -medoids clustering | More robust to outliers and noise                                                                        | Need to specify $k$<br>More processing time                                                                       |
| Hierarchical clustering      | Suitable for problems involving point linkage<br>Easy to select $k$<br>Can deal with any attribute types | Poor scaling                                                                                                      |
| DBSCAN clustering            | Handles arbitrary-shaped clusters<br>Handles noise well<br>Need to initialize density parameters         | Poor cluster descriptors<br>Sensitive to input parameters<br>Doesn't do well with clusters of different densities |
| SOM clustering               | Good for vector quantization, speech recognition                                                         | Sensitive to initial weight vector and parameters                                                                 |
| EM clustering                | Easy and simple<br>Efficient for iterative computations                                                  | Converges to local minima                                                                                         |
| Fuzzy clustering             | Less sensitive to local minima                                                                           | Need to select membership functions                                                                               |

dimensions the analyst is looking for. There are options available in the form of different algorithms, although *K*-means is usually what users end up applying. EWKM can be interesting for data with different scales, although *K*-means can be run after re-scaling. Hierarchical clustering has been suggested as a way to identify the optimal value of  $K$ , but it is a relatively slow method that is complex to analyze, and it isn't that hard to run multiple runs for different  $K$  values and seeing which best provides the insight desired by the analyst.

Other sources are given in the references (Witten and Frank 2005; Zume1 and Mount 2020).

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# Chapter 6

## Time Series Forecasting



**Keywords** Time series · ARIMA · OLS regression · Box-Jenkins models

This chapter discusses time series forecasting. A wide variety of forecasting techniques are available. Regression modeling is an obvious approach if appropriate statistical data is available. Some of the many statistical considerations of using regression for forecasting are presented. The Box-Jenkins technique can often be very useful when cyclical data (often encountered in economic activities) is present.

### 6.1 Time Series Forecasting Example

Emergency departments around the world are taxed with overcrowding at some times. Demand for emergency service is highly variable. Accurate service demand forecasts would enable much better allocation of scarce emergency resources. Tuominen et al. (2022) analyzed daily arrivals at a university hospital emergency department for the period from June 2015 through June 2019. Traditionally emergency room demand forecasting models use univariate time series models. Tuominen et al. (2022) applied time series analysis in the form of autoregressive integrated moving average (ARIMA) models and regression with ARIMA errors versus models utilizing additional explanatory variables. They obtained 158 potential explanatory variables to include weather and calendar variables as well as lists of local public events, website visits, numbers of available hospital beds in the area, and Google searches. Simulated annealing and other approaches were utilized to select these explanatory variables. Those explanatory variables retained were calendar variables, loads at secondary care facilities, and local public events.

ARIMA models are widely used in time series forecasting. When additional independent variables are added to univariate historical data, the model is called regression with ARIMA errors, or ARIMAX. For seasonal data, time tags of known seasonality are often applied in seasonal ARIMA models. For comparative purposes, a random forest (decision tree) model was applied to the selected explanatory

variables. The study also included naïve (use the latest observed value as the prediction) and seasonal naïve (use the latest observed value a season ago) models.

Model errors were measured by mean absolute percentage error (MAPE). Significance was measured against the seasonal naïve model as a base. The best fit was from an ARIMA model supplemented with selected seasonal and weather variables. However, this was not significantly more accurate than the univariate ARIMA model, or recursive least squares models using variables obtained from simulated annealing or from a floating search technique.

There is a great benefit of utilizing univariate models in that the infinite number of potential variables do not have to be examined. Seasonality and day of the week, on the other hand, are obvious variables that might be utilized to improve forecasts of medical service demand.

## 6.2 Classes of Forecasting Techniques

Economic data often includes cycles, whether you are measuring things related to the national economy or to the sales of a particular business. Many economic activities are seasonal. There are also changes in economic conditions, for example unemployment rate and inflation. Sometimes there are also detectable trends or relationships.

A broad range of forecasting tools are available. These can vary substantially, depending upon the characteristics of the forecasting problem, as well as available information upon which the forecast can be based.

## 6.3 Time Series Forecasts

In time series data, only historical data on the variable to be predicted is required. A wide range of time series forecasting techniques exist, from simple moving average calculations through very advanced models which are incorporating seasonality, trend cycles, and other elements. The simplest approach is to fit a straight line through the past few observations in order to identify trend. That is what an ordinary least squares regression of the dependent (predicted) variable versus time provides. However, there are usually other complications involved with time series data. Each of the time series methods requires data in the form of measured response of the variable being forecast over time. A fundamental assumption is that the passage of time and the other components included (such as seasonality and trend) explain all of the future change in the variable to be predicted. An advantage of strict time series forecasting is that they do not require a theory of causation.

Another simple approach is the **moving average** (MA) method. The concept is that the next period's predicted value will be the average of the last  $n$  observations, which would be  $n$ -terms moving average forecast. This can be modified by

weighting these  $n$  observations in any way the analyst or user wants, which would be weighted moving average (WMA). While this method is very simple, it has proven to be useful in stable environments, such as inventory management.

Another relatively simple approach that has proven highly useful is **exponential smoothing**. With exponential smoothing, the forecast for the next period equals the forecast for the last period, plus a portion ( $0 \leq \alpha \leq 1$ ) of last period's forecast error. The parameter  $\alpha$  can be manipulated to change the model's response to changes. An  $\alpha$  of 0 would simply repeat last period's forecast. An  $\alpha$  of 1 would forecast last period's actual demand. The closer  $\alpha$  is to 1, the more the model responds to changes. The closer  $\alpha$  is to 0, the less it is affected by changes.

There are many variations to exponential smoothing, allowing more complex adaptation to trend changes or seasonal factors. Because of its very simple computational requirements, exponential smoothing is popular when many forecasts need to be computed regularly.

Trends can be identified through regression of the variable to be predicted versus time. But the degree of fit of this model is often not very good, and more accurate information is usually available.

**ARIMA** models (AutoRegressive Integrated Moving Average) provide a means to fit a time series model incorporating cycles and seasonality in addition to trend. Box-Jenkins models are of this class. ARIMA models have up to three parameters: autocorrelation terms, differencing terms, and moving average terms. These will be discussed in the Box-Jenkins section. Exponential smoothing is a special case of the Box-Jenkins technique. However, while exponential smoothing is very computationally efficient, ARIMA models require large amounts of computation time. Furthermore, because so many parameters are used, larger amounts of data are required for reliable results. ARIMA works very well when time series data has high degrees of autocorrelation, but rather poorly when this condition does not exist. It usually is a good idea to test for autocorrelation and compare the fit of the ARIMA model with linear regression against time, or some other forecasting model. A regression of the variable to be forecasted versus time is a special case of ARIMA (0 autocorrelation terms, 0 moving average terms).

Other more advanced techniques to forecast time series exist. One of these is X-11, developed by the Census Bureau ([www.abs.gov.au](http://www.abs.gov.au)). That technique decomposes a time series into seasonal, trend, cycles, and irregular components. As a rule, the more sophisticated the technique, the more skill is required to use it to get reliable results.

## 6.4 Forecasting Models

There are a variety of model approaches available to aid forecasting. A simple and widely used method is moving average. For a  $q$ -moving average, simply take the prior  $q$  observations and average them.

$$Y = \frac{\sum Y_{t-1} + Y_{t-2} + \dots + Y_{t-q}}{n}$$

Exponential smoothing is a similar easy time series forecasting model, but we will only demonstrate moving average. Another popular model is regression analysis.

### 6.4.1 Regression Models

Regression models are a basic data-fitting tool in data mining. Essentially, they fit a function to the data minimizing some error metric, usually sum of squared errors. Regression models are applied to continuous data (with dummy, or 0–1, variables allowed). When the dependent variable is continuous, ordinary least squares (OLS) is applied. When dependent variables are binary (or categorical) as they often are in classification models, logistic regression is used. Regression models allow you to include as many independent variables as you want. In traditional regression analysis, there are good reasons to limit the number of variables. The spirit of exploratory data mining, however, encourages examining a large number of independent variables. Here we are presenting very small models for demonstration purposes. In data mining applications, the assumption is that you have very many observations, so that there is no technical limit on the number of independent variables.

Regression can be used to obtain the relationship given below,

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \epsilon$$

(which can be extended by adding more input variables, i.e.,  $X$  variables) and then to use this as a formula for prediction. Given you know (or have estimates for)  $X_1$  and  $X_2$ , your regression model gives you a formula to estimate  $Y$ .

### 6.4.2 Coincident Observations

Establishing relationships is one thing—forecasting is another. For a model to perform good in forecasting, you have to know the future values of the independent variables. Measures such as  $r^2$  assume absolutely no error in the values of the independent variables you use. The ideal way to overcome this limitation is to use independent variables whose future values are known.

### 6.4.3 Time

Time is a very attractive independent variable in time series forecasting because you will not introduce additional error in estimating future values of time. About all we know for sure about next year's economic performance is that it will be next year. And models using time as the only independent variable have a different philosophical basis than causal models. With time, you do not try to explain the changes in the dependent variable. You assume that whatever has been causing changes in the past will continue to do so at the same rate in the future.

### 6.4.4 Lags

Another way to obtain known independent variable values is to lag them. For example, instead of regressing a dependent variable value for 1995 against the independent variable observation for 1995, regress the dependent variable value for 1995 against the *1994 value of* independent variable. This would give you one year of known independent variable values with which to forecast. If the independent variable is a leading indicator of the dependent variable,  $r^2$  of your model might actually go up. However, usually lagging an independent variable will lower  $r^2$ . Additionally, you will probably lose an observation, which in economic data may be a high price. But at least you have perfect knowledge of a future independent variable value for your forecast.

That is not to say that you cannot utilize coincident models (coincident models include variables that tend to change direction at the same time). These models in fact give decision makers the opportunity to play "what if" games. Various assumptions can be made concerning the value of the independent variables. The model will quickly churn out the predicted value of the dependent variable. Do not, however, believe that the  $r^2$  of the forecast reflects all the accuracy of the model. Additional errors in the estimates of the independent variables are not reflected in  $r^2$ .

### 6.4.5 Nonlinear Data

Thus far we have only discussed linear relationships. Life usually consists of nonlinear relationships. Straight lines do not do well in fitting curves and explaining these nonlinearities. There is one trick to try when forecasting obviously nonlinear data. For certain types of curves, logarithmic transformations fall back into straight lines. When you make a log transform of the dependent variable, you will need to retransform the resulting forecasts to get useful information.

### 6.4.6 Cycles

In the data collection chapter, we commented that most economic data is cyclical. We noted above that models with the single independent variable of time have some positive features. There is a statistical problem involved with OLS regressions on cyclical data. The error terms should be random, with no pattern. A straight-line fit of cyclical data will have very predictable patterns of error (autocorrelation). This is a serious problem for OLS regression, warping all the statistical inferences.

Autocorrelation can occur in causal models as well, although not as often as in regressions versus time. When autocorrelation occurs in causal models, more advanced statistical techniques are utilized, such as second-stage least squares. However, when autocorrelation occurs in regressions where time is the only independent variable, Box-Jenkins models are often very effective. Box-Jenkins forecasting takes advantage of the additional information of the pattern in error terms to give better forecasts.

## 6.5 OLS Regression

Ordinary least squares regression (OLS) is a model of the form:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n + \varepsilon$$

where

$Y$  is the dependent variable (the one being forecast)

$X_n$  are the  $n$  independent (explanatory) variables

$\beta_0$  is the intercept term

$\beta_n$  are the  $n$  coefficients for the independent variables

$\varepsilon$  is the error term

OLS regression is nothing more than the straight line (with intercept and slope coefficients  $\beta_n$ ) which minimizes the error terms  $\varepsilon_i$  over all  $i$  observations. The idea is that you look at past data to determine the  $\beta$  coefficients which worked best, and given knowledge of the  $X_n$  for future observations, the most likely future value of the dependent variable will be what the model gives you. This approach assumes a linear relationship, and error terms that are normally distributed around zero without patterns. While these assumptions are often unrealistic, regression is highly attractive because of the existence of well-developed computer packages as well as highly developed statistical theory. Statistical packages provide the probability that estimated parameters differ from zero.

## 6.6 Tests of Regression Models

### 6.6.1 Sum of Squared Residuals (SSR)

The accuracy of any forecasting model can be assessed by calculating the sum of squared residuals (SSR). All that means is that you obtain a model which gives you a forecasting formula, then go back to the past observations and see what the model would have given you for the dependent variable for each of the past observations. Each observation's residual (error) is the difference between actual and predicted. The sign doesn't matter, because the next step is to square each of these residuals. The more accurate the model is, the lower its SSR. An SSR doesn't mean much by itself. But it is a very good way of comparing alternative models, if there are equal opportunities for each model to have error.

#### R-Squared

SSR can be used to generate more information for a particular model.  $r^2$  is the ratio of explained squared-dependent variable values over total squared values. Total squared value is defined as explained squared-dependent variable values plus SSR. To obtain  $r^2$ , square the forecast values of the dependent variable values, add them up (yielding MSR), and divide MSR by (MSR + SSR). This gives the ratio of change in the dependent variable explained by the model.  $r^2$  can range from a minimum of 0 (the model tells you absolutely nothing about the dependent variable) to 1.0 (the model fits the data perfectly).

#### Adjusted R-Squared

Note that in the OLS model, you were allowed an unlimited number of independent variables. The fact is that adding an independent variable to the model will always result in  $r^2$  equal to or greater than  $r^2$  without the last independent variable. This is true despite the probability that one or more of the independent variables have very little true relationship with the dependent variable. To get a truer picture of the worth of adding independent variables to the model, adjusted  $r^2$  penalizes the  $r^2$  calculation for having extra independent variables.

$$\text{Adjusted } r^2 = 1 - \frac{\text{SSR}(i-1)}{\text{TSS}(i-n)}$$

where

SSR = sum of squared residuals

MSR = sum of squared predicted values

$$TSS = SSR + MSR$$

$i$  = number of observations

$n$  = number of independent variables

While these measures provide some idea of how well a model fits past data, it is more important to know how well the model fits data to be forecast. A widely used approach to measuring how well a model accomplishes this is to divide the dataset into two parts (for instance, the first two thirds of the data used to develop the model, and then test this model on the last one third of the dataset). An idea of model accuracy can be obtained by developing a prediction interval. The upper bound of this prediction interval can be obtained by

$$\text{Forecast} + 2\left(\sqrt{\text{mean square forecast error}}\right)$$

and the lower bound by

$$\text{Forecast} - 2\left(\sqrt{\text{mean square forecast error}}\right).$$

If the forecast errors are independent and identically normally distributed with a mean of zero, then the future observation should fall within these bounds about 95% of the time.

## 6.7 Causal Models

So far we have referred to a general regression model, with any number of independent variables. This type of model seeks to explain changes in the dependent variable by changes in the independent variables.

It must be recognized that a good fit in a model says nothing about causation. The real relationship may be due to the dependent variable causing changes in the independent variable(s). Regression models wouldn't know any better.

Models with more than one independent variable introduce added complications. In general, from a statistical viewpoint, it is better to have as simple a model as possible.

One rational way to begin constructing a multivariate causal model is to collect data on as many candidate independent variables (plus of course the dependent variable) as possible. Independent variables should make some sense in explaining the dependent variable (you should have some reason to think changes in the independent variable cause changes in the dependent variable). Then run a correlation analysis. **Correlation between the dependent variable and a candidate independent variable should be high.**

### 6.7.1 Multicollinearity

The primary complication arising from the use of multiple independent variables is the potential for multicollinearity. What that means can be explained as two or more independent variables are likely to contain overlapping information, i.e., presenting high correlation. The effect of multicollinearity is that the  $t$  tests are drastically warped, and bias creeps into the model. This has the implication that as future information is obtained, the estimates of the  $\beta$  coefficients will likely change drastically, because the model is unstable. Multicollinearity can be avoided by *NOT* including independent variables that are highly correlated with each other. How much correlation is too much is a matter of judgment. Note that the sign of correlation simply identifies if the relationship is positive or negative. In a positive relationship, if one variable goes up, the other variable tends to go up. A negative correlation indicates that as one variable goes up, the other variable tends to go down.

To demonstrate this concept, assume you have a correlation matrix in Table 6.1, giving you the correlations between the dependent variable  $Y$  and candidate independent variables  $A$ ,  $B$ ,  $C$ , and  $D$ .

Note that a first priority should be the existence of a theoretical relationship between independent variables and the dependent variable. You should have a reason for expecting  $A$ ,  $B$ ,  $C$ , and  $D$  to have some impact upon  $Y$ . Correlations can be used to verify the relationship among a pair of variables. In this matrix, variables  $D$ ,  $B$ , and  $C$  have some identifiable relationship with  $Y$ .  $D$  has a direct relationship (as one goes up, the other tends to go up).  $B$  and  $C$  have inverse relationships with  $Y$  (as one goes up, the other tends to go down). The regression model  $Y = f(B, D)$  is likely to be multicollinear, because  $B$  and  $D$  contain much of the same information.

### 6.7.2 Test for Multicollinearity

A variance inflation measure provides some measure of multicollinearity in a regression. In SAS, the option VIF can be included in the model line. If the variance inflation measure is below 10, the rule of thumb is that you don't reject the evidence of collinearity. However, this is a very easy test limit to pass. The first priority would be to select variables that would make sense. Secondly, it is best to design models without overlapping information.

**Table 6.1** Correlation matrix

|     | $Y$  | $A$  | $B$  | $C$  | $D$  |
|-----|------|------|------|------|------|
| $Y$ | 1.0  | -0.1 | -0.8 | -0.6 | 0.9  |
| $A$ | -0.1 | 1.0  | 0.2  | 0.2  | -0.2 |
| $B$ | -0.8 | 0.2  | 1.0  | 0.2  | -0.8 |
| $C$ | -0.6 | 0.2  | 0.2  | 1.0  | -0.7 |
| $D$ | 0.9  | -0.2 | -0.8 | -0.7 | 1.0  |

## 6.8 Regression Model Assumptions

The basic simple regression model is:

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

where

$Y_i$  is the  $i$ th observed value of the dependent variable,  
 $X_i$  is the  $i$ th observed value of the independent variable,  
 and  $\varepsilon$  is a normally distributed random variable with a mean of zero and a standard deviation of  $s_\varepsilon$ . The error term is assumed to be statistically independent over observations.

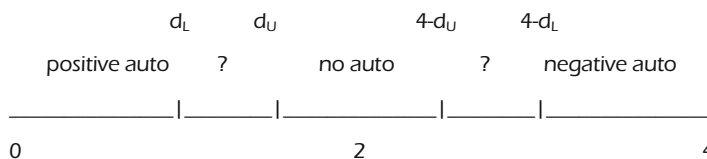
### 6.8.1 Autocorrelation

Autocorrelation exists in a regression model if there is correlation between error  $\varepsilon_i$  and some prior error of a given lag  $\varepsilon_{i-j}$ . For  $j$  of 1 (first degree autocorrelation) to be significant, an apparent influence of the immediately preceding error on the current error would exist. Second-degree autocorrelation is the correlation between error in a given time period and the error two time periods prior. Autocorrelation can be of any degree up to one less than the number of observations, although larger degrees of autocorrelation are less likely to exist and estimating them is more difficult because there are fewer instances to observe.

Autocorrelation can often occur in time series data involving cycles. OLS regression seeks to force a straight line through wavy data. Therefore, there may well be a relationship between error in a given time period and the error one period prior. If you are at the high side of a cycle, and the cycle is longer than the period between observations, you are more likely to be on the high side of the regression line in the next observation. This would be positive autocorrelation, as the sign of the error is likely to be the same. Negative autocorrelation exists when there is a significant tendency for error in the following period to have an opposite sign.

Over the long run, autocorrelation does not affect the bias of model estimates (in the short-run, it can make it erratic). However, autocorrelation in a model results in underestimation of the standard errors of the  $\beta$  coefficients (you get misleading  $t$  scores, biased the wrong way).

The Durbin-Watson test provides an estimate of autocorrelation in a regression model. The null hypothesis of this test is that there is no autocorrelation in a regression model. Durbin-Watson statistics can range between 0 and 4. The ideal measure indicating no autocorrelation is 2. Value for the lower and upper Durbin-Watson limits at the  $-0.95$  level are given in most statistics books. You need a computer regression package to obtain  $d$ , the estimate of first order autocorrelation.



**Fig. 6.1** Durbin-Watson scale

Then obtain  $d_L$  and  $d_U$  from a Durbin-Watson table ( $k'$  is the number of non-intercept independent variables, and  $n$  is the number of observations).

To test for positive autocorrelation ( $\varepsilon_i$  is directly related to  $\varepsilon_j$ )

if  $d$  is less than  $d_L$ , reject the null (conclude there is positive autocorrelation)

if  $d$  is greater than  $d_L$  but less than  $d_U$ , there is no conclusion relative to positive autocorrelation

if  $d$  is greater than  $d_U$ , accept the null (conclude no positive autocorrelation exists)

To test for negative autocorrelation ( $\varepsilon_i$  is inversely related to  $\varepsilon_j$ )

if  $d$  is less than  $4-d_U$ , accept the null (conclude no negative autocorrelation exists)

if  $d$  is greater than  $4-d_U$  and less than  $4-d_L$ , there is no conclusion relative to negative autocorrelation

if  $d$  is greater than  $4-d_L$ , reject the null (conclude the existence of negative autocorrelation)

There is a continuum for the evaluation of  $d$  (given in Fig. 6.1):

If autocorrelation exists in a regression against TIME ( $Y = f\{\text{time}\}$ ), this feature can be utilized to improve the forecast through a Box-Jenkins model. The second-stage least squares is an alternative approach, which runs the OLS regression, identifies autocorrelation, then adjusts the data to eliminate the autocorrelation, regresses on the data, and replaces the autocorrelation. As you can see, second-stage least squares is rather involved. (The SAS syntax for second-stage least squares regression is given at the end of the chapter.)

To summarize autocorrelation, the error terms are no longer independent. One approach (Box-Jenkins) is to utilize this error dependence to develop a better forecast. The other approach (second-stage least squares) is to wash the error dependence away.

## 6.8.2 Heteroskedasticity

The regression model statistics and associated probabilities assume that the errors of the model are unbiased (the expected mean of the errors is zero), the error terms are normally distributed, and that the variance of the errors is constant. Heteroskedasticity is the condition that exists when error terms do not have constant

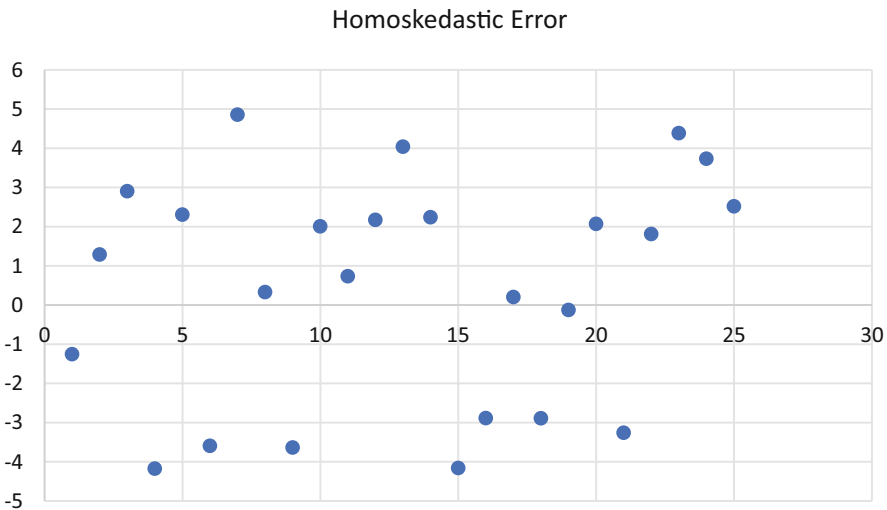


Fig. 6.2 Homoskedastic error



Fig. 6.3 Heteroskedastic error

(or relatively constant) variance over time. If errors are homoskedastic (the opposite of heteroskedastic), they would look like Fig. 6.2:

Heteroskedasticity would look like Fig. 6.3:

This plot of heteroskedastic error implies that the variance of the errors is a function of a model’s independent variable. If we were dealing with a time series, and the errors got worse with time, this should lead us to discount the goodness of fit

of the model, because if we were going to use the model to forecast, it would be more and more inaccurate when it was needed most.

Of course, if the opposite occurred, and the errors got smaller with time, that should lead us to be more confident of the model than the goodness of fit statistics would indicate. This situation is also heteroskedastic but would provide improving predictors.

There is no easy way to test for heteroskedasticity. About the best quick test is to plot the errors versus time and apply the eyeball test.

## 6.9 Box-Jenkins Models

Box-Jenkins models were designed for time series with:

No trend

Constant variability

Stable correlations over time

Box-Jenkins models have a great deal of flexibility. You must specify three terms:

1. **P**—the number of autocorrelation terms
2. **D**—the number of differencing elements
3. **Q**—the number of moving average terms

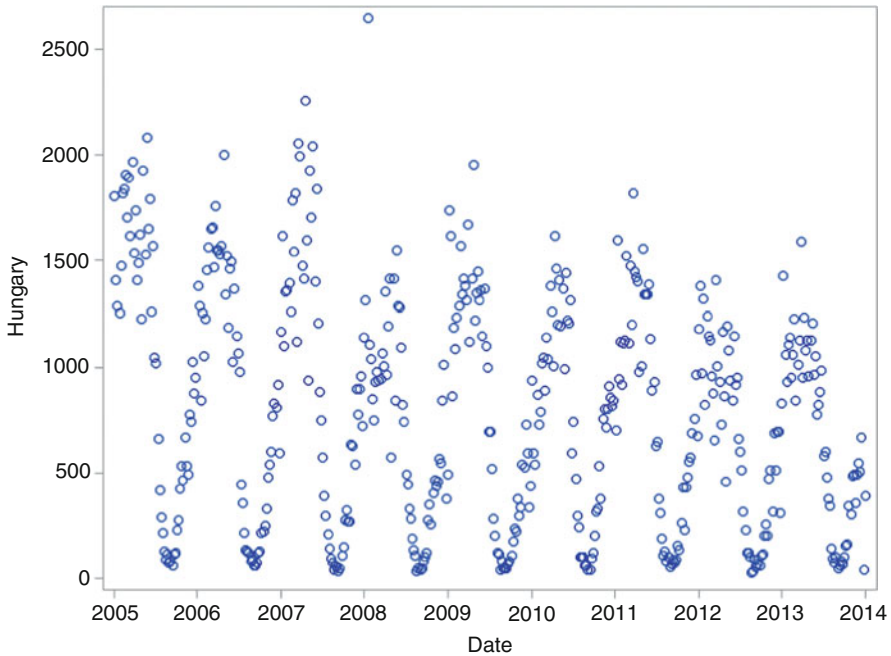
The **P** term is what makes a Box-Jenkins model work, taking advantage of the existence of strong autocorrelation in the regression model  $Y = f(\text{time})$ .

The **D** term can sometimes be used to eliminate trends. **D** of 1 will work well if your data has a constant trend (it's linear). **D** of 2 or 3 might help if you have more complex trends. Going beyond a **D** value of 3 is beyond the scope of this course. If there is no trend to begin with, **D** of 0 works well.

The model should also have constant variability. If there are regular cycles in the data, moving average terms equal to the number of observations in the cycle can eliminate these. Looking at a plot of the data is the best way to detect cyclical data. One easily recognized cycle is seasonal data. If you have monthly data, a moving average term **Q** of 12 would be in order. If you have quarterly data, **Q** of 4 should help. If there is no regular pattern, **Q** of 0 will probably be as good as any.

**D** and **Q** terms are used primarily to stabilize the data. **P** is the term which takes advantage of autocorrelation. The precise number of appropriate autocorrelation terms (**P**) to use can be obtained from the computer package. **P** is the number of terms significantly different from 0. Significance is a matter of judgement. Since Box-Jenkins models are often exploratory, you will want to try more than one model anyway, to seek the best fit (lowest mean square forecasting error).

Box-Jenkins models tend to be volatile. They are designed for datasets of at least 100 observations. You won't always have that many observations. We are looking at them as an alternative to time series, especially when autocorrelation is present in a



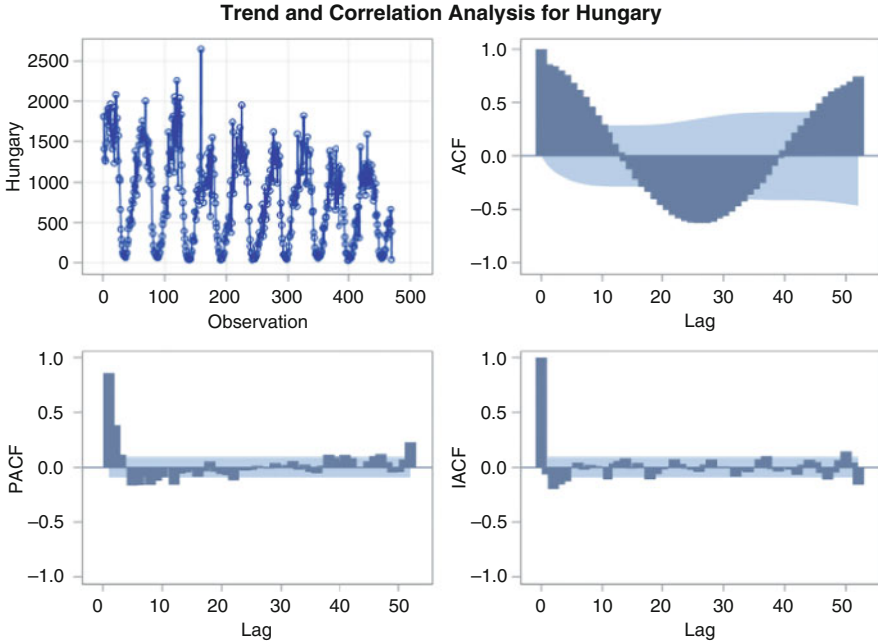
**Fig. 6.4** Scatterplot of Hungarian chickenpox over time

regression versus time. So the idea is to compare different models and select the best one.

Box-Jenkins models require a computer package for support. There are a number available. IDA has been mentioned, and quick and dirty commands are given at the end of the chapter. Minitab and SAS are other sources. Specific operating instructions require review of corresponding manuals. In general, IDA is very good for diagnosing a data series before running Box-Jenkins models. SAS requires less parameter settings, but is a little more rigid. Minitab commands for Box-Jenkins are very easy, also given at the end of the chapter.

Now that we have seen some of the techniques available for forecasting, we will demonstrate with a time series of Hungarian chickenpox. We use a dataset of weekly counts of chickenpox cases in Hungarian counties, taken from Rozemberczki et al. (2021). This dataset is a time series of 521 rows of 20 counties (plus the sum to represent the country). The time period covered is from year 2005 to 2015. Figure 6.4 shows a plot of the training set (years 2005 through 2014). The SAS syntax is given, followed by the resulting plot (you could obtain a similar plot very easily in Excel or R):

```
Proc sgplot data=train;
Scatter y=Hungary x=date;
Run;
```



**Fig. 6.5** SAS ARIMA initial output

Figure 6.4 shows a very distinct cycle, of 52 weeks. Clearly chickenpox is seasonal, peaking in February and March, and becoming quite rare in the summer months. There also appears to be a slight downward trend. The next step is to try models. We will run a 3-period moving average and an OLS regression against time (a trend model) in Excel. The trend model fits Fig. 6.4 with a straight line, which obviously is a pretty bad fit. We will take the relative average for each of the 52 weeks and put it back in the trend line to get a more reasonable forecasting model (which actually is part of the way to an ARIMA model). Finally, we compare with an ARIMA model from SAS using the syntax:

```
Proc arima data = train
Identify var. = Hungary nlag = 52;
Run;
```

SAS ARIMA modeling yields Fig. 6.5:

Figure 6.4 indicated a very distinct cycle of 52 weeks. The Q implied is 52. Given the severe cycle, the time plot indicates a consistent trend, indicating a D of 1. The PACF plot shows three significant autocorrelation terms, verified by SAS output. Thus the best likely Box-Jenkins model would be (3,1,52). To enter the difference term D, we need to reenter the data in the following syntax:

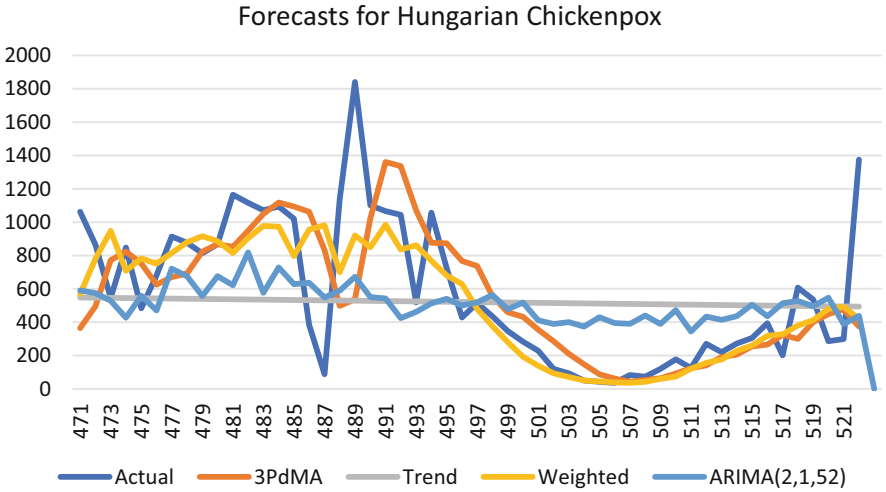


Fig. 6.6 Model plots versus actual

Table 6.2 Model MSEs

| Model                      | MSE     |
|----------------------------|---------|
| 3-period moving average    | 118,557 |
| OLS trend model            | 176,134 |
| OLS trend plus seasonality | 85,571  |
| ARIMA(2,1,52)              | 138,056 |
| ARIMA(3,1,52)              | 136,547 |

```
Identify var=Hungary(1);
Estimate p=3 q=52;
Run;
Forecast lead=52 interval=week id=Date out=results;
Run;
```

Figure 6.6 shows the results of models. A better test is to measure something like mean square error (MSE). Table 6.2 shows the results of these model, including an ARIMA(2,1,52) model run as a check for P-values. The two ARIMA models were very similar, both with instability warnings as there is limited data for all of the model parameters included. But the 3-period partial autocorrelation model had a slight advantage. The 3-period moving average model turned out to be quite a bit better, as can be seen in Fig. 6.6. The moving average models will always lag the actual data, but provide a decent fit. The OLS trend model (the straight line in Fig. 6.6) was the worst but adding back the relative seasonality gave by far the best model here. In effect, that model is an ARIMA(0,1,52) model.

## 6.10 Conclusions

Time series forecasting is very important, as there are many real-life applications. We have looked at some basic models, but they can each be effective in particular contexts, depending upon data behavior. Straight-line trend models are simple regressions, which only work well with very consistent data. Moving average models (and exponential smoothing) are good for short-term forecasts when there is cyclical behavior. ARIMA models are useful to pick up complex patterns involving autocorrelation. There are many other time series forecasting models as well, each useful in some context.

In the Hungarian chickenpox data, there was evidently not enough datapoints to make ARIMA work well. What worked best was a simple OLS trend model modified by seasonality averages. In forecasting time series data, a good start is to plot the data over time, looking for trends and cycles.

## References

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# Chapter 7

## Classification Models



**Keywords** Classification models · Random forest · Extreme boosting · Logistic regression · Decision trees · Support vector machines · Neural networks

Classification is a major data mining application. It applies to cases with a finite number of outcomes (usually two), with the idea of predicting which outcome will occur for a given set of circumstances (survival or death for a medical event, such as surgery; presence of a disease or not, such as monkey pox).

### 7.1 Basic Classification Models

This chapter will cover some basic classification tools.

#### 7.1.1 Regression

**Regression** models fit a function through the data minimizing some error metric. You can include as many independent variables as you want, but in traditional regression analysis, there are good reasons to limit the number of variables. The spirit of exploratory data mining, however, encourages examining a large number of independent variables. In data mining applications, the assumption is that you have very many observations, so that there is no technical limit on the number of independent variables.

### 7.1.2 *Decision Trees*

**Decision trees** are models that process data to split it in strategic places to divide the data into groups with high probabilities of one outcome or another. They are widely used because the resulting model is easy to understand. Decision trees consist of nodes, or splits in the data defined as particular cutoffs for a particular independent variable, and leaves, which are the outcome. It is especially effective at data with finite categorical outcomes, but can also be applied to continuous data, such as time series (but the results are limited as it can only predict a finite number of continuous outcomes). For categorical data, the outcome is a class. For continuous data, the outcome is a continuous number, usually some average measure of the dependent variable. Application of decision tree models to continuous data is referred to as regression trees.

### 7.1.3 *Random Forest*

**Random forest** models are an ensemble of un-pruned decision trees. Essentially they consist of a melding of many decision tree runs. They are often used when there are large training datasets available with many input variables. They tend to be robust to variance and bias, and thus more reliable than single decision trees.

### 7.1.4 *Extreme Boosting*

**Extreme boosting** builds a series of decision tree models and associates a weight with each dataset observation. Weights are increased (boosted) if a model incorrectly classifies the observation. Along with random forests, they tend to fit data quite well.

### 7.1.5 *Logistic Regression*

**Logistic regression** is a regression with a finite number of dependent variable values (especially 0 and 1). The data is fit to a logistic function. The purpose of logistic regression is to classify cases into the most likely category.

Logistic regression provides a set of  $\beta$  parameters for the intercept (or intercepts in the case of ordinal data with more than two categories) and independent variables, which can be applied to a logistic function to estimate the probability of belonging to a specified output class. The formula for probability of acceptance of a case  $i$  to a stated class  $j$  is:

$$P_J = \frac{1}{1 + e^{(-\beta_0 - \sum_{i=1}^n \beta_i x_i)}}$$

where  $\beta$  coefficients are obtained from logistic regression.

Probit models are an alternative to logistic regression. Both estimate probabilities, usually with similar results, but probit models tend to have smaller coefficients and use probit function instead of logit function.

### 7.1.6 Support Vector Machines

**Support vector machines** (SVMs) are supervised learning methods that generate input–output mapping functions from a set of labeled training data. The mapping function can be either a classification function (used to categorize the input data) or a regression function (used to estimation of the desired output). For classification, nonlinear kernel functions are often used to transform the input data (inherently representing resenting highly complex nonlinear relationships) to a high-dimensional feature space in which the input data becomes more separable (i.e., linearly separable) compared to the original input space. Then, the maximum-margin hyperplanes are constructed to optimally separate the classes in the training data. Two parallel hyperplanes are constructed on each side of the hyperplane that separates the data by maximizing the distance between the two parallel hyperplanes. An assumption is made that the larger the margin or distance between these parallel hyperplanes, the better the generalization error of the classifier will be.

### 7.1.7 Neural Networks

**Neural network** models are applied to data that can be analyzed by alternative models. The normal data mining process is to try all alternative models and see which works best for a specific type of data over time. But there are some types of data where neural network models usually outperform alternatives, such as regression or decision trees. Neural networks tend to work better when there are complicated relationships in the data, such as high degrees of nonlinearity. Thus, they tend to be viable models in problem domains where there are high levels of unpredictability. Each node is connected by an arc to nodes in the next layer. These arcs have weights, which are multiplied by the value of incoming nodes and summed. The input node values are determined by variable values in the dataset. Middle layer node values are the sum of incoming node values multiplied by the arc weights. These middle node values in turn are multiplied by the outgoing arc weights to successor nodes. Neural networks “learn” through feedback loops. For a given input, the output for starting weights is calculated. Output is compared to target

values, and the difference between attained and target output is fed back to the system to adjust the weights on arcs. This process is repeated until the network correctly classifies the proportion of learning data specified by the user (tolerance level). Ultimately a set of weights might be encountered that explain the learning (training) dataset very well. The better the fit that is specified, the longer the neural network will take to train, although there is really no way to accurately predict how long a specific model will take to learn. The resulting set of weights from a model that satisfies the set tolerance level is retained within the system for application to future data. The neural network model is a black box. Output is there, but it is too complex to analyze.

There are other models that have been applied to classification. Clustering has been used but is not really appropriate for classification—it is better at initial analysis trying to identify distinct groups. Clustering requires numeric data. Naïve Bayes models have also been applied, but only apply to categorical data.

We will demonstrate classification models with a medical dataset involving employee attrition.

## 7.2 Watson Healthcare Data

IBM created Watson, an artificial intelligence agent that was successful at the game Jeopardy and was then applied to healthcare. One of these applications was to nursing turnover in a northeastern US healthcare facility. A masked dataset based upon this application has been posted to the [www.kaggle.com](https://www.kaggle.com/datasets/jpmiller/employee-attrition-for-healthcare) website (<https://www.kaggle.com/datasets/jpmiller/employee-attrition-for-healthcare>) which posts many datasets for users to apply (<https://www.kaggle.com/datasets/jpmiller/employee-attrition-for-healthcare>). There are 33 variables, with changes made from real numbers for public consumption without risking revealing private information. Table 7.1 lists the variables:

The target variable is Attrition, which is binary. Of the categorical data, we can check the relative attrition to determine if we want to pursue that variable further (by splitting the data). Table 7.2 gives percentages of attrition by category:

Review of Table 7.2 shows a lower attrition rate in neurology. There is a much greater difference in position, with nurses having much higher attrition rates. Further, single employees had a much higher attrition rate, as did those with frequent travel. Some of these variables are ordinal in some sense, but it is dangerous to tag them with numbers. It is better to split the data and run separate models, especially for nurses, single employees, and frequent travelers.

Table 7.3 ranks variables by correlation with Attrition showing those variables with correlation  $\geq 0.1$ . For these variables, cross-correlations  $\geq 0.5$  are also shown. Pairing variables with high cross-correlation is to be avoided.

Table 7.1 Watson healthcare data variables

| Variable                | Type        | Variable        | Type        | Variable      | Type        |
|-------------------------|-------------|-----------------|-------------|---------------|-------------|
| Age                     | Numeric     | HourlyRate      | Numeric     | PerfRat       | Rating      |
| Attrition               | Binary      | JobInvolvement  | Numeric     | RelSatis      | Rating      |
| BusinessTravel          | Categorical | JobLevel        | Numeric     | StandardHours | Constant    |
| DailyRate               | Numeric     | JobRole         | Categorical | Shift         | Categorical |
| Department              | Categorical | JobSatisfaction | Rating      | WorkYr        | Numeric     |
| DistanceFromHome        | Numeric     | MaritalStatus   | Categorical | TrainTime     | Numeric     |
| Education               | Numeric     | MonthlyIncome   | Numeric     | WLBalance     | Rating      |
| EducationField          | Categorical | MonthlyRate     | Numeric     | YrComp        | Numeric     |
| EmployeeCount           | Constant    | Companies       | Numeric     | YrRole        | Numeric     |
| EnvironmentSatisfaction | Rating      | OT              | Binary      | YrPromo       | Numeric     |
| Gender                  | Binary      | %SalHike        | Numeric     | YrCurMgr      | Numeric     |

**Table 7.2** Attrition by category

| Department       | Total | Attrition | Percentage |
|------------------|-------|-----------|------------|
| Maternity        | 796   | 98        | 12.3       |
| Cardiology       | 531   | 74        | 13.9       |
| Neurology        | 349   | 27        | 7.7        |
| Totals           | 1676  | 199       | 11.9       |
| Position         | Total | Attrition | Percentage |
| Nurse            | 822   | 107       | 13.0       |
| Therapist        | 189   | 4         | 2.1        |
| Administrative   | 131   | 1         | 0.8        |
| Other            | 534   | 87        | 16.3       |
| Totals           | 1676  | 199       | 11.9       |
| Degree           | Total | Attrition | Percentage |
| Life Sciences    | 697   | 84        | 12.1       |
| Medical          | 524   | 51        | 9.7        |
| Marketing        | 189   | 28        | 14.8       |
| Technical Degree | 149   | 22        | 14.8       |
| Human Resources  | 29    | 6         | 20.7       |
| Other            | 88    | 8         | 9.1        |
| Totals           | 1676  | 199       | 11.9       |
| Marital Status   | Total | Attrition | Percentage |
| Married          | 777   | 61        | 7.9        |
| Single           | 522   | 114       | 21.8       |
| Divorced         | 377   | 24        | 6.4        |
| Totals           | 1676  | 199       | 11.9       |
| Travel           | Total | Attrition | Percentage |
| Rare             | 1184  | 126       | 10.6       |
| Frequent         | 320   | 57        | 17.8       |
| Non-Travel       | 172   | 16        | 9.3        |
| Totals           | 1676  | 199       | 11.9       |

**7.2.1 Initial Decision Tree**

We run the decision tree algorithm at complexity 0.1 on the 70% training data, obtaining Table 7.4:

We obtain the 18 rules shown, using 11 variables. Model fit had a specificity of 0.950, sensitivity of 0.567 m accuracy of 0.905, and area under the curve of 0.872.

**7.2.2 Variable Selection**

We used decision tree models with different complexity levels to select a set of variables, as well as stepwise regression. The decision tree complexity levels used were 0.05 (generating a set of 3 variables) and 0.01 (generating a set of 10 variables),

Table 7.3 Correlation with attrition

| Variable                | Attrition | WorkYr | JobLevel | MonthlyInc | YrComp | MonthlyInc | YrPromo | YrCurMgr |
|-------------------------|-----------|--------|----------|------------|--------|------------|---------|----------|
| OT                      | 0.337     |        |          |            |        |            |         |          |
| Age                     | -0.240    | 0.693  | 0.518    | 0.511      |        |            |         |          |
| WorkYr                  | -0.234    |        | 0.781    |            | 0.622  | 0.773      |         |          |
| YrRole                  | -0.208    |        |          |            | 0.759  |            | 0.548   | 0.722    |
| JobLevel                | -0.208    |        |          | 0.952      | 0.533  |            |         |          |
| YrComp                  | -0.201    |        |          | 0.514      | 0.617  |            |         | 0.771    |
| YrCurMgr                | -0.201    |        |          |            |        |            |         |          |
| MonthlyIncome           | -0.194    |        |          |            |        |            |         |          |
| JobInvolvement          | -0.166    |        |          |            |        |            |         |          |
| Shift                   | -0.158    |        |          |            |        |            |         |          |
| DistanceFromHome        | 0.106     |        |          |            |        |            |         |          |
| EnvironmentSatisfaction | -0.101    |        |          |            |        |            |         |          |

**Table 7.4** Decision tree 70% training ( $n = 1173$ )

|               |                        |                      |                         |                          |  |  |                        |                      | $n$ | Outcome | Prob<br>{0} |
|---------------|------------------------|----------------------|-------------------------|--------------------------|--|--|------------------------|----------------------|-----|---------|-------------|
| OF < 0.5      | YrComp $\geq 1.5$      |                      |                         |                          |  |  |                        |                      | 715 | 0       | 0.9776      |
|               | YrComp < 1.5           | Age $\geq 33$        |                         |                          |  |  |                        |                      | 57  | 0       | 0.9649      |
|               |                        | Age < 33             | RelSat $\geq 3.5$       |                          |  |  |                        |                      | 16  | 0       | 0.8750      |
|               |                        |                      | RelSat < 3.5            | JobInvolve $\geq 2.5$    |  |  | EnvSat $\geq 2.5$      |                      | 16  | 0       | 0.9375      |
|               |                        |                      |                         |                          |  |  | EnvSat < 2.5           |                      | 12  | 1       | 0.4167      |
|               |                        |                      |                         | JobInvolve < 2.5         |  |  |                        |                      | 12  | 1       | 0.0833      |
| OT $\geq 0.5$ | MonthlyInc $\geq 3924$ | DistHome < 21.5      |                         |                          |  |  |                        |                      | 18  | 0       | 0.9086      |
|               |                        | DistHome $\geq 21.5$ | Age $\geq 34.5$         |                          |  |  |                        |                      | 24  | 0       | 0.7500      |
|               |                        |                      | Age < 34.5              |                          |  |  |                        |                      | 8   | 1       | 0.0000      |
|               | MonthlyInc < 3924      | Age $\geq 33.5$      | DailyRate $\geq 1383.5$ |                          |  |  |                        |                      | 9   | 0       | 1.0000      |
|               |                        |                      | DailyRate < 1383.5      | Shift $\geq 1.5$         |  |  |                        |                      | 7   | 1       | 0.0000      |
|               |                        |                      |                         | Shift < 1.5              |  |  | MonthlyInc $\geq 2420$ | TrainTime $\geq 2.5$ | 11  | 0       | 0.8182      |
|               |                        |                      |                         |                          |  |  |                        | TrainTime < 2.5      | 4   | 1       | 0.4000      |
|               |                        |                      |                         |                          |  |  | MonthlyInc < 2420      |                      | 10  | 1       | 0.2000      |
|               |                        | Age < 33.5           | JobInvolve $\geq 2.5$   | MonthlyInc $\geq 2194.5$ |  |  | Shift $\geq 0.5$       |                      | 7   | 0       | 0.6316      |
|               |                        |                      |                         |                          |  |  | Shift < 0.5            |                      | 3   | 1       | 0.2143      |
|               |                        |                      |                         | MonthlyInc < 2194.5      |  |  |                        |                      | 8   | 1       | 0.0000      |
|               |                        |                      |                         |                          |  |  |                        |                      | 28  | 1       | 0.0357      |

while stepwise regression settings used were `slentry` of 0.8 and `slstay` of 0.7 (generating 18 variables). Table 7.5 shows the variables selected:

Looking back at Table 7.3, the decision tree-generated datasets pair Age with JobLevel, a pair of variables that have a degree of multicollinearity. We will proceed with the variable sets as given. Note that stepwise regression has no variable pairs showing high cross-correlation in Table 7.3.

Results are given in terms of sensitivity (true positive divided by (true positive plus false negative)), specificity (true negative over (true negative plus false positive)), accuracy (true over total), and area under the curve (Table 7.6):

The original decision tree model, used to identify variables, had a worse specificity and area under the curve than any of these five models. It was better on sensitivity than the random forest, SVM, and neural network models, and better on accuracy than the neural network model. We only used the original decision tree model to identify variables and excluded it from other comparisons.

Looking at the five model results when using all 25 variables, it appears that linear regression might be the best. It had the highest sensitivity. All five models had similar accuracy, although the neural network model was a bit lower. But the neural network model was degenerate (it called all cases not attriting). While neural network models can be fine-tuned, that extra work may not be worth the effort given the results from the other four model types.

Results from the models with reduced variables are shown in Table 7.7:

Table 7.7 numbers are displayed in Fig. 7.1:

The results for the model using variables selected by stepwise regression are shown in Fig. 7.2:

The results for the 18-variable model generated through stepwise regression are very similar to that of the full model. The highest accuracy came from the SVM model, although it was only slightly better than that of the random forest model, logistic regression, or extreme boosting. The extreme boosting and logistic regression models were best at sensitivity, predicting attriting nurses. Overall, it seems that there is just as much if not better results from eliminating the seven variables reduced by using stepwise regression than using the full set of variables.

Figure 7.3 displays error metrics for the model using the 10 variables generated with a decision tree using complexity of 0.01:

The 10 variables identified by the decision tree model using complexity of 0.01 yielded similar results to that of the stepwise model with 18 variables. One improvement was for the neural network model, which was no longer degenerate (it identified some cases as leaving). Overall accuracy was very similar to the models using 25 or 18 variables, on average even better by a small amount.

Figure 7.4 shows model results for the model using the 3 variables generated by a decision tree model with a complexity setting of 0.005:

The 3-variable model did slightly worse than the 10-variable model in terms of sensitivity, accuracy, and area under the curve. The overall conclusion we draw is that it pays to reduce variables, but not too far. We next examine if there were differences in results for the subset of nurses as opposed to the above models applied to data from all employees.

**Table 7.5** Variable selection

|                    |                    |                 |
|--------------------|--------------------|-----------------|
| DT Complexity 0.05 | DT Complexity 0.01 | Stepwise        |
| OT                 | OT                 | OT              |
| Age                | Age                | Age             |
| JobLevel           | JobLevel           |                 |
|                    | Shift              | Shift           |
|                    | JobInvolvement     | JobInvolvement  |
|                    | EnvSatisfaction    | EnvSatisfaction |
|                    | DistanceHome       | DistanceHome    |
|                    | WeekYr             | WeekYr          |
|                    | YrPromo            | YrPromo         |
|                    | HourlyRate         | HourlyRate      |
|                    |                    | YrRole          |
|                    |                    | Companies       |
|                    |                    | WLBalance       |
|                    |                    | YrCurMgr        |
|                    |                    | DailyRate       |
|                    |                    | MonthlyRate     |
|                    |                    | RelSat          |
|                    |                    | HourlyRate      |
|                    |                    | TrainTime       |
| 3 variables        | 10 variables       | 18 variables    |

7.2.3 Nurse Data

The initial decision tree model yielded 13 rules using 8 variables, slightly less than the full model. The metrics for this model were specificity of 0.991, sensitivity 0.375, accuracy of 0.911, and area under the curve 0.965. The area under the curve actually improved over the model using all employees, but sensitivity declined. The decision tree is given in Table 7.8:

The difference between nurses and the other employees is fairly clear. Decision trees have a strong advantage in being clearly transparent (once you sift out the rules). Nurses with low overtime and older than 31 years tend to stay—those younger are sensitive to their daily rate of pay. Nurses with monthly income over 3200 tend to stay, but if their monthly income is less than 6200, there is some attrition (although that is based on a very low sample size). Nurses with higher overtime and monthly income over 3200 tend to stay with small exceptions based on small sample sizes. If income is less than 3200, the sample sizes are small and results dependent on a complex function of distance from home. Error metrics for this model are given in Table 7.9:

Figure 7.5 displays the relative errors for the nurses data using all 25 variables, followed by Figs. 7.6, 7.7, and 7.8 for the other models:

**Table 7.6** Full model coincident charts and metrics

|                     | Full     | All 25 variables |       |             |             |          |       |
|---------------------|----------|------------------|-------|-------------|-------------|----------|-------|
|                     | Actual 0 | Actual 1         | Total | Specificity | Sensitivity | Accuracy | AuC   |
| Extreme Boost       |          |                  |       |             |             |          |       |
| Model 0             | 426      | 18               | 444   | 0.959       |             |          |       |
| Model 1             | 22       | 37               | 59    |             | 0.627       |          | 0.927 |
| Totals              | 448      | 55               | 503   |             |             | 0.920    |       |
| Random Forest       |          |                  |       |             |             |          |       |
| Model 0             | 435      | 9                | 444   | 0.980       |             |          |       |
| Model 1             | 32       | 27               | 59    |             | 0.458       |          | 0.924 |
| Totals              | 467      | 36               | 503   |             |             | 0.918    |       |
| SVM                 |          |                  |       |             |             |          |       |
| Model 0             | 432      | 12               | 444   | 0.973       |             |          |       |
| Model 1             | 35       | 24               | 59    |             | 0.407       |          | 0.936 |
| Totals              | 467      | 36               | 503   |             |             | 0.907    |       |
| Logistic Regression |          |                  |       |             |             |          |       |
| Model 0             | 429      | 15               | 444   | 0.966       |             |          |       |
| Model 1             | 23       | 36               | 59    |             | 0.610       |          | 0.942 |
| Totals              | 452      | 51               | 503   |             |             | 0.924    |       |
| Neural Net          |          |                  |       |             |             |          |       |
| Model 0             | 444      | 0                | 444   | 1.000       |             |          |       |
| Model 1             | 59       | 0                | 59    |             | 0.000       |          | —     |
| Totals              | 503      | 0                | 503   |             |             | 0.883    |       |

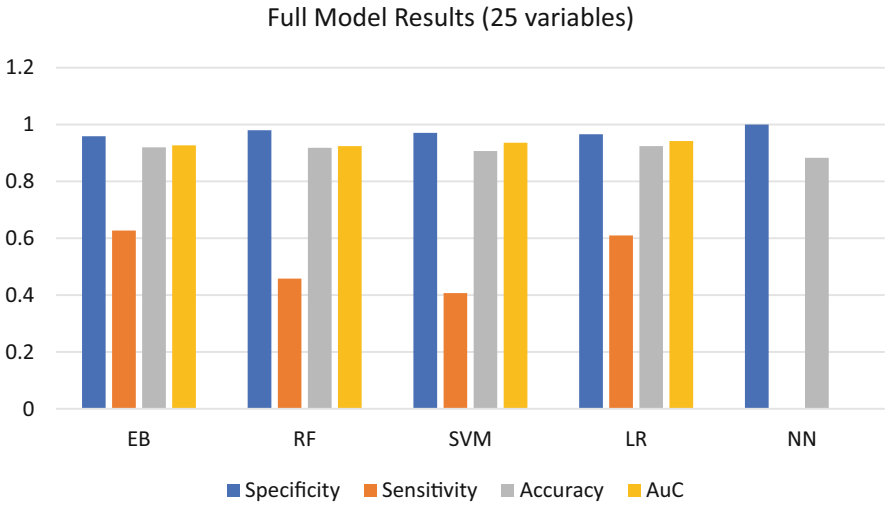
## 7.3 Example Case

The US civilian healthcare system has been found to have a significant correlation between stroke patient outcomes and race. The US civilian healthcare system has wide variations in care access and quality of care. The US military healthcare system conversely has universal access. Holtkamp (2020) collected data over the years 2010 through 2015 to compare race with healthcare morbidity outcomes for acute ischemic stroke, with secondary outcomes of hospitalization cost and length of hospital stay.

The data involved 3623 patients. Independent variables by factor are demographic factors (age, rank, gender, marital status), predisposition (comorbidities of hypertension, diabetes mellitus, and atrial fibrillation), and hospital care (hospital size, specialty, time of admission). Age and length of stay were continuous, while all other variables were binary. Outcome measures were evaluated with a multivariable binary logistic regression model. Models were run for white, African American, Asian, and other (not identified) subjects. In logistic regression, an odds ratio is the constant effect of an independent variable on the likelihood of an outcome for binary variables. Table 7.10 gives outcomes in terms of odds ratios and  $p$ -values by racial group for the first three outcomes, and the beta coefficient reflecting change from average for LOS and Full cost (which are continuous variables).

**Table 7.7** Metrics for models

|              | EB    | RF    | SVM   | LR    | NN    |
|--------------|-------|-------|-------|-------|-------|
| Full (25)    |       |       |       |       |       |
| Specificity  | 0.959 | 0.98  | 0.971 | 0.966 | 1     |
| Sensitivity  | 0.627 | 0.458 | 0.407 | 0.61  | 0     |
| Accuracy     | 0.92  | 0.918 | 0.907 | 0.924 | 0.883 |
| AuC          | 0.927 | 0.924 | 0.936 | 0.942 | 0     |
| Step (18)    |       |       |       |       |       |
| Specificity  | 0.959 | 0.986 | 0.986 | 0.964 | 1     |
| Sensitivity  | 0.559 | 0.424 | 0.441 | 0.559 | 0     |
| Accuracy     | 0.913 | 0.92  | 0.922 | 0.917 | 0.883 |
| AuC          | 0.935 | 0.935 | 0.946 | 0.948 | 0     |
| DT 0.01 (10) |       |       |       |       |       |
| Specificity  | 0.962 | 0.973 | 0.98  | 0.973 | 0.882 |
| Sensitivity  | 0.576 | 0.525 | 0.373 | 0.441 | 0.441 |
| Accuracy     | 0.917 | 0.92  | 0.909 | 0.911 | 0.913 |
| AuC          | 0.916 | 0.92  | 0.908 | 0.914 | 0.882 |
| DT 0.005 (3) |       |       |       |       |       |
| Specificity  | 0.968 | 0.973 | 0.968 | 0.977 | 0.968 |
| Sensitivity  | 0.441 | 0.424 | 0.424 | 0.39  | 0.424 |
| Accuracy     | 0.907 | 0.909 | 0.905 | 0.909 | 0.905 |
| AuC          | 0.851 | 0.822 | 0.739 | 0.841 | 0.856 |



**Fig. 7.1** Comparison of results full model

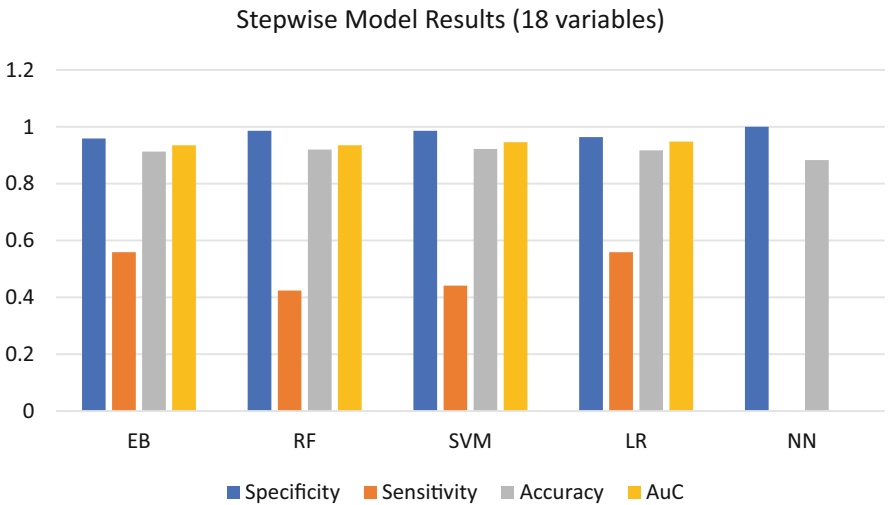


Fig. 7.2 Comparison of results stepwise model

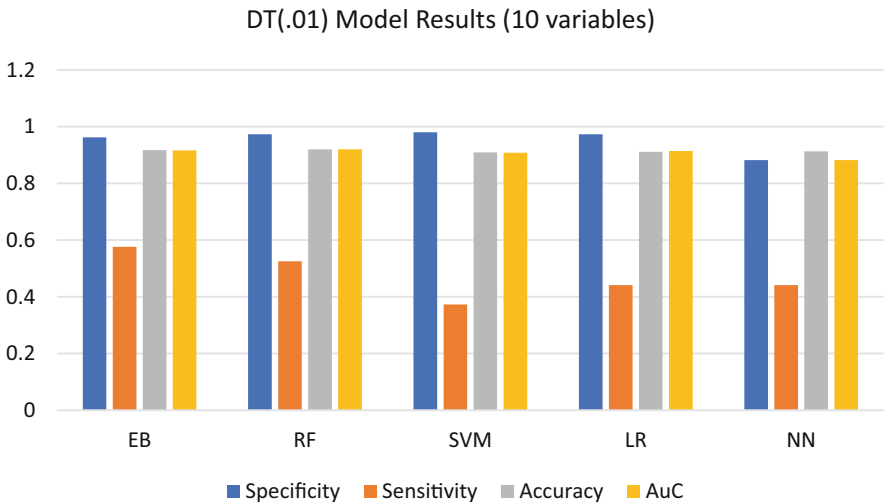
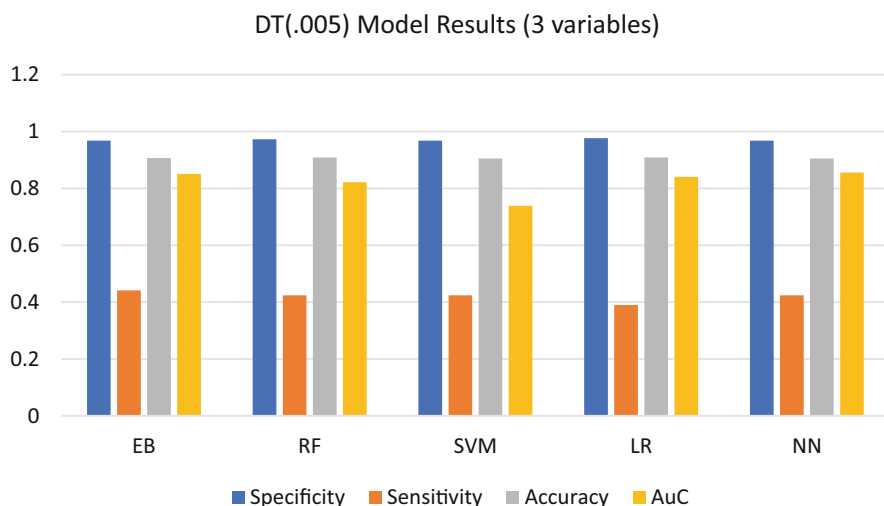


Fig. 7.3 Comparison of results DT(0.01) model

IV-tPA administration was similar among all racial groups, with no significant differences. Negative outcomes and mortality were slightly higher for whites (17.5% negative outcome, 2.9% in-hospital mortality) than for African Americans (15.3% negative outcomes, 1.3% in-hospital mortality) or for Asians (16.7% negative outcomes, 5.5% in-hospital mortality). There were no significant differences in these outcomes. Asians had longer length of stay at 6.42 days compared to 4.11 days for whites and 4.47 days for African Americans. After adjustment for baseline



**Fig. 7.4** Comparison of results DT(0.005) model

demographics and clinical characteristics, there was no statistical significance across races for length of stay. Full costs were highest for Asians at \$22,244 compared to \$15,377 for whites and \$17,676 for African Americans. After adjusting for baseline demographics and clinical characteristics, only African American costs were statistically significant (higher).

The case demonstrates the use of logistic and OLS regression on healthcare data. The odds ratio is useful to interpret logistic regression output.

## 7.4 Summary

We review what we see as essential points in classification modeling:

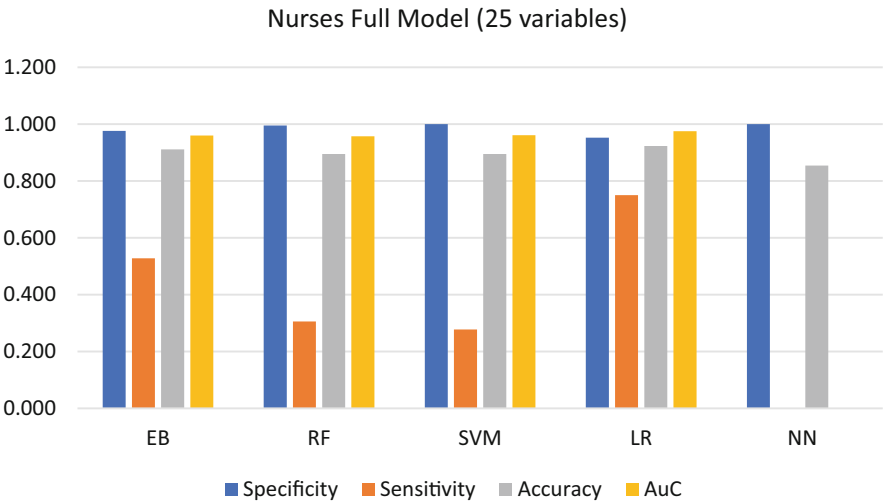
1. The process of training, testing, and applying the model is best done if you have independent training and test sets. In this case, you do not want to partition the data. If you need to skip the test set, you can partition, selecting the proportion of the dataset you want the software to hold out for testing.
2. Classification is for categorical target variables. Prior chapters have covered techniques for continuous target variables. Ordinary least squares is the common regression model for continuous variables. Logistic regression (or probit) works for categorical targets. Decision trees work best for categorical targets. You can run regression trees on continuous targets, but as we discussed in that chapter, the finite number of possible predictions make regression trees less believable.
3. Models can be controlled. Regression is controlled primarily through variable selection. Using all available input variables will always yield a higher  $r$ -squared

**Table 7.8** Decision tree for the full model (Complexity 0.01)

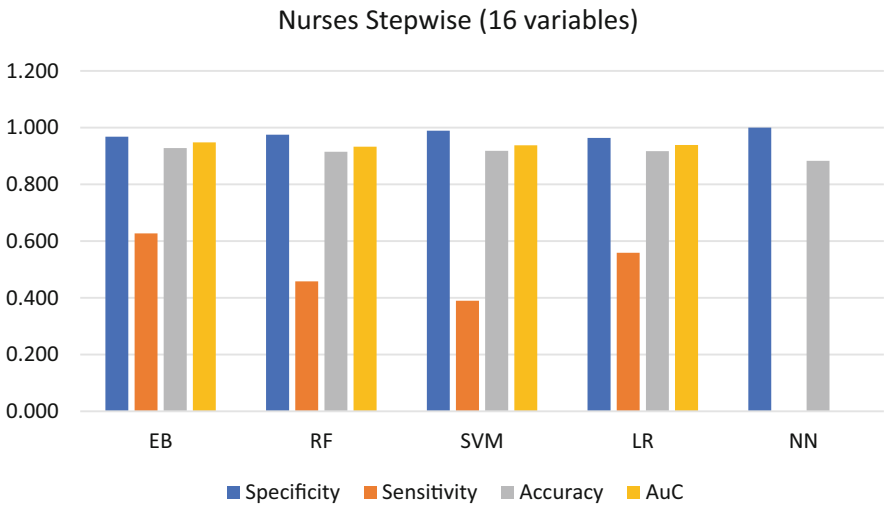
|          |                   |                      |                   |                   |                   |     |   |        |
|----------|-------------------|----------------------|-------------------|-------------------|-------------------|-----|---|--------|
| OT < 0.5 | YrComp ≥ 2.5      |                      |                   |                   |                   | 341 | 0 | 0.9765 |
|          | YrComp < 2.5      | Age ≥ 31.5           |                   |                   |                   | 52  | 0 | 0.9615 |
| OT ≥ 0.5 |                   | Age < 31.5           | DailyRate ≥ 554.5 |                   |                   | 24  | 0 | 0.7083 |
|          |                   |                      | DailyRate < 554.5 |                   |                   | 8   | 1 | 0.1250 |
|          | MonthlyInc ≥ 3191 | Shift ≥ 0.5          |                   |                   |                   | 68  | 0 | 0.9625 |
|          |                   | Shift < 0.5          | MonthlyInc < 5221 |                   |                   | 12  | 0 | 0.9167 |
|          |                   |                      | MonthlyInc ≥ 5221 | DailyRate < 437.5 | MonthlyInc ≥ 6184 | 18  | 0 | 0.7778 |
|          |                   |                      |                   |                   | MonthlyInc < 6184 | 8   | 1 | 0.2500 |
|          |                   |                      |                   | DailyRate ≥ 437.5 |                   | 7   | 1 | 0.1429 |
|          |                   |                      |                   | DistHome ≥ 9      |                   | 7   | 0 | 0.8571 |
|          | MonthlyInc < 3191 | MonthlyRate ≥ 6041.5 | DistHome < 21.5   | DistHome < 9      |                   | 14  | 1 | 0.4286 |
|          |                   |                      | DistHome ≥ 21.5   |                   |                   | 7   | 1 | 0.0000 |
|          |                   | MonthlyRate < 6041.5 |                   |                   |                   | 9   | 1 | 0.0000 |

**Table 7.9** Nurses full model coincident charts and metrics

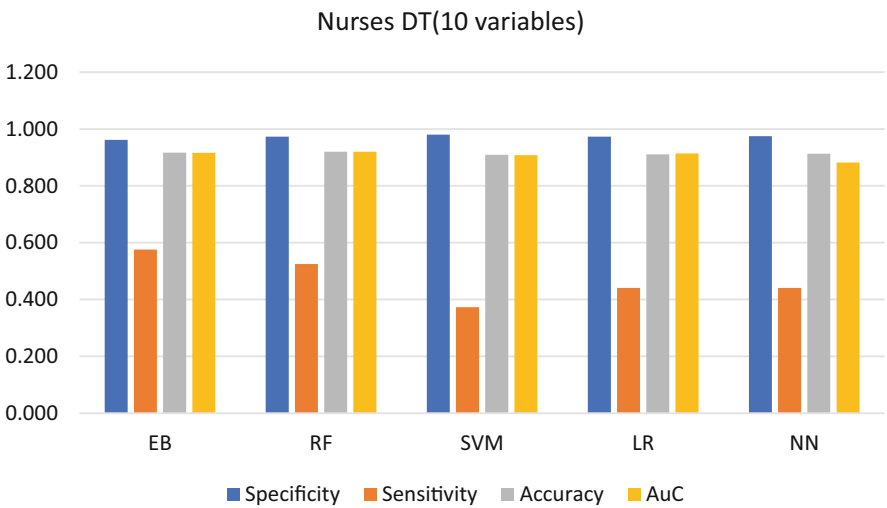
|                     | Full     | All 25 Variables |       |             |             |          |       |
|---------------------|----------|------------------|-------|-------------|-------------|----------|-------|
|                     | Actual 0 | Actual 1         | Total | Specificity | Sensitivity | Accuracy | AuC   |
| Extreme Boost       |          |                  |       |             |             |          |       |
| Model 0             | 108      | 0                | 108   | 0.976       |             |          |       |
| Model 1             | 6        | 10               | 16    |             | 0.528       |          | 0.960 |
| Totals              | 114      | 10               | 124   |             |             | 0.911    |       |
| Random Forest       |          |                  |       |             |             |          |       |
| Model 0             | 108      | 0                | 108   | 0.995       |             |          |       |
| Model 1             | 9        | 7                | 16    |             | 0.306       |          | 0.957 |
| Totals              | 117      | 7                | 124   |             |             | 0.895    |       |
| SVM                 |          |                  |       |             |             |          |       |
| Model 0             | 108      | 0                | 108   | 1.000       |             |          |       |
| Model 1             | 12       | 4                | 16    |             | 0.278       |          | 0.961 |
| Totals              | 120      | 4                | 124   |             |             | 0.895    |       |
| Logistic Regression |          |                  |       |             |             |          |       |
| Model 0             | 104      | 4                | 108   | 0.953       |             |          |       |
| Model 1             | 6        | 10               | 16    |             | 0.750       |          | 0.975 |
| Totals              | 110      | 14               | 124   |             |             | 0.923    |       |
| Neural Net          |          |                  |       |             |             |          |       |
| Model 0             | 108      | 0                | 108   | 1.000       |             |          |       |
| Model 1             | 16       | 0                | 16    |             | 0.000       |          | —     |
| Totals              | 124      | 0                | 124   |             |             | 0.854    |       |



**Fig. 7.5** Errors for the nurse data model using all 25 variables

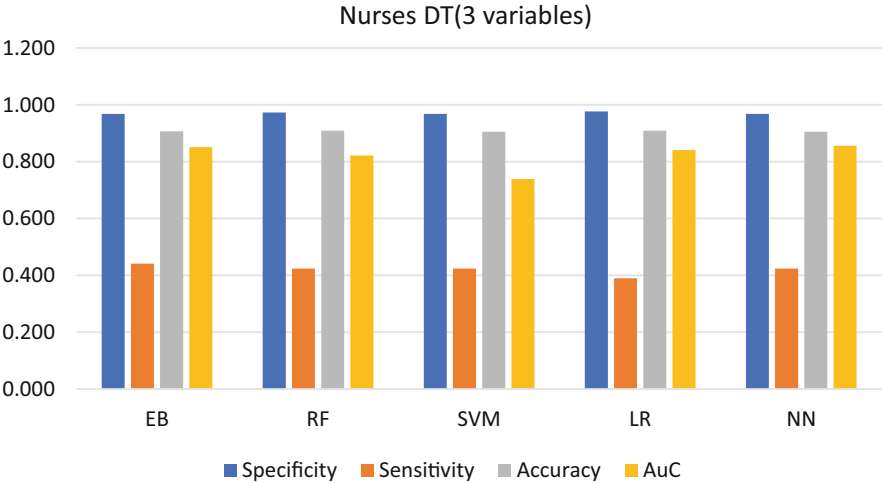


**Fig. 7.6** Errors for nurse data using stepwise generated 16 variables



**Fig. 7.7** Errors for nurse data using decision tree generated 10 variables

but will usually involve overlapping information that makes the beta coefficients unstable. Predicting isn't a problem as long as you have inputs for all independent variables, but trimmed models (using fewer but better variables) are stabler. Decision trees are controlled by setting minimum support and minimum confidence. An easier control within Rattle is to use the Complexity parameter. A lower complexity will yield models with fewer rules, which are easier to implement. Neural network models can be controlled by adjusting hidden layers, although "control" is probably too clear a term in this case.



**Fig. 7.8** Errors for nurse data using decision tree generated 3 variables

**Table 7.10** Holtkamp (2020) logistic regression results

| Outcome     | Odds ratio<br>White | Odds ratio<br>African | Odds ratio<br>Asian | <i>p</i> -value<br>White | <i>p</i> -value<br>African Amer. | <i>p</i> -value<br>Asian |
|-------------|---------------------|-----------------------|---------------------|--------------------------|----------------------------------|--------------------------|
| IV-tPA      | 0.83                | 0.61                  | 0.41                | 0.54                     | 0.26                             | 0.17                     |
| Neg outcome | 0.94                | 0.85                  | 0.79                | 0.61                     | 0.28                             | 0.22                     |
| Mortality   | 0.82                | 0.55                  | 1.80                | 0.49                     | 0.17                             | 0.09                     |
| LOS         | −0.32               | −0.42                 | 0.40                | 0.11                     | 0.10                             | 0.22                     |
| Full cost   | \$933               | −\$1404               | \$3270              | 0.33                     | 0.02                             | 0.43                     |

IV-tPA is an intravenous procedure. Neg outcome is death or sending the patient to another care facility. LOS is length of stay in days

4. There are alternative software products available. Commercial software products are very good, but often very expensive. Rattle, i.e., an R package for datamining applications, and WEKA are open source. There are quite a few other open-source software products. WEKA is very interesting from a research perspective, with many optional modeling techniques. Rattle is a GUI for R, which makes it commercially viable and scalable.

Reference

Holtkamp MD (2020) Does race matter in universal healthcare? Stroke cost and outcomes in US military health care. *Ethn Health* 25(6):888–896

# Chapter 8

## Applications of Predictive Data Mining in Healthcare



**Keywords** Applications · Predictive models · Ethics

The era of big data offers many opportunities for data mining in the healthcare domain. This chapter will look at some of the predictive data mining applications in healthcare management. We will first talk about where such data is found. We present a demonstrative application of classification modeling, then review applications by data mining modeling for a number of diseases. After comparing methods, we review a study of ethics in predictive analytics.

### 8.1 Healthcare Data Sources

The medical field has always generated a great deal of data. Knowledge bases in the biomedical domain are usually sparse, redundant, and incomplete relative to those typically found in other fields. They contain non-topological domain information such as gene expressions, chemical structures, and disease descriptions that involve complex interconnections. An example of such data was covered by Zheng et al. (2021) in the field of pharmaceuticals. This scientific domain deals with complex biological systems and pathologies, which have become critical in medical practice. They developed a multi-relational biomedical knowledge graph system of more than 500,000 interconnections between genes, drugs, and diseases. It was constructed from six public databases. Trivial entities were filtered out. Knowledge graphs are multi-relational directed graphs with nodes representing entities and edges representing their relations. Entities and relationships were embedded in the knowledge graph by distance-based scoring functions, semantic matching scoring functions, or neural network-based scoring functions. Case studies were reported on an analysis of drug repurposing for Alzheimer's disease and for Parkinson's disease. A major limitation noted was the relative lack of negative samples in the database.

Contemporary society has seen developments that have led to additional sources of healthcare data. Electronic health records (EHR) have been a goal of government

for over 30 years, with the primary goal of improving management of patient health-related information. They improve healthcare service quality by:

- Enabling data sharing;
- Allowing access to data in real-time across healthcare providers;
- Better tracking of data over time;
- Automation of clinician workflow;
- Timely reminders for patient screenings and preventive checkups.

EHRs also facilitate research, especially in the form of data mining. Applications include tracking disease progression, health risk prediction, and detection of adverse event. EHRs have been extensively used for clinical research and for improvement of health practice, collecting a rich set of demographic, medical history, medication, laboratory test results, and diagnoses. The diverse types of data contained, some structured, some unstructured in the form of notes or media such as X-rays, create many challenges. The major issues are inaccurate and inconsistent data, along with privacy and ethical concerns. Data mining has been extensively used (Sarwar et al. 2022) for clinical and research applications to include phenotype extraction, precision medicine, evaluation of interventions, prediction of disease, and monitoring of disease progression. Challenges include diverse data types and characteristics.

In the past 20 years or so there also have been great advances on the Internet of Things (IoT), connecting everything linked to the Internet. This impacts healthcare by collecting data from wearable devices that provide information relative to physical, behavioral, and psychological health. Millions of sensors continuously monitor the temperature, blood pressure, heartbeat, respiration, and blood glucose of patients with the primary purpose of informing healthcare providers of real-time issues. The processing layer analyzes data that is gathered, makes decisions, and raises alarms and notifications if indicated. These wearables and other IoT healthcare devices generate big data, which has been mined to knowledge extraction and for making inferences and predictions. The presence of big data calls for artificial intelligence and machine learning using data mining models to predict the presence of diseases that can be customized for each individual patient. Li et al. (2021) gave a comprehensive review of machine learning techniques for big data analysis in healthcare, evaluating strengths and weaknesses of techniques. These systems have been effective in Cloud-connected systems monitoring elderly patients as well as patients with critical conditions. Some of the challenges involved include limited computational and processing power of devices such as smart phones, microcontrollers, RFIDs, and gateways. Not all systems are completely interoperable, and the aggregate impact on the energy system grows. The data generated becomes redundant, creating a challenge for predictive data mining models.

## 8.2 Example Predictive Model

Neonatal death is a key health indicator. About two thirds of infant deaths occur in the neonatal period. The neonatal mortality rate is a major issue globally. Better public health management and policy-making benefits from accurate prediction of neonatal deaths and identification of related risk factors. This will enable development of better care policies and aid physicians in providing early treatment. Sheikhtaheri et al. (2021) applied machine learning models to analyze and predict neonatal problems.

The study began with a literature review that identified 21 important risk factors. Data was collected from a Tehran neonatal registry database for the period May 2018 through July 2018. There were 138 deaths and 1624 survivors. Of these, four were not recorded in the data available, leaving 17. Univariate statistical analysis in the form of Mann-Whitney and chi-squared tests assessed variable significance, which eliminated three of the 17 factors.

1. Birth weight (quantitative);
2. Gestational age (quantitative);
3. Preterm birth (binary);
4. Small for gestational age (binary);
5. Prenatal care (binary);
6. Mother disease (binary);
7. Respiratory distress syndrome (binary);
8. Steroid therapy (binary);
9. Surfactant administration (binary);
10. Pulmonary hemorrhage (binary);
11. Congenital malformation (binary);
12. Necrotizing enterocolitis (binary);
13. Sepsis (binary);
14. Intraventricular hemorrhage (binary);
15. Asphyxia (binary);
16. Intubation (binary);
17. Ventilation (binary).

The machine learning algorithms applied were Bayesian network, decision trees (random forests, C5.0, CHAID), neural networks, and support vector machines. An ensemble approach was used, meaning that model results were treated as votes. Because the dataset was highly imbalanced, initial models had poor specificity. Therefore, oversampling was applied in the form of synthetic minority oversampling technique (SMOTE) and adaptive synthetic (ADASYN) using R software. Four levels of oversampling were applied with SMOTE, and one with ADASYN yielding roughly equal outcomes. The dataset was split into 70% for training and 30% for testing. Each algorithm was executed ten times with different randomly selected training and test sets.

Results were evaluated in terms of accuracy (correct classifications divided by total), precision (positive predictive value), specificity (true negative rate), sensitivity (true positive rate), F-score, and area under the curve. The outcomes using the smallest number of SMOTE additions were slightly better for most models. Trying a number of different models, 17 variables were included with the best results in terms of accuracy (0.86), sensitivity (0.86), *F*-score (0.91), and area under the curve (0.92) for neural network models. The C5.0 decision tree model had the best precision (0.97) and specificity (0.89). Of the input variables, only intubation was important in all predictive models (birth weight and gestational age were important in four models).

There are thus a variety of sources of healthcare management data. We will now look at some predictive data mining applications in the healthcare domain.

## 8.3 Applications

### 8.3.1 *General Hospital System Management*

Traditional healthcare systems relying on manual data entry have been found to be highly inefficient with respect to allocation of medical resources to patient needs. Skylakha et al. (2020) reported a decision-support system implemented in India to better serve the population. Their system collected data from various government directorates allowing analysis to predict resource needs, to include better allocation of medical doctors. Input information included the number of patients arriving at facilities, newborn mortality rates, government insurance claims, and admissions for particular diseases. Data analytic tools such as logistic regression and decision tree models enabled prediction of hospital conditions by region, enabling better facility planning, suggestion of preventive actions, and resource allocation. The data from reports and publications was used to improve preventive measures. The system provided alerts and notifications of disease and epidemics. Decision-support dashboards generated health profiles to improve public planning. Predictive analytics had the benefits of identifying trends, understanding patients, and improving operations and clinical performance. The effectiveness of the system, outpatient arrival was reduced to controllable levels.

### 8.3.2 *Disease-specific Applications*

As life expectancy has increased along with economic growth, society is aging, with health problems related to dietary change, decreased physical activity, and increases in certain chronic diseases. These chronic diseases in longer-living patients call for healthcare management. Typical dietary habits over time have led to increases in cardiovascular diseases and diabetes, often associated with lifestyle habits such as

smoking and lack of exercise. These and other diseases have been widely analyzed in many predictive analytic studies. Machine learning applications over the years have involved identification of heart disease, classification of soft tissue tumors into benign or malignant classes, identification of diabetics, classification of lung cancer patients, prognosis and prediction of cancer, survivor prediction for ALS patients, and prediction of cardiovascular risk (Bahra and Wiese 2020). We now review recent specific applications of predictive modeling involving diseases.

Disease detection at early stages vastly improves likelihood of long-term patient survival. Mukhopadhyay et al. (2021) developed linear programming models to set boundaries on functions of symptoms. The approach was proposed for disease detection and tested on breast cancer and diabetes datasets. The linear programming model results were compared with decision tree and neural network data mining models. Four datasets were used in the study. The breast cancer dataset consisted of 699 patients with nine biomarkers, 34.5% of which were malignant. A second breast cancer dataset had 286 cases, 85 involving breast cancer recurrence. The other two datasets involved diabetes. The first of these had 768 cases (of which 268 tested positive for diabetes) with eight attributes. The second had 1151 cases 611 of which were positive, with 19 attributes to predict diabetic retinopathy. The linear programming model was found to perform at least well, if not better, than the data mining algorithms.

Premature ventricular contraction (PVC) is a common form of cardiac arrhythmia signaling life threats to the heart. Cardiac arrhythmia involves complexities due to a number of factors, with high levels of inter-patient variation. Mastoi et al. (2021) reported a computer-assisted predictive system consisting of five classification models (multiple logistic regression, random forest, support vector machines, K-nearest neighbor clustering, and naïve Bayes). The system improved the efficiency of medical practitioners in accurately identifying cardiac arrhythmia. Inaccurate PVC identification causes biased diagnoses that are harmful to patients. Accurate prediction requires accurate selection of appropriate model variables. Mastoi et al. (2021) developed a method to extract long compensatory pauses from the electrocardiogram signal, to identify problems more accurately.

Yoo et al. (2021) also dealt with cardiovascular disease analysis, specifically myocardial infarction and angina pectoris. They applied logistics regression on personal health record data including multimedia such as text, images, and audio. This data included lifestyle patterns and eating habits. Unstructured data was preprocessed with association rules and correlations to identify knowledge patterns. This included text data, parsed from a medical dictionary and analysis of wordclouds. The modeling approach estimated the proximity to patient groups with high similarity (using Minkowski distance). They applied multivariate analysis to analyze 210 variables, selecting 27 to generate an effective and efficient predictive system with the ability to customize for personal patient characteristics. It was found that there was a tradeoff in that too short a similarity distance led to variable results due to there being too small a subject pool, while too wide a similarity distance led to too many subjects in a similarity group.

Diabetic patients' glucose levels are critical to avoid short-term issues such as coma and long-term issues such as cardiovascular disease, from either glucose levels below 70 mg/dL or above 180 mg/dL. Predictive models such as logistic regression or decision trees are very easy to interpret, but tend to be less accurate than more complex models such as random forests or gradient boosting machines, which require more data and are difficult to interpret. De Bois et al. (2021) built a neural network model considering past glucose values as well as insulin infusions and carbohydrate intakes to predict glucose values 15 min to 2 h ahead. Because of high variability, models needed to be personalized to each patient, which considerably reduced the amount of available data. This was found to lead to overfitting training data. Thus, the study trained on source patients and then fine-tuned models to target patients. To evaluate performance, results were compared with random forest and gradient boosting machine models. The model enabled interpretation of the importance of each of the three input variables (glucose, carbohydrate intakes, and insulin infusions) for each patient.

The liver is a critical vital organ performing over 500 functions to include waste removal, producing bile for digestion, processing food, purifying blood, creating new proteins and amino acids, and clotting blood. Excess alcohol consumption, or exposure to contaminated food or water, viruses, high cholesterol, autoimmune disorders, or excessive use of medicines for prolonged period cause liver problems. Early diagnosis of liver diseases is important as there are few early symptoms. Achariya and Ahmed (2022) gathered data from 583 Indian patients, of whom 416 had liver issues while 167 did not. There were ten explanatory variables available, each categorized into two to four levels. Besides demographic variables of age and gender, variables were total bilirubin, direct bilirubin, alkaline phosphate, alanine aminotransferase, aspartate aminotransferase, total proteins, albumin, and ratio of albumin to globulin. Feature selection was conducted using a bat-inspired algorithm, yielding seven features for retention: age, gender, total bilirubin, alanine aminotransferase, aspartate aminotransferase, albumin, and the albumin/globulin ratio. Data was partitioned into 321 cases in the training set (234 with liver disease, 87 not) and 262 (193 with liver disease, 69 not) retained for the test set. A rough set analysis yielded 32 rules for the categorical data. A preset cutoff of 65% accuracy was set to evaluate rules, which attained 76.83% accuracy, better than a baseline decision tree model.

Bahra and Wiese (2020) applied neural network models to predict and classify uric acid cases, seeking to identify the presence of either gout or leukemia. While higher levels of uric acid are found in both cases, different treatments are called for, making it important to accurately predict disease. They selected final hospital admissions involving uric acid measurements having either leukemia or gout, but not both. Time series data for these patients were incomplete, inconsistent, sparse, and noise, so linear regression was applied to smooth the data. They developed neural network models, experimenting with different number of layers and cost functions. They reduced the training set data size, finding a tradeoff between higher data quality with a reduced dataset size versus overall lower data quality with a larger

dataset. The reduced dataset size gave them the greatest benefit. The linear programming processing gave better accuracy on the reduced dataset size.

### **8.3.3 *Genome Research***

The genome project has revolutionized the field of medicine. There are many challenges in this developing field, including reliable detection of short genomic insertions or deletions (indels). Several genetic disorders to include cystic fibrosis, trinucleotide repeat disorders, Mendelian disorders, and Bloom syndrome are linked to deleterious indels. Next generation sequencing has become the standard tool to discover disease variants. Short indels are assumed to cause cancers such as acute myeloid leukemia and lung cancer. Current indel technology has an accuracy around 60%, one of many opportunities for research. The most common modeling approach is neural network based, but the results of these models are difficult to interpret. Bhuyan et al. (2021) reported a gradient boosting model with a probabilistic filtering scheme to reliably identify false short indel calls. Genomic features were grouped into eight categories based on similarities. The gradient boosting algorithm gave a core useful to identify how useful each feature was in constructed boosted decision trees. The most useful feature was level of conservation, followed by local physical DNA structure and DNA flexibility. In Bhuyan et al. (2021) gradient boosting algorithm provided strong accuracy as measured by area under the precision recall curve. Because interpretability is very important in biology, it has the additional relative advantage of better interpretability.

### **8.3.4 *Internet of Things Connectivity***

The age of connectivity has enabled tracking daily lifestyle of individuals. Sensors, found in smartphones, can be used to track body motion. Directing these activities to the healthy area can reduce the risk of diseases such as diabetes, obesity, and heart or neurological diseases. Related non-medical applications include predicting terrorist activity or monitoring employee activities. Mukherjee et al. (2020) developed an ensemble of neural network models to analyze patterns in human activity sensor data.

Sensor technologies have led to rapid evaluation of applications, to include behavioral and physiological monitoring systems. One use of sensor technology is to monitor students who live in isolated areas during quarantine conditions. Souri et al. (2020) presented a machine learning healthcare monitoring model drawing upon Internet of Things connection. The system consisted of a device layer, a cloud layer, and a health monitoring layer. Decision tree, random forest, support vector machine, and neural network models were applied to predict problem conditions. Data on eighteen variables were collected, to include age, gender, and weight as well

as fourteen measures, leading to prediction of health status as either sensitive or not. In the device layer, historical and real-time data were combined and fed into the database in the cloud. In the health monitoring layer, data mining models were applied. Based on test results, the support vector machine model did much better on all metrics tested.

### 8.3.5 *Fraud Detection*

Predictive models have also been applied to the healthcare management problem of fraud detection. Medicare in the USA provides healthcare insurance for many, especially the elderly population. The claims process involves a physician performing procedures and submitting claims to Medicare for payment. Fraud has been known to occur in the form of double billing, upcoding, or other incorrect coding. Some studies identified about 10% of US medical claims being fraudulent (Joudaki et al. 2016). Herland et al. (2020) presented a system based on logistic regression model for each Medicare specialty in 14 groups, each with three to six members. Models also were run by location. Residuals were generated and input into a multinomial naïve Bayes to generate the probability of a case being an outlier, which indicated the presence of fraud. Support vector machine and random forest models were compared, but the best baseline model proved to be logistic regression. Overall accuracy was relatively low for both type I and type II error, but the modeling system provided a tool to identify cases calling for more detailed analysis by authorities.

## 8.4 Comparison of Models

Table 8.1 lists the predictive applications covered with the algorithms used. Rough sets and text mining don't deal with numerical data, but the others do. We see a wide variety of models applied to aid identification of various. Each article tends to focus on a particular model because the authors were developing a particular algorithm, but generally the practice is to try all models appropriate to the data, and either vote or use the model that fits the data best. Not all applications were disease diagnosis. One study dealt with research into genomes, others with classifying human activities, student health, predicting needed hospital resources, and another with fraud detection. The point is that there are many healthcare management applications of predictive modeling.

There are some ethical considerations of predictive modeling (and data mining in general) that have been expressed. The next section reviews that issue.

**Table 8.1** Applications with predictive methods

| Authors             | Year | Application                     | DT | GBM | LR | NB | NN | RF | RS | SVM | T |
|---------------------|------|---------------------------------|----|-----|----|----|----|----|----|-----|---|
| Acharjya et al.     | 2022 | Liver disease                   |    |     |    |    |    |    | x  |     |   |
| Bahra et al.        | 2020 | Gout, leukemia classification   |    |     |    |    | x  |    |    |     |   |
| Bhuyan et al.       | 2021 | Genome analysis                 |    | X   |    |    |    |    |    |     |   |
| De Bois et al.      | 2021 | Glucose prediction—diabetes     |    |     |    |    | x  |    |    |     |   |
| Herland et al.      | 2018 | Fraud detection                 |    |     | x  | x  |    | x  |    | x   |   |
| Mastoi et al.       | 2021 | Cardiac arrhythmia detection    |    |     | x  | x  | x  | x  |    | x   |   |
| Mukherjee et al.    | 2020 | Human activity recognition      |    |     |    |    | x  |    |    |     |   |
| Mukhopadhyay et al. | 2021 | Breast cancer, diabetes         | x  |     | x  |    | x  |    |    |     |   |
| Sheikhtaheri et al. | 2021 | Neonatal death                  | x  |     | x  | x  | x  | x  |    | X   |   |
| Souri et al.        | 2020 | Student health                  | x  |     |    |    | x  | x  |    | x   |   |
| Skylakha et al.     | 2020 | Predict hospital resource needs | x  |     | x  |    |    |    |    |     |   |
| Yoo et al.          | 2019 | Cardiac disease prediction      |    |     | x  |    |    |    |    |     | x |

*DT* decision tree, *GBM* gradient boosting machine, *LR* logistic regression, *NB* naïve Bayes, *NN* neural network, *RF* random forest, *RS* rough set, *SVM* support vector machine, *T* text mining

## 8.5 Ethics

Machine learning offers potential benefits, leading to new and unexpected knowledge. But it also raises ethical concerns that Char et al. (2020) outlined in a conceptual model following the normal data mining process of conception, development, implementation, evaluation, and oversight. One issue is transparency. While some data mining algorithms can be clearly analyzed, others like neural networks are black boxes, not understood by users or developers. Other issues that have raised problems are that large repositories of private data have been sold, training data has been known to introduce biases, ownership of data is an issue, as is accountability for failings. Char et al. (2020) identified five significant challenges:

1. Uncertain impact of emerging technologies
2. Exceptionalism—machine learning provides means to more expeditiously sell and publish data, but the ethical issues in health decision-making prior to machine learning still exist with machine learning
3. Breadth of applications—machine learning is applied to a wide range of health analyses
4. Allure of highly restricted focus—the focus on fairness and equity may not be wide enough
5. Diverse stakeholders—perspectives of patients, practitioners, entrepreneurs, and others need to be considered

Char et al. (2020) also gave a process for machine learning applied to healthcare data mining:

**Conception:** The pipeline of machine learning begins with conception. Transparent goals of studies are needed to provide auditability, aligned with human decision-making processes tied to clinical evidence. Locked machine learning applications are fixed upon development. There are other applications where learning is continuous, which makes it harder to track what is happening, especially with respect to training data not matching patient data, risking inaccurate conclusions.

**Development:** Latent biases in training data may be perpetuated or amplified. Machine learning also requires a large amount of data leading to questions of data ownership, pricing, and protection.

**Calibration:** Models inherently involve some degree of inaccuracy, of different kinds (such as Type I and Type II errors). Risk of harm needs to be considered.

**Implementation, Evaluation, and Oversight:** True errors tend to be worse than test set measures. Thus, results need to be monitored. Unintended uses can also arise, and their appropriateness is a matter for oversight.

## 8.6 Summation

As we opened the chapter, big data offers many opportunities for data mining in the healthcare domain. We looked at data sources, which add big data elements such as electronic health records, Internet of Things components, and cloud-connected systems to the traditional medical record systems. These developments have exploded the amount of data available, confusing many users, but also offering the opportunity to mine massive amounts of data. There are some ethical aspects to that data mining, but a great deal of value has been attained. We presented a review of one predictive model application in some detail in order to demonstrate the typical data mining process in a medical context. We then reviewed other applications of predictive analytics to healthcare management. Predictive data mining is proving to be a very useful tool in advancing medical research and practice.

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# Chapter 9

## Decision Analysis and Applications in Healthcare



**Keywords** Decision analysis · Probability tree structure · Selection criteria

In the healthcare industry, decision analytic tools are used extensively. As decision-making problems emerge in different forms from financial analysis to clinical decision-making, decision analysis tools are found extremely useful. Decision tree analysis has been long used in healthcare for identifying cost-effective policies in public health (Araz et al. 2013), evaluating alternatives in healthcare operations management (Ozcan 2005) and evaluating treatment options in clinical decision-making (Kros and Brown 2013) and in many other applications.

Decision analysis problems consist of states of nature, alternatives, and possible outcomes.

*State of Nature:* All the possible outcomes that can occur in the future form the state of nature.

*Decision Alternatives:* The set of possible decisions the decision maker, i.e., the healthcare professional, can make form the set of decision alternatives.

*Outcomes:* Given the state and the decision the resulting observation can be defined as the outcome. The outcomes can also be called the *payoff* (Kros and Brown 2013).

### 9.1 Selection Criteria

In this chapter we will demonstrate how basic decision-making evaluations can be performed using MS Excel. These evaluations include maximax, maximin, minimax regret, equal likelihood, Hurwitz criterion, and expected value criterion. Then we will demonstrate a decision-making problem using a decision tree analysis tool.

**Maximax Strategy:** The alternative that provides the highest return in all optimistic scenarios, i.e., maximum payoff, is identified as the decision, which is called the Maximax strategy. This is the most optimistic approach in decision-making.

**Table 9.1** Hurwitz strategy

| Alternatives\Scenarios | $S_1$                | $S_2, \dots$         | $S_n$                |
|------------------------|----------------------|----------------------|----------------------|
| $A_1$                  | Payoff <sub>11</sub> | Payoff <sub>12</sub> | Payoff <sub>1n</sub> |
| $A_2$                  | Payoff <sub>21</sub> | Payoff <sub>22</sub> | Payoff <sub>2n</sub> |
| $\dots$                |                      |                      |                      |
| $A_n$                  | Payoff <sub>n1</sub> | Payoff <sub>n2</sub> | Payoff <sub>nn</sub> |

**Table 9.2** Demonstration example payoffs

| Alternatives\Scenarios             | Low demand | Medium demand | High demand |
|------------------------------------|------------|---------------|-------------|
| Doubling the ventilator capacity   | −300       | 400           | 1500        |
| Downsizing to half of the capacity | 400        | 20            | −30         |
| No change on the capacity          | 30         | 40            | 80          |

**Maximin Strategy:** For each alternative, the worst outcome is identified under all scenarios and the strategy that returns the maximum payoff out of these alternatives is called Maximin strategy. This is a pessimistic approach in decision-making.

**Equal Likelihood Strategy:** Average returns are calculated for each alternative under all scenarios and the one that provides the highest expected return is called the equal likelihood strategy. This is also called the Laplace Strategy.

**Minimax Regret Strategy:** The highest possible opportunity cost is calculated for each alternative and the alternative that provides the minimum regret is called the Minimax Regret Strategy.

**Hurwitz Strategy:** Considering the most pessimistic strategy (maximin) and the most optimistic strategy, Hurwitz strategy allows (Table 9.1) taking the weighted average of returns given the level of optimism or pessimism of the decision maker, i.e., the weights used.

All these strategies are demonstrated in an example. A major hospital is planning ventilator capacity and stockpiling decisions after the COVID-19 pandemic for potential demand shifts and surges in the ICU care. The administration is evaluating potential scenarios for low, medium, and high demand cases and considering three decision alternatives: doubling the ventilator capacity, downsizing to half of the capacity after the pandemic, and lastly keeping the current capacity. The finance department estimated the costs and returns for each decision alternative under each scenario and presented in Table 9.2 for the administration as predicted payoffs.

**Maximax strategy:** The administration is optimistic for possible outcome and would choose the best alternative that would bring the highest payoff under each scenario, i.e., identify the highest yield for each decision alternative, i.e., max in each row, and then find the best decision alternative that provides the highest return among these. The maximum payoff for doubling the capacity decision is \$1500, downsizing to half is \$400 and keeping the status quo is \$80. Thus, the best decision alternative for Maximax strategy would be doubling the ventilator capacity. This would be the optimistic administration's decision.

**Maximin strategy:** In case the administration is pessimistic who would consider potentially the worst possible realization of the demand for their services, they would make decisions based on the minimum payoffs they would achieve and make

**Table 9.3** Equal likelihood results

| Alternatives\Scenarios                  | Low demand | Medium demand | High demand | Expected value (Laplace) |
|-----------------------------------------|------------|---------------|-------------|--------------------------|
| <b>Doubling the ventilator capacity</b> | −300       | 400           | 1500        | <b>533.33</b>            |
| Downsizing to half of the capacity      | 400        | 20            | −30         | 130                      |
| No change on the capacity               | 30         | 40            | 80          | 50                       |

decision to maximize the payoff given these returns under worst case. Now, sorting each alternative’s worst possible payoff given the scenarios doubling capacity would return −\$300 (overall net cost of \$300), downsizing the capacity to half would return −\$30 and the current capacity would return \$30 as the minimum expected payoff given all the scenarios. The administrator would go with the decision of keeping the current ventilator capacity as \$30 is the maximum of the minimum returns for all alternatives.

**Equal likelihood strategy:** This strategy assigns equal probability of realization to each scenario and computing the expected return for each alternative under equal likelihood of scenarios. It is also called the Laplace Strategy with “insufficient reason principle” (Ozcan 2005). Expected return for each decision alternative, i.e.,  $A_i$ , would be calculated as following:

$$E(A_i) = \sum_j \text{prob}_j \text{Payoff}_{ij} \text{ where the prob}_j \text{ is the probability of scenario } j \text{ and}$$

$\text{Payoff}_{ij}$  is the estimated payoff of decision alternative  $i$  under scenario  $j$ . Given these computations for each alternative in Table 9.3, the best alternative would be doubling the ventilator capacity.

**Minimax Regret Strategy:** In many cases, the decision makers would be interested in minimizing the maximum regret, or in other words “the opportunity loss.” This would be a situation hospital administrator would face when he/she makes the decision for a scenario and other cases would be realized. This can be computed as the difference between the maximum possible payoff under scenario  $j$  and the payoff that would be obtained by going with alternative  $i$  under scenario  $j$ . For this computation we need to form the regrets table, i.e., opportunity loss table as given below. But before let’s identify the best payoffs under each scenario. Under each scenario the best possible payoffs that can be achieved are presented in Table 9.4 with \*.

For example, the regrets under low demand scenario are computed as following:

The best possible outcome under this scenario is \$400 that can be achieved by downsizing decision. Thus, the regret for this alternative would be 0. However, if the hospital administration decided to double the capacity and realized a low demand, the regret would be  $\$400 - (-\$300) = \$700$ . Similarly, if the decision was to keep the same ventilator capacity, then the regret for this decision under the low demand scenario would be  $\$400 - \$30 = \$370$ . For other scenarios the regrets are computed as given in Table 9.4. Now, assessing the alternatives given the regret table, the worst regret (maximum regret) for capacity doubling would be \$700, downsizing to

**Table 9.4** Maximum regret results

| Alternatives\Scenarios                  | Low demand             | Medium demand | High demand |
|-----------------------------------------|------------------------|---------------|-------------|
| Doubling the ventilator capacity        | −300                   | 400*          | 1500*       |
| Downsizing to half of the capacity      | 400*                   | 20            | −30         |
| No change on the capacity               | 30                     | 40            | 80          |
| Regret Values (Opportunity Loss)        |                        |               |             |
| Alternatives\Scenarios                  | Low Demand             | Medium Demand | High Demand |
| <b>Doubling the ventilator capacity</b> | $400 - (-300) = 700^*$ | 0             | 0           |
| Downsizing to half of the capacity      | 0                      | 380           | 1530        |
| No change on the capacity               | 370                    | 360           | 1420        |

**Table 9.5** Hurwitz value computation for  $w = 0.1$ 

| Hurwitz parameter value of optimism ( $w$ ) = 0.1 |            |               |             |               |
|---------------------------------------------------|------------|---------------|-------------|---------------|
| Alternatives\Scenarios                            | Low demand | Medium demand | High demand | Hurwitz value |
| Doubling the ventilator capacity                  | −300       | 400           | 1500        | −120          |
| Downsizing to half of the capacity                | 400        | 20            | −30         | 13            |
| <b>No change on the capacity</b>                  | 30         | 40            | 80          | 35*           |

half would be \$1530 and for no capacity change would be \$1420. To minimize the maximum regret the hospital administration would choose doubling the capacity as it is the MIN (\$700, \$1530, \$1420) = \$700\*.

**Hurwitz Strategy:** In many applications, the decision makers may need to assess the scenarios between the optimistic and pessimistic outcomes. Hurwitz methodology allows decision makers to weigh optimism with a weight, i.e.,  $w$ , which can take values between 0 and 1,  $0 \leq w \leq 1$ . The Hurwitz strategy takes the weighted average of best and worst possible outcomes under each decision alternative and then suggests the alternative with the maximum weighted average value. For example, in Table 9.4 the Hurwitz values are computed for each decision alternative given the optimism parameter value of 0.1. For doubling the capacity,  $\text{MAX}(-300, 400, 1500) * 0.1 + \text{MIN}(-300, 400, 1500) * (1 - 0.1) = -120$ . For downsizing decision,  $\text{MAX}(400, 20, -30) * 0.1 + \text{MIN}(400, 20, -30) * (1 - 0.1) = 13$ . Finally for No Change decision,  $\text{MAX}(-300, 400, 1500) * 0.1 + \text{MIN}(-300, 400, 1500) * (1 - 0.1) = 35$ . Thus, the Hurwitz decision given the  $w = 0.1$  would be no change in ventilator capacity Table 9.5. Next, we demonstrate the assessment for  $w = 0.9$  in Table 9.6. In this case the Hurwitz decision would be doubling the ventilator capacity.

**Table 9.6** Hurwitz value computation for  $w = 0.9$ 

| Hurwitz parameter value of optimism ( $w$ ) = 0.9 |            |               |             |               |
|---------------------------------------------------|------------|---------------|-------------|---------------|
| Alternatives\Scenarios                            | Low demand | Medium demand | High demand | Hurwitz value |
| <b>Doubling the ventilator capacity</b>           | −300       | 400           | 1500        | 1320*         |
| Downsizing to half of the capacity                | 400        | 20            | −30         | 357           |
| No change on the capacity                         | 30         | 40            | 80          | 75            |

## 9.2 Decision Tree Analysis

Decision Tree Analysis process is generally summarized with the following steps (McNeil and Pauker 1984) which we will also demonstrate over an example.

1. Defining and structuring the problem on hand, gathering data for decision alternatives
2. Defining and computing the probability for each possible outcome
3. Measuring and estimating the relative return or value (utility) for each potential outcome
4. Calculating best alternative using the expected value computations, i.e., average out and fold back principle, over the tree
5. Performing several sensitivity analyses over a set of values for various parameters in the decision tree model.

The relations of these decision tree components are demonstrated in Fig. 9.1:

Using the ventilator stockpiling example we are going to utilize a decision tree analysis approach to make decisions under uncertainty with probabilities given for each scenario. The tree structure is constructed as given in Fig. 9.2, with the probabilities,  $\text{prob}_j$ , assigned for each scenario  $j$  as the occurrence likelihood of each scenario, of which the sum should be equal to 1, i.e.,

$\sum_j \text{prob}_j = 1$ . In the example below, probabilities are estimated and utilized as

following: high demand occurrence probability is 0.2, probability of medium demand is 0.3, and probability of low demand is 0.5 as given in the decision tree (Fig. 9.2).

Using the rollback procedure for evaluating the expected value, i.e., expected value (EV) model, we can compute the expected return for each alternative as following:

$$\text{EV}_{\text{doubling}} = 0.2*(1500) + 0.3*(400) + 0.5*(-300) = 270^*$$

$$\text{EV}_{\text{downsizing}} = 0.2*(-30) + 0.3*(20) + 0.5*(400) = 200$$

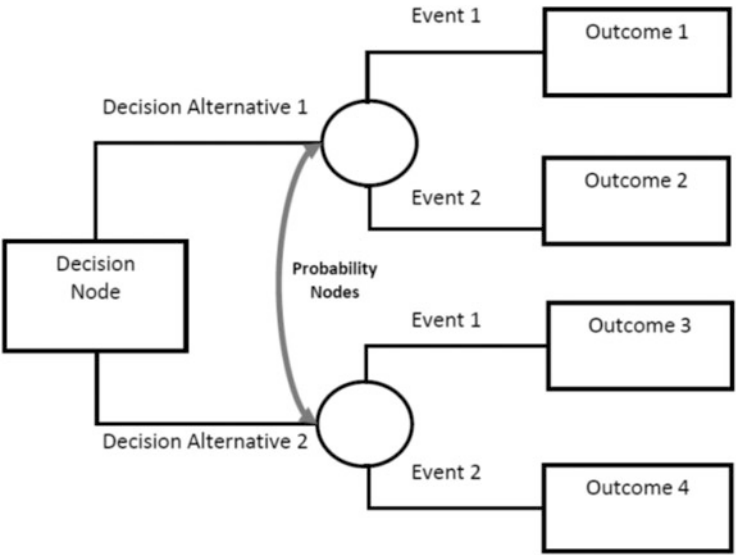


Fig. 9.1 Components of decision tree analysis

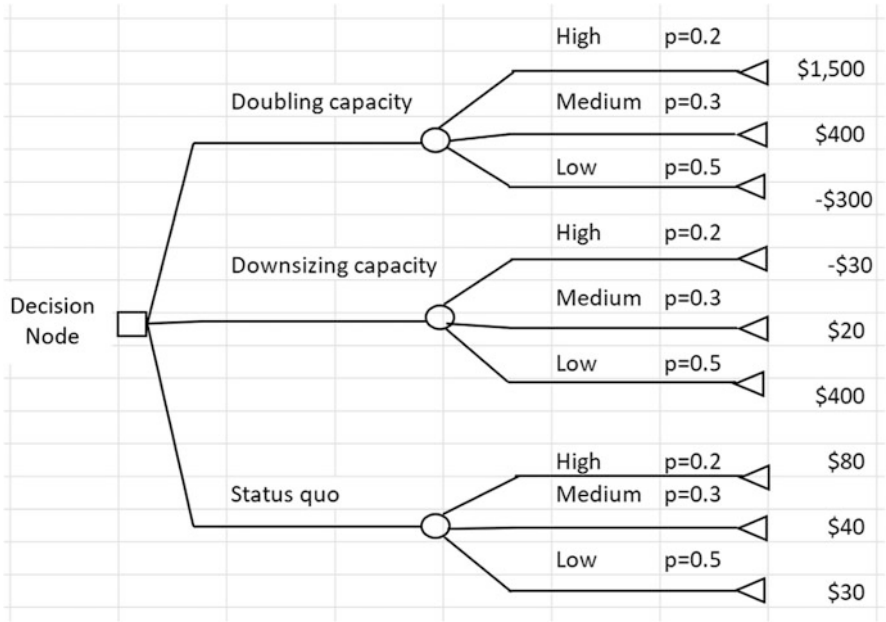


Fig. 9.2 Decision tree for ventilator stockpiling

$$EV_{\text{status quo}} = 0.2*(80) + 0.3*(40) + 0.5*(30) = 43$$

Given the expected payoff values of the alternative decisions, the doubling the stockpiling ventilator capacity would be the decision.

In the following sections we analyze the applications of decision analysis applications in public health, medical decision-making, and healthcare operations.

### 9.3 Decision Analysis in Public Health and Clinical Applications

Decision analysis techniques have been widely used in public health and medical applications. McNeil and Pauker (1984) had presented systematic guidelines for using the decision analysis tools, including decision trees to support policy making in public health. They described the model building exercise for identifying the optimal strategy under uncertainty. In public health, the most common applications of decision analysis are seen in disease screening, e.g., cancer screening, diabetes, and other chronic disease screening applications, in addition infectious disease screenings. Willis and Moore (2014) presented an application of decision tree models and value-of-information analysis for assessing the value of public health bio-surveillance systems for improving the public health preparedness decision-making. Another application area is on developing vaccination policies against infectious diseases or informing individuals for cost-benefits of vaccine intake. Vardavas et al. (2007) presented a decision analysis model informed with an individual level human cognition and behavior which is memory and adaptability based that individuals use when making influenza vaccination decisions. In an earlier study Zalkind and Shachtman (1980) presented an application of decision analysis tools for individual level decision-making for swine influenza vaccination decisions. More recent applications would include HPV screening for cervical cancer prevention (Sroczynski et al. 2011; Kim et al. 2018). Another application of decision analysis was presented by Long and Brandeau (2009) for evaluating the strategies of HPV prevention, specifically for preventing the transmission from mothers to children. Rafia et al. (2016) presented an economic evaluation of hepatitis B vaccination using incentives-based strategies. These examples use decision analysis tools with simulation modeling and more recently this application is applied in COVID-19 vaccination studies, e.g., Kohli et al. (2021). Araz et al. (2022) also presented a decision analytic approach for cost–benefit analysis of screening and treatment of Chlamydia patients for assessing the community benefits of several policies.

In clinical settings, Esnaola et al. (2001) used decision tree analysis methodology coupled with a Markov model for treatment strategy development for patients with papillary thyroid cancer. Vickers (2008) demonstrates how decision analytic tools and decision tree analysis are used for prostate cancer biopsy and testing. Similarly,

in an earlier study Plevritis (2005) presented a decision analytic framework to inform surgery decision-making and present the value of radiological imaging given the accuracy statistics about the diagnostic tests.

## 9.4 Decision Analysis in Healthcare Operations Management

In healthcare operations management, many operational problems emerge, and cost–benefit analysis would be needed to support the decision-making. These problems can be formulated for evaluating alternatives for purchasing and/or outsourcing various medical devices in certain units in a hospital, e.g., MRI as given in Ozcan (2005) or evaluating suppliers for a hospital, or making decisions about if certain services should be provided by the hospital or not given the demand and resources that hospital can dedicate, e.g., nutritional assistances, mental clinics, etc. and many of these problems can be formulated as a decision tree medical device purchasing decision-making.

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# Chapter 10

## Analysis of Four Medical Datasets



**Keywords** Variable selection · Comparative algorithm performance

In this chapter we discuss the variable selection process in healthcare data mining applications and present an overview of 4 medical datasets which are available on-line. These datasets are commonly used in demonstration of healthcare applications of data mining and predictive modeling techniques in research.

### 10.1 Variable Selection

Frequently, data mining analysis involves data with a large number of variables. But including all available variables in a model can be counterproductive in some cases. Often some variables contribute more noise than value in predicting the dependent variable. Chan et al. (2007) identified several applications where too many variables generated noise, and ignoring redundant variables improved prediction. Thus, reducing the set of independent variables can result in better prediction. This chapter presents model analyses of datasets showing that reducing the number of variables simplifies analyses and may improve accuracy.

Independent variables may be highly correlated, containing overlapping information which may distort regression coefficients. Usually, a small subset of these explanatory (independent) variables provides the bulk of the predictive power of a model. The reasons to be able to trim larger sets of variables in data mining to smaller subsets include:

1. Easier analysis
2. Shorter training time
3. Avoiding the curse of dimensionality, which makes problems more complex
4. Reducing overfitting, by reducing variance

A parsimonious model with good predictive power that is easy to implement is preferable (Cui et al. 2020).

The ideal way to select variables is based on deep understanding of the problem. This is the approach of classical statistics, where the ideal regression model is based on selecting variables known to have strong relationships with the dependent variable. This amounts to understanding the system and selecting independent variables the human analyst expects to have a strong relationship with the dependent variable. Machine learning, conversely, uses statistical measures to select variables. Decision trees have a complexity parameter based on entropy levels to select variables. Another machine learning method for variable reduction is stepwise regression, which is adding variables by their contribution to explaining variance in the dependent variable (Foster 2004). This is done with partial correlation, which considers overlapping content of potential independent variables. The process can start with selecting that independent variable with the highest correlation with the dependent variable. Then, given use of that variable, the partial contributions of independent variables are analyzed to iteratively proceed until added improvement in fit falls below some stated level. Some control is obtained by using entry and retention levels. We will experiment with four medical datasets obtained from the Kaggle website for which the links to references are provided below with explanations of these datasets.

## 10.2 Pima Indian Diabetes Dataset

The National Institute of Diabetes and Digestive and Kidney Diseases originated the dataset with the objective to diagnose whether or not a patient has diabetes (<https://www.kaggle.com/datasets/uciml/pima-indians-diabetes-database>). All patients were of Pima Indian heritage, 21 years of age or older. The dataset contains 768 observations. It was first published academically by Smith et al. (1988). Predictor variables are shown in Table 10.1.

We present the correlation matrix below in Table 10.2:

Correlations with Outcome indicate which measures are most likely to predict the presence of diabetes. Here the highest correlation is with Glucose concentration,

**Table 10.1** Pima Indian Diabetes variables

| Variable          | Definition                                        | Metric             |
|-------------------|---------------------------------------------------|--------------------|
| Pregnancies       |                                                   | Count              |
| Glucose           | Plasma glucose concentration at 2 hours—oral test |                    |
| Blood pressure    | Diastolic blood pressure                          | mm Hg              |
| Skin thickness    | Triceps skin fold thickness                       | mm Hg              |
| Insulin           | 2-h serum insulin                                 | mu U/ml            |
| BMI               | Body mass index                                   | [wgt(kg)/hgt(m)]^2 |
| Diabetes pedigree |                                                   |                    |
| Age               |                                                   | Years              |
| Outcome           |                                                   | 0—no; 1—diabetes   |

Table 10.2 Correlations—Pima Indian Diabetes data

|                          | Pregnancies | Glucose | BloodPressure | SkinThickness | Insulin | BMI   | DPF   | Age   | Outcome |
|--------------------------|-------------|---------|---------------|---------------|---------|-------|-------|-------|---------|
| Pregnancies              | 1.000       |         |               |               |         |       |       |       |         |
| Glucose                  | 0.129       | 1.000   |               |               |         |       |       |       |         |
| BloodPressure            | 0.141       | 0.153   | 1.000         |               |         |       |       |       |         |
| SkinThickness            | −0.082      | 0.057   | 0.207         | 1.000         |         |       |       |       |         |
| Insulin                  | −0.074      | 0.331   | 0.089         | 0.437         | 1.000   |       |       |       |         |
| BMI                      | 0.018       | 0.221   | 0.282         | 0.393         | 0.198   | 1.000 |       |       |         |
| DiabetesPedigreeFunction | −0.034      | 0.137   | 0.041         | 0.184         | 0.185   | 0.141 | 1.000 |       |         |
| Age                      | 0.544       | 0.264   | 0.240         | −0.114        | −0.042  | 0.036 | 0.034 | 1.000 |         |
| Outcome                  | 0.222       | 0.467   | 0.065         | 0.075         | 0.131   | 0.293 | 0.174 | 0.238 | 1.000   |

**Table 10.3** Logistic regression—Pima Indian Diabetes data

| Variable      | Coefficient | <i>P</i> -value | Significance  |
|---------------|-------------|-----------------|---------------|
| Intercept     | −8.4047     | 2.00E-16        | <0.001        |
| Pregnancies   | 0.1232      | 0.0001          | <0.001        |
| Glucose       | 0.0352      | 2.00E-16        | <0.001        |
| BloodPressure | −0.0133     | 0.0111          | <0.05         |
| SkinThickness | 0.0006      | 0.9285          | Insignificant |
| Insulin       | −0.0012     | 0.1861          | Insignificant |
| BMI           | 0.0897      | 0.0000          | <0.001        |
| DPF           | 0.9452      | 0.0016          | <0.01         |
| Age           | 0.0149      | 0.1112          | Insignificant |

Pseudo *R*-Square (optimistic): 0.5725

**Table 10.4** Significant relationships

| Significant positive relationships (at 0.01 significance) | Significant negative relationships (at 0.05 significance) |
|-----------------------------------------------------------|-----------------------------------------------------------|
| Glucose concentration                                     | Blood pressure                                            |
| BMI                                                       |                                                           |
| Pregnancies                                               |                                                           |
| DPF                                                       |                                                           |

followed by BMI, Age, and Pregnancies. We run a logistic regression to display all variables and get further clues as to predictive value. Table 10.3 presents the estimated coefficients of variables in the logit model with *p*-values.

Glucose concentration, BMI, and Pregnancies do indeed show up as significant, although Age does not. Looking closer at Table 10.2, we can see that Age is highly correlated with Pregnancies. Table 10.4 lists the variables which are significant with positive and negative relationship.

Following Olson and Chae (2022), we selected variables for predictive models using decision trees at various complexity levels, using stepwise regression at different levels of entering and staying, and by logistic regression significance (all using a training set of 80% of the data, retaining 20% for testing). Five models were applied (extreme boosting—EB, random forests—RF, support vector machines—SVM, logistic regression—LR, and neural networks—NN). Table 10.5 shows the variables selected by logistic regression significance or by decision tree models using various complexity levels. Stepwise regression was also applied but generated the same variable set as the decision tree model with 0.02 complexity parameter. Table 10.5 also gives sensitivity, specificity, accuracy, and area under the curve measures for each of the five predictive models applied.

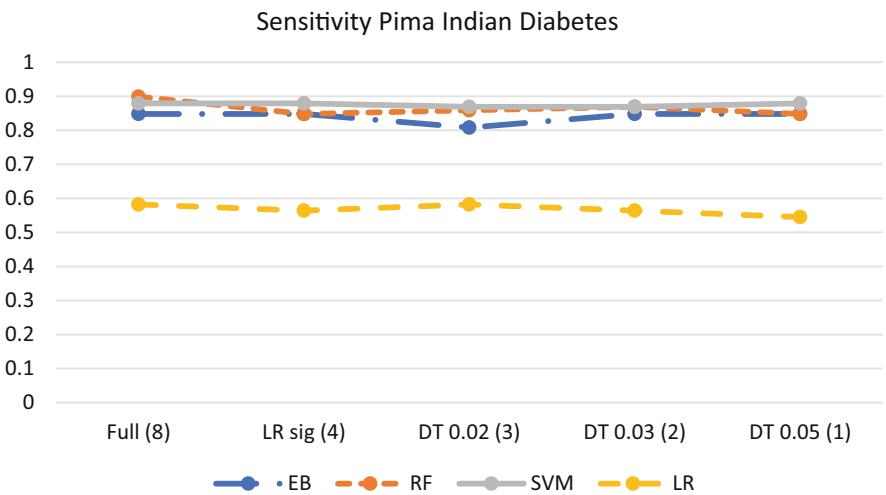
From Table 10.5, the best accuracy and area under the curve were obtained using all eight variables, despite three of them measuring insignificant. Here, more variables performed better, although that will not always be the case. As to relative model fit, extreme boosting and random forest models performed better, as often is the case. Note that neural network models can be fine-tuned to perform better, but often their output is degenerate (calling everything the majority outcome), which is

**Table 10.5** Pima Indian Diabetes model results

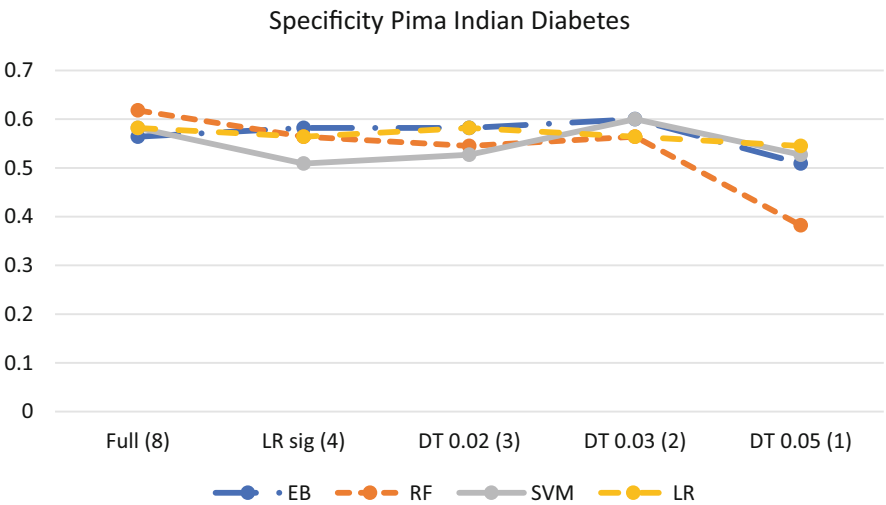
|                 |              |                 |           |         |         |
|-----------------|--------------|-----------------|-----------|---------|---------|
| Method          |              | Variables       |           |         |         |
| LR significance |              | Pregnancy       | Glucose   | BMI     | DBF     |
| DT 0.02         |              | Pregnancy       | Glucose   | BMI     |         |
| DT 0.03         |              |                 | Glucose   | BMI     |         |
| DT 0.05         |              |                 | Glucose   |         |         |
| Method          |              | LR significance | DT 0.02   | DT 0.03 | DT 0.05 |
| Variables       | All 8        | Pregnancy       | Pregnancy |         |         |
|                 |              | Glucose         | Glucose   | Glucose | Glucose |
|                 |              | BMI             | BMI       | BMI     |         |
|                 |              | DBF             |           |         |         |
| EB              |              |                 |           |         |         |
| Sensitivity     | 0.848        | 0.848           | 0.808     | 0.848   | 0.848   |
| Specificity     | 0.564        | 0.582           | 0.582     | 0.600   | 0.509   |
| Accuracy        | 0.746        | 0.753           | 0.727     | 0.760   | 0.727   |
| AUC             | <b>0.893</b> | 0.857           | 0.836     | 0.834   | 0.804   |
| RF              |              |                 |           |         |         |
| Sensitivity     | 0.899        | 0.848           | 0.859     | 0.869   | 0.848   |
| Specificity     | 0.618        | 0.564           | 0.545     | 0.564   | 0.382   |
| Accuracy        | <b>0.799</b> | 0.746           | 0.747     | 0.759   | 0.681   |
| AUC             | 0.857        | 0.869           | 0.836     | 0.838   | 0.789   |
| SVM             |              |                 |           |         |         |
| Sensitivity     | 0.879        | 0.879           | 0.869     | 0.869   | 0.879   |
| Specificity     | 0.582        | 0.509           | 0.527     | 0.600   | 0.527   |
| Accuracy        | 0.773        | 0.747           | 0.746     | 0.772   | 0.753   |
| AUC             | 0.855        | 0.837           | 0.847     | 0.871   | 0.750   |
| LR              |              |                 |           |         |         |
| Sensitivity     | 0.879        | 0.869           | 0.859     | 0.879   | 0.848   |
| Specificity     | 0.582        | 0.564           | 0.582     | 0.564   | 0.545   |
| Accuracy        | 0.779        | 0.759           | 0.760     | 0.766   | 0.740   |
| AUC             | 0.850        | 0.842           | 0.831     | 0.830   | 0.818   |
| NN              |              |                 |           |         |         |
| Sensitivity     | 1            | 1               | 1         | 1       | 1       |
| Specificity     | 0            | 0               | 0         | 0       | 0       |
| Accuracy        | 0.643        | 0.643           | 0.643     | 0.643   | 0.643   |
| AUC             | degen        | degen           | degen     | degen   | degen   |

not useful. Figures 10.1, 10.2, 10.3, and 10.4 display relative algorithm performance in terms of sensitivity (ability to identify positive cases), specificity (ability to avoid calling negative cases positive), accuracy (a combination of sensitivity and specificity—correct classification of test cases over opportunities), and area under the curve, a metric of predictive accuracy.

The figure for sensitivity shows that logistic regression model had much lower sensitivity than the other three models (neural network output was not plotted as it



**Fig. 10.1** Pima Indian Diabetes dataset sensitivity



**Fig. 10.2** Pima Indian Diabetes dataset specificity

was degenerate in most cases). Specificity, on the other hand, was relatively similar for all four models. Note that having fewer variables did not hurt much until the last model, with only one variable. Accuracy combines sensitivity and specificity and is quite similar for all models. The random forest model showed better performance for the full model, declining with the last model. Accuracy plotted for the neural network models gives a baseline where all cases were called diabetic. The area under the curve plot shows a general decline in fit with fewer variables, although the random forest model did better with four variables, and the support vector machine

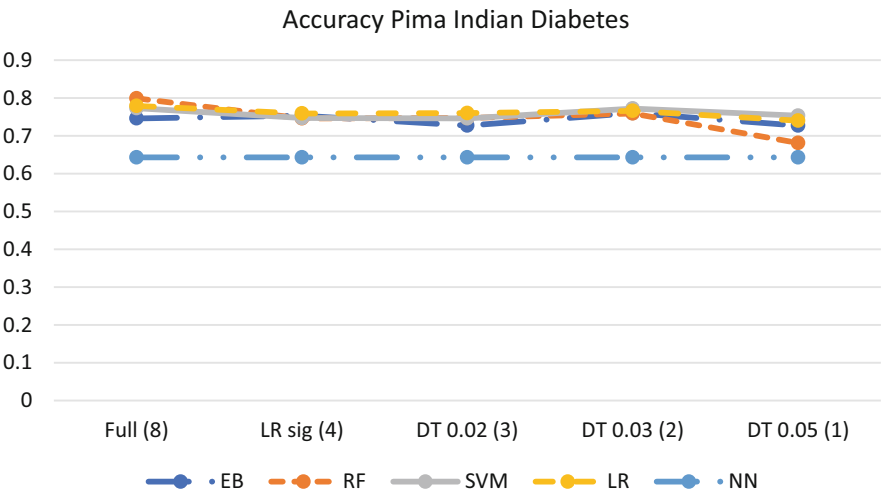


Fig. 10.3 Pima Indian Diabetes dataset accuracy

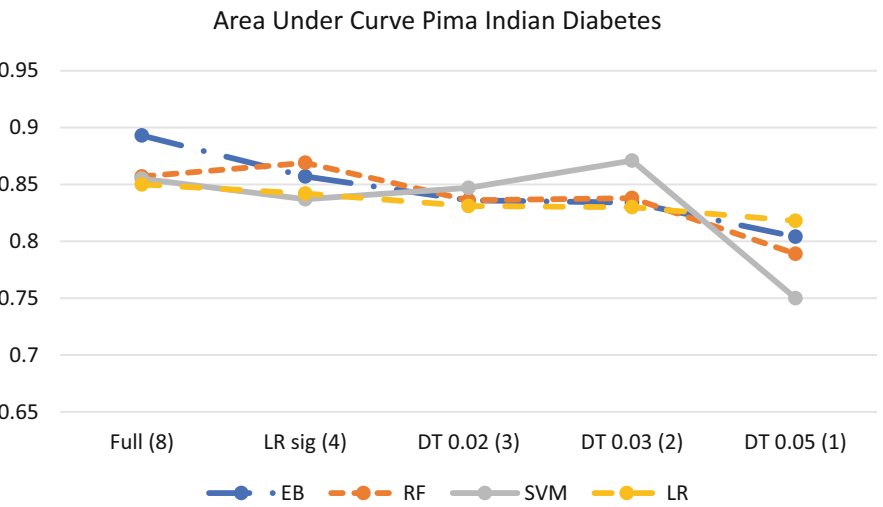


Fig. 10.4 Pima Indian Diabetes dataset area under the curve

models varied in an anomalous manner. In general, the extreme boosting model using all variables gave the best fit.

### 10.3 Heart UCI Data

This data relates to detection of heart disease, and more information can be found in the following link: <https://www.kaggle.com/datasets/redwankarimsony/heart-disease-data>. The dataset contains 1024 observations. The thirteen independent variables plus the target are shown in Table 10.6:

Table 10.7 gives correlation of these variables:

From the bottom row of Table 10.7 we see that the strongest relationship with Target is 0.439, which is moderate. There are four independent variables with correlations with Target greater than 0.4, and another three greater than 0.3. A logistic regression model is run using all predictors and the estimated predictor coefficients are presented in Table 10.8.

Comparing the numbers in Table 10.8 with those in Table 10.7, it is seen that all of those variables with correlations over 0.4 were highly significant, while the variable slope, despite a relatively high correlation with Target of 0.382, is less significant. That is due to cross-correlation with other independent variables. Table 10.9 shows the variable sets generated with decision trees, logistic regression significance, and stepwise regression, with headings for the full set of variables, the variables selected by the decision tree model, the variables selected by the logistic regression model with significance set at 0.05 and 0.01, variables selected by stepwise regression, and variables selected by logistic regression with significance of 0.001.

**Table 10.6** Heart UCI data variables

| Variable | Definition                     | Measures     |
|----------|--------------------------------|--------------|
| Age      |                                | Years        |
| Sex      |                                | M/F          |
| cp       | Chest pain type                | Categories   |
| trestbps | Resting blood pressure         |              |
| chol     | Cholesterol                    |              |
| fbs      | Fasting blood sugar            | Categories   |
| Restecg  | Rest electrocardiogram         | Categories   |
| Thalach  | Max heart rate                 |              |
| Exang    | Exercise-induced angina        | Y/N          |
| Oldpeak  |                                |              |
| Slope    | Of peak exercise               | 3 categories |
| ca       | Vessels colored by fluoroscopy |              |
| Thal     | Blood disorder                 |              |
| Target   | Heart disease diagnosis        | 0 no; 1 yes  |

Table 10.7 Heart UCI correlations

|          | Age    | Sex    | cp     | trestbps | chol   | fbs    | restecg | thalach | Exang  | oldpeak | Slope  | ca     | thal   |
|----------|--------|--------|--------|----------|--------|--------|---------|---------|--------|---------|--------|--------|--------|
| Age      | 1      |        |        |          |        |        |         |         |        |         |        |        |        |
| Sex      | -0.104 | 1      |        |          |        |        |         |         |        |         |        |        |        |
| cp       | -0.073 | -0.042 | 1      |          |        |        |         |         |        |         |        |        |        |
| trestbps | 0.271  | -0.081 | 0.037  | 1        |        |        |         |         |        |         |        |        |        |
| chol     | 0.22   | -0.198 | -0.083 | 0.128    | 1      |        |         |         |        |         |        |        |        |
| fbs      | 0.121  | 0.027  | 0.079  | 0.181    | 0.027  | 1      |         |         |        |         |        |        |        |
| restecg  | -0.133 | -0.057 | 0.044  | -0.125   | -0.147 | -0.104 | 1       |         |        |         |        |        |        |
| thalach  | -0.391 | -0.048 | 0.306  | -0.04    | -0.024 | -0.009 | 0.05    | 1       |        |         |        |        |        |
| exang    | 0.088  | 0.139  | -0.403 | 0.06     | 0.067  | 0.049  | -0.066  | -0.382  | 1      |         |        |        |        |
| oldpeak  | 0.208  | 0.083  | -0.176 | 0.187    | 0.065  | 0.011  | -0.051  | -0.35   | 0.311  | 1       |        |        |        |
| Slope    | -0.169 | -0.025 | 0.132  | -0.12    | -0.015 | -0.062 | 0.088   | 0.395   | -0.267 | -0.575  | 1      |        |        |
| ca       | 0.271  | 0.111  | -0.177 | 0.104    | 0.075  | 0.137  | -0.079  | -0.208  | 0.108  | 0.221   | -0.073 | 1      |        |
| thal     | 0.072  | 0.197  | -0.163 | 0.059    | 0.102  | -0.042 | -0.022  | -0.096  | 0.198  | 0.202   | -0.093 | 0.149  | 1      |
| Target   | -0.229 | -0.278 | 0.436  | -0.139   | -0.101 | -0.041 | 0.137   | 0.422   | -0.439 | -0.438  | 0.345  | -0.382 | -0.337 |

**Table 10.8** Logistic regression model for full data

| Variable  | Coefficient | <i>P</i> -value | Significance  |
|-----------|-------------|-----------------|---------------|
| Intercept | 3.6902      | 0.0085          | <0.001        |
| Age       | −0.0082     | 0.5155          | Insignificant |
| Sex       | −1.8465     | 6.16e-13        | <0.001        |
| Cp        | 0.8546      | 2e-16           | <0.001        |
| Trestbps  | −0.0182     | 0.0012          | <0.01         |
| Chol      | −0.0057     | 0.0058          | <0.01         |
| FBS       | −0.1012     | 0.7225          | Insignificant |
| Restecg   | 0.4132      | 0.0288          | <0.05         |
| Thalach   | 0.0236      | 3.21e-05        | <0.001        |
| Exang     | −0.9908     | 9.98e-06        | <0.001        |
| Oldpeak   | −0.5707     | 8.66e-07        | <0.001        |
| Slope     | 0.5341      | 0.0047          | <0.01         |
| CA        | −0.7545     | 2.45e-13        | <0.001        |
| Thal      | −0.8861     | 1.24e-08        | <0.001        |

Pseudo *R*-Square (optimistic): 0.74930748

**Table 10.9** Variable sets selected

|         |         |          |          |         |           |
|---------|---------|----------|----------|---------|-----------|
| Full    | DT      | SigLR.05 | SigLR.01 | STEP    | SigLR.001 |
| CA      | CA      | CA       | CA       | CA      | CA        |
| CP      | CP      | CP       | CP       | CP      | CP        |
| Exang   | Exang   | Exang    | Exang    | Exang   | Exang     |
| OldPeak | OldPeak | OldPeak  | OldPeak  | OldPeak | OldPeak   |
| Sex     | Sex     | Sex      | Sex      | Sex     | Sex       |
| Thal    | Thal    | Thal     | Thal     | Thal    | Thal      |
| Thalach | Thalach | Thalach  | Thalach  | Thalach | Thalach   |
| RestBP  | RestBP  | RestBP   |          | RestBP  |           |
| Age     | Age     |          |          |         |           |
| Chol    | Chol    |          |          |         |           |
| Slope   |         | Slope    | Slope    |         |           |
| FBS     |         |          |          |         |           |
| RestECG |         |          |          |         |           |

Table 10.10 gives the results of running these models, plotted in Figs. 10.5, 10.6, 10.7, and 10.8:

With respect to sensitivity, the extreme boosting model performed very well. The random forest model was a close second. The other three model types did less well, although the neural network model did very well for models with few independent variables.

Specificity is the other side of the coin relative to sensitivity. The neural network model performed poorly when few variables were used for model construction. The other models did relatively well, although extreme boosting and random forest models were best.

**Table 10.10** Heart UCI model results

|             | Full  | DT    | SigLR.05 | SigLR.01 | STEP  | SigLR.001 |
|-------------|-------|-------|----------|----------|-------|-----------|
| Variables   | 13    | 10    | 9        | 8        | 8     | 7         |
| EB          |       |       |          |          |       |           |
| Sensitivity | 1     | 1     | 1        | 1        | 1     | 1         |
| Specificity | 0.972 | 0.972 | 0.972    | 0.972    | 0.972 | 0.972     |
| Accuracy    | 0.985 | 0.985 | 0.985    | 0.985    | 0.985 | 0.985     |
| AuC         | 1     | 1     | 0.997    | 0.999    | 1     | 1         |
| RF          |       |       |          |          |       |           |
| Sensitivity | 1     | 1     | 1        | 0.979    | 0.958 | 0.969     |
| Specificity | 0.972 | 0.972 | 0.972    | 0.954    | 0.945 | 0.954     |
| Accuracy    | 0.985 | 0.985 | 0.985    | 0.966    | 0.951 | 0.961     |
| AuC         | 1     | 1     | 1        | 0.999    | 0.998 | 1         |
| SVM         |       |       |          |          |       |           |
| Sensitivity | 0.938 | 0.885 | 0.875    | 0.917    | 0.833 | 0.875     |
| Specificity | 0.908 | 0.917 | 0.936    | 0.917    | 0.899 | 0.908     |
| Accuracy    | 0.922 | 0.903 | 0.908    | 0.917    | 0.868 | 0.893     |
| AuC         | 0.975 | 0.975 | 0.978    | 0.976    | 0.977 | 0.974     |
| LR          |       |       |          |          |       |           |
| Sensitivity | 0.843 | 0.812 | 0.802    | 0.823    | 0.802 | 0.812     |
| Specificity | 0.917 | 0.917 | 0.936    | 0.890    | 0.954 | 0.908     |
| Accuracy    | 0.883 | 0.868 | 0.874    | 0.858    | 0.883 | 0.863     |
| AuC         | 0.957 | 0.952 | 0.953    | 0.951    | 0.950 | 0.949     |
| NN          |       |       |          |          |       |           |
| Sensitivity | 0.740 | 0.760 | 0.885    | 0.823    | 1     | 1         |
| Specificity | 0.725 | 0.697 | 0.817    | 0.890    | 0     | 0         |
| Accuracy    | 0.731 | 0.727 | 0.849    | 0.858    | 0.469 | 0.469     |
| AuC         | 0.829 | 0.827 | 0.912    | 0.951    | degen | degen     |

Accuracy, as the average of sensitivity and specificity, yields the same results. Extreme boosting and random forest models were very good.

Area under the curve matches the prior results. Except for neural networks, all four models performed similarly.

## 10.4 India Liver Data

The India Liver dataset has been widely used, obtained from the Kaggle site. As the name implies, it involves analysis of data on liver patients from an Indian medical establishment. The dataset (<https://www.kaggle.com/datasets/jeevannagaraj/indian-liver-patient-dataset>) consists of 467 liver patients and 167 who are not ( $n = 634$ ). The variables are:

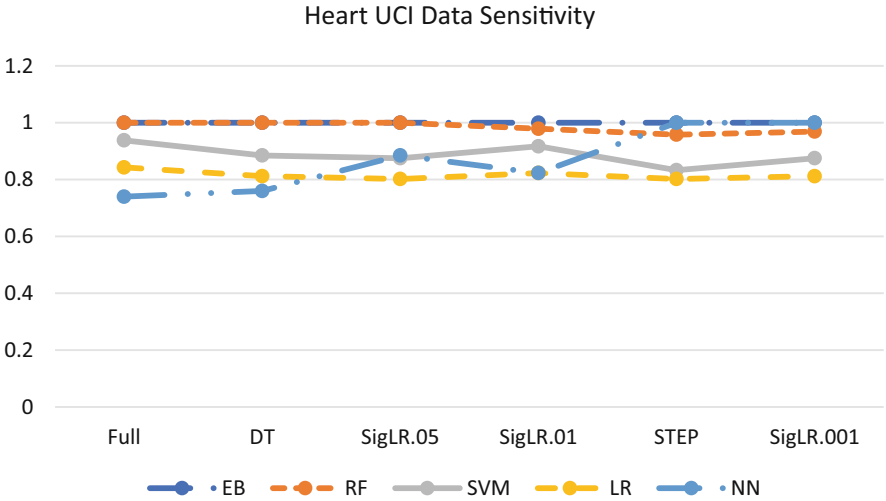


Fig. 10.5 Heart UCI dataset sensitivity

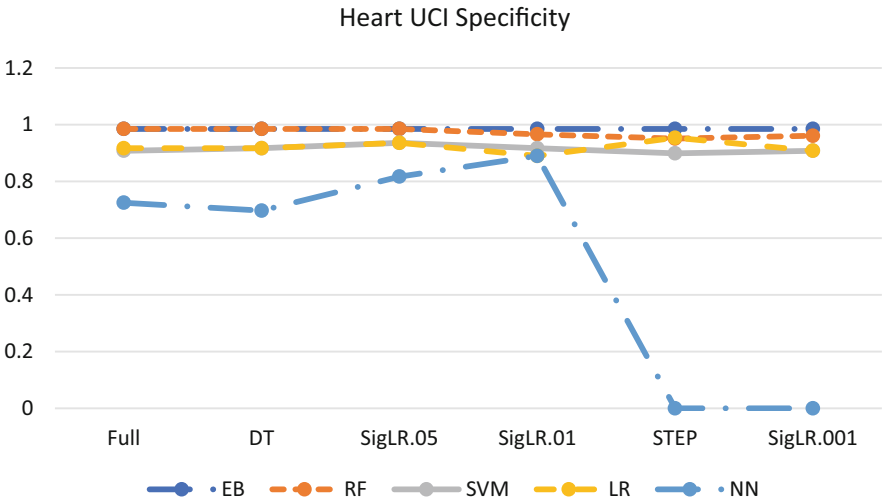


Fig. 10.6 Heart UCI dataset specificity

1. Age
2. Gender
3. Total Bilirubin (TB)
4. Direct Bilirubin (DB)
5. Alkaline Phosphatase (Alkphos)
6. Alanine Aminotransferase (SGPT)
7. Aspartate Aminotransferase (SGOT)

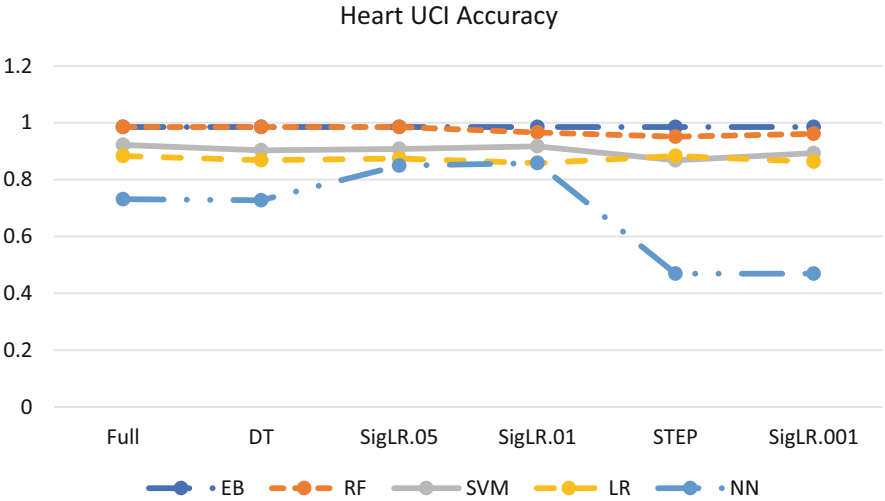


Fig. 10.7 Heart UCI dataset accuracy

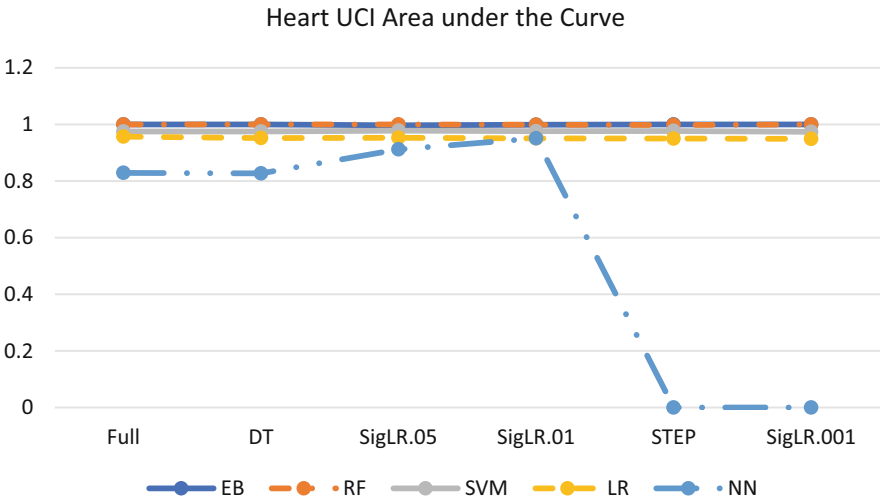


Fig. 10.8 Heart UCI dataset area under the curve

- 8. Total Proteins (TP)
- 9. Albumin (Alb)
- 10. Albumin/Globulin ratio (AG)
- 11. Liver patient or not (target)

Table 10.11 reports correlations for this dataset.

Table 10.11 India liver correlations

|         | Age    | GenF0M1 | TB     | DB     | Alkphos | Sgpt   | Sgot   | TP    | Alb   | AG    | Group |
|---------|--------|---------|--------|--------|---------|--------|--------|-------|-------|-------|-------|
| Age     | 1.000  |         |        |        |         |        |        |       |       |       |       |
| GenF0M1 | 0.056  | 1.000   |        |        |         |        |        |       |       |       |       |
| TB      | 0.011  | 0.088   | 1.000  |        |         |        |        |       |       |       |       |
| DB      | 0.006  | 0.100   | 0.874  | 1.000  |         |        |        |       |       |       |       |
| Alkphos | 0.079  | -0.029  | 0.206  | 0.234  | 1.000   |        |        |       |       |       |       |
| Sgpt    | -0.088 | 0.082   | 0.213  | 0.233  | 0.125   | 1.000  |        |       |       |       |       |
| Sgot    | -0.021 | 0.080   | 0.237  | 0.257  | 0.166   | 0.792  | 1.000  |       |       |       |       |
| TP      | -0.186 | -0.096  | -0.007 | 0.001  | -0.027  | -0.042 | -0.025 | 1.000 |       |       |       |
| Alb     | -0.264 | -0.097  | -0.222 | -0.228 | -0.163  | -0.028 | -0.084 | 0.783 | 1.000 |       |       |
| AG      | -0.216 | -0.005  | -0.206 | -0.199 | -0.234  | -0.001 | -0.069 | 0.233 | 0.688 | 1.000 |       |
| Group   | -0.132 | -0.083  | -0.220 | -0.246 | -0.183  | -0.163 | -0.151 | 0.032 | 0.156 | 0.159 | 1.000 |

**Table 10.12** India liver dataset variable significances

| Variable  | Coefficient | P-value | Significance  |
|-----------|-------------|---------|---------------|
| Intercept | 3.6046      | 0.0067  | <0.01         |
| Age       | −0.0183     | 0.0041  | <0.01         |
| GenF0M1   | −0.0310     | 0.8941  | Insignificant |
| TB        | −0.0112     | 0.8972  | Insignificant |
| DB        | −0.4497     | 0.0626  | <0.1          |
| Alkphos   | −0.0013     | 0.1159  | Insignificant |
| Sgpt      | −0.0119     | 0.0193  | <0.05         |
| Sgot      | −0.0025     | 0.4280  | Insignificant |
| TP        | −0.9274     | 0.0150  | <0.05         |
| Alb       | 1.6951      | 0.0235  | <0.05         |
| AG        | −1.8081     | 0.1130  | Insignificant |

Pseudo R-Square (optimistic): 0.42809836

**Table 10.13** Variables in different India liver data

| Full          | SigLR.01 | SigLR.05 | STEP    | SigLR.1 |
|---------------|----------|----------|---------|---------|
| Age           | Age      | Age      | Age     | Age     |
| Sgpt          | Sgpt     | Sgpt     | Sgpt    | Sgpt    |
| Alb           | Alb      | Alb      |         |         |
| TB            | TB       |          |         |         |
| Alkphos       | Alkphos  |          | Alkphos |         |
| DB            | DB       |          | DB      |         |
| Sgot          | Sgot     |          |         |         |
| TP            |          | TP       |         |         |
| Gender        |          |          |         |         |
| AlamineAT     |          |          |         |         |
| Alb/GlobRatio |          |          |         |         |

Note that the correlation with the target variable (group) is much lower than it was with the heart data in the prior section. Variable significances are given in Table 10.12.

Table 10.13 shows the variables generated by the different methods. Table 10.14 gives the results for these models, also displayed in Figs. 10.9, 10.10, 10.11, and 10.12.

The models had relatively consistent performance relative to predicting the presence of liver problems. Here the support vector machine did the best. Extreme boosting did relatively poorly with respect to sensitivity. Neural network models’ performances were highly variable.

While sensitivity measures were good, there was much more difficulty with respect to specificity. All models were variable, although the extreme boosting model was relatively steady (compensating for its relatively poor sensitivity outcomes).

Accuracy was affected by the high variability in specificity. Overall, the support vector machine models did relatively well, as did logistic regression. This was

**Table 10.14** Fits of India liver models

|             | Full  | DT    | STEP  | SigLR.1 | SigLR.05 |
|-------------|-------|-------|-------|---------|----------|
| Variables   | 10    | 7     | 4     | 4       | 2        |
| EB          |       |       |       |         |          |
| Sensitivity | 0.831 | 0.819 | 0.783 | 0.819   | 0.771    |
| Specificity | 0.273 | 0.333 | 0.394 | 0.394   | 0.333    |
| Accuracy    | 0.673 | 0.681 | 0.672 | 0.698   | 0.647    |
| AuC         | 0.785 | 0.763 | 0.772 | 0.747   | 0.685    |
| RF          |       |       |       |         |          |
| Sensitivity | 0.855 | 0.843 | 0.819 | 0.867   | 0.831    |
| Specificity | 0.242 | 0.303 | 0.455 | 0.303   | 0.394    |
| Accuracy    | 0.681 | 0.689 | 0.715 | 0.707   | 0.707    |
| AuC         | 0.786 | 0.784 | 0.797 | 0.742   | 0.718    |
| SVM         |       |       |       |         |          |
| Sensitivity | 1     | 0.988 | 0.964 | 1       | 0.988    |
| Specificity | 0.030 | 0.030 | 0.182 | 0.030   | 0.152    |
| Accuracy    | 0.725 | 0.716 | 0.742 | 0.725   | 0.75     |
| AuC         | 0.723 | 0.683 | 0.746 | 0.715   | 0.711    |
| LR          |       |       |       |         |          |
| Sensitivity | 0.892 | 0.928 | 0.928 | 0.94    | 0.952    |
| Specificity | 0.313 | 0.212 | 0.212 | 0.273   | 0.182    |
| Accuracy    | 0.724 | 0.724 | 0.724 | 0.750   | 0.733    |
| AuC         | 0.818 | 0.815 | 0.815 | 0.802   | 0.776    |
| NN          |       |       |       |         |          |
| Sensitivity | 1     | 0.783 | 1     | 1       | 0.916    |
| Specificity | 0     | 0.455 | 0     | 0       | 0.182    |
| Accuracy    | 0.716 | 0.689 | 0.716 | 0.716   | 0.707    |
| AuC         | degen | 0.742 | degen | degen   | 0.769    |

counter to the norm, but highlights the fact that different models do well for different datasets. Thus, it is a normal practice to run all five and see which fits.

The area under the curve results only show that neural network models were bad here, in part due to degeneracy in some models. While the support vector machine models did well on accuracy, they surprisingly performed relatively poorly as measured by area under the curve.

## 10.5 Watson Healthcare Data

IBM developed an artificial intelligence software, Watson, that they successfully applied to the Jeopardy game show. In an effort to show its artificial intelligence capabilities in useful terms, they have applied it to healthcare data. The Watson Healthcare dataset was presented in Chap. 7 (<https://www.kaggle.com/datasets/jpmiller/employee-attribution-for-healthcare>), including 25 independent variables as

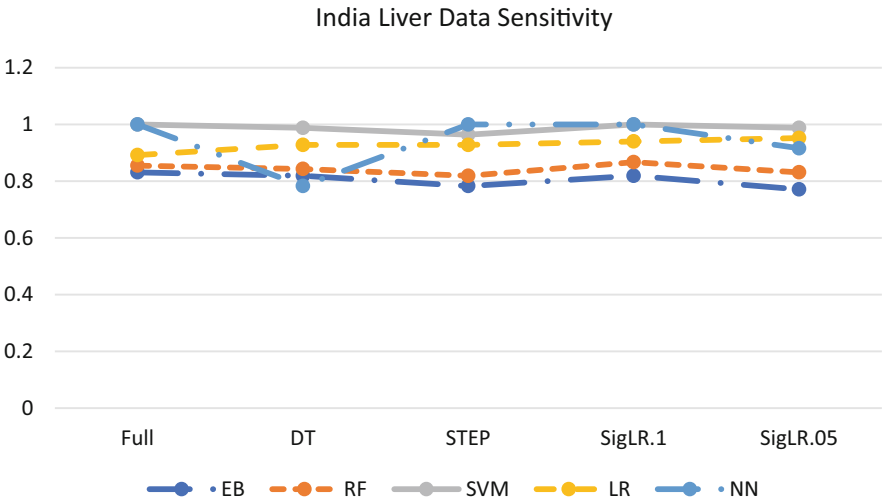


Fig. 10.9 India liver dataset sensitivity

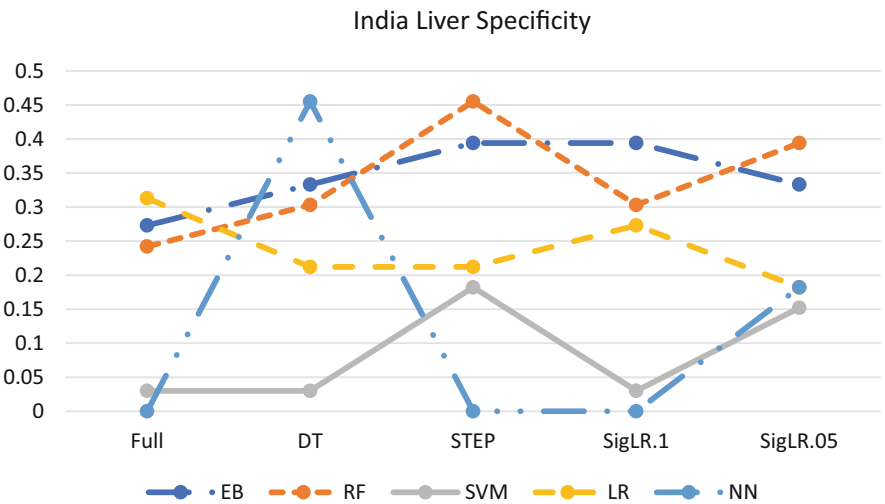


Fig. 10.10 India liver dataset specificity

shown in Table 7.1. Table 10.15 gives the correlations of these variables with the target variable (Attrition).

Table 10.16 gives the results of a logistic regression using all observations.

Eight models were generated by logistic regression significance, decision trees, and stepwise regression. Table 10.17 gives the measures of fit for these models, displayed in Figs. 10.13, 10.14, 10.15, and 10.16:

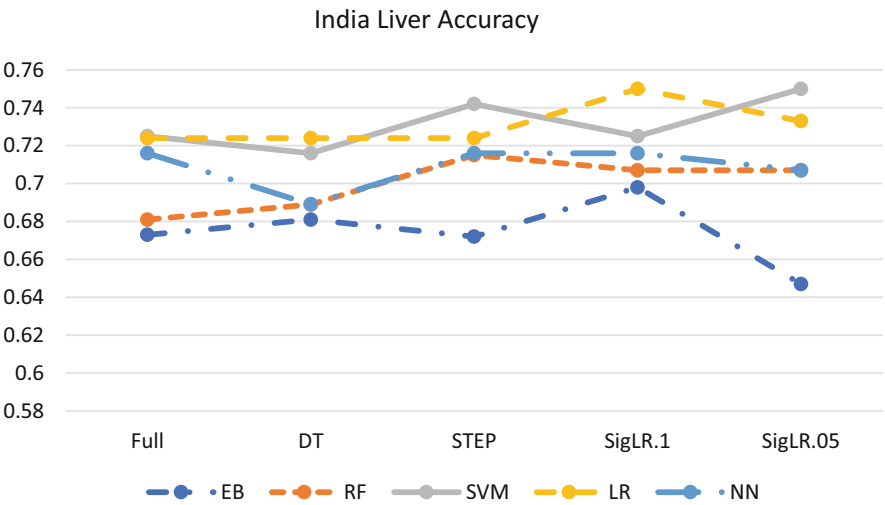


Fig. 10.11 India liver dataset accuracy

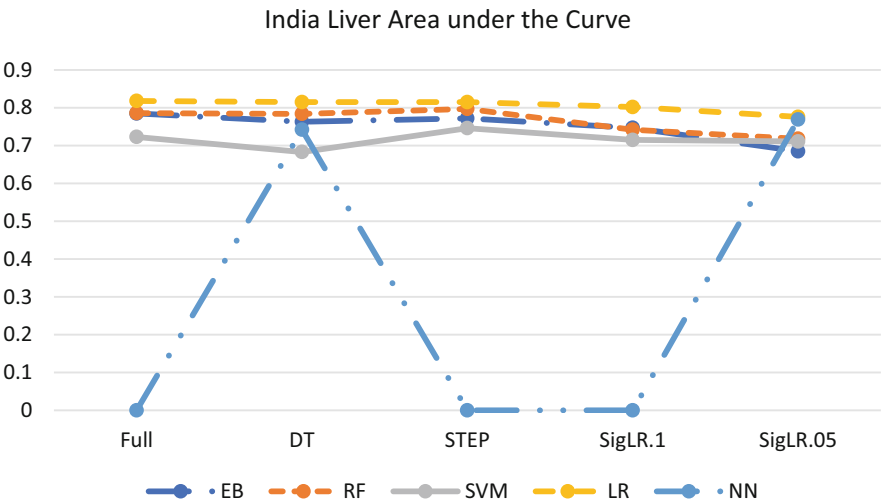


Fig. 10.12 India liver dataset area under the curve

As with the India Liver data, sensitivity results were fairly good, while specificity measures were highly variable. The extreme boosting and random forest models proved to be the most reliable overall, despite having relatively lower sensitivity.

Overall accuracy, however, was relatively poor for extreme boosting and random forest models. Support vector machine and logistic regression did relatively well on this dataset.

**Table 10.15** Correlations with attrition in Watson healthcare data

|                | Attrition |
|----------------|-----------|
| Age            | −0.240    |
| DailyRate      | −0.054    |
| DistanceHome   | 0.106     |
| Education      | −0.039    |
| EnvSat         | −0.101    |
| F0M1           | −0.021    |
| HourlyRate     | −0.036    |
| JobInvolvement | −0.166    |
| JobLevel       | −0.208    |
| JobSat         | −0.082    |
| MonthlyIncome  | −0.194    |
| MonthlyRate    | 0.046     |
| Companies      | 0.017     |
| OT             | 0.337     |
| PerSalHike     | 0.003     |
| PerfRat        | 0.011     |
| RelSat         | −0.020    |
| Shift          | −0.158    |
| WorkYr         | −0.234    |
| TrainTime      | −0.055    |
| WLBalance      | −0.091    |
| YrComp         | −0.201    |
| YrRole         | −0.208    |
| YrPromo        | −0.086    |
| YrCurMgr       | −0.201    |

Area under the curve measures were best for logistic regression, pointing to that model as being the best fit for this dataset. Again, this demonstrates that models’ performances differ with different types of data. It is a good practice to run all five and select that model working best for that dataset (or maybe voting with predictions).

10.6 Conclusions

Our results found that in general, the more variables available, the less error in fit, although variable selection for trimming few variables improved accuracy performance. The main benefit of balancing was to avoid degenerate models that were obtained with neural network and SVM models (and one case with logistic regression). The analyses we have presented in this chapter found little added benefit from balancing to error being more than 10% of total cases. Among methods, we found extreme boosting to be the most beneficial, with random forest models close in

**Table 10.16** Model beta coefficients with significances

| Variable                | Coefficient | <i>P</i> -value | Significance  |
|-------------------------|-------------|-----------------|---------------|
| Intercept               | 8.8208      | 3.93e-08        | <0.001        |
| Age                     | −0.0898     | 4.71e-07        | <0.001        |
| DailyRate               | −0.0004     | 0.1159          | Insignificant |
| DistanceFromHome        | 0.0689      | 3.97e-07        | <0.001        |
| Education               | 0.1546      | 0.1502          | Insignificant |
| EnvironmentSatisfaction | −0.5884     | 4.16e-09        | <0.001        |
| FOM1                    | −0.0339     | 0.8785          | Insignificant |
| HourlyRate              | −0.0066     | 0.2282          | Insignificant |
| JobInvolvement          | −0.9928     | 9.74e-11        | <0.001        |
| JobLevel                | 0.0050      | 0.9890          | Insignificant |
| JobSatisfaction         | −0.4636     | 3.87e-06        | <0.001        |
| MonthlyIncome           | −0.0001     | 0.2707          | Insignificant |
| MonthlyRate             | 0.00002     | 0.1650          | Insignificant |
| Companies               | 0.2537      | 1.04e-07        | <0.001        |
| OT                      | 3.5062      | 2e-16           | <0.001        |
| SalHike                 | −0.0458     | 0.3276          | Insignificant |
| PerfRat                 | 0.1868      | 0.6981          | Insignificant |
| RelSatis                | −0.2348     | 0.0213          | <0.05         |
| Shift                   | −1.0051     | 3.09e-11        | <0.001        |
| WorkYr                  | −0.1326     | 0.0004          | <0.001        |
| TrainTime               | −0.1417     | 0.1086          | Insignificant |
| WLBalace                | −0.6066     | 3.68e-05        | <0.001        |
| YrComp                  | 0.0098      | 0.8781          | Insignificant |
| YrRole                  | −0.2339     | 0.0005          | <0.001        |
| YrPromo                 | 0.2659      | 7.72e-06        | <0.001        |
| YrCurMgr                | −0.1157     | 0.0891          | <0.1          |

Pseudo *R*-Square (optimistic): 0.713665

relative accuracy. These are models that internally manipulate multiple models. SVM models were next in performance, followed by linear regression and neural networks, the latter two methods yielding very similar results. We note that SVM, linear regression, and neural networks can be fine-tuned to specified datasets, which we did not do, but this fine-tuning would take significant computational effort which is not needed by extreme boosting and random forest models.

Our basic conclusions can be itemized:

1. Balancing highly imbalanced datasets has advantage, especially in avoiding degenerate models. However, the results of our experiments presented in this chapter indicate that complete balancing is not needed—10% balancing gains most of the advantage of balancing.
2. Extreme boosting and random forest models were clearly more accurate in our results. Support vector machines had some advantage over linear regression and

Table 10.17 Watson healthcare data model fits

|             | Full  | LRsig.1 | LRsig.05 | LRsig.01 | DT.01 | Lrsig.001 | STEP8 | DT.01 | STEP6 |
|-------------|-------|---------|----------|----------|-------|-----------|-------|-------|-------|
|             | 25    | 15      | 13       | 12       | 12    | 11        | 8     | 6     | 6     |
| EB          |       |         |          |          |       |           |       |       |       |
| Sensitivity | 0.987 | 0.983   | 0.983    | 0.983    | 0.983 | 0.980     | 0.960 | 0.960 | 0.960 |
| Specificity | 0.444 | 0.472   | 0.500    | 0.472    | 0.472 | 0.472     | 0.432 | 0.528 | 0.486 |
| Accuracy    | 0.929 | 0.929   | 0.932    | 0.929    | 0.929 | 0.926     | 0.902 | 0.914 | 0.908 |
| AuC         | 0.951 | 0.941   | 0.945    | 0.927    | 0.885 | 0.935     | 0.893 | 0.848 | 0.867 |
| RF          |       |         |          |          |       |           |       |       |       |
| Sensitivity | 0.990 | 0.987   | 0.983    | 0.983    | 0.983 | 0.983     | 0.970 | 0.980 | 0.970 |
| Specificity | 0.482 | 0.472   | 0.472    | 0.500    | 0.472 | 0.417     | 0.405 | 0.417 | 0.378 |
| Accuracy    | 0.935 | 0.932   | 0.929    | 0.932    | 0.929 | 0.923     | 0.908 | 0.920 | 0.905 |
| AuC         | 0.929 | 0.941   | 0.944    | 0.947    | 0.919 | 0.936     | 0.903 | 0.875 | 0.879 |
| SVM         |       |         |          |          |       |           |       |       |       |
| Sensitivity | 0.993 | 0.993   | 0.990    | 0.983    | 0.990 | 0.987     | 0.970 | 0.983 | 0.967 |
| Specificity | 0.417 | 0.389   | 0.417    | 0.417    | 0.361 | 0.361     | 0.351 | 0.333 | 0.351 |
| Accuracy    | 0.932 | 0.929   | 0.929    | 0.923    | 0.923 | 0.920     | 0.902 | 0.914 | 0.899 |
| AuC         | 0.947 | 0.944   | 0.938    | 0.937    | 0.881 | 0.922     | 0.910 | 0.790 | 0.808 |
| LR          |       |         |          |          |       |           |       |       |       |
| Sensitivity | 0.983 | 0.982   | 0.983    | 0.983    | 0.987 | 0.987     | 0.963 | 0.983 | 0.963 |
| Specificity | 0.528 | 0.500   | 0.492    | 0.444    | 0.472 | 0.444     | 0.486 | 0.333 | 0.378 |
| Accuracy    | 0.935 | 0.929   | 0.929    | 0.926    | 0.932 | 0.929     | 0.911 | 0.914 | 0.899 |
| AuC         | 0.925 | 0.920   | 0.920    | 0.927    | 0.899 | 0.912     | 0.909 | 0.861 | 0.890 |
| NN          |       |         |          |          |       |           |       |       |       |
| Sensitivity | 1.000 | 1.000   | 1.000    | 0.977    | 1.000 | 1.000     | 0.933 | 0.967 | 0.950 |
| Specificity | 0.000 | 0.000   | 0.000    | 0.472    | 0.000 | 0.000     | 0.541 | 0.444 | 0.486 |
| Accuracy    | 0.893 | 0.893   | 0.893    | 0.923    | 0.893 | 0.893     | 0.890 | 0.911 | 0.899 |
| AuC         | degen | degen   | degen    | 0.931    | degen | degen     | 0.889 | 0.860 | 0.884 |

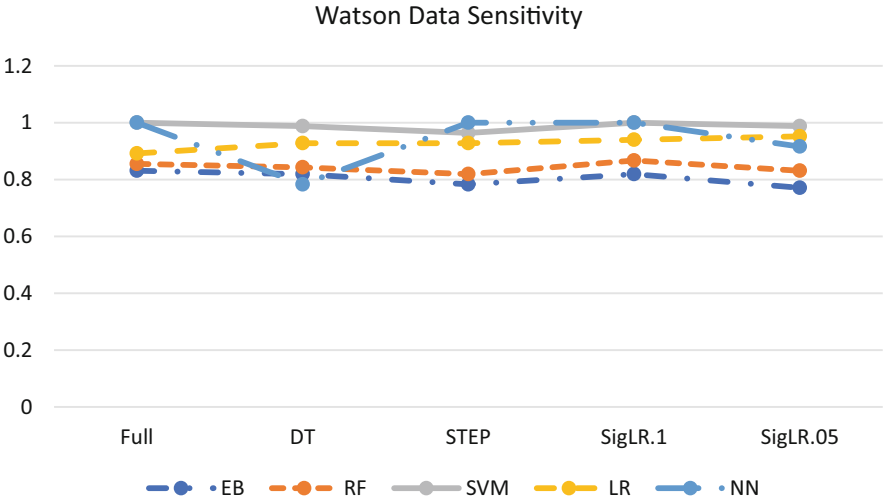


Fig. 10.13 Watson dataset sensitivity

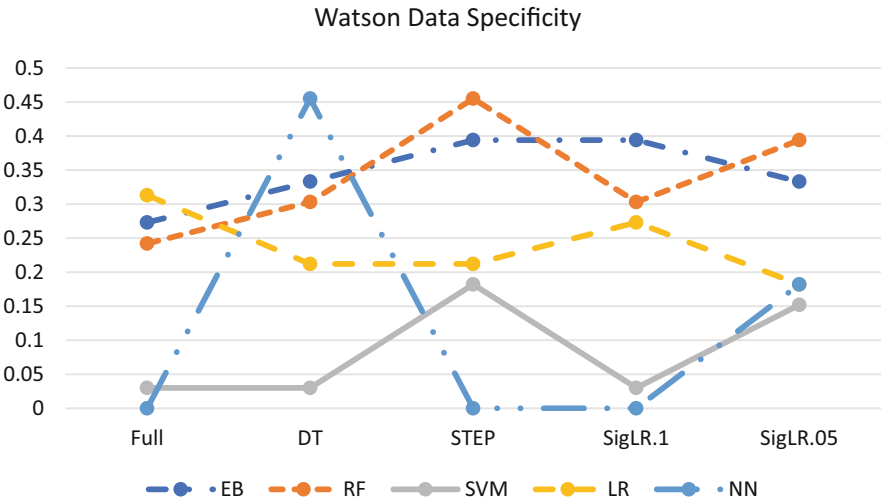


Fig. 10.14 Watson dataset specificity

- neural networks, recognizing that we did not fine-tune these last three models. To do so, however, would create more computational burden.
3. Variable selection has some benefit, although there is a slight cost in reduced accuracy. The smaller number of variables reduced from stepwise variable selection improved accuracy slightly. The benefit of trimming datasets is that results are much more focused and clearer to apply, at a small cost in accuracy.

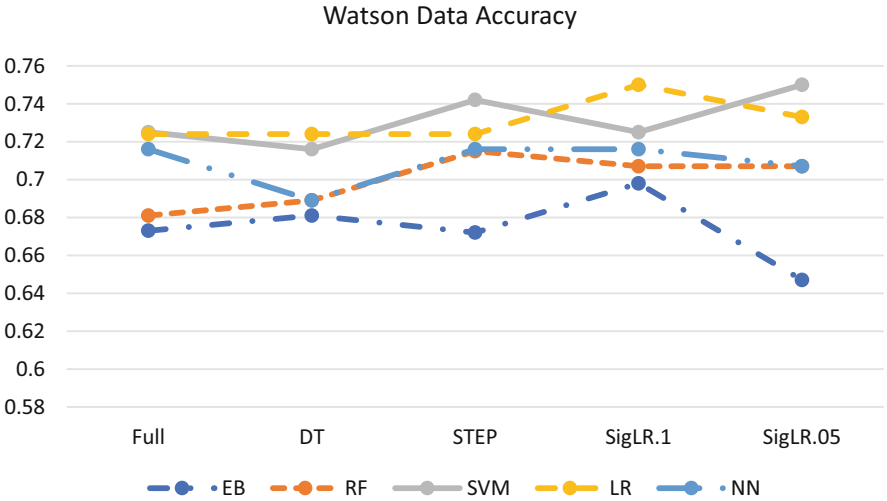


Fig. 10.15 Watson dataset accuracy

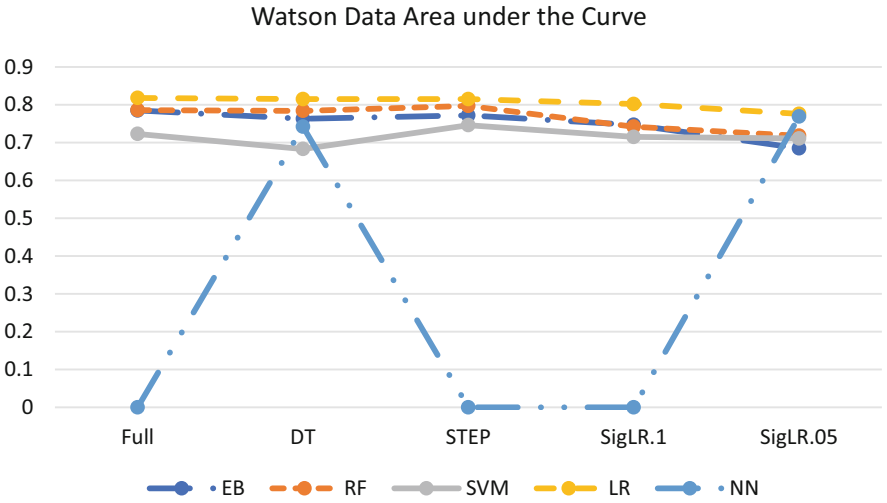


Fig. 10.16 Watson dataset area under the curve

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# Chapter 11

## Multiple Criteria Decision Models in Healthcare



**Keywords** Multiple criteria analysis · Utility · Analytic hierarchy process · TOPSIS

Medical decision-making usually involves tradeoffs of at least two criteria. A number of multiple criteria modeling tools have been applied to healthcare, more recently with big data. This chapter reviews six such applications, seeking to demonstrate the variety of techniques, as well as the healthcare decisions that can be supported.

### 11.1 Invasive Breast Cancer

Ayvaci et al. (2018) gave a modeling framework considering probabilistic inputs in a risk-sensitive context of selecting mammography exam treatments. It applied utility theory in a Markov decision process.

Almost 250,000 women were diagnosed with invasive breast cancer in 2016, over 20,000 of them dying. The motivation is to detect cancer before it becomes life threatening. The most economical means to detect breast cancer at early stages is screening mammography, X-ray examination of the breast. Radiologists assess the X-ray images and select one of the three options:

1. Biopsy to confirm whether specific lesions are cancerous;
2. Follow-up treatment with another mammography screening after 6 months;
3. Routine mammography at regular intervals.

Mammography is imperfect, with a risk of false positives (as well as false negatives). Patients, with doctor guidance, have different risk aversion levels, which need to be considered in selecting among the three options above. Risk averse patients might feel that it is better to risk an unnecessary biopsy if it reduces the odds of missing an undetected cancer. Others find biopsies to be painful and scary, so might select one of the two repeated mammography regimes.

Ayvaci et al. (2018) applied a Markov decision process model to consider patient risk preference. Data was collected on false negative and false positive rates. This data consisted of 65,892 consecutive mammography records collected from 18,270 patients between April 1999 and February 2004. The cancer registry was used to identify whether or not the patient actually had cancer. This data was used to estimate the probability of cancer as well as transition probability measures over time for the Markov model. The data was highly imbalanced in that the vast majority of patients did not have cancer. If a patient was diagnosed with cancer at the time the mammogram was taken, the bias was high that cancer was present.

Risk averse and risk seeking utility functions were developed using exponential, power/logarithmic, and logistics utility functions.

Given  $y$  as the survival duration,  $\gamma$  the level of risk aversion (or risk seeking if negative) and  $u(y)$  the utility of survival duration  $y$ , the exponential utility function was defined as (Ayvaci et al. 2018):

$$u(y) = y \text{ if } \gamma = 0$$

$$u(y) = \frac{1 - e^{-\gamma \times y}}{1 - e^{-\gamma}} \text{ otherwise.}$$

Positive  $\gamma$  characterized short-range behavior with increased propensity to focus on the short term, worrying less about long-term survival while avoiding current pain and discomfort. Emphasis on the short-run (risk aversion) favors options with higher changes of survival in the short term. In the context of lung cancer, it has been found that radiation therapy with superior short-term survival was preferred by risk averse patients to surgery which has better long-term survival (McNeil et al. 1978). For patients emphasizing the short-term that had lower risk, avoiding biopsy is preferred to avoid loss of utility from anxiety and the increased possibility of complications. Risk preference had the highest impact at age 40 (the youngest case modeled), then slowly diminishing to approach risk neutrality at older ages where lotteries had smaller payoff due to the increased probability of death from non-cancer causes. The switch in the tradeoff between avoiding discomfort and avoiding death was around age 70.

The power/logarithmic function assumes a constant willingness to exchange short-term and long-term survival. Given  $c$  as an individual's willingness to exchange long-term and short-term survival, the power/logarithmic utility function was defined by McNeil et al. (1978) as:

$$u(y) = -y^{1-c} \text{ if } c > 1$$

$$u(y) = \ln(y) \text{ if } c = 1$$

$$u(y) = y^{1-c} \text{ if } c < 1$$

As with exponential utility functions, the thresholds to biopsy and to follow-up were non-decreasing as preferences became more risk averse (emphasizing

short-term). The difference was that the magnitude of impact from risk behavior was constant by age with power utility, where valuation of lotteries depends on relative value of life years at stake. Older patients have higher biopsy thresholds than younger patients independent of risk aversion.

The logistics utility function has an S-shaped curve, proposing that risk behavior depends upon whether outcomes were gains or losses. Utility is a function of aspiration level (how many more years of survival expected).

$$u(y) = \frac{\rho}{(1 + \alpha/y^\beta)}$$

This yields an s-shaped utility function, convex at smaller  $y$  values and concave at the inflection point or aspiration level. As aspiration level increased, there was greater far-sighted behavior. At younger ages, with expected survival close to 40 years, shortsighted behavior was higher than the risk neutral case. At middle-age and older age groups, thresholds were lower than the risk-neutral case.

The study concluded that optimal policies in the selection of treatment were sensitive to risk preferences under all three utility models.

## 11.2 Colorectal Screening

Colorectal screening, like breast cancer, involves a number of treatment options. Sava et al. (2020) modeled patient choices with an analytic hierarchy process (AHP) approach considering personal patient preferences. Sensitivity was accomplished in part by extending to analytic network processing.

The authors gave a list of multiple screening options available to patients and healthcare to screen for colorectal cancer. Non-invasive alternatives were:

- Annual fecal occult blood test with sensitivity of 20%
- Annual fecal occult blood test with sensitivity of 40%
- Annual immunochemical fecal occult blood test
- Fecal DNA test every 5 years
- CT colonography

Invasive alternatives were:

- Flexible sigmoidoscopy every 5 years
- Double contrast barium enema
- Colonoscopy every 10 years

Two alternatives were available either as invasive or non-invasive:

- Annual fecal occult blood test and flexible sigmoidoscopy every 5 years
- Annual immunochemical fecal occult blood test and flexible sigmoidoscopy every 10 years

**Table 11.1** Final AHP scores (Sava et al. 2020)

| Alternative                                                                    | Category     | AHP score | Rank |
|--------------------------------------------------------------------------------|--------------|-----------|------|
| Colonoscopy every 10 years                                                     | Invasive     | 0.4842    | 1    |
| Flexible sigmoidoscopy every 5 years                                           | Invasive     | 0.1715    | 2    |
| Annual immunochemical fecal occult blood test & flexible sigmoidoscopy 5 years | Non-invasive | 0.1404    | 3    |
| Double contrast barium enema                                                   | Invasive     | 0.0757    | 4    |
| CT colonography                                                                | Non-invasive | 0.0669    | 5    |
| Annual immunochemical fecal occult blood test                                  | Non-invasive | 0.023     | 6    |
| Annual fecal occult blood test and flexible sigmoidoscopy every 5 years        | Both         | 0.0184    | 7    |
| Fecal DNA test every 5 years                                                   | Non-invasive | 0.0084    | 8    |
| Annual fecal occult blood test with 20% sensitivity                            | Non-invasive | 0.0077    | 9    |
| Annual fecal occult blood test with 40% sensitivity                            | Non-invasive | 0.0033    | 10   |

Six criteria were considered:

1. Cancer prevention, measured as the level of accuracy in detection
2. Avoidance of side effects measured in number of side effects avoided
3. Minimization of false positives
4. Procedure frequency
5. Procedure preparation measured by duration of the procedure
6. Procedure complexity measured by invasiveness

AHP was applied from a medical perspective, yielding scores shown in Table 11.1:

From the medical perspective, the emphasis was clearly on invasive procedures. The authors conducted sensitivity analysis, in part using analytic network process (AHP with feedback). The sensitivity focused on the sensitivity and stability of decision to preference weight changes. The fastest change in preference involved the criteria of avoiding side effects, minimizing false positives, and procedure frequency. If the weights of all three of these criteria were increased, a switch in preference from colonoscopy every 10 years to flexible sigmoidoscopy every 5 years (original second choice).

## 11.3 Senior Center Site Selection

As populations around the world increase, the need for senior centers has become a large issue. Zhuang et al. (2019) combined AHP with TOPSIS to aid a fuzzy multi-attribute group decision involving selection of a senior center.

A literature review was used to identify fourteen criteria of importance in selection of senior center alternatives. Input data was obtained from seniors through a questionnaire, yielding relative performances of four alternative senior center sites over the fourteen sub-criteria (which were grouped into five primary issues). Table 11.2 gives criteria performances.

Price (in new Taiwan dollars per month) was to be minimized. All other input scores were obtained from raw data. The last four columns convert input scores into normalized outputs.

Assessments by three seniors were used to assess each alternative center's performance on each of the five major criteria using AHP pairwise comparisons. Table 11.3 gives decision maker assessments of criteria weights:

In Table 11.3, the Geometric line gives the geometric means of the three decision maker inputs on weights, while the Normalized line gives the AHP weights. These were both applied to normalized scores for the four alternatives in Table 11.4:

The relative rankings of the four alternatives were the same for weights generated by geometric mean and by AHP, with Alternative 1 holding a very slight advantage over Alternative 3, followed by Alternative 2 and lastly Alternative 4.

**Table 11.2** Senior center performance scores

| Criteria    | Alt1   | Alt2   | Alt3   | Alt4   | #1    | #2    | #3    | #4    |
|-------------|--------|--------|--------|--------|-------|-------|-------|-------|
| FINANCE     |        |        |        |        |       |       |       |       |
| Price       | 20,900 | 19,000 | 16,000 | 12,800 | 0.401 | 0.456 | 0.541 | 0.633 |
| Support     | 50     | 0      | 50     | 0      | 0.707 | 0.000 | 0.707 | 0.000 |
| ENVIRONMENT |        |        |        |        |       |       |       |       |
| Neatness    | 100    | 100    | 100    | 100    | 0.500 | 0.500 | 0.500 | 0.500 |
| Space use   | 30     | 16     | 22     | 23.17  | 0.643 | 0.343 | 0.472 | 0.497 |
| Safety      | 100    | 80     | 80     | 80     | 0.585 | 0.468 | 0.468 | 0.468 |
| Bedroom     | 50     | 75     | 50     | 50     | 0.436 | 0.655 | 0.436 | 0.436 |
| FACILITIES  |        |        |        |        |       |       |       |       |
| Ventilation | 100    | 100    | 100    | 100    | 0.500 | 0.500 | 0.500 | 0.500 |
| In-room     | 83     | 83     | 100    | 67     | 0.494 | 0.494 | 0.595 | 0.399 |
| Leisure     | 86     | 100    | 71     | 86     | 0.498 | 0.579 | 0.411 | 0.498 |
| SERVICE     |        |        |        |        |       |       |       |       |
| Health      | 100    | 100    | 60     | 80     | 0.577 | 0.577 | 0.411 | 0.498 |
| Staff       | 100    | 100    | 83     | 83     | 0.544 | 0.544 | 0.452 | 0.452 |
| Food        | 75     | 75     | 100    | 50     | 0.487 | 0.487 | 0.649 | 0.324 |
| CONVENIENCE |        |        |        |        |       |       |       |       |
| Transport   | 75     | 100    | 100    | 50     | 0.447 | 0.596 | 0.596 | 0.298 |
| Living      | 100    | 80     | 100    | 100    | 0.524 | 0.419 | 0.524 | 0.524 |

**Table 11.3** Relative weights

|            | Finance | Environment | Facilities | Service | Convenience |
|------------|---------|-------------|------------|---------|-------------|
| DM1        | 0.418   | 0.170       | 0.170      | 0.170   | 0.170       |
| DM2        | 0.151   | 0.354       | 0.267      | 0.172   | 0.056       |
| DM3        | 0.433   | 0.156       | 0.095      | 0.280   | 0.037       |
| Geometric  | 0.301   | 0.211       | 0.163      | 0.202   | 0.053       |
| Normalized | 0.324   | 0.227       | 0.175      | 0.217   | 0.057       |

**Table 11.4** Weights \* normalized performance scores

|      | Finance | Environment | Facilities | Service | Convenience | Geo    | AHP    |
|------|---------|-------------|------------|---------|-------------|--------|--------|
| Alt1 | 0.554   | 0.541       | 0.497      | 0.536   | 0.486       | 0.4959 | 0.5334 |
| Alt2 | 0.228   | 0.491       | 0.524      | 0.536   | 0.508       | 0.3928 | 0.4223 |
| Alt3 | 0.624   | 0.469       | 0.502      | 0.482   | 0.560       | 0.4957 | 0.5332 |
| Alt4 | 0.317   | 0.475       | 0.465      | 0.413   | 0.411       | 0.3766 | 0.4049 |

The paper also applied intuitionist fuzzy numbers where intensity of membership as well as intensity of non-membership was elicited from the decision makers for alternative performance on criteria. The results yielded the same rank ordering as with AHP. The TOPSIS (Shih 2016) method for group decision makers was also applied, again yielding the same rankings.

### 11.4 Diabetes and Heart Problem Detection

Diaz Ochoa et al. (2021) developed a methodology including neural network model for Pakistani patient populations facing diabetes and/or heart problems. The model considered treatment options of beta-blockers, angiotensin-converting enzyme inhibitors (ACE), angiotensin II receptor blockers (ARB), or aspirin. The database was taken from a publicly available patient population in Pakistan. Variables included are shown in Table 11.5:

When a doctor queried the system, data was drawn and fed into an artificial neural network (ANN) model in an R environment. The process was to draw training data and run an ANN for each treatment. Model I was run on data drawn from the total patient population, while Model II was run on data drawn from patients with positive outcomes. The data was divided into 80% for training and 20% for testing. ANN parameters were adjusted to obtain better fit, adjusting both the number of neurons per layer and squashing activation functions. If Model I and Model II matched, a high confidence was inferred. Accuracy of Model I was 63.49%, and 64.00% for Model II.

A visual display was presented to the physician requesting advice. For each patient, a preferred therapy was identified, along with the confidence outcome

**Table 11.5** Pakistani diabetic patient variables

| Input Variable           | Parameter Type | Output Variable | Parameter Type          |
|--------------------------|----------------|-----------------|-------------------------|
| Sex                      | Binary         | Time (feedback) | Continuous              |
| Age                      | Continuous     | Treatment       | Discrete (death or not) |
| Creatinine phosphokinase | Continuous     |                 |                         |
| Ejection fraction        | Continuous     |                 |                         |
| High blood pressure      | Binary         |                 |                         |
| Platelets                | Continuous     |                 |                         |
| Serum creatinine         | Continuous     |                 |                         |
| Serum sodium             | Continuous     |                 |                         |
| Smoking                  | Binary         |                 |                         |
| Anemia                   | Binary         |                 |                         |

(high or low), and a categorical survival probability assessment (less than or greater than 1 year).

11.5 Bolivian Healthcare System

Pinazo et al. (2021) modeled a healthcare system in Bolivia for the purpose of disease management. There was a need to prioritize interventions and actions to address neglected diseases in lower income locations. Health programs in Latin American countries were reviewed, and a list of almost 200 interventions were generated. Experts narrowed this list yielding specific criteria used in the study. Surveys were then conducted to obtain outcome and complexity criteria weights following AHP.

An iterative process was used to generate improved options. Modeling applied in R statistical programming language to implement AHP with other multi-criteria methods.

The World Health Organization includes four main outcome criteria (improved health, responsiveness, financial risk protection, and improved efficiency). In order to improve performance on these criteria, supportive actions in six areas were proposed:

1. Service Delivery—management of health centers to deliver primary and secondary levels of care, mechanisms to communicate with and deliver help to patients.
2. Health Workforce—training and support plan to develop healthcare workforce, provision of training materials and programs.
3. Information Systems—software to monitor and control data.
4. Access to Treatment—supply needed drugs and medical devices.
5. Leadership and Governance—coordinate with regional and national governments and sponsor local activities.
6. Financing—coordinate government funding, seek grants, and implement a contribution system.

Interventions supporting each action were assessed. This led to identification of specific interventions. These mechanisms were in support of the following recommended actions:

- Prevention and Control—fumigation to combat infestations, community education to improve reporting.
- Diagnostic Interventions—screening of pregnant women to identify suspected acute cases.
- Treatment Interventions—protocol to support insolvent patients, monitoring diet, develop a treatment record system, follow-up after treatment.

While the details of the AHP ratings were not published in the article, that application is straightforward. Data envelopment analysis (DEA—Cooper and Seiford 2006) was applied to identify non-dominated actions as measured in terms of outcomes and complexity.

## 11.6 Breast Cancer Screening

Krohn et al. (2021) applied a mixed-integer linear programming model to increase expected participation in a breast cancer screening campaign. Preventive care facilitates early detection and aids in treatment to retard cancer development. The study objective was to develop a methodology to organize a limited number of facilities to maximize patient participation. Quality of care and waiting time (defined as the time between patient requesting an appointment and treatment date) were both modeled.

Random utility theory was used to model client behavior. Assumptions included the concept that higher demand increases both quality of care and waiting time. As quality of care increases, it was assumed that utility increases. As waiting time increases, it was assumed that utility decreases. To close the loop, demand was assumed to be a function of utility.

High quality care was measured as exceeding a given number of screenings accomplished. Also important is the ratio of the likelihood of a true positive over a true negative diagnosis. Waiting time is a queuing phenomenon, where length of waiting will increase with demand.

A mixed-integer nonlinear preventive healthcare facility location planning model was developed reflecting demand nodes and facility nodes. The objective was to maximize the product of the number of eligible clients in the network times the choice probability of clients in a particular demand node going to a facility node. Nodes were either open or closed, providing the integer aspect of the model.

The tradeoff between minimizing the number of open facilities subject to constraining participation level was applied. This enabled identification of the tradeoff between cost and expected participation. The model was applied to real data of women's preferences for breast cancer screening in greater Sydney, Australia. Publicly available geographic and census data from Queensland governmental agencies. The number of women between the ages of 50 and 74 (the eligible

**Table 11.6** Studies, big data features, decisions, and models

| Study             | Big Data           | Health Decision   | MCDM Model       |
|-------------------|--------------------|-------------------|------------------|
| Ayvaci et al.     | Breast cancer      | Screening choice  | Utility theory   |
| Sava et al.       | Colorectal cancer  | Treatment choice  | AHP (ANP)        |
| Zhuang et al.     | Senior center      | Selection         | AHP TOPSIS       |
| Diaz Ochoa et al. | Diabetes Heart     | Therapy selection | Neural network   |
| Pinazo et al.     | Disease management | System selection  | AHP DEA          |
| Krohn et al.      | Breast cancer      | Facility location | Mixed-integer LP |

pool for breast screening) was obtained for selected statistical areas. Centroids of these areas were used for demand node locations. The locations of 20 existing screening facilities and 14 mobile van locations were obtained from the web site of a New South Wales agency. Real distances were generated by Manhattan distance, converted to expected travel time. Connections with expected travel time greater than 2 h were excluded.

### 11.7 Comparison

The chapter intended to demonstrate some multiple criteria decision-making models and to show healthcare decisions supported by these tools. The studies presented involve different features and decisions, applying different models. These are recapped in Table 11.6:

Some of the basic MCDM models were included, but there are quite a number of other models. All provide some value in analyzing the inherent tradeoffs in healthcare management problems. There are many other health issues that have been supported. These focused on analysis of particular diseases. But multiple criteria analysis can be applied to administrative and policy issues as well.

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# Chapter 12

## Naïve Bayes Models in Healthcare



**Keywords** Bayesian models · Naïve Bayes

As with logistic regression, naïve Bayes models produce probability estimates of a case belonging to a class. Bayesian networks can be applied to prediction models involving conditional independence.

Bayesian modeling has been widely applied in healthcare analytics. Most of the published applications go beyond simple naïve Bayes modeling, which we attribute to the need for journal articles to have a certain degree of complexity. Thus, published models tend to have features dealing with special problems such as variable selection or variable interaction.

### 12.1 Applications

Nonetheless, there have been many applications. Shameer et al. (2018) applied Bayesian logistic regression to predict patients at risk of readmission, a feature that government reimbursement is designed to discourage. Therefore, being able to predict readmission allows hospitals to be more careful in releasing at-risk patients. Hassan's (2016) dissertation thesis dealt with the same problem, applying multi-objective Bayesian network modeling. Lin et al. (2017) presented Bayesian multi-task learning to coordinate a set of baseline models allowing healthcare providers to profile risks more accurately. Ajorlou et al. (2015) presented a generalized linear model in a Bayesian framework to predict workload in medical homes at both the primary care and non-primary care levels.

Many Bayesian decision models have been applied to healthcare problems. Cooper et al. (2002) gave an early Bayesian decision model applied to the use of prophylactic antibiotics in cesarean section incidents. They evaluated their model using Markov Chain Monte Carlo simulation. Dag et al. (2016) applied Bayesian Belief Networks to include interactions among predictor variables to identify conditional probability of survival in heart transplant operations. Topuz et al. (2018b)

also used Bayesian Belief Networks, along with variable selection techniques in predicting graft survival in kidney transplants. Liu et al. (2020) used a hierarchical Bayesian framework to incorporate domain knowledge as priors in assessing risk in diabetes mellitus cases. Mueller et al. (2020) used a Bayesian machine learning analytics platform to construct generalized linear models to predict outcomes in hypoglycemia risk with type 2 diabetes.

## 12.2 Bayes Model

Naïve Bayes classifiers provide a simple means of classifying categorical data. It seeks to model each class of the training data by probability density functions. Items are assigned to the most probable class. The method assumes a number of explanatory variables and assumes them to be independent. It is considered to have the relative advantage that it does not require a large training dataset for effective classification. Naïve Bayes is a conditional probability model. For a vector of  $n$  factors (independent variables), for total of  $K$  possible outcomes or classes, you seek:

$$\text{Prob}\{C_k|x_1, x_2, \dots x_n\}$$

for each of  $k$  possible outcomes or classes. It would be accurate to enumerate all combinations of classes, but this can become unwieldy if  $n$  is large, and in the simple example to follow, the sample size might be quite low. Naïve Bayes offers a simple alternative. Bayes theorem can be applied to estimate this probability:

$$\text{Prob}\{C_k|x_1, x_2, \dots x_n\} = \frac{\text{Prob}\{C_k\}\text{Prob}\{x_1, x_2, \dots x_n|C_k\}}{\text{Prob}\{x_1, x_2, \dots x_n\}}$$

Restating this,

$$\text{Posterior} = \frac{\text{prior} \times \text{likelihood}}{\text{evidence}}$$

The evidence denominator doesn't depend on  $C_k$ , and the vector of independent variable values  $x_1, x_2, \dots x_n$  are given to the denominator is a constant. A formal statement of the method is:

Probability{Membership in Class  $k$ } is given by the formula:

$$\text{Prob}\{C_k\} \prod_{i=1}^n \text{Prob}\{x_i|C_k\}$$

To explain this, look at the categorical data from Kaggle that we used in the association rule chapter.

### 12.2.1 *Demonstration with Kaggle Data*

Intensive Care Units (ICUs) often lack verified medical histories for incoming patients. A patient in distress or a patient who is brought in confused or unresponsive may not be able to provide information about chronic conditions such as heart disease, injuries, or diabetes. Medical records may take days to transfer, especially for a patient from another medical provider or system. Knowledge about chronic conditions can inform clinical decisions about patient care and ultimately improve patient's survival outcomes. The global open-source severity of illness score (GOSSIS) data (Raffa et al. 2019) is found at [https://journals.lww.com/ccmjournal/Citation/2019/01001/33\\_THE\\_GLOBAL\\_OPEN\\_SOURCE\\_SEVERITY\\_OF\\_ILLNESS.36.aspx](https://journals.lww.com/ccmjournal/Citation/2019/01001/33_THE_GLOBAL_OPEN_SOURCE_SEVERITY_OF_ILLNESS.36.aspx) site. Variables are shown in the Rattle screen (Fig. 12.1):

Note that variable Readmit had only one value, so Rattle disregarded it. Models were run with one disease at a time. The results for AIDS are shown in Table 12.1.

Of the 91,690 patients at the clinic, 793 had AIDS, of whom 95 died. The rates indicate a higher proportion in Afro-Americans, followed by Hispanics, Native Americans, and Caucasians. Asians had the lowest rate. The death rates, on the other hand, were highest for Hispanics (or for the seven cases where the patient was unknown). Males had a higher death rate than females. Compiled results are given in Table 12.2:

Table 12.2 shows that diabetes impacted nearly one fourth of the patients at this clinic. Other diseases had much lower occurrences, with AIDS being less than 1%. The deaths during clinic treatment are shown in Table 12.3:

Table 12.3 shows that diabetic patients, although by far the most common, had the lowest death rate. Table 12.4 gives death rates contingent upon having the disease.

Table 12.4 results indicate one out of roughly 12 patients at the clinic died during the measured period. The highest rate in Table 12.4 were those with solid tumors with metastasis, the lowest those with cirrhosis or diabetes mellitus.

Bayesian prediction amounts to identifying the base subset of cases and counting. The formula for Bayesian probability is:

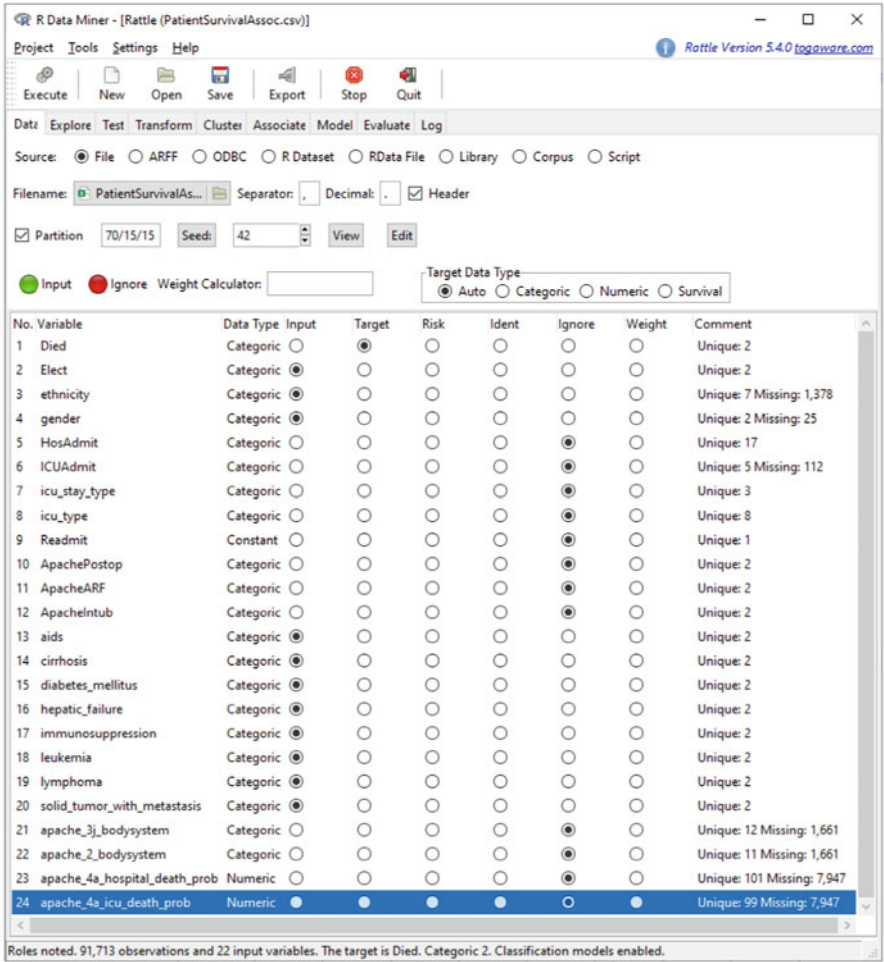


Fig. 12.1 Rattle screen for patient survival data

$$\text{Prob}\{A \mid B\} = \frac{\text{Prob}\{B \mid A\} \times \text{Prob}\{A\}}{\text{Prob}\{B\}}$$

For instance, to find the probability of a Caucasian male dying of AIDS, from Table 12.1 we have the following values:

|                                               |              |
|-----------------------------------------------|--------------|
| Prob{B   A} = Prob{died} given Caucasian Male | = 45/3257    |
| Prob{A} = Prob{died}                          | = 3257/38283 |
| Prob{B} = Prob{Caucasian Male with AIDS}      | =329/38283   |

Thus,

**Table 12.1** AIDS results

| Ethnicity | Gender | Total  | Died | AIDS | Daids | RateD | RateA | IfAidDie |
|-----------|--------|--------|------|------|-------|-------|-------|----------|
| Afro      | F      | 4725   | 375  | 54   | 3     | 0.079 | 0.011 | 0.056    |
|           | M      | 4820   | 375  | 60   | 6     | 0.078 | 0.012 | 0.100    |
| Asian     | F      | 531    | 35   | 1    | 0     | 0.066 | 0.002 | 0.000    |
|           | M      | 598    | 58   | 2    | 0     | 0.097 | 0.003 | 0.000    |
| Cauc      | F      | 32,399 | 2911 | 260  | 27    | 0.090 | 0.008 | 0.104    |
|           | M      | 38,283 | 3257 | 329  | 45    | 0.085 | 0.009 | 0.137    |
| Hispan    | F      | 1786   | 183  | 22   | 6     | 0.102 | 0.012 | 0.273    |
|           | M      | 2011   | 193  | 15   | 3     | 0.096 | 0.007 | 0.200    |
| Native    | F      | 383    | 35   | 3    | 0     | 0.091 | 0.008 | 0.000    |
|           | M      | 405    | 35   | 4    | 0     | 0.086 | 0.010 | 0.000    |
| Other     | F      | 1810   | 144  | 17   | 1     | 0.080 | 0.009 | 0.059    |
|           | M      | 2555   | 203  | 19   | 2     | 0.079 | 0.007 | 0.105    |
| Unknown   | F      | 587    | 48   | 2    | 0     | 0.082 | 0.003 | 0.000    |
|           | M      | 797    | 57   | 5    | 2     | 0.072 | 0.006 | 0.400    |
| Afro      |        | 9547   | 750  | 114  | 9     | 0.079 | 0.012 | 0.079    |
| Asian     |        | 1129   | 93   | 3    | 0     | 0.082 | 0.003 | 0.000    |
| Cauc      |        | 70,682 | 6168 | 589  | 72    | 0.087 | 0.008 | 0.122    |
| Hispan    |        | 3798   | 376  | 37   | 9     | 0.099 | 0.010 | 0.243    |
| Native    |        | 788    | 70   | 7    | 0     | 0.089 | 0.009 | 0.000    |
| Other     |        | 4374   | 347  | 36   | 3     | 0.079 | 0.008 | 0.083    |
| Unknown   |        | 1395   | 107  | 7    | 2     | 0.077 | 0.005 | 0.286    |
| F         |        | 42,221 | 3731 | 359  | 37    | 0.088 | 0.009 | 0.103    |
| M         |        | 49,469 | 4178 | 434  | 58    | 0.084 | 0.009 | 0.134    |
| Total     |        | 91,690 | 7909 | 793  | 95    | 0.086 | 0.009 | 0.120    |

$$\text{Prob}\{A \mid B\} = \frac{\frac{45}{3257} \times \frac{3257}{38283}}{\frac{329}{38283}} = \frac{45}{329} = 0.137$$

Actually, given Table 12.1 you could find the contingent row (there were 329 Caucasian males with AIDS in the dataset, 45 of whom died of AIDS, giving the probability of 0.137).

## 12.3 Naïve Bayes Analysis of Watson Turnover Data

In Chap. 7, we looked at healthcare personnel turnover data (<https://www.kaggle.com/datasets/jpmiller/employee-attrition-for-healthcare>) and ran predictive models for the continuous data. We can apply Naïve Bayes analysis to some of the categorical data. We can focus on the difference between nurse turnover and turnover of all other employees (see Table 12.5):

Table 12.2 Compiled results

| Ethnicity | Gender | Total  | AIDS  | Cirrhosis | Diabetes | Hepatic | Immuno | Leukemia | Lymphoma | Tumor |
|-----------|--------|--------|-------|-----------|----------|---------|--------|----------|----------|-------|
| Afro      | F      | 4725   | 54    | 82        | 1329     | 83      | 162    | 73       | 53       | 137   |
|           | M      | 4820   | 60    | 88        | 1210     | 80      | 150    | 70       | 54       | 137   |
| Asian     | F      | 531    | 1     | 7         | 130      | 9       | 18     | 4        | 6        | 20    |
|           | M      | 598    | 2     | 14        | 217      | 10      | 13     | 5        | 3        | 15    |
| Cauc      | F      | 32,399 | 260   | 640       | 6917     | 579     | 1157   | 451      | 391      | 986   |
|           | M      | 38,283 | 329   | 969       | 8770     | 858     | 1311   | 620      | 480      | 1065  |
| Hispan    | F      | 1786   | 22    | 53        | 436      | 51      | 56     | 29       | 26       | 57    |
|           | M      | 2011   | 15    | 77        | 539      | 61      | 59     | 33       | 15       | 53    |
| Native    | F      | 383    | 3     | 34        | 129      | 30      | 13     | 6        | 3        | 7     |
|           | M      | 405    | 4     | 36        | 142      | 31      | 13     | 6        | 6        | 8     |
| Other     | F      | 1810   | 17    | 57        | 523      | 35      | 49     | 25       | 22       | 33    |
|           | M      | 2555   | 19    | 73        | 710      | 59      | 75     | 26       | 23       | 50    |
| Unknown   | F      | 587    | 2     | 4         | 84       | 3       | 7      | 4        | 3        | 10    |
|           | M      | 797    | 5     | 9         | 116      | 8       | 13     | 7        | 6        | 15    |
| Afro      |        | 9547   | 114   | 170       | 2539     | 163     | 312    | 143      | 107      | 274   |
| Asian     |        | 1129   | 3     | 21        | 347      | 19      | 31     | 9        | 9        | 35    |
| Cauc      |        | 70,682 | 589   | 1609      | 15,687   | 1437    | 2468   | 1071     | 871      | 2051  |
| Hispan    |        | 3798   | 37    | 130       | 975      | 10      | 115    | 62       | 41       | 110   |
| Native    |        | 788    | 7     | 70        | 271      | 61      | 26     | 12       | 9        | 15    |
| Other     |        | 4374   | 36    | 130       | 1234     | 94      | 124    | 51       | 45       | 83    |
| Unknown   |        | 1395   | 7     | 13        | 200      | 11      | 20     | 11       | 9        | 25    |
| F         |        | 42,221 | 359   | 877       | 9548     | 790     | 1462   | 592      | 504      | 1250  |
| M         |        | 49,469 | 434   | 1266      | 11,704   | 1107    | 1634   | 767      | 587      | 1343  |
| Total     |        | 91,690 | 793   | 2143      | 21,252   | 1897    | 3096   | 1359     | 1091     | 2593  |
| %         |        |        | 0.86% | 2.34%     | 23.18%   | 2.07%   | 3.38%  | 1.48%    | 1.19%    | 2.83% |

Table 12.3 Deaths by disease

| Ethnicity | Gender | Died   | AIDS   | Cirrhosis | Diabetes | Hepatic | Immuno | Leukemia | Lymphoma | Tumor  |
|-----------|--------|--------|--------|-----------|----------|---------|--------|----------|----------|--------|
| Afro      | F      | 375    | 3      | 9         | 93       | 6       | 30     | 10       | 5        | 19     |
|           | M      | 375    | 6      | 13        | 79       | 11      | 21     | 14       | 5        | 24     |
| Asian     | F      | 35     | 0      | 0         | 4        | 1       | 2      | 1        | 2        | 4      |
|           | M      | 58     | 0      | 4         | 13       | 2       | 2      | 0        | 0        | 1      |
| Cauc      | F      | 2911   | 27     | 101       | 573      | 91      | 161    | 67       | 48       | 158    |
|           | M      | 3257   | 45     | 150       | 691      | 136     | 215    | 88       | 71       | 184    |
| Hispan    | F      | 183    | 6      | 9         | 38       | 9       | 9      | 8        | 6        | 10     |
|           | M      | 193    | 3      | 13        | 46       | 14      | 4      | 8        | 4        | 9      |
| Native    | F      | 35     | 0      | 11        | 18       | 8       | 0      | 0        | 0        | 1      |
|           | M      | 35     | 0      | 8         | 11       | 7       | 1      | 1        | 0        | 1      |
| Other     | F      | 144    | 1      | 5         | 37       | 5       | 6      | 2        | 1        | 4      |
|           | M      | 203    | 2      | 5         | 62       | 6       | 12     | 3        | 4        | 11     |
| Unknown   | F      | 48     | 0      | 1         | 6        | 0       | 3      | 0        | 0        | 3      |
|           | M      | 57     | 2      | 4         | 8        | 3       | 3      | 2        | 2        | 3      |
| Afro      |        | 750    | 9      | 22        | 172      | 17      | 51     | 24       | 10       | 43     |
| Asian     |        | 93     | 0      | 4         | 17       | 3       | 4      | 1        | 2        | 5      |
| Cauc      |        | 6168   | 72     | 251       | 1264     | 227     | 376    | 155      | 119      | 342    |
| Hispan    |        | 376    | 9      | 22        | 84       | 5       | 13     | 16       | 10       | 19     |
| Native    |        | 70     | 0      | 19        | 29       | 15      | 1      | 1        | 0        | 2      |
| Other     |        | 347    | 3      | 10        | 99       | 11      | 18     | 5        | 5        | 15     |
| Unknown   |        | 107    | 2      | 5         | 14       | 3       | 6      | 2        | 2        | 6      |
| F         |        | 3731   | 37     | 136       | 769      | 120     | 211    | 88       | 62       | 199    |
| M         |        | 4178   | 58     | 197       | 910      | 179     | 258    | 116      | 86       | 233    |
| Cases     |        | 91,690 | 793    | 2143      | 21,252   | 1897    | 3096   | 1359     | 1091     | 2593   |
| Died      |        | 3257   | 95     | 333       | 1679     | 299     | 469    | 204      | 148      | 432    |
| %         |        |        | 11.98% | 15.54%    | 7.90%    | 15.76%  | 15.15% | 15.01%   | 13.57%   | 16.66% |

Table 12.4 Death rates by disease

| Ethnicity | Gender | AIDS  | Cirrhosis | Diabetes | Hepatic | Immuno | Leukemia | Lymphoma | Tumor |
|-----------|--------|-------|-----------|----------|---------|--------|----------|----------|-------|
| Afro      | F      | 0.056 | 0.110     | 0.070    | 0.072   | 0.185  | 0.137    | 0.094    | 0.139 |
|           | M      | 0.100 | 0.148     | 0.065    | 0.138   | 0.140  | 0.200    | 0.093    | 0.175 |
| Asian     | F      | 0     | 0         | 0.031    | 0.111   | 0.111  | 0.250    | 0.333    | 0.200 |
|           | M      | 0     | 0.286     | 0.060    | 0.200   | 0.154  | 0        | 0        | 0.067 |
| Cauc      | F      | 0.104 | 0.158     | 0.083    | 0.157   | 0.139  | 0.149    | 0.123    | 0.160 |
|           | M      | 0.137 | 0.155     | 0.079    | 0.159   | 0.164  | 0.142    | 0.148    | 0.173 |
| Hispan    | F      | 0.273 | 0.170     | 0.087    | 0.176   | 0.161  | 0.276    | 0.231    | 0.175 |
|           | M      | 0.200 | 0.169     | 0.085    | 0.230   | 0.068  | 0.242    | 0.267    | 0.170 |
| Native    | F      | 0     | 0.324     | 0.140    | 0.267   | 0      | 0        | 0        | 0.143 |
|           | M      | 0     | 0.222     | 0.077    | 0.226   | 0.077  | 0.167    | 0        | 0.125 |
| Other     | F      | 0.059 | 0.088     | 0.071    | 0.143   | 0.122  | 0.080    | 0.045    | 0.121 |
|           | M      | 0.105 | 0.068     | 0.087    | 0.102   | 0.160  | 0.115    | 0.174    | 0.220 |
| Unknown   | F      | 0     | 0.250     | 0.071    | 0.071   | 0.071  | 0.071    | 0.071    | 0.071 |
|           | M      | 0.400 | 0.444     | 0.069    | 0.069   | 0.069  | 0.069    | 0.069    | 0.069 |
| Afro      |        | 0.079 | 0.129     | 0.068    | 0.104   | 0.163  | 0.168    | 0.093    | 0.157 |
| Asian     |        | 0     | 0.190     | 0.049    | 0.158   | 0.129  | 0.111    | 0.222    | 0.143 |
| Cauc      |        | 0.122 | 0.156     | 0.081    | 0.158   | 0.152  | 0.145    | 0.137    | 0.167 |
| Hispan    |        | 0.243 | 0.169     | 0.086    | 0.500   | 0.113  | 0.258    | 0.244    | 0.173 |
| Native    |        | 0     | 0.271     | 0.107    | 0.246   | 0.038  | 0.083    | 0        | 0.133 |
| Other     |        | 0.083 | 0.077     | 0.08     | 0.071   | 0.145  | 0.098    | 0.111    | 0.181 |
| Unknown   |        | 0.286 | 0.385     | 0.070    | 0.069   | 0.300  | 0.182    | 0.222    | 0.240 |
| F         |        | 0.103 | 0.155     | 0.081    | 0.071   | 0.144  | 0.149    | 0.123    | 0.159 |
| M         |        | 0.134 | 0.156     | 0.078    | 0.069   | 0.158  | 0.151    | 0.147    | 0.173 |
| Total     |        | 0.120 | 0.155     | 0.079    | 0.070   | 0.151  | 0.150    | 0.136    | 0.167 |

**Table 12.5** Watson healthcare turnover data

|            |       |      |       |       |            |       |      |       |       |
|------------|-------|------|-------|-------|------------|-------|------|-------|-------|
| Nurses     | Leave | Stay | Total | Left  | Others     | Leave | Stay | Total | Left  |
|            | 107   | 715  | 822   | 0.130 |            | 92    | 762  | 854   | 0.108 |
| Travel     | Leave | Stay | Total | Left  | Travel     | Leave | Stay | Total | Left  |
| Non        | 8     | 85   | 93    | 0.086 | Non        | 8     | 71   | 79    | 0.101 |
| Rare       | 74    | 493  | 567   | 0.131 | Rare       | 52    | 565  | 617   | 0.084 |
| Frequent   | 25    | 137  | 162   | 0.154 | Frequent   | 32    | 126  | 158   | 0.203 |
| Department | Leave | Stay | Total | Left  | Department | Leave | Stay | Total | Left  |
| Cardiology | 42    | 315  | 357   | 0.118 | Cardiology | 32    | 142  | 174   | 0.184 |
| Maternity  | 56    | 329  | 385   | 0.145 | Maternity  | 42    | 369  | 411   | 0.102 |
| Neurology  | 9     | 71   | 80    | 0.113 | Neurology  | 18    | 251  | 269   | 0.067 |
| Ed Level   | Leave | Stay | Total | Left  | Ed Level   | Leave | Stay | Total | Left  |
| 1          | 10    | 78   | 88    | 0.114 | 1          | 17    | 91   | 108   | 0.157 |
| 2          | 21    | 142  | 163   | 0.129 | 2          | 14    | 145  | 159   | 0.088 |
| 3          | 50    | 270  | 320   | 0.156 | 3          | 39    | 296  | 335   | 0.116 |
| 4          | 26    | 200  | 226   | 0.115 | 4          | 21    | 200  | 221   | 0.095 |
| 5          | 0     | 25   | 25    | 0     | 5          | 1     | 30   | 31    | 0.032 |
| Gender     | Leave | Stay | Total | Left  | Gender     | Leave | Stay | Total | Left  |
| Female     | 42    | 270  | 312   | 0.135 | Female     | 44    | 322  | 366   | 0.120 |
| Male       | 65    | 445  | 510   | 0.127 | Male       | 48    | 440  | 488   | 0.098 |
| Marital    | Leave | Stay | Total | Left  | Marital    | Leave | Stay | Total | Left  |
| Single     | 61    | 202  | 263   | 0.093 | Single     | 53    | 206  | 259   | 0.205 |
| Married    | 34    | 344  | 378   | 0.106 | Married    | 27    | 372  | 399   | 0.068 |
| Divorced   | 12    | 169  | 181   | 0.178 | Divorced   | 12    | 184  | 196   | 0.061 |

Nurse turnover was greater than for other employees, with 13% of nurses leaving as opposed to just less than 11% for other employees. Travel seemed to affect all employees, other employees even more than nurses. Nurse turnover was greatest in the Maternity Department, while other employees had more turnover in the Cardiology Department. Nurses left at a higher rate than did other employees in the Neurology Department. Education level seemed to have the greatest turnover at the intermediate level, for both nurses and other employees. Gender did not seem to play a different role across nurses and other employees. Female employees left at a slightly higher rate than did male employees. Divorced nurses had high turnover, while other employees had highest turnover with single employees.

We can demonstrate the Bayes rule with a case of a married female nurse in the cardiology department who travels frequently.

The Bayesian formulas would involve:

|                                                                         |           |
|-------------------------------------------------------------------------|-----------|
| Prob{Married Female Cardiology Nurse who travels frequently given Left} | = 2/199   |
| Prob{Left}                                                              | =199/1676 |
| Prob{Married Female Cardiology Nurse who travels frequently}            | = 15/1676 |

$$\text{Target probability} = \frac{\frac{2}{199} \times \frac{199}{1676}}{\frac{15}{1676}} = 0.133$$

The easy way is to go to Excel and use the COUNTIFS function in the WatsonHealthcareoriginal.csv file:

$$\begin{aligned} &= \text{COUNTIFS}\left(\text{R2 : R1677, "Married", L2 : L1677, "Female",}\right. \\ &\quad \text{F2 : F1677, "Cardiology", P2 : P1677, "Nurse",} \\ &\quad \left. \text{D2 : D1677, "Travel_Frequently", C2 : C1677, "No"}\right) \end{aligned}$$

This yields a count of 15.

$$\begin{aligned} &= \text{COUNTIFS}\left(\text{R2 : R1677, "Married", L2 : L1677, "Female",}\right. \\ &\quad \text{F2 : F1677, "Cardiology", P2 : P1677, "Nurse",} \\ &\quad \left. \text{D2 : D1677, "Travel_Frequently", C2 : C1677, "Yes"}\right) \end{aligned}$$

This yields a count of 2. Thus, the probability of leaving is  $2/15 = 0.133$ .

## 12.4 Association Rules and Bayesian Models

Association rules will yield the same results as naïve Bayesian models. For the GOSSIS data (Raffa et al. 2019) we looked at earlier, we ran association rules on Rattle with minimum support set at a very low 0.001, and confidence at 0.10. This yielded the association rules shown in Table 12.6. Note that the confidence column provides the same results as Table 12.1, although there were 138 rules to wade through. Table 12.6 was sorted to focus on Caucasian males with AIDS.

The key row to answer the question of how many Caucasian males with AIDS died in this clinic is the sixth row, where confidence is 0.915 for NOT dying. This matches Table 12.1's probability of dying for Caucasian males with AIDS of 0.085.

## 12.5 Example Application

Topuz et al. (2018a) presented a model to predict no-shows in a pediatric clinic. This prediction model enabled the clinic to apply an overbooking policy to more efficiently schedule appointments. Missed appointments waste medical resources and

also may have negative impact on patient health. No-shows mitigation policies include penalties, although this is especially hard on low income patients. Reminders and education have limited impact. Thus overbooking and short lead time scheduling have been applied, but these approaches require accurate information on no-show probabilities.

Traditional studies apply logistic regression models. However, logistic regression models usually do not consider predictor variable interactions. Bayesian belief networks (BBN) apply machine learning to apply naïve Bayes networks. Such networks are limited by datasets with irrelevant and noisy variables that increase network complexity. Therefore, a heuristic feature selection algorithm (elastic net variable selection) was applied to eliminate redundant variables using correlations among predictor variables.

The data for a pediatric primary care clinic with about 20 pediatric and internal medicine residents and ten general pediatricians were obtained for the period 1 June 2004 and 31 December 2010. The clinic's scheduling system stored patient demographic information as well as date and time of appointments, and status of those appointments. There were 105,343 appointments involving 16,345 patients. In addition to the binary outcome (show or no-show), there were 18 prediction variables. The R package "glmnet" was used to apply elastic net variable selection.

Bayesian belief networks (BBN) are directed acyclic graphs representing the probabilistic relationship between variables with nodes corresponding to variables and arcs representing conditional dependencies between variables. They have relative advantages when uncertainty and/or complex nonlinear interactions are present. Construction of BBNs is much harder with larger network size. Naïve Bayes classifiers are simple models assuming conditional independence between predictor variables. BBNs provide more accurate models at the expense of model complexity.

Variable selection by elastic net (EN) variable reduction using different parameters was compared with results from stepwise regression. EN selected 11 variables, as opposed to 14 obtained from stepwise regression. Table 12.7 shows variables selected by method:

The error metrics reported in Table 12.8 are the traditional area under the curve (AUC) and accuracy (correct classifications over total classifications). The G-mean takes the geometric mean of sensitivity and specificity, thus considering the balance of the type of error involved. Here both models were reasonably close. Models were also compared to applying BBN without variable selection, as well as decision tree, logistic regression, and neural network models. Table 12.8 compares results:

The proposed BBN with EN yielded the best measures for AUC and Accuracy, although all models were within reason. Typically, these models work well with some data and slightly less well with others. Ensemble methods tend to do better yet, although at the cost of more complexity. BBN models seem to have been imbalanced, and the three traditional models performed better by G-mean measure. Using EN for variable selection provided a slight improvement for the BBN model.

Interactions were indicated by the tree model. The BBN tree showed that Appointment Time impacted Appointment Lead Time, which in turn impacted Appt After Holiday and ApptReason. ApptReason had an impact on ApptMonth,

**Table 12.6** Association rules obtained

| lhs                                                 | Rhs                        | Support      | Confidence   | Coverage     | Lift         | Count         |
|-----------------------------------------------------|----------------------------|--------------|--------------|--------------|--------------|---------------|
| {ethnicity = Caucasian}                             | => {gender = M}            | 0.417        | 0.542        | 0.771        | 1.004        | 38,283        |
| {ethnicity = Caucasian}                             | => {aids = no}             | 0.764        | 0.992        | 0.771        | 1.000        | 70,095        |
| {ethnicity = Caucasian}                             | => {Died = no}             | 0.703        | 0.913        | 0.771        | 0.999        | 64,516        |
| {ethnicity = Caucasian,gender = M}                  | => {aids = no}             | 0.414        | 0.991        | 0.417        | 1.000        | 37,954        |
| {ethnicity = Caucasian,gender = M}                  | => {Died = no}             | 0.382        | 0.915        | 0.417        | 1.001        | 35,026        |
| <b>{ethnicity = Caucasian,gender = M,aids = no}</b> | => <b>{Died = no}</b>      | <b>0.379</b> | <b>0.915</b> | <b>0.414</b> | <b>1.002</b> | <b>34,742</b> |
| {ethnicity = Caucasian,aids = no}                   | => {Died = no}             | 0.698        | 0.913        | 0.764        | 0.999        | 63,999        |
| {gender = M}                                        | => {aids = no}             | 0.535        | 0.991        | 0.539        | 1.000        | 49,035        |
| {gender = M}                                        | => {Died = no}             | 0.494        | 0.916        | 0.539        | 1.002        | 45,293        |
| {gender = M,aids = no}                              | => {Died = no}             | 0.490        | 0.916        | 0.535        | 1.003        | 44,917        |
| {Died = yes}                                        | => {aids = no}             | 0.085        | 0.988        | 0.086        | 0.997        | 7820          |
| {Died = yes}                                        | => {ethnicity = Caucasian} | 0.067        | 0.779        | 0.086        | 1.011        | 6168          |
| {Died = yes,ethnicity = Caucasian}                  | => {aids = no}             | 0.066        | 0.988        | 0.067        | 0.997        | 6096          |
| {Died = yes}                                        | => {gender = M}            | 0.046        | 0.528        | 0.086        | 0.978        | 4176          |
| {Died = yes,gender = M}                             | => {aids = no}             | 0.045        | 0.986        | 0.046        | 0.995        | 4118          |
| {Died = no,ethnicity = Caucasian}                   | => {gender = M}            | 0.382        | 0.543        | 0.703        | 1.007        | 35,026        |
| {Died = yes,ethnicity = Caucasian}                  | => {gender = M}            | 0.036        | 0.528        | 0.067        | 0.979        | 3257          |
| {Died = yes,ethnicity = Caucasian,gender = M}       | => {aids = no}             | 0.035        | 0.986        | 0.036        | 0.995        | 3212          |
| {aids = no}                                         | => {Died = no}             | 0.906        | 0.914        | 0.991        | 1.000        | 83,100        |

**Table 12.7** Variables selected

| Variable                       | Elastic net | Stepwise regression |
|--------------------------------|-------------|---------------------|
| Appointment start time         | Selected    | Selected            |
| Patient age                    | Selected    | Selected            |
| Appointment lead time          | Selected    | Selected            |
| Appointment duration           | Selected    | Selected            |
| Appointment reason             | Selected    | Selected            |
| Family no-show history         | Selected    | Selected            |
| Number of patient visits       | Selected    | Selected            |
| Number of times with provider  | Selected    | Selected            |
| Appointment after holiday      | Selected    | Selected            |
| Appointment month              | Selected    | Selected            |
| Appointment day                | No          | No                  |
| Geographical poverty level     | No          | No                  |
| Appointment with sibling       | No          | Selected            |
| Appointment before holiday     | No          | No                  |
| Number of patient appointments | No          | Selected            |
| Number of providers served     | No          | Selected            |
| Distance to clinic             | Selected    | No                  |
| Gender                         | No          | Yes                 |
| Area under curve               | 0.691       | 0.687               |
| Accuracy                       | 0.739       | 0.731               |
| G-mean                         | 0.464       | 0.491               |

**Table 12.8** Model performance

| Measure  | BBN with EN | BBN without selection | Decision tree | Logistic regression | Neural network |
|----------|-------------|-----------------------|---------------|---------------------|----------------|
| AUC      | 0.691       | 0.686                 | 0.560         | 0.686               | 0.684          |
| Accuracy | 0.743       | 0.733                 | 0.741         | 0.720               | 0.721          |
| G-mean   | 0.648       | 0.629                 | 0.690         | 0.667               | 0.670          |

Appt Duration, and AgeatAppt. AgeatAppt had an impact on NumberOfVisits, which impacted NumberofTimesSeenbyThatProv and FamilyHistoryofNoShow. FamilyHistoryofNoShow in turn impacted EDistancetoClinic.

Sensitivity of the BBN model was measured by entropy. This enabled identification of those variables providing the best information about no-shows. The list by variable with entropy measure of mutual information was:

|                               |       |
|-------------------------------|-------|
| 1. Appointment lead time      | 0.033 |
| 2. Family history of no-show  | 0.029 |
| 3. Appointment reason         | 0.022 |
| 4. Patient age at appointment | 0.009 |
| 5. Appointment duration       | 0.004 |
| 6. Distance to clinic         | 0.003 |

(continued)

|                                          |       |
|------------------------------------------|-------|
| 7. Number of visits                      | 0.002 |
| 8. Appointment start time                | 0.002 |
| 9. Number of times seen by that provider | 0.001 |
| 10. Appointment month                    | 0.001 |
| 11. Appointment after holiday            | 0.000 |

The study revealed that acute patients tend to no-show if the lead time was more than 2 days. Appointment month and appointment after a holiday had the least explanatory power.

## 12.6 Conclusions

Naïve Bayes models are straightforward, providing a clear descriptive analytic tool. We have demonstrated using a publicly available dataset from [Kaggle.com](https://www.kaggle.com). The results obtained from naïve Bayes modeling matches that obtained from association rules. But the naïve Bayes model is much more straightforward in identifying results, as there are many association rules generated, most spurious.

The chapter began with a review of published applications of naïve Bayes modeling in healthcare management. The journal articles reviewed delve into some advanced topics of interest, especially the idea of variable selection.

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# Chapter 13

## Summation



**Keywords** Healthcare management decisions · CRISP

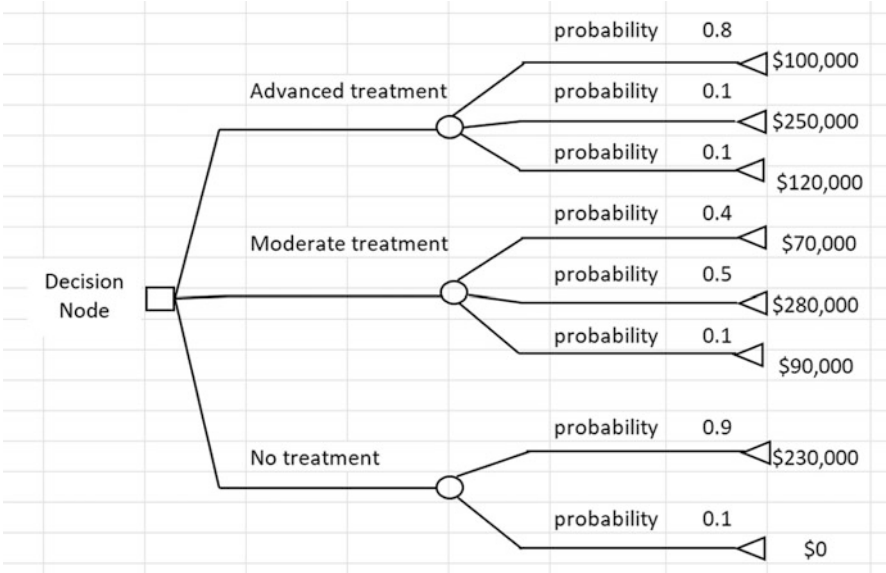
Healthcare management involves many decisions. Physicians typically have a variety of potential treatments to select from. There also are important decisions involved in resource planning, such as identifying hospital capacity and hiring requirements. Healthcare decisions discussed in the book include:

- Selection of patient treatment
- Design of facility capacity
- Utilization of resources
- Personnel management
- Analysis of hospital stay
- Annual disease progression
- Comorbidity patterns
- Patient survival rates
- Disease-specific analysis
- Overbooking analysis

These decisions arose in covering analytic tools. The book began with a discussion of healthcare issues, discussed knowledge management, and then focused on analytic tools useful in healthcare management.

### 13.1 Treatment Selection

We demonstrate a common healthcare decision-making in this section. Decision tree models often involve Bayesian estimation, covered in Chap. 12. Selection of treatment options can be supported with a number of models. Decision trees discussed in Chap. 9 help frame such decisions. To provide an example, assume a treatment decision involving two options, in addition to doing nothing. The first option is more expensive with a higher probability of providing a cure. The second option is less



**Fig. 13.1** Treatment selection decision tree

expensive but involves less probability of curing severe conditions. Of course, the need for treatment is probabilistic—here we will assume that tests indicate a 40% probability of a severe case, 50% probability of a mild case, and 10% probability that the patient will not actually need treatment for this condition. The decision has three options:

1. Advanced treatment, 80% probability of successful treatment with expected cost of \$100,000, 10% probability of complications with expected cost of \$250,000, and 10% probability of failure with expected cost of \$120,000.
2. Scaled back treatment with 40% probability of successful treatment with expected cost of \$70,000, 50% probability of complications with expected cost of \$280,000, and 10% probability of failure with expected cost of \$90,000.
3. No treatment, with 90% probability of complications with expected cost of \$230,000, and 10% probability of no problems with zero cost.

The decision tree displayed in Fig. 13.1 provides a visual display of this decision: At every circle node, expected values can be calculated as the sum product of probability times cost. These come out:

- Advanced treatment  $0.8 \times \$100,000 + 0.1 \times \$250,000 + 0.1 \times \$120,000 = \$117,000$
- Moderate treatment  $0.4 \times \$70,000 + 0.5 \times \$280,000 + 0.1 \times \$90,000 = \$177,000$
- No treatment  $0.9 \times \$230,000 + 0.1 \times 0 = \$207,000$

Based on monetary value (in dollars) alone, it appears advanced treatment would be the most cost-effective.

There are of course quality-of-life (QoL) issues to be considered as well. You could use some patient utility function combining cost with expected quality-of-life (which could also include cost impact to the patient) to replace the dollar estimates used above:

- Successful advanced treatment quality-of-life—0.5
- Complications from advanced treatment quality-of-life—0.1
- Failure of advanced treatment quality-of-life—0
- Successful scaled back treatment quality-of-life—0.8
- Complications from advanced treatment quality-of-life—0.4
- Failure of advanced treatment quality-of-life—0
- Complications from no treatment quality-of-life—0
- No problems with no treatment quality-of-life—1.0

Replacing dollars with quality-of-life yields:

- Advanced treatment  $0.8 \times 0.5 + 0.1 \times 0.1 + 0.1 \times 0 = 0.41$
- Moderate treatment  $0.4 \times 0.8 + 0.5 \times 0.4 + 0.1 \times 0 = 0.52$
- No treatment  $0.9 \times 0 + 0.1 \times 1.0 = 0.10$

These decisions also involve tradeoffs, especially those involving probability of treatment success versus quality-of-life impact. Multiple criteria analysis in healthcare was discussed in Chap. 11.

Analysis needed to identify probabilities needed by these models can be estimated through regression analysis (covered in Chap. 5) or classification models (Chap. 7).

Data mining is exploratory data analysis. Masses of data are generated from current medical computer systems, in a variety of forms to include accounting data, many text sources, and analog data from many forms of radiology and other physical measures. Data mining tools need to be versatile, scalable, capable of accurately predicting responses between actions and results, and capable of automatic implementation. Versatile refers to the ability to apply to a wide variety of problem models. Scalable implies that tools working on small datasets also are capable of analyzing large datasets. Automation is useful, but its application is relative. Some analytic functions are often automated, but a great deal of human setup prior to implementation is required. In fact, analyst judgment is critical to successful implementation. Proper selection of data is critical. Data transformation is often required. Too many variables produce too much output and confuse matters, while too few can overlook key relationships.

## 13.2 Data Mining Process

The process of data mining to support healthcare can be applied following a number of systems, one of which is the Cross-Industry Standard Process for Data Mining (CRISP-DM). This system involves six phases intended as a cyclical process:

**Business understanding:** This includes determining objectives, assessing the current situation, establishing data mining goals, and developing a project plan. In the context of healthcare, this may be a doctor's selection of treatment option, an administrator's decision of a clinic's capacity.

**Data understanding:** Since data mining is task-oriented, different tasks require different sets of data. The first stage of a data mining analysis is to obtain data in as accurate and appropriate a form as possible.

**Data preparation:** The purpose of preprocessing is to clean selected data for better data quality. Since selected data may have different formats, either the data needs to be transformed to a common type (continuous numbers for clustering; binary outcomes for classification; categorical data for Bayesian analysis) or different analytics techniques need to be applied.

**Modeling:** Software tools such as visualization (plotting data and establishing relationships) and cluster analysis (to identify observation groupings) are useful for initial analysis. Association rules may be applicable to identify pairs of circumstances that coexist. Once greater data understanding is obtained, more detailed forecasting or classification models can be applied.

**Evaluation:** Model results should be evaluated in the context of healthcare objectives established in the business understanding phase. This will lead to identification of other needs, frequently reverting to prior CRISP-DM phases.

**Deployment:** Data mining can be used to verify previously held hypotheses, or for knowledge discovery (identification of unexpected and useful relationships). This can lead to sound models that can be applied to healthcare decisions of many kinds, to include prediction or identification of key situations. Models need to be monitored for changes in conditions, because relationships may change over time.

### 13.3 Topics Covered

This book has discussed analytic tools useful for some of these decision types in healthcare.

Chapter 1 introduced the issue of healthcare urgency. The presence of big data is caused by electronic systems applied by governments, hospitals, clinics, and healthcare professionals. This situation calls for analytics techniques to deal with many healthcare management decisions related to patient treatment, facility utilization, personnel utilization, and other critical issues. Healthcare analytics provides cost reduction, more effective treatment, and faster delivery of healthcare.

Chapter 2 considered knowledge management relative to healthcare. Information technology has been widely applied, in the form of electronic healthcare data systems and other forms of business intelligence. Data is not only available from these electronic data records, but also from social media platforms and machine-to-machine sources. Medical data comes in many forms to include biometric and text data. Automated knowledge management systems can sometimes provide real-time response to decision-making in fast-moving crises.

The book then moved into analytic tools, beginning with visualization in Chap. 3. Excel visualization tools were demonstrated on hospital stay data, while time series visualization was demonstrated with a Hungarian chickenpox dataset.

Chapter 4 presented another descriptive analytics tool in the form of association rules. Healthcare decisions supportable with this model include identification of comorbidity patterns in patients, monitoring diseases by geographical area and age group to assess doctor or nurse shortages, and to identify abnormalities detected by real-time sensor data. The algorithm was demonstrated using patient survival data. Two published reports of usage were reviewed.

Chapter 5 described cluster analysis, another descriptive analytics tool. The hospital stay data was used again to demonstrate model use. Two published reports of clustering usage were reviewed involving barriers to mental health and nursing home capacity analysis.

Chapter 6 moved on to time series forecasting models. The Hungarian chickenpox dataset was again used for demonstration, this time using SAS software.

Classification modeling was covered in Chap. 7. Data from IBM's application of Watson artificial intelligence to healthcare was used for demonstration. Chapter 8 presented reviews of published applications of classification models in general hospital system management, disease-specific applications, genome research, and connectivity in the Internet-of-Things era.

Chapter 9 presented decision tree modeling in healthcare analytics.

Chapter 10 demonstrated predictive modeling on four medical datasets, focusing on methods of variable reduction.

Chapter 11 covered applications focusing on multiple criteria analysis. Six applications covering decisions such as breast cancer diagnosis, colorectal screening, senior center selection, diabetic/heart treatment, disease management in Bolivia, and breast cancer screening campaign analysis.

Chapter 12 reviewed application of naïve Bayes models, applicable to categorical data. The global open-source severity of illness score dataset was used to demonstrate. Bayesian models were compared with association rule modeling. A published report of application to a pediatric clinic's problem of overbooking was reviewed.

## 13.4 Summary

Healthcare is flooded with data of many types. These types include numbers, text, and more complex data forms such as videos, MRI output, CAT scans, etc. Some of this data is generated internally through clinical information systems or similar software. Data mining provides tools to utilize this data, which can be applied using the CRISP-DM data mining process described in Sect. 13.3.

# Name Index

## A

Abdulkareem, K.H., 109, 113  
Abdul Wahab, A., 64, 65  
Abramoff, M.D., 114  
Abrams, K.R., 161  
Achariya, D.P., 110  
Agrawal, R., 36  
Aguinis, H., 36  
Ahmad, A.M., 111, 113  
Ahmad Bahuri, H., 64, 65  
Ahmed, P.K., 110  
Ahsen, M.E., 151, 152, 159  
Aidelsburger, P., 123  
Ajlou, S., 161  
Akbari, V., 9  
Aksakalli, V., 106  
Alagoz, O., 151, 152, 159  
Alas, V., 162  
Alexander, C.A., 3, 4, 10  
Almehmi, A., 161, 170  
Almerares, A.A., 3  
Al Turjman, F., 109, 113  
Amjad, T., 7  
Ammi, M., 110, 113  
Anand Muthu, B., 50  
Anastasia, E.A., 60, 66  
Arab, M., 112  
Arunselvan, K.S., 108, 113  
Ayvaci, M.U.S., 151, 152, 159

## B

Bachar, R., 161  
Badawi, O., 24, 163, 170

Bahra, G., 109, 110, 113  
Balamurali, S., 50  
Bauder, R.A., 112, 113  
Becker, D., 123  
Benkatesulu, M., 50  
Berhanu, P., 162  
Bhattacharjee, D., 111, 113  
Bhuyan, M.S.I., 111, 113  
Blower, S., 123  
Boakye-Dankwa, E., 62  
Bouchard, J., 162  
Brandeau, M.L., 123  
Brebner, R., 123  
Brennan, A., 123  
Bridges, A.J., 60, 66  
Brown, E.C., 117  
Brown, R.A., 161  
Buluir, S., 161  
Buneman, P., 36  
Burger, E.A., 123  
Burnside, E.S., 151, 152, 159  
Byford, S., 123

## C

Cantor, S.B., 123  
Carrasco-Labra, A., 3  
Cavedon, L., 106  
Cavell, T.A., 60, 66  
Celi, L.A., 24, 163, 170  
Chae, B., 130  
Chai, Y., 106  
Chan, A.B., 127  
Chan, H.K., 7

Chan, J., 106  
 Chang, S.C., 155, 159  
 Char, D.S., 114  
 Chen, H., 161  
 Choi, T.-M., 7  
 Chung, K., 13, 45, 48, 49  
 Cidoncha, A., 157, 159  
 Cooper, N.J., 161  
 Cooper, W.W., 158  
 Cruz-Aponte, M., 123  
 Csiszár, O., 156, 159  
 Cui, H., 127

## D

Dag, A., 161  
 De Bois, M., 110, 113  
 De Moor, B., 49, 50  
 Diaz Ochoa, J.G., 156, 159  
 Dickson, W.C., 128  
 Dodd, P.J., 123  
 Dolan, J.G., 153, 154, 159  
 Dudley, J.T., 161  
 Dueweke, A.R., 60, 66

## E

Eckert, G., 3  
 Elder, K., 162  
 El Yacoubi, M.A., 110, 113  
 Esnaola, N.F., 123  
 Evans, D.B., 123  
 Everhart, J.E., 128

## F

Fang, E.F., 105  
 Faria, B.M., 13  
 Farooq Ahmad, H., 7  
 Ferenci, T., 31, 82  
 Feudtner, C., 114  
 Fontana, M., 3  
 Forcum, L.E., 36  
 Foster, D.P., 128  
 Fowler, J.W., 11, 117  
 Frank, E., 67

## G

Galets, P., 1  
 Garcia, M.J., 3  
 Gascon, J., 157, 159  
 Geraili, B., 112  
 Ghafour, M.Y., 111, 113  
 Ghosh, S., 162

Glicksberg, B.S., 161  
 Gonçalves, J., 13  
 Goodman, W., 161  
 Gopal, G., 157, 159  
 Gore, R., 62, 67  
 Gregus, S.J., 60, 66  
 Groves, P., 2, 3  
 Gupta, S., 1

## H

Haase, K., 158, 159  
 Hadzi, T., 162  
 Han, S., 109  
 Hanisch, B.W., 123  
 Hassan, A., 161  
 He, Y., 10  
 Hellier, J., 123  
 Herland, M., 112, 113  
 Hernandez, I., 4  
 Hitchcock, C., 162  
 Holtkamp, M.D., 97, 104  
 Hope, V.D., 123  
 Hoseyninezhad, M., 111, 113  
 Househ, M., 3  
 Hsu, C.-H., 50  
 Hu, J., 162  
 Huang, S., 11  
 Hudson, I., 106  
 Huttenen, H.J., 69

## I

Imieliński, T., 36  
 Imran, M., 1, 5

## J

Jajodia, S., 36  
 Jan, S.R.U., 106  
 Jebakaran, J., 161  
 Jehn, M., 11, 117  
 Johannes, R.S., 128  
 Johnson, A., 24, 163, 170  
 Johnson, K.W., 161  
 Joo, H., 36  
 Joudaki, H., 112  
 Jovanović Milenković, M., 3, 5  
 Jung, H., 13

## K

Kadry, S., 50  
 Kasarskis, A., 161  
 Katsaliaki, K., 1

Katz, B., 3  
 Kauffman, R.J., 2  
 Kavita, V.G., 106  
 Kayyali, B., 2, 3  
 Kemani, F., 107, 113  
 Khalil, I., 162  
 Khan, F., 106  
 Khoshgoftaar, T.M., 112  
 Kim, J.-C., 44, 48, 49  
 Kim, J.J., 123  
 Kishwer, S., 7  
 Kiss, O., 31, 82  
 Knott, E.D., 2, 3  
 Knowler, W.C., 128  
 Kohli, M., 123  
 Kovatch, P., 161  
 Krohn, R., 158, 159  
 Kros, J.F., 117

## L

Lakhan, A., 109, 113  
 Lambert, J.H., 7  
 Lanckriet, G.R., 127  
 Lang, K., 123  
 Lant, T., 11, 117  
 Lee, D.D., 38  
 Lee, J.E., 123  
 Li, H., 11  
 Li, L., 161  
 Li, S.-H., 161  
 Li, W., 106  
 Li, X., 106  
 Li, Y., 106  
 Lim, S., 161  
 Lin, Y.-K., 161  
 Liu, B., 162  
 Liu, L., 11  
 Liu, N., 2, 10  
 Lomio, F., 69  
 Long, E.F., 123  
 Luna, D., 3

## M

Mahmoodi, M., 112  
 Manogaran, G., 50  
 Margalit, R.S., 123  
 Maschio, M., 123  
 Mastoi, Q., 109, 113  
 May, J.H., 153, 154, 159  
 Mayan, J.C., 3  
 McNeil, B.J., 121, 123, 152  
 Megahed, F.M., 161  
 Meier, P.S., 123

Memon, S., 109  
 Menon, V.G., 106  
 Metrebian, N., 123  
 Milenković, D., 3, 5  
 Miller-Wilson, L.-A., 162  
 Minaei-Bidgoli, B., 112  
 Mohammed, M.A., 109, 113  
 Mondal, R., 111, 113  
 Moore, M., 123  
 Moradi, R., 107, 113  
 Moriana, R.S., 157, 159  
 Mount, J., 54, 67  
 Mueller, L., 162  
 Mühlberger, N., 123  
 Mukherjee, D., 111, 113  
 Muller, S., 158, 159

## N

Nasiri, M., 112  
 Navaneetha Krishnan, M., 50  
 Nazirun, N.N., 64, 65  
 Ncube, F., 123  
 Ng, K., 162  
 Nikoliv, G., 49, 50  
 Niu, Z., 105

## O

Oksala, N., 69  
 Olson, D.L., 8, 12, 16, 130  
 Omar, N., 64, 65  
 Ozcan, Y.A., 117, 119, 124  
 Oztekin, A., 161, 170

## P

Palomäki, A., 69  
 Pannett, L., 62, 67  
 Pauker, S.G., 121, 123  
 Paulo Reis, L., 13  
 Pe'er, I., 111, 113  
 Peltonen, J., 69  
 Perez-Rodriguez, M.M., 161  
 Petrovic, S., 9  
 Pilcher, D., 24, 163, 170  
 Pinazo, M.-J., 157, 159  
 Plevritis, S.K., 124  
 Pollard, T., 24, 163, 170

## Q

Qabulio, M., 109, 113  
 Qadir, J., 7

**R**

Raffa, J., 163, 170  
 Rafia, R., 123  
 Rahman, M.S., 111, 113  
 Rajagopalan, S., 127  
 Ramasamy, L., 50  
 Rao, J., 105  
 Rashidian, A., 112  
 Razzak, M.I., 1, 5  
 Readhead, B., 161  
 Regan, C., 123  
 Reich, D.E.L., 161  
 Ritchie, H., 7  
 Rocha, Á., 13  
 Roine, A., 69  
 Roser, M., 7  
 Rozemberczki, B., 31, 82

**S**

Safara, F., 111, 113  
 Sakthivel, P., 108, 113  
 Salman, F.S., 8, 16  
 Sarkar, R., 31, 82  
 Sarwar, T., 106  
 Sava, M.G., 153, 154, 159  
 Scarpa, J., 161  
 Scharff, R.L., 10  
 Scherer, P., 31, 82  
 Schimper, T., 156, 159  
 Schnell-Inderst, P., 123  
 Seifollahi, S., 106  
 Seiford, K.T., 158  
 Seung, H.S., 38  
 Shachtman, R.H., 123  
 Shafqat, S., 7  
 Shameer, K., 161  
 Shams, I., 161  
 Sheikhtaheri, A., 107, 113  
 Sherman, S.I., 123  
 Shi, X., 49, 50  
 Shih, H.S., 156  
 Singh, P.K., 111, 113  
 Skylakha, S., 108, 113  
 Smith, J.W., 128  
 Smith, M.R., 161  
 Song, Y., 105  
 Sornalakshmi, M., 50  
 Sour, A., 111, 113  
 Spallek, H., 3  
 Sroczynski, G., 123  
 Strang, J., 123  
 Su, C.R., 155, 159  
 Sun, Z., 162

Sutton, A.J., 161  
 Swami, A., 36  
 Sy, S., 123

**T**

Tatonetti, N.P., 161  
 Teeple, E., 62  
 Thai, N., 162  
 Tie, H., 123  
 Tippong, D., 9  
 Topuz, K., 161, 170  
 Torrico, F., 157, 159  
 Tuominen, J., 69

**U**

Under, H., 161, 170  
 Ur Rasool, R., 7

**V**

Van der Akker, M., 49  
 Van Kuiken, S., 2, 3  
 Van Pottelbergh, G., 49, 50  
 Vardavas, R., 123  
 Varg, M.G., 153, 154, 159  
 Vasconcelos, N., 127  
 Verma, S., 106  
 Verspoor, K., 106  
 Vickers, A.J., 123  
 Vijayam, B., 64, 65  
 Villalobos, B.T., 60, 66  
 Vos, R., 49, 50  
 Vukmirović, A., 3, 5

**W**

Wang, L., 3, 4, 10  
 Ward, A.R., 127  
 Wasem, J., 123  
 Weaver, T., 123  
 Weichselbaum, R., 152  
 Weinstein, M.C., 123  
 Willis, H.H., 123  
 Wilson, F.A., 123  
 Witten, I.H., 67  
 Wu, J., 11

**X**

Xiao, X., 105  
 Xu, G., 1, 5

**Y**

Yamini, A., 111, 113  
Yang, H-J., 161  
Yang, K., 161  
Yang, Y., 105  
Yildirim, M.B., 161, 170  
Yoo, H., 113  
Yue, X., 7

**Z**

Zalkind, D.L., 123

Zarkesh, M.R., 107, 113  
Zengul, F.D., 161, 170  
Zhang, J., 105  
Zhang, X., 106  
Zhang, Y., 4  
Zheng, H., 11  
Zheng, S., 105  
Zhuang, Z-Y., 155, 159  
Zumel, N., 54, 67  
Zweifel, P., 7

# Subject Index

## A

Accuracy, 14, 15, 35, 73, 75, 76, 92, 95–98, 102, 108, 110–112, 124, 127, 130–132, 137, 139, 141, 142, 144–148, 154, 156, 171, 173  
Adjusted r-squared, 75–76  
Affinity analysis, 35  
Algorithms, 5, 14, 15, 34–38, 51, 54–59, 64, 65, 67, 92, 107, 109–112, 114, 131, 171, 181  
Analytic hierarchy process (AHP), 153–159  
Analytics, ix, 1–4, 7–13, 15–18, 21, 31–33, 108, 117, 123, 124, 153, 154, 161, 162, 177, 179–181  
Analytic strategies, 9, 12–13, 16, 17  
Applications, 1, 4, 8–12, 16–18, 35, 36, 44–51, 60, 64, 72, 85, 87, 88, 90, 104–106, 108, 109, 111–115, 117, 120, 123, 127, 151, 158, 161–162, 170–174, 179, 181  
Applications of predictive modeling, 109, 112  
Apriori algorithm, 35–37, 45, 48, 50, 51  
Area under the curve (AUC), 65, 92, 95, 96, 108, 130–132, 137, 142, 145, 171, 173  
Association rules, 2, 11, 34–40, 42, 49–51, 163, 170, 172, 174, 180, 181  
Autocorrelation, 71, 74, 78–79, 81, 83–85  
Autoregressive integrated moving average (ARIMA), 32, 69–71, 83–85

## B

Bayesian models, 170, 181  
Belgian comorbidity study, 49–51

Big data, 1–3, 5, 7–9, 11–13, 15–17, 36, 45, 51, 105, 106, 115, 151, 159, 180  
Big data analytics, 1–3, 5, 7, 8, 11, 12  
Blockchain, 9, 13, 18  
Bolivian healthcare system, 157–158  
Box-Jenkins analysis, 71, 74, 79, 81, 83  
Box Jenkins models, 71, 74, 79, 81–84  
Breast cancer screening, 158–159, 181

## C

Causal models, 73, 74, 76–77  
Chickenpox data, 30–32, 34, 85, 181  
Classification models, 65, 72, 90, 109, 179–181  
Cluster analysis, 11, 54, 61, 64, 66, 180, 181  
Coincidence charts, 72, 73, 97, 102  
Coincident observations, 72  
Colorectal screening, 153–154, 181  
Comparative algorithm performance, 24, 131  
Complexity, 7, 11, 92, 95, 101, 103, 109, 128, 130, 154, 157, 158, 161, 171  
Confidence, 35–37, 39, 40, 42, 45, 46, 50, 103, 156, 170, 172  
CRISP, 179–181  
Cycles, 31, 70, 71, 74, 78, 81, 83, 85

## D

Data cleaning, 45  
Data mining process, 89, 114, 115, 179–181  
Data sources, 11, 12, 115  
DBSCAN clustering, 67  
Decision analysis, 117, 123

Decision tree models, 88, 92, 95, 96, 108, 110, 121, 123, 130, 134, 177  
 Decision trees, 5, 15, 69, 88, 89, 92, 95, 96, 100, 103, 109–111, 113, 117, 121, 123, 124, 128, 130, 134, 143, 171, 173, 177, 178, 181  
 Descriptive analytics, 11, 12, 174, 181  
 Diabetes and heart problem detection, 156–157  
 Diabetes mellitus cases, 63–65, 162  
 Diagnostic analytics, 11  
 Disaster management, 8, 9, 16–18  
 Disease specific applications, 108–111, 181  
 Distance metrics, 54  
 Durbin Watson statistic, 78

## E

Equal likelihood criterion, 117–119  
 Ethics, 105, 114  
 EWKM clustering, 54, 55, 59, 67  
 Excel graphics, 21, 33  
 Exponential smoothing, 71, 72, 85  
 Extreme boosting, 88, 95, 130, 134, 136, 137, 141, 144–146

## F

Food safety, 8–10, 16, 17  
 Forecasting, 2, 32, 69–75, 81, 82, 85, 180  
 Forecasting models, 69, 71–75, 83, 85, 181  
 Fraud detection, 112, 113  
 Fuzzy clustering, 67

## G

General hospital system management, 181  
 Genome research, 111, 181  
 Global open source severity of illness (GOSSIS) dataset, 163, 170, 181

## H

Healthcare, ix, 1–5, 7–11, 13, 14, 17, 21, 33–36, 45, 55, 60, 90, 97, 100, 105, 106, 108, 111, 112, 114, 115, 117, 123, 127, 142–145, 151–159, 161–174, 177, 179–181  
 Healthcare decisions, ix, 151, 159, 177, 180, 181  
 Healthcare Hospital Stay database, 22, 34, 55  
 Healthcare management decisions, 180  
 Heart UCI dataset, 138, 139  
 Heteroskedasticity, 79–81

Hierarchical clustering, 65, 67  
 Homoskedasticity, 80  
 Hospital stay database, 181  
 Hungarian Chickenpox database, 30, 33, 34, 82, 85, 181  
 Hurwitz criterion, 117

## I

India liver dataset, 137–142  
 Information systems management, 9, 12  
 Internet of Things (IoT), 13, 15, 44, 106, 111–112, 115, 181  
 Invasive breast cancer analysis, 151–153

## K

Kaggle data, 38–42, 163–165  
 K-means clustering, 55, 60, 62, 65, 67  
 Knowledge management, ix, 8, 9, 12–14, 17, 177, 180  
 Korean healthcare study, 44–49

## L

Lags, 73, 78, 84  
 Lift, 35–37, 44, 46, 50, 172  
 Logistic regression (LR), 72, 88, 89, 95, 97, 98, 100, 102, 108–110, 112, 113, 130, 131, 134, 136, 137, 141–145, 147, 161, 171, 173

## M

Machine learning, 4, 9, 35–37, 42, 106, 107, 109, 111, 114, 128, 162, 171  
 Market basket analysis, 36  
 Maximin criterion, 117, 118  
 Mental health care example, 60, 61  
 Minimax criterion, 117–119  
 Moving average, 70–72, 81, 83–85  
 Multicollinearity, 77, 95  
 Multiple criteria analysis, 159, 179, 181

## N

Naïve Bayes, 14, 15, 90, 109, 112, 113, 161–174, 181  
 Neural network models, 89, 90, 95, 103, 108, 110, 111, 130, 132, 136, 141, 142, 156, 171  
 Neural networks (NN), 5, 89, 98, 107, 113, 114, 130, 131, 137, 142, 146–148

Nonlinear data, 73

Non-negative matrix factorization (NMF), 38

Nursing home service quality, 62–63

## O

OLS regressions, 74, 78, 79, 83, 100

Optimization, 8, 9, 11, 16, 17, 50, 51

Ordinary least squares regression (OLS), 70, 72, 74, 75, 84, 85

## P

Patient survival database, 25, 28, 34, 38, 39, 109, 164, 181

Personal healthcare, 13

Pima Indian dataset, 128–134

Predictive analytics, 4, 11, 12, 16, 17, 105, 108, 109, 115

Predictive analytics process, 4, 11, 12, 16, 105, 108, 109, 115

Predictive models, 2, 14, 108, 110, 112, 115, 130, 165

Prescriptive analytics, 11, 12, 16, 17

Probability tree structure, 121

Public health, 1, 4, 8–10, 16–18, 44, 107, 117, 123

## Q

Quality-of-life, 179

## R

Random forest models, 88, 95, 112, 130, 132, 136, 137, 144–146

Rattle software, 37, 39, 42, 44, 54, 55, 58, 103, 104, 163, 164, 170

Regression model assumptions, 78–81

Risk assessment, 8, 9, 17, 18

Risk management, 8–10, 16, 17

Risk mitigation, 8, 9

## S

Senior center site selection, 155–156

Sensitivity, 92, 95–98, 102, 108, 121, 130–132, 136, 137, 141, 142, 144, 147, 153, 154, 171, 173

Simulation, 2, 9, 123, 161

Social welfare, 8, 9, 11, 16–18

SOM clustering, 67

Specificity, 92, 95–98, 102, 107, 108, 130–132, 136, 137, 141, 142, 144, 147, 171

Stepwise regression, 92, 95, 128, 130, 134, 143, 171, 173

Support, 2–4, 7, 10, 12, 13, 17, 35–37, 39, 40, 42, 46, 48, 50, 82, 103, 108, 123, 124, 130, 155, 157, 158, 170, 172, 179

Support vector machines (SVMs), 5, 14, 15, 89, 95, 97, 98, 102, 107, 109, 111–113, 130–132, 137, 141, 142, 144–147

## T

Tests of the regression model, 75

Time series, 30–32, 34, 69–73, 78, 80–82, 85, 88, 110, 181

TOPSIS, 155, 156, 159

## U

Utility, 121, 151–153, 158, 159, 179

## V

Variable selection, 100, 127–128, 145, 148, 161, 162, 171, 174

Visualization, 8–11, 23, 24, 33, 50, 180, 181

Visualization tools, 4, 21, 181

## W

Watson healthcare dataset, 142